

Nano-scale events with macroscopic effects in PP/silica nanocomposites : effect of polymer adsorption on processibility and properties

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Nano-scale events with macroscopic effects in PP/silica nanocomposites

Effect of polymer adsorption on processibility and properties

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prof.dr. P.J. Lemstra

Copromotoren:

dr.ir. J.G.P. Goossens

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To my parents, my brother, my inspiration...

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Summary

The properties of polymeric materials are not solely determined by the chemical structure of the polymer chain molecules but equally well by the processing step, viz. the orientation of the molecules in the final product. The majority of synthetic polymers is processed via the melt, the so-called thermoplastics. Generally, the properties of the product such as toughness and strength increase with increasing molecular weight but the melt-viscosity also strongly increases with increasing molecular weight. The (zero-shear) viscosity η_0 of polymer melts scales with the weight average molecular weight, M_w , to the power 3.4, above a certain threshold value for the molecular weight M_c , which implies that on doubling the molecular weight, the viscosity increases more than a decade. Consequently, processing synthetic polymers into useful products is often a compromise between properties and processibility with an optimum molecular weight.

Next to chemistry and processing, the final product properties can be tuned by additives. In fact, a plastic product is often a polymer plus additive, especially in engineering applications. In the current era of nano-technology we encountered an intriguing phenomenon, the melt-viscosity of polymers could be reduced by more than a decade in the presence of only a minute fraction, less than 1 %, of nano-sized filler particles.

In the present study polypropylene (PP) reactor powders were used as the model substance. The nascent morphology of these PP powders, obtained by the reactor granular technology, is characterized by granular structures with porous channels. Sol-gel reactions of silica precursors were used to incorporate nano-sized silica particles in the PP powder. Two types of nanocomposites were prepared, respectively with and without chemical interaction between the fillers and the PP matrix. For the first type, PP was first grafted with vinyl triethoxy silane (VTES) followed by the preparation of silica nano-particles in the pores of PP via a sol-gel reaction of tetraethoxy orthosilicate (TEOS). For the second type, only the last step was performed. Chemical and structural characterization showed that micro- and meso-porous silica nano-particles were formed in-situ with some hydroxyl group on the surface. In the grafted samples, VTES was found grafted not only on the surface, but also inside PP powder particles with an uniform distribution. The grafted VTES was found to be integrated in the in-situ synthesized silica particles. The dispersion of the silica particles in grafted and non-grafted samples was found to be uniform.

The thus prepared silica nano-particles demonstrated an unique effect, an *enhanced flow* and a *homogeneous distribution of properties*. The rheology of these especially prepared nanocomposites was studied. The micro- and meso-porous nature of in-situ prepared silica and high surface energy results in preferential adsorption of high molecular weight chains. This results in demixing with low molecular weight chains forming matrix and high molecular weight chains forming a shell around silica particles giving core-shell-type morphology. This selective physisorption of long PP chains on silica particles leads to a drop in the viscosity in the lower silica concentration regime, where the inter-particle distance is larger than the radius of gyration of polymer chains.

The mechanical properties of semi-crystalline polymers are governed by the lamellar thickness, crystallinity and entanglement density of the amorphous phase. Therefore, the crystallization process is of major influence. The non-isothermal crystallization kinetics of PP/silica nanocomposites was studied using a combined Avrami-Ozawa approach and showed a two-stage crystallization process: the primary stage characterized by nucleation and spherulitic growth and the secondary stage characterized by crystal perfectioning. The nucleating efficiency based on the scale as proposed by Lotz and coworkers of the in-situ prepared silica particles was found to be 20% in the low concentration range. This is higher than other nano-fillers reported in literature.

Solid-state modification of PP with VTES resulted in a non-random grafting of VTES on PP, while solution grafting resulted in random grafting. This was illustrated by an increase in elasticity due to the formation of clusters of grafted VTES, which act as cross-links. Two crystal populations with different lamellar thicknesses were observed due to the non-random grafting of VTES, which lead to a segregation of PP chains with different chemical compositions, i.e. placement of the grafted VTES along the PP chains. The segregation and clustering of the VTES groups grafted on PP decreases the mobility and hence diffusivity of chains. Especially the secondary crystallization process was influenced, which implies that the amorphous phase is more entangled. Rheological measurements showed that when silica particles are connected to multiple PP chains via grafted VTES network formation occurs.

The flow-induced crystallization (FIC) behavior was studied using in-situ small-angle X-ray scattering (SAXS) and wide-angle X-ray diffraction (WAXD). It is well known that flow-induced crystallization is governed by the high molecular weight part, but the drop in the viscosity indicates a low molecular weight matrix. The absence of high molecular weight chains in the matrix along with lower shear stresses experienced due to the low viscosity results in less orientation under flow. Therefore, these nanocomposites have a more uniform and isotropic structure and morphology in a final product along with the reduction in viscosity. The X-ray study showed that the orientation is minimal at a silica concentration of 0.5 wt%, when the reduction in viscosity is maximal. PP with higher silica concentrations (above 1 wt%) showed a higher flow-induced orientation. However, TEM and FT-IR dichroic measurements on FIC and injection molded bars showed a uniform and isotropic structure and random growth of lamellae. To explain this behavior, a three-stage crystallization process is proposed: 1) formation of row nuclei due to molecular, stress-induced orientation; 2) relaxation of row nuclei or destruction of the oriented structures into a number of primary point nuclei; 3) growth from these point nuclei, disoriented nuclei in random directions.

Finally, the mechanical properties of the PP-silica nanocomposites and the effect of chemical interaction between filler and polymer matrix were explored as a function of the silica concentration. The benefits of easier flowability and a homogeneous (isotropic) structure development during processing on the reinforcing effect of nano-particles were investigated. The improvement in the mechanical properties viz. yield stress, strain hardening modulus, Young's modulus and high speed tensile impact strength were discussed as a function of the silica concentration. PP/silica nanocomposites exhibit three different filler concentration regimes based on the morphology formed due to physisorption of the high molecular weight PP-chains on the silica particles. In the lower concentration regime (below 0.8 wt%), where particles are non-interacting, the yield stress and strain hardening modulus decreased slightly, whereas the increase in

the Young's modulus is marginal. In the intermediate concentration (between 0.8 and 2 wt%), the particles start to interact and the viscosity increases. The yield stress and strain hardening modulus along with the Young's modulus increased. At still higher concentrations, the aggregation of silica particles resulted in a steady increase of yield stress, strain hardening modulus and Young's modulus and the materials exhibited brittle failure. The effect of chemical interaction between filler and matrix proved detrimental for the properties.

The technological consequences of the newly prepared systems could be enormous, but cannot be predicted at this point in time. The decrease in viscosity is especially advantageous for injection-molding operations: lower injection and times are required or, alternatively, the processing temperature can be lowered with higher dimensional stability. It is expected that the developed materials fulfill future demands in precision molding of thin-walled products with high demands on dimensional stability. From a scientific point of view, the effect of lowering the melt viscosity with nano-sized particles is, in our opinion, the first true "nano-effect" observed in the area of synthetic polymers.

Chapter 1

General Introduction

Preamble

The properties of synthetic polymer products are not only determined by their chemical structure but equally well by the processing step, viz. the orientation of the polymer molecules. A striking example in this respect is the most simple synthetic polymer polyethylene (PE) which can be shaped into flexible containers and films but also processed into fibers stronger and stiffer than steel, e.g. the Dyneema fiber produced by DSM.

Next to chemistry and processing, the properties of polymers can be tuned with additives. In fact, a plastic (product) is often a synthetic polymer + additive. There are numerous additives such as anti-oxidants, colorants, processing aids and, notably to enhance the stiffness for engineering applications, mineral additives are used in the form of particulate or fibrous matter. Well known example is the use of glass fibers to reinforce polymers such as polypropylene and polyamides in engineering applications.

In the current era of “Nano-technology”, expectations are running high concerning the use of nano-sized additives in order to obtain synergistic effects at low loadings. Examples are nano-clays and carbon nano-tubes (CNT). Pioneering work was done at Toyota [1] to disperse nano-clays fully exfoliated in nylon-6 via in-situ polymerization of caprolactam. Many attempts were reported to disperse nano-clays and CNTs directly in polymer melts but without success. Due to agglomeration, and in the case of CNTs even entangling [2], the melt-viscosity increases and hence processibility is decreased. At the IUPAC meeting in Paris (July, 2004), it was concluded after a forum discussion [3] that real beneficial “nano-effects” in the polymer domain are still to be expected, if any.

In this thesis, a realistic “nano-effect” is reported using nano-sized silica particles in standard polymers such as polyethylene and polypropylene. The presence of only a minute amount, < 1 wt%, of nano-sized silica particles, generated in-situ in the polymer powder before processing, could lower the melt-viscosity by a factor of 10-20 without sacrificing the product properties, in fact enhancing the product properties such as dimensional stability. These findings reported in the thesis could have a major impact in the world of polymer processing. In general, the properties such as strength and toughness increase with increasing molecular weight but also the melt-viscosity strongly increases. If the melt-viscosity is lowered by the presence of only a minute amount of nano-silica, a breakthrough can be realized in the structure vs. performance of polymers since higher molecular weight polymers can be processed. Even intractable polymers such as ultra-high molecular weight polyethylene might become processible with these nano-sized additives.

1.1 Introduction

Polymers have played an important role since the beginning of life. Well-known polymers in Nature are DNA, RNA, proteins and polysaccharides, the so-called natural polymers, and they play a crucial role in plant and animal life. Natural polymers have been exploited as materials for providing clothing, weapons, shelter, writing materials and other requirements. In the nineteenth century, natural polymers were modified to meet technical requirements and classical examples are nitrocellulose, cellophane and natural rubber. The fact that natural rubber (cis-polyisoprene) could be cross-linked (vulcanized) with sulphur (Goodyear 1847) stimulated the development of the rubber industry with useful products such as tires for cars. The lack of supply of natural rubber during World War II from Indonesia, Malaysia, etc. to the USA and Europe, triggered in fact the start of the synthetic rubber industry, based on synthetic polymer molecules such as SBR (Styrene-Butadiene Rubber). After Staudinger introduced the concept of polymers as chain molecules in the 1920s, synthetic polymer molecules have been made in the first half of the 20th century, e.g. the polyamides (nylons) by Carothers at DuPont, but with limited commercial success. The synthetic polymer (plastic) industry took really off the ground in the 1950s when Ziegler and Natta invented catalyst systems to produce stereoregular polymers such as (isotactic) polypropylenes and (linear) polyethylenes. The impact of their invention has been major as evidenced by a production volume of approx. 35 million tonnes/annum of polypropylene, viz. 7 kg per capita per annum!

Polymers can broadly be divided in thermoplastics and thermosetting resins. Thermoplastics (or simply plastics) account for 70 percent of all polymers produced. Depending on the intrinsic properties, applications, volume and use, thermoplastics are further classified into:

Commodity Plastics

Comprising the four major large volume plastics: poly(ethylene) (PE), poly(propylene) (PP), the combination of the latter two are often referred to as polyolefins, poly(vinyl chloride) (PVC) and poly(styrene) (PS).

Engineering Plastics

Comprising the polymers possessing usually a better high temperature performance compared with the commodity plastics such as polyamides (Nylons), polyesters such as poly(ethylene terephthalate) (PET) and poly(butylene terephthalate) (PBT), poly(carbonate) (PC), poly(methyl methacrylates) (PMMA), poly(phenylene ether) (PPE) etc.

Specialty Polymers

High performance engineering (HPE) thermoplastics such as poly(ether sulfone) (PES) and poly(ether ether ketone) (PEEK), intrinsic conducting polymers, etc.

The division of current polymers in the three categories mentioned above is, however, subject to change. The strong growth of e.g. PET in the past decades, notably in bottle applications, gives this polymer a commodity status. The same applies to some extent to poly(carbonate). In general Engineering Plastics show a trend towards *commoditization*, both in terms of volume and market price.

At the turn of the century several promising projects in industry concerning the development of novel plastics, e.g. the polyketones at Shell, were terminated. It becomes increasingly difficult to introduce new polymers in the market in view of the general trend of *commoditization* with a strong focus on cost reduction by increasing the scale of operations, e.g. polyolefin plants with a capacity up to 500 kilotonnes per annum. The world of plastics has seemingly become mature and no new polymers with market potential for engineering applications are expected in the coming decades [3]. Consequently, the focus is on exploring the ultimate performance of existing polymers.

As stated above, the properties of polymers are not determined by the chemical structure alone but equally well by the processing step and the use of additives to tune or boost the performance further. The use of smart additives will be the prime topic for the coming decades and the use of nano-sized silica particles with macroscopic effects, as described in the thesis, is a nice example in this respect.

The high molecular weight imparts properties to polymers. It is well understood that with increasing molecular weight the viscosity of polymer melt increases strongly. This increase in viscosity affects the processibility of polymers. For easy processing low molecular weights are required. This leads to a compromise between properties and processing due to the competing effects of molecular weight. The final product's properties are the resultant of not only the chemical composition, but also of the processing or shaping step, in many cases underestimated or even neglected. During the shaping process, polymers are subjected to complex flow fields with large temperature and pressure gradients in short times, resulting in large variations in properties throughout the entire product. Numerous efforts have been made to improve the performance of polymers by balancing the processibility and properties.

At present, the area of additives in polymers is ill-defined and often lacks a proper scientific understanding. In the late 1980s, Wu [4] proposed, for nylon/rubber composites, that the matrix ligament thickness [5] is the single parameter in determining the improvement in mechanical properties of plastics, especially in semi-crystalline plastics. However, later it was proven that the cavitations and oriented crystalline layer around particles are also important parameters. In the mid-1990s, Argon and co-workers proposed a mechanism based on stress delocalization and debonding which facilitate plastic stretching for improving both the toughness as well as E-modulus for inorganic hard-particle (CaCO_3) filled PP and PE [6,7], considered for some time to be the holy grail. However, the effect of flow was neglected in these studies. The presence of filler particles affects the structure development during flow (injection molding). Recently, Schrauwen [16] showed that the improvement in CaCO_3 -filled PE and PP as observed by Argon and co-workers is mainly dependent on the flow history and the explanation put forth by Argon and co-workers does not hold.

In fact the studies concerning the effect of particulate fillers in the last two decades of the 20th century were performed on the micrometer-sized fillers and at high loadings, up to 20 vol%. At such high loadings, the processibility invariably deteriorates.

In the beginning of the 21st century, expectations are running high concerning the use of nanometer-sized additives aiming at major improvements at low loadings. Well-known examples of additives in the nanometer range are the (exfoliated) nano-clays and carbon nano-tubes (CNT). Siegel and

Mayo [8] described how mesoscopic physics can change the macroscopic properties of materials. However, no real improvements in the processing and mechanical properties have been observed yet. In reality, the mechanical properties achieved in the best clay/polymer nanocomposites are much lower than conventional fiber-reinforced composites with a high fiber volume fraction [9]. Carbon nano-tube (CNT)-based composite systems suffer significant challenges to real success. Given the costs of the materials and processes involved, it is not sufficient to obtain marginal improvements in properties over traditional micron-scale composites or virgin matrices. Yet, with few exceptions, notable enhancements have not been observed [10]. Recently, Robert et al. [11] observed an unusual decrease in the viscosity. Mackay et al. [12] investigated such a behavior by suspending polystyrene (PS) nanoparticles in linear PS and observed a non-Einstein like decrease in viscosity with a change in free volume introduced by the nano-particles. Even though only a few studies have successfully utilized the benefits offered by nano-particles, it is still a viable option when considering the selection of filled or blend polymer systems.

1.2 Effect of flow and processibility

Polymers without additives (fillers)

It is known that polymer chains orient under flow. Various studies have been performed on the structure and morphology of flow-oriented semi-crystalline polymers [13]. Although shear flow is considered as a “weak” flow, the crystallization behavior is affected together with the resultant morphologies; it accelerates the overall crystallization rate [14] and shish-kebab structures [15] have been observed. The increase in crystallization rate is attributed to the increase in the nucleation rate caused by orientation-induced row nuclei. Injection molding is one of the most commonly used polymer processing method. Very complex shear and elongational stresses on the polymer melt are encountered during injection molding. This in turn results in anisotropy in the injection molded products. This non-uniformity in the structure of injection-molded polypropylene (PP) developed during flow can be seen in the optical micrograph in Figure 1.1. This shows a layered structure formed during processing, each exhibiting different properties which collectively determine the final properties of the product. Schrauwen [16] recently showed that orientation due to flow has large effects on the properties. The strength and toughness of such semi-crystalline systems are interdependent, due to several phenomena affecting cavitation, plasticity, orientation and flow history [17].

Micrometer-sized particulate-filled polymers

In general, the viscosity of polymer filled with particles increases as predicted by the classical Einstein equation for non-interacting hard particles. With increasing filler content, the effective local shear stresses increase. At such high stresses, a strong elongational flow is generated between the separating particles; other highly stretched regions occur when two particles approach each other; there is also a weak elongational flow region between two particles aligned in the flow direction [18]. The presence of elongational flows between separating particles leads to highly non-uniform and anisotropic micro-structures. This is illustrated in Figure 1.2 as studied by Hwang et al. [18]. These stretched molecules enhance the nucleation and crystallization process for semi-crystalline polymers [19]. In essence, the effect of flow and effect of fillers on improvement of mechanical properties are two inter-related, yet independent phenomena.

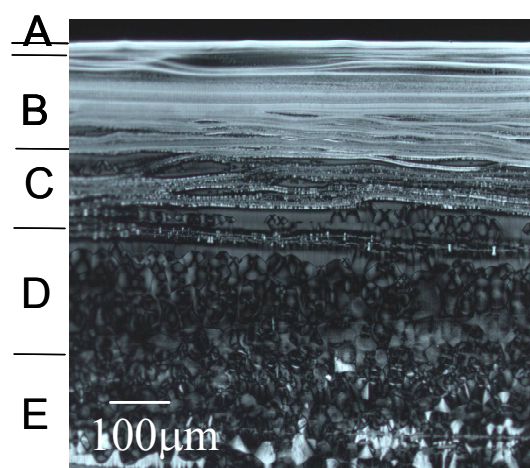


Figure 1.1: Optical micrograph of an injection molded bar of PP showing the various orientation layers identified as “skin layer” (A), “shear layer” (B), “fiber-like layer” (C), “fine-grained layer” (D), and “isotropic core” (E).

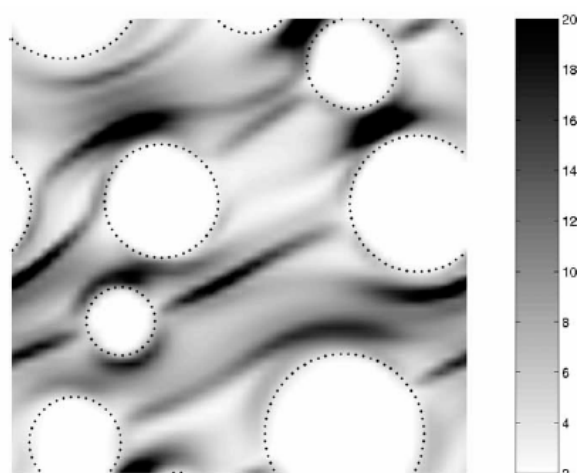


Figure 1.2: The effect of particle loading in a polymer melt: High shear stress, high orientation. The scale on right shows the stress level. Reproduced from Hwang et al.[18].

1.3 Mechanical behavior of polymers

The relationship between mechanical properties and micro-structures which strongly depend on the processing conditions was studied by Schrauwen et al. [16]. For tensile properties mainly, the yield stress was found to be proportional to the lamellar thickness, whereas the strain hardening modulus was related mainly on the crystallization conditions and the resultant entanglement density of the amorphous phase. It was also observed that the crystallinity does not influence the strain hardening modulus. This was concluded based on the fact that the crystals yield at large deformation allowing chain transfer and slip without giving a network contribution. However, the state of the entangled amorphous phase originates from the crystallization process, since it allows the rearrangement of chains during folding and perfection. This “reeling in” of polymer chains during crystallization leads to disentanglement of the chains in the melt or amorphous phase [20,21]. The effect is more pronounced for lower cooling rates, resulting in a lower entanglement density and, consequently, a lower strain hardening modulus. A qualitative comparison of the intrinsic parameters with the macroscopic tensile behavior shows that brittle failure can either be introduced by increasing the yield stress, a decrease in strain hardening modulus, or a decrease in tensile strength. The decrease in strain hardening modulus and fracture strength as a result of loss on entanglements and tie molecules is considered to be the reason for brittle failure in slowly crystallized semi-crystalline polymers [16].

Various mechanisms for improving the toughness of semi-crystalline polymers by adding micrometer-sized fillers have been proposed in literature. The brittle-to-ductile transition (BDT) of polymer composites depends on both intrinsic and extrinsic properties. Intrinsic variables include mechanical properties of matrix, interfacial adhesion, and morphological parameters such as volume fraction, size and shape of dispersed particles. Extrinsic variables include specimen geometry, strain rate, and temperature. As proposed by Argon et al. [22,23], the inorganic hard-

fillers can improve the impact toughness of semi-crystalline polymers at the same time keeping the high modulus, related to the debonding processes, which is the separation of the filler particles from the polymer matrix, is thus affected in these particle-filled composites [22]. As mentioned above, however, the effect of flow cannot be neglected [16].

1.4 Adsorption phenomena

Nano-fillers have a very large surface area and high molecular weight polymers are highly surface-active molecules. Even a weak interaction between a single monomeric unit and a surface magnifies into a powerful attraction or repulsion in case of polymers. O'Shaughnessy and Vavylonis [24] reviewed the adsorption of polymers on surfaces in dilute, semi-dilute solutions and melts in detail. Most studies in the past were performed on polymer solutions. It is reported that when nanoparticles strongly adsorb the surrounding polymer chains, the polymer becomes a non-continuum medium [25]. Individual polymer chains can physically adsorb to multiple particles even at low particle concentrations, resulting in a strong bridging effect and forming deformable colloidal structures [26]. Also, adsorbed polymer molecules enhance the effective filler volume fraction. In this respect, filler particles therefore can be regarded as core-shell particle with hard core surrounded by an immobilized soft polymer shell of comparable size [27].

Theoretical background

An increase in the concentration of the solute in the interfacial region is generally called **adsorption**. When adsorption involves the formation of a chemical bond this is usually denoted as *chemisorption*; the term *physisorption* is used when only physical interactions play a role. Physisorption is a time and temperature-dependent reversible process and is greatly influenced by the effect of flow. The opposite case, a reduction in solute concentration near the interface is often called *negative adsorption* or *depletion*. Whether adsorption or depletion is observed depends on the net adsorption energy experienced by the polymer segments, i.e. on the difference between the free energy of segment/surface contacts and that of solvent/surface contacts.

The adsorption excess energy for polymer chains may be positive or negative, depending on the net strength of interaction between a polymer segments and surface, since a positive adsorption energy implies partial unfolding of the molecule (which costs some conformational energy); this net interaction energy has to exceed a certain threshold value. The kinetics of the "sticking" process is largely unknown. A common assumption is that the attachment is very rapid on a bare surface and then slows down proportionally with increasing coverage. For uncharged homopolymers the attachment is so fast that the adsorption rate is constant up to about 80% overcharge.

1.5 Choice of system and experimental approach

The toughness mechanisms, effect of flow in particle-filled polymer composites and the advantages offered by nanocomposites as described above indicate the necessity of nanocomposites with good particle dispersion without chemical interaction with the polymer matrix, while at the same time these nanocomposites should offer the possibility of systematically varying the degree of chemical interactions between fillers and matrix. Therefore, a novel approach by combining solid-state modification (SSM) with sol-gel reactions has been developed in this thesis to make PP/silica

nanocomposites with varying degree of chemical interaction [28]. Solid-state modification (SSM) is an advantageous modification method for grafting on PP due to its inherent simplicity, low temperature and solvent-free reaction conditions. SSM of PP essentially involves grafting of polar monomers, which can be incorporated in *in-situ* made inorganic particles by a sol-gel reaction thus providing chemical interaction between matrix and fillers.

Polypropylene was chosen as matrix for various reasons:

- It is a commercially very important and well-studied semi-crystalline polymer
- Various studies related to mechanism of improvement in the mechanical properties have been performed on hard-particle filled PP systems
- An important basis for SSM of PP is the porous morphology of the nascent powder from the polymerization reactor based on reactor granular technology [29]. Nano-particles can be synthesized in-situ in these porous structures.

The hard inorganic particles as filler were chosen based on earlier studies [6,7,22]. The several reasons for the use of silica particles as fillers are listed below:

- Argon and co-workers [6] claimed that use of hard fillers may improve both the toughness and modulus, although the mechanism proposed does not hold.
- Control over the synthesis of the inorganic hard fillers, the size and shape and the ease with which they can be made in-situ.
- The most prevalent and widely used technique for preparing inorganic particles is by sol-gel chemistry [30]. The sol-gel chemistry utilizes mild synthetic conditions, which allow versatile access to chemically designed organo-inorganic hybrid materials. Silicon alkoxides, e.g. tetraethyl orthosilicate (TEOS) and tetramethyl orthosilicate (TMOS), are commonly used metal alkoxides. Silica nano-particles can be prepared using this process [31].

It is anticipated that this new approach of combining SSM and sol-gel chemistry would prove effective and instrumental in making ideal model systems to study the improvement in mechanical properties of particle filled semi-crystalline materials.

1.6 Highlights of the thesis

The nano-sized silica particle-filled PP nanocomposites were prepared in-situ via the combination of SSM and sol-gel reactions. Using these specific nano-sized silica particles at very low concentrations (less than 1%), the melt viscosity of polymers could be lowered dramatically up to two decades (see Figure 1.3) due to selective physisorption of high molecular weight chains of PP on the silica surface. The consequence of this lowered melt viscosity could be a breakthrough in the field of synthetic polymers.

The practical implication of the decrease in the viscosity and orientation under flow can be observed in the injection molding process. Usually, a very strong orientation of the polymer molecules occurs as a result of the stress field around the particles (see Figure 1.2), which leads to a strong flow-induced nucleation and highly oriented structures. However, preliminary results on the new systems showed isotropic crystalline structures.

The OM picture shown in Figure 1.4 is part of an injection molded impact test bar of PP with 0.8 wt% silica. The layered structure developed during flow in PP as shown in Figure 1.1 has disappeared in Figure 1.4. This illustrates the effect of physisorption of polymer of silica nanoparticles which reduced viscosity and hence orientation under flow. The decrease in orientation reduces the anisotropy in the final product. These effects result in a more homogeneous distribution of the mechanical properties within an injection molded part, not only in the thickness direction, but also in the flow direction.

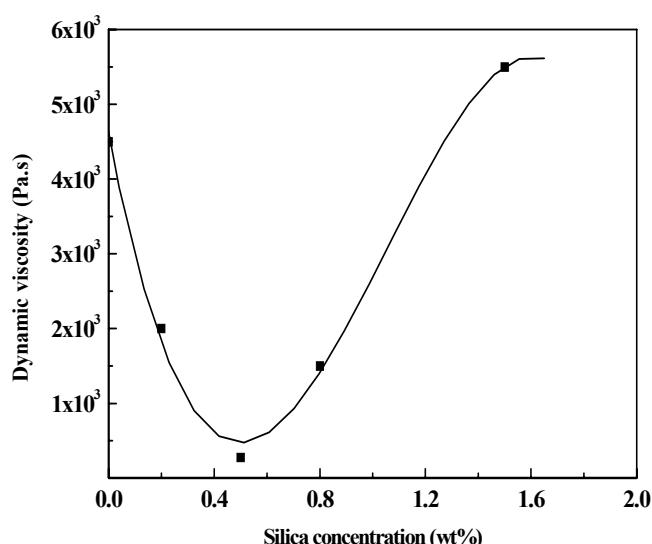


Figure 1.3: Decrease in dynamic viscosity as a function of silica concentration at a particular frequency and strain rate.

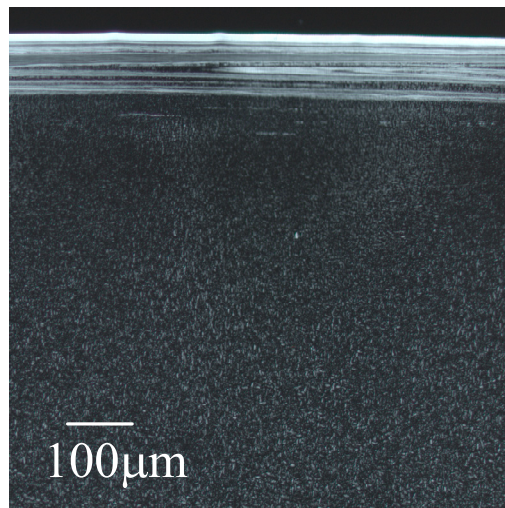


Figure 1.4: 0.8% silica filled poly(propylene) showing a reduction of the thickness of the oriented layer and a smaller size distribution of spherulites. Compare with Figure 1.1.

1.7 Outline of the thesis

The original goal of the thesis was to understand the underlying mechanisms of toughening in rigid particle-filled semi-crystalline polymers and fill the gap between established knowledge in nanocomposite technology and understanding the mechanism of the enhancement effect in nanocomposites. The rheological behavior which affects the processability, crystallization kinetics and structures is also studied which affects the mechanical properties in several aspects.

In **Chapter 2**, a new route developed by combining solid-state modification (SSM) with a sol-gel method to prepare PP/silica nanocomposites with varying degree of chemical interactions between filler and matrix is described. The synthetic aspects, chemical and structural characterization of PP/silica nanocomposites are reported. Grafting of vinyl triethoxy silane (VTES) on PP via SSM and effects thereof are also discussed in detail.

The effect of nanoscale interactions between fillers and polymer matrix on the rheology is investigated in **Chapter 3**. The hypothesis of selective physisorption of long polymer chains on the filler surface is discussed to explain the observed drop in the viscosity of PP/silica nanocomposites. This particular rheological behavior is accompanied by strong effects on the crystallization during

flow. The rheological behavior of VTES-grafted PP/silica nanocomposite is also reported in this chapter.

Chapter 4 gives insight on the effect of in-situ prepared silica nano-particles on the non-isothermal crystallization behavior of PP. The length scale of silica particles and radius of gyration of polymer chains are of the same order, thus the effect of particles is different than conventional fillers. Further, the influence of polymer adsorption on the overall crystallization is investigated. The crystallization rate is accelerated due to silica, which in turn affects the amorphous phase (entanglement density).

In **Chapter 5**, the effect of grafting on the crystallization of the VTES-grafted PP/silica nanocomposites is investigated. The random grafting in solution and non-random grafting via SSM are discussed in view of their effect on the crystallization and melting behavior. The effect of non-random grafting of VTES via SSM on PP greatly influences the crystallization behavior and hence the structures formed thereof are discussed in this chapter. Further, the similar non-isothermal crystallization analysis is performed as in Chapter 4.

The preferential adsorption of long polymer chains on the particles, as discussed in Chapter 3, can have a strong effect on flow-induced crystallization. Flow orients molecules, especially the long chains, and this orientation depends on the strength of the flow, the molecular weight (distribution) and on the temperature. This is described in **Chapter 6**.

The ultimate effects of these nano-scale phenomena on the final mechanical properties of PP/silica nanocomposites are addressed in **Chapter 7**. Two concentration regimes of silica were identified depending on the inter-particle distance and physisorption effect which exhibits different mechanical properties similar to the rheological behavior. The intrinsic deformation behavior in these two regimes is explained based on the viscosity and orientation (and flow-induced crystallization) described in the earlier chapters.

In **Appendix A**, the main emphasis is on the effect of chemical interaction between filler and matrix. The outcome of the results described in this appendix gives a clear insight in the mechanism of toughness improvement.

Finally, the commercial importance of this new route to prepare nanocomposites is compared and discussed. This approach can be extended to make nanocomposites with different inorganic mesoporous fillers which can be synthesized in-situ by sol-gel or similar method. As an example, the hypothesis of “selective physisorption of long polymer chains” applied to nanocomposites of PP/silica hollow fibers and UHMPE/silica particles prepared by the same approach are described.

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Chapter 2

Synthesis and characterization of polypropylene/silica nanocomposites^{*}

Abstract

A new route has been developed to produce PP/silica nanocomposites starting from porous PP reactor powder and making use of sol-gel chemistry. Silica-like nano-sized particles were prepared in the pores of the PP reactor powder with a controlled degree of adhesion between PP and silica. Two so-called solid state modification (SSM) reactions have been explored. In route 1, tetraethoxy orthosilicate (TEOS) was added to PP reactor powder and, subsequently, via hydrolysis and condensation nano-sized silica particles were generated in-situ. In this method chemical interaction was deliberately not introduced between the PP matrix and the silica particles. In route 2, chemical interaction was promoted by grafting vinyl triethoxysilane (VTES) on PP and subsequently silica particles were generated in-situ. Magic-angle spinning (MAS) ^{29}Si NMR spectra showed that the chemical building blocks of the silica-like clusters are of Q^3 and Q^4 -type. For VTES-grafted PP/silica nanocomposites, VTES was grafted via SSM in porous PP particles. Bulk polymerization of VTES under similar experimental conditions but in the absence of PP resulted in homopolymerization of VTES. However, SEC and NMR experiments showed that VTES was grafted on PP as a single monomeric unit in the amorphous phase with the possibility of VTES-polymer grafting during SSM. Subsequently, silica particles were prepared by sol-gel method in the VTES-grafted PP. MAS ^{29}Si NMR and FT-IR spectroscopy showed that the grafted VTES becomes the part of the in-situ formed silica particles. No decrease in molecular weight of PP was observed, indicating that chain scission is marginal compared to melt modification. The morphology of the nanocomposites as observed by ATR-FTIR microscopy showed a uniform dispersion of grafted VTES as well as in-situ formed silica.

2.1 Introduction

Inorganic particle-filled polymer nanocomposites, notably (exfoliated) nano-clays, have attracted great interest over the last decade [1,2,3,4]. However, to disperse nano-particles homogeneously in thermoplastics via conventional blending techniques is a major problem, the lack of commercially available and thermally stable organoclays and the lack of understanding of the mechanism of the improvement in properties are some of the difficulties faced in the development of clay/polymer nanocomposites [5]. In the context of optimizing the dispersion of fillers, the interfacial properties and understanding the mechanism of improvement in the mechanical properties, it is important to

^{*} Part of this chapter is reproduced from: Jain, S.; Goossens, J.G.P.; Picchioni, F.; van Duin, M.; Magusin, P.; Mezari, B. "Synthetic aspects and characterization of polypropylenes-silica nanocomposites prepared via solid-state modification and sol-gel reactions" Polymer, accepted.

synthesize inorganic particle-filled polypropylene (PP) nanocomposites in which the interfacial strength can be tuned.

Melt compounding [6,7], solution blending [8], melt intercalation, *in-situ* intercalative polymerization [9] are well-studied conventional methods to prepare polymer nanocomposites. Despite many efforts in the past, these conventional routes for preparing inorganic particle-filled PP nanocomposites have their deficiencies and could not be exploited commercially. Moreover, during melt compounding, oxidation and β -scission lead to degradation of PP. Solvent removal is another problem faced in some of these methods. These problems add extra parameters, apart from the dispersion of fillers and the interfacial strength, which affect the morphology and the final properties of the PP nanocomposites prepared by these methods.

In view of these problems, a new strategy to prepare PP nanocomposites was developed by combining *solid-state modification* (SSM) of PP reactor powder and the *sol-gel method* for making organic-inorganic hybrid materials. SSM is a solvent-free method of PP modification largely suppressing oxidation and β -scission [10]. SSM is advantageous due to its inherent simplicity, low temperature and solvent-free reaction conditions. Although the absence of solvent is one of the main advantages of SSM, small quantities may be used as diffusion aid or promoter to increase the grafting efficiency [11,12]. In this study, SSM of PP essentially involves the grafting of polar monomers, which can be subsequently incorporated in *in-situ* made inorganic particles, thus providing chemical interaction between matrix and filler. Some fundamental aspects of solid-state grafting onto PP were studied by Rätzsch et al. [13]. An important parameter for SSM of PP is the morphology of nascent PP powder from the polymerization reactor [14]. The actual morphology of such PP powders prepared by the reactor granular technology is shown in Figure 2.1. In general, the porous macroparticles are composed of many sub-particles, which are also porous. The PP macroparticles are approx. 300 μm . The sub-particles are in the order of 50 μm . The irregular channels between these sub-particles are typically 2-3 μm , while the pore sizes are 2 μm at the surface to 500 nm inside the particle.

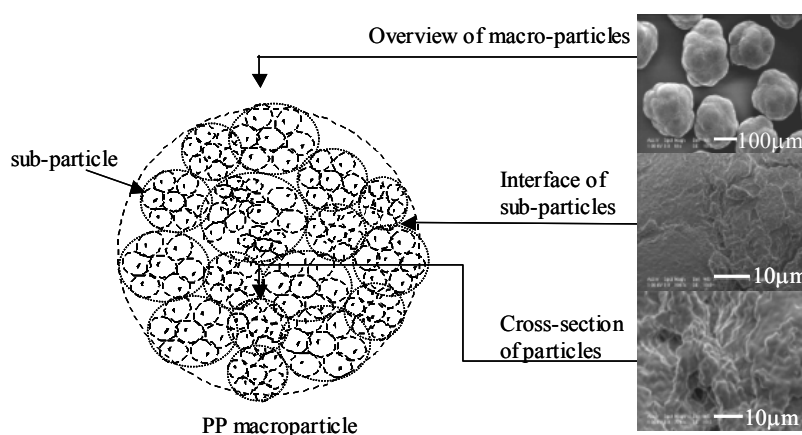
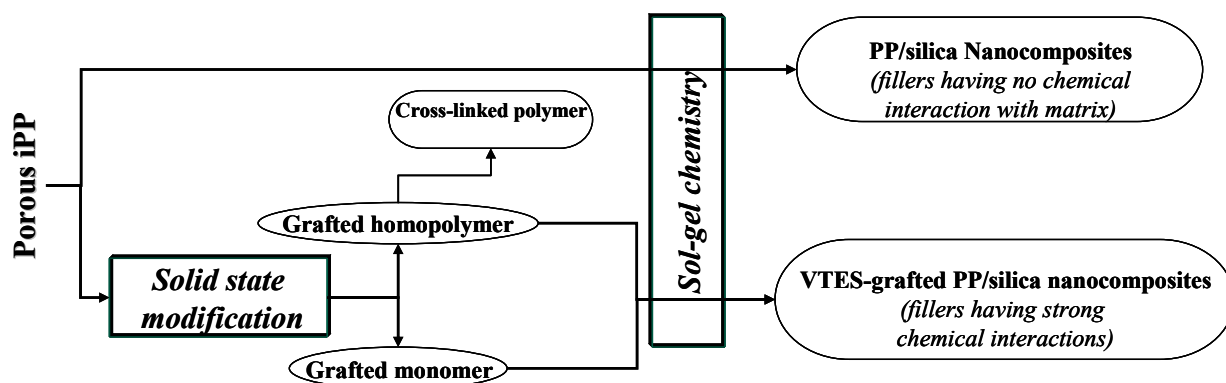


Figure 2.1: Schematic representation of double grain model and SEM micrographs showing the morphology of the PP powder prepared by reactor granular technology.

Sol-gel chemistry is a method to produce nano-scale inorganic particles in organic matrices with a good dispersion of the filler in the matrix [15]. The most prevalent and widely used hybridization technique for preparing inorganic particles, like silica, is a sol-gel reaction [16,17]. The sol-gel method utilizes mild synthetic conditions, which allow versatile access to chemically designed

organo-inorganic hybrid materials. The process involves the hydrolysis of metal alkoxides, followed by a condensation reaction to produce metal oxides [18].

The experimental procedure is shown in Scheme 1. Two type of PP nanocomposites were prepared using the following approach: 1) PP/silica nanocomposites with physical (weak) interactions; there are no chemical bonds between the filler and the PP matrix; 2) (VTES-grafted PP)/silica nanocomposites with controllable chemical interactions between the organic PP and the inorganic silica phase; the filler particles are chemically attached to the PP chains. Solid-state grafting of vinyl triethoxysilane (VTES) on PP enables the interfacial interaction between filler and matrix. Subsequently, the silica-like inorganic nano-particles can be prepared *in-situ* via the sol-gel method. In this chapter the synthetic aspects of inorganic particle-filled PP nanocomposites are discussed with the focus on the chemical and structural characterization and on the morphology. Grafting of VTES and its influence on the *in-situ* preparation of PP/silica nanocomposites is investigated and discussed, taking into account the possibility of homopolymerization of VTES, which has been neglected in previous studies [10].



Scheme 1: Experimental scheme for the complete nanocomposite synthesis followed during this study.

2.2 Experimental

2.2.1 Materials

Porous, isotactic polypropylene (PP) possessing a porosity of $S_{BET(Kr)} = 0.0678 \text{ m}^2/\text{g}$ was supplied in powder form by Euro-Sabic, The Netherlands. The number-average and weight-average molecular weight of PP were 60 and 380 kg/mol respectively. The crystallinity of the PP determined from DSC measurements is approx. 60%. The powder was free of antioxidants and stabilizers and was stored at -10°C in the dark. Vinyl triethoxy silane (VTES) and tetraethoxy orthosilicate (TEOS) were obtained from Aldrich Chemicals. Ammonium hydroxide (NH_4OH) (28% NH_3 in water) obtained from Aldrich Chemicals was used as catalyst for sol-gel reactions. Various free-radical initiators were selected for grafting based on their chemical structures and half-life times. Di-tert-butyl peroxide (DTBP) and α,α' -azoisobutyronitrile (AIBN) from Merck Schuchardt, Germany, dibenzoyl peroxide (BPO) from Fluka Chemicals and lauroyl peroxide (LPO) from Aldrich Chemicals were used as initiators. The half-life time of decomposition at 100°C for each initiator is: DTBP: 8400 min, BPO: 22 min, AIBN: 7 min, and LPO: 6 min. All the chemicals and polymers were used as purchased without any further purification or modification.

2.2.2 Reactor set-up

All reactions were carried out in a double-skinned SSM reactor as shown in Figure 2.2 [19,20]. Hot oil circulated through the double-skinned jacket to maintain the reaction temperature. The reactor was equipped with a water condenser and a mechanical stirrer. Reactions were carried out under a nitrogen flow to have inert atmospheric conditions and to remove traces of oxygen, which otherwise could trigger the oxidation of

PP. A peristaltic pump was attached to the reactor to feed the reactive mixture at the desired rate. The feeding rate was calculated from the density of the reactive mixture and the addition time. The flow rate is expressed in pph/min, i.e. parts per hundred of reactive mixture with respect to PP per minute. All the reactions were carried at temperatures below the melting temperature of PP. The nanocomposites and modified PP were obtained from the reactor in powder form. The final products were immediately recovered from the reactor and stored at -4°C before further characterization.

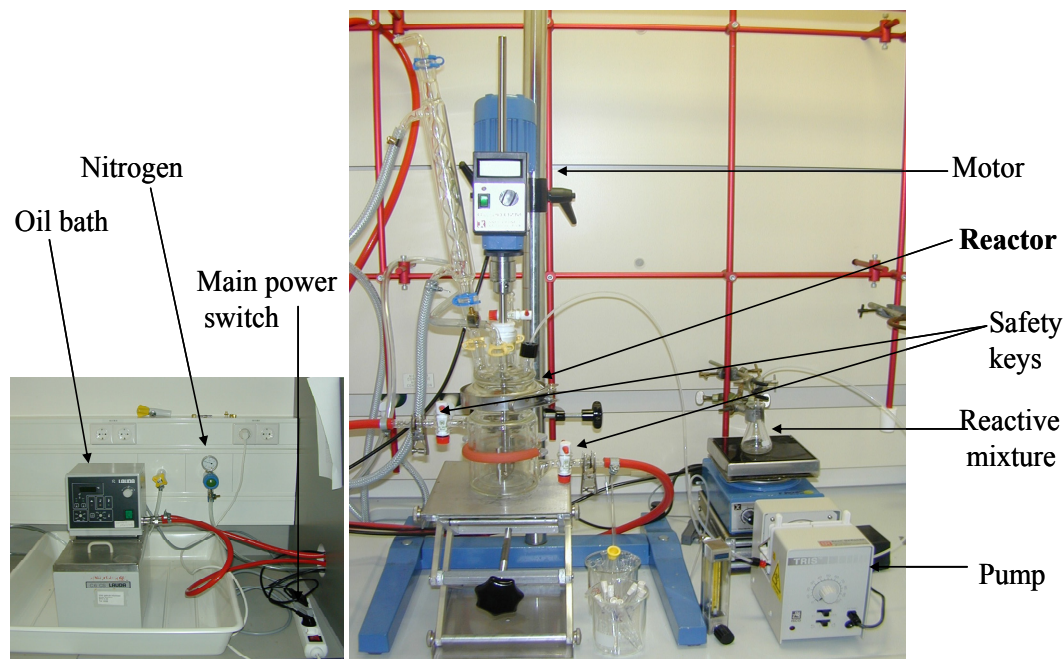


Figure 2.2: Reactor set-up for the solid-state modification and the synthesis of nanocomposites.

2.2.3 Infrared (FT-IR) spectroscopy

Biorad Infrared Excalibur 3000 and FTS 6000 series infrared spectrometers were used for Infrared analysis. The FTS 6000 series spectrometer coupled with an UMA500 microscope was used to characterize individual particles in ATR mode.

Attenuated total reflectance mode (ATR): Samples in powder form were characterized in the reflectance mode with both instruments in the range of $4000\text{--}650\text{ cm}^{-1}$ with a resolution of 2 cm^{-1} co-adding 100 scans. To characterize the inner part, particles were cut into two halves and selectively the outer and inner surfaces were measured using the microscope. The penetration depth of the evanescent wave is about $2\text{--}3\text{ }\mu\text{m}$, while the thickness of a cut powder particle is about $500\text{ }\mu\text{m}$. So, it can be assumed that the spectra correspond to the measured surfaces. The same argument holds for the outer surface characterization.

Transmission mode: To get an impression of grafting in bulk the powder samples were melt pressed to $50\text{ }\mu\text{m}$ films at 180°C for about 10 min using a Collin Compression Press. Melt-pressed nanocomposites were characterized in transmission mode in the range of $4000\text{--}400\text{ cm}^{-1}$ with a spectral resolution of 4 cm^{-1} and 32 scans were co-added.

The calibration curve for quantitative analysis was constructed by recording FT-IR and solution ^1H NMR spectra of VTES-grafted PP with varying degrees of grafting. The degree of grafting (wt%) of VTES is calculated assuming that the grafted VTES neither hydrolyzed nor homopolymerized. At least 10 different specimen of the same sample were measured to ensure the repeatability and uniformity of the analysis.

2.2.4 Solution ^1H NMR spectroscopy

A Varian Gemini 300 MHz NMR spectrometer was used for solution ^1H NMR to construct the calibration curve for quantitative measurements using FT-IR spectroscopy. It was also useful in determining whether grafted monomers homopolymerize or form cross-links. Samples were prepared by dissolving grafted PP powder ($\sim 11.5\text{ mg}$) in ($\sim 1\text{ ml}$) of deuterated tetrachloroethane (with 1 mg DBPC stabilizer per 10 ml) at around 140°C until a clear solution was obtained. All the measurements for modified PP were performed at 130°C . In case of VTES and homopolymerized VTES, carbon tetrachloride was used as the solvent and reference, and measurements were done at room temperature. The number of moles of grafted VTES to PP was calculated by taking the ratio of the H-atoms of the ethoxy group ($-\text{O}-\text{CH}_2-\text{CH}_3$) of VTES to that of PP.

2.2.5 Magic-angle-spinning (MAS) ^{29}Si solid-state NMR spectroscopy

Magic-angle-spinning (MAS) ^{29}Si NMR was carried out on a Bruker DMX500 spectrometer operating at a ^{29}Si NMR frequency of 99 MHz and equipped with a 7-mm MAS probe head and the sample rotation rate was 4 kHz (2 kHz for liquids). About 2000 scans were accumulated. MAS ^{29}Si NMR was performed on nanocomposites powder to characterize the structure and the molecular connectivity of the cross-linked VTES and *in-situ* formed silica. The two-pulse sequence $90^\circ\text{-}\tau\text{-}180^\circ\text{-}\tau$ was used with a 90° pulse of 5 μs , an echo time 2τ of 15 μs , and an interscan delay of 180 s. The value of 180 s was chosen for practical reasons given the limited amount of measuring time. However, even at this relatively long interscan delay the ^{29}Si NMR spectra are not quantitative, as was evident from the line shape changes observed in a few test experiments at still longer interscans. The peak assignments for the structure of silica is denoted by $Q^n = (\text{RO})_{4-n}\text{Si}(\text{OSi})_n$, as explained by Young et al. [21]. Assignments for various degrees of Si-atom substitution of the SiO_4 tetrahedra are reported elsewhere [22,23,24]. The peaks are located within the following range of chemical shifts relative to TMS [$\text{Si}(\text{Me})_4$]: $Q^4 = -106$ to -120 ppm, $Q^3 = -91$ to -101 ppm, $Q^2 = -74$ to -93 ppm and $Q^1 = -68$ to -83 ppm. The chemical shift of the Si-atom next to the C=C bond in VTES is -60 ppm. It represents the T^0 structure, where T^x corresponds to $\text{Si}(\text{CH}=\text{CH}_2)(\text{OR})_{3-x}(\text{O}-\text{Si}\equiv)_x$.

2.2.6 Size Exclusion Chromatography (SEC)

Molecular weights of VTES and its homopolymers were measured by size exclusion chromatography (SEC) at 40°C using a Waters GPC equipped with a Waters model 510 pump, a model 410 differential refractometer (40°C), a Waters WISP 712 auto-injector (50 μL injection volume), a PL gel (5 μm particles) 50×7.5 mm guard column and a set of two mixed bed columns (Mixed-C, Polymer Laboratories, 300×7.5 mm, 5 μm bead size, 40°C). THF was used as the eluent at a flow rate of 1.0 mL/min. Calibration was carried out using narrow MWD polystyrene (PS) standards ranging from 600 to 7×10^6 g/mol. The molecular weight was calculated relative to the PS standards. Data acquisition and processing were performed using Waters Millennium 32 software.

2.2.7 High-temperature Size Exclusion Chromatography (HT-SEC)

The molecular weight of PP in the nanocomposites was determined using high temperature SEC on a PL-GPC 210 with 4 PL gel Mixed A Columns, a RALLS (light scattering) detector (Precision Detector, PD2040 at 800 nm), a H502 Viscometer (Viscotek), a refractive detector and DM400 data manager (Viscotek). The measurements were performed at 140°C , using 1,2,4-trichlorobenzene as solvent and a narrow molecular weight PS standard as reference. The average of two independent measurements is reported.

2.2.8 Thermo-gravimetric Analysis (TGA)

A Perkin Elmer Pyris-6 TGA was used for quantitative determination of the *in-situ* formed silica and the grafted VTES on PP. It was also used to study the thermal stability of nanocomposites. Samples were heated under an oxygen atmosphere at 10K/min up to 900°C and hold for 15 min to ensure complete oxidation of organic material. The residue is assumed to be silica due to complete oxidation. The moles of Si are calculated based on silica residue, from which moles of VTES or *in-situ* formed silica or grafted VTES-silica were determined.

2.2.9 Scanning Electron Microscopy (SEM)

A Philips Field Emission Gun (FEG) Environmental Scanning Electron Microscope (ESEM) was used to determine the shape and size of *in-situ* formed silica particles. A qualitative measure of the particle size distribution was obtained from the SEM micrographs. In some cases (as described on the micrographs), the back-scattered emission (BSE) mode was used to get better contrast. For better focusing, a high electron voltage (viz. 1 eV) was used and samples were coated with carbon or platinum to avoid charging.

2.2.10 Transmission Electron Microscopy (TEM)

A Jeol JEM 2000 FX Transmission Electron Microscope (TEM) was used to characterize the morphology of the nanocomposites. The TEM was operated at 80 kV. Ultrathin sections of 50-70 nm were obtained at low temperatures (-40°C) using a Reichert Ultracut E microtome, equipped with a diamond knife. The sections were not stained, since the electron density of the filler particles (silica) was much higher than matrix (PP). The effect of processing, such as extrusion or injection molding, on the morphology, especially the dispersion of silica in the PP, was also investigated using TEM.

2.3 Results and discussion

2.3.1 Synthesis of nanocomposites

2.3.1.1 PP/silica nanocomposites (route 1: no chemical interactions)

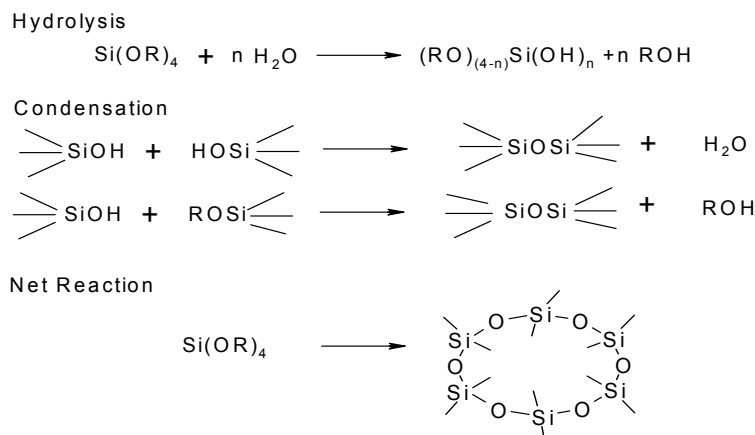
PP/silica nanocomposites were prepared by the sol-gel method. TEOS was added to PP in the reactor and stirred for approx. 30 min at 60°C. After the dispersion of TEOS, a mixture of water and ammonium hydroxide (NH₄OH) was added slowly at a controlled rate to the reactor with continuous stirring. The molar ratio of TEOS/H₂O was 1:5 and NH₄OH was 1 wt% based on TEOS. The sol-gel reaction was carried out in the reactor at 60°C for 3 hrs with continuous stirring. The thus formed sol in PP is then gelled at 80°C for 5 hrs and dried under vacuum at 120°C for 24 hrs. The samples were physically mixed with antioxidants using a powder blender before any further treatment and dried under vacuum at 100°C before chemical and structural characterization.

2.3.1.1.1 Sol-gel reaction (ex-situ PP)

The sol-gel reaction is an important step for making inorganic nanofillers in the pores of PP. Scheme 2 shows the sol-gel reaction mechanism. The rate of formation and size of silica hybrid particles was controlled by variation of the pH of the HCl/water (acidic) or NH₄OH/water (basic) systems, which trigger the hydrolysis and condensation reactions. At high pH, solubility of small particles is higher than at low pH, resulting in larger particles (5-10 nm). Under basic conditions, particles are negatively charged and hence repel each other. Therefore, they do not collide and are stable as such, while under acidic conditions the charge of repulsion is reduced and aggregation occurs, resulting in a branched network of silica particles of (2-4 nm) [25,26]. Figure 2.3 shows the SEM micrographs of the silica particles prepared under different catalytic conditions in the absence of PP. It is evident that the size of silica particles formed is in close agreement with the results from literature. In view of this, base catalyst conditions are used for *in-situ* preparation of silica in the pores of PP.

2.3.1.1.2 Silica synthesis in PP (in-situ PP powder)

Silica nano-clusters were prepared *in-situ* in the pores of PP through the sol-gel method. The FT-IR spectra depicted in Figure 2.4 are of pure silica (in the absence of PP) prepared by the sol-gel method and of PP/silica nanocomposites under the same reaction conditions. The strong and broad peaks around 1000-1100 cm⁻¹ can be attributed to Si-O-Si linkages. Fidalgo et al. [38] discussed the structural defects of sol-gel derived silica in the presence of polymers in detail. The intermediate frequency band at ~ 800 cm⁻¹ involves the symmetric stretching vibration of the Si-O-Si atoms (ν_s Si-O-Si). The dominant band near 1080 cm⁻¹ can be assigned to asymmetric stretching vibrations of Si-O-Si atoms (ν_{as} Si-O-Si) of nearly perfect silica structures with cyclic Si-O-Si (see also Table 2.1). This dominant band at 1080 cm⁻¹ can be observed in both spectra, which implies the formation of silica clusters in the pores of PP. The bands are slightly shifted in the presence of PP. The 1080 cm⁻¹ peak is usually split into two components as a result of long range coupling interactions: a transverse optical (TO) and a longitudinal optical (LO). The shoulder associated with this band, centered at ~1150 cm⁻¹, is usually related to the LO component and the band nearing 1070 cm⁻¹ to TO component [27,28].



Scheme 2: Reaction mechanism of sol-gel reaction of TEOS ($R \sim \text{C}_2\text{H}_5$).

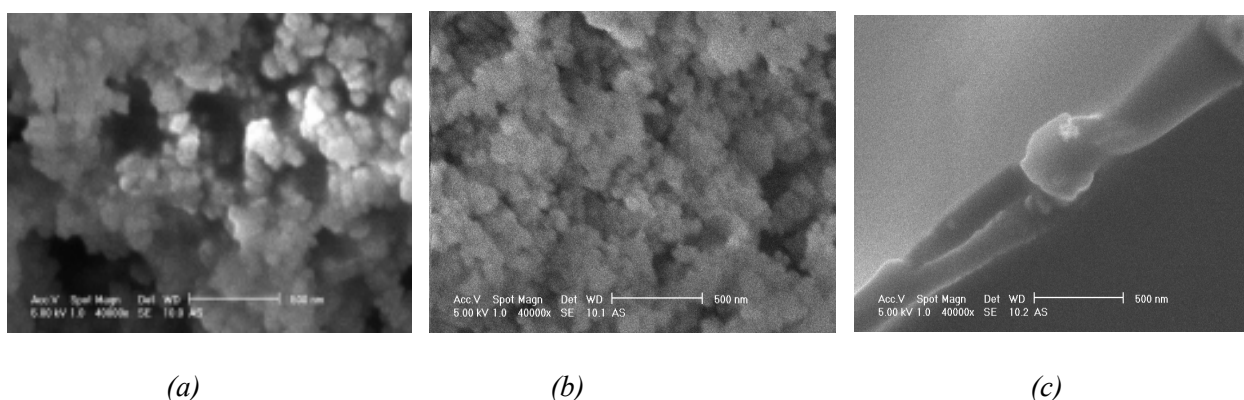


Figure 2.3: SEM micrographs of silica prepared by the sol-gel method: (a) no catalyst (~ 80 nm), (b) base catalyst: $\text{pH}=10$ (~ 50 nm), (c) acid catalyst: $\text{pH}=2$ (≤ 10 nm).

There is no appreciable increase in relative intensities of the peaks at $\sim 1207 \text{ cm}^{-1}$ and 1090 cm^{-1} , which indicates that the *in-situ* formed silica is similar to that formed by the sol-gel method without PP. Moreover, the relative intensity of the $\nu_{\text{Si-OH}}$ band ($\sim 950 \text{ cm}^{-1}$) in the case of pure silica and *in-situ* formed silica in PP is comparable, indicating a similar extent of condensation in both cases. These peaks also imply that some hydroxy groups are still present on the outer surface of the silica particles.

The MAS ^{29}Si NMR spectra of the sol-gel derived silica and *in-situ* formed silica-clusters in PP are shown in Figure 2.5. The band at -112 ppm, assigned to the Q^4 structure, indicates fully condensed silica (SiO_2), while the peak observed at -104 ppm is assigned to a Q^3 structure with a Si-atom connected to three Si and one H or C atom in the second coordination sphere. The high intensity of the Q^3 band suggests a large surface area with $-\text{OH}$ or $-\text{OR}$ groups which is in agreement with FT-IR results. There are no peaks related to Q^1 and Q^2 structures in the spectra, indicating that the condensation reaction during the sol-gel reaction is fairly complete. The structure of *in-situ* formed silica (Figure 2.5b) is close to the separately prepared silica (Figure 2.5a), suggesting that the presence of PP does not affect the structure formation of silica, as in agreement with the FT-IR measurements. The extremely long ^{29}Si spin-lattice relaxation time of Q^4 structures makes a quantitative analysis difficult, especially when the silica particles are enclosed in the PP matrix.

Table 2.1: Assignments of Infrared bands of the FT-IR spectra [35, 43].

Assignment	Wavenumber (cm ⁻¹)
Stretching ν Si-O-C	1120
Stretching ν_{as} Si-O-Si	1080
ν_1 (LO)	~1207
ν_2 (LO)	~1160
ν_3 (TO)	~1085
ν_4 (TO)	~1068
Stretching ν_s Si-O-R (R~ H; C ₂ H ₅)	950 - 980
Stretching ν_s Si-O-Si	800
Rocking ρ -CH ₃ of PP	899
Stretching ν -O-H	2500-3800

LO- Longitudinal optical; TO- transverse optical

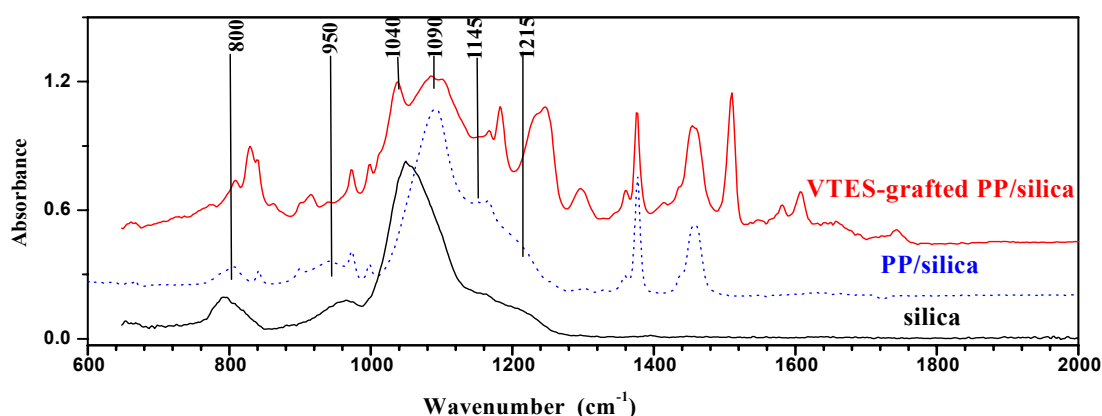
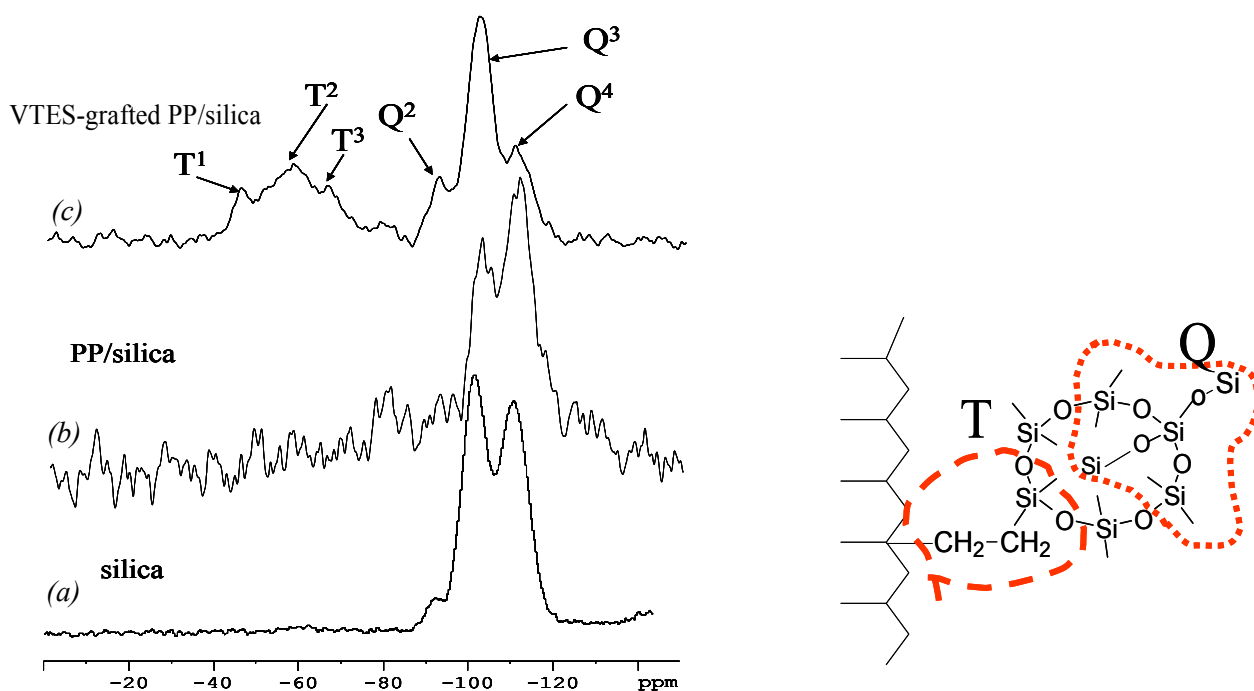


Figure 2.4: FT-IR spectra of silica made by the sol-gel method and in-situ formed silica in PP and VTES-grafted PP.

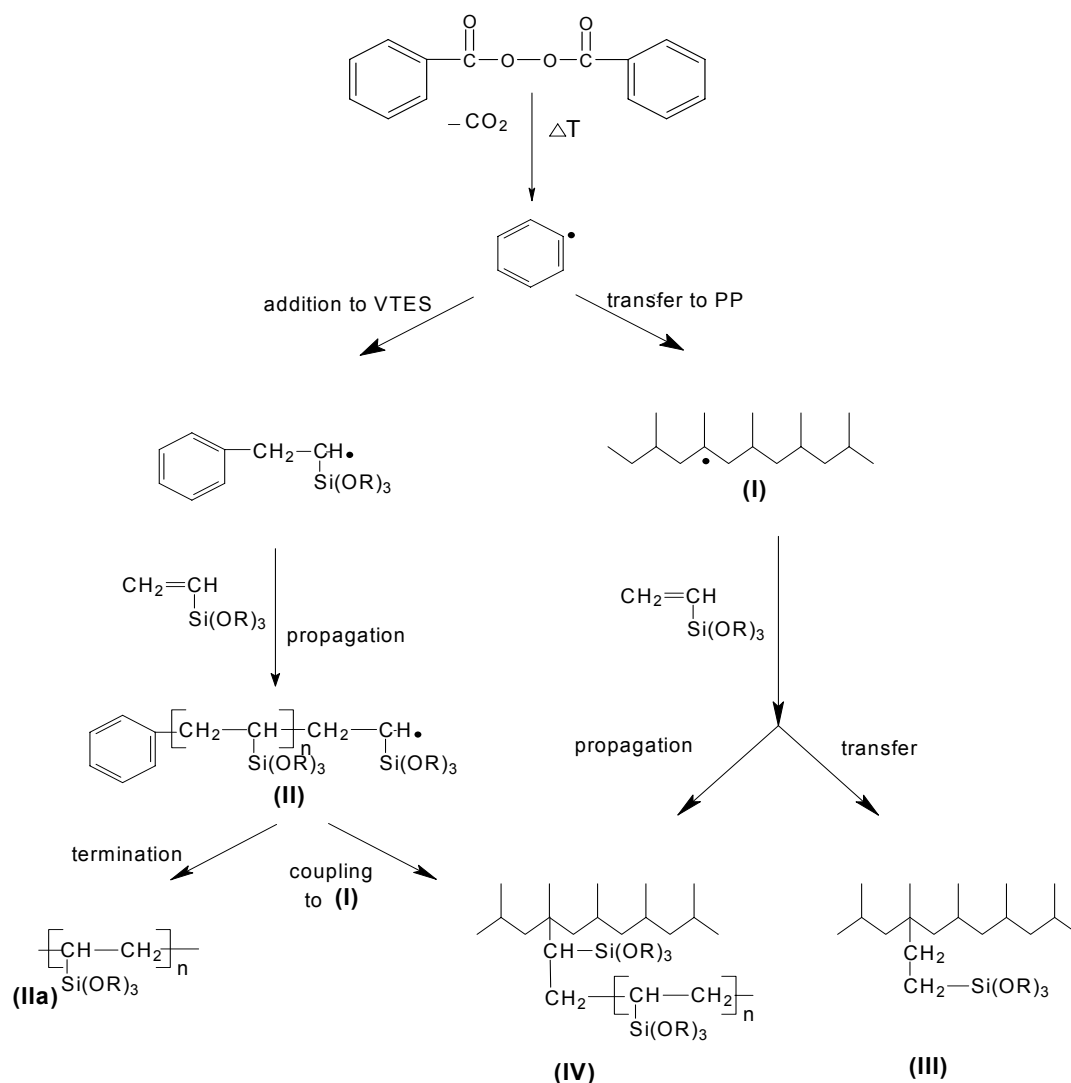
Figure 2.5: MAS ²⁹Si NMR spectra of (a) silica, (b) PP/silica, (c) (VTES-grafted PP)/silica nanocomposites. Figure on the right shows the T and Q type of structures, details of which are given in the experimental section.

2.3.1.2 (VTES-grafted PP)/silica nanocomposites (route 2: with chemical interactions)

The first step in the synthesis of (VTES-grafted PP)/silica nanocomposites was grafting of VTES on PP via SSM as shown in Scheme 3. The free-radical initiator decomposes into free radicals, which initiate the reactions (i) by transferring the radical to the PP backbone and (ii) by addition to the monomer. Part of the monomer may homopolymerize and eventually might graft by a coupling reaction to a PP radical.

Solid-state modification (SSM)

The grafting of VTES is an important aspect of the synthesis carried out via SSM. It controls the degree of adhesion between the matrix and the fillers. The following section discusses the reaction conditions for SSM and the reaction mechanism involved in the VTES grafting on PP, with special attention to the possibility of homopolymerization of VTES during grafting.



Scheme 3: Possible reactions proposed in this study for radical grafting of VTES on PP by initiation with peroxide during solid-state modification.

2.3.1.2.1 Reaction of vinyl silane with initiators (without PP)

A number of monomers, such as maleic anhydride [29,30], methacrylic acid [31,32] and vinylsilane (VTES) [32,33], have been used in previous SSM studies. These monomers upon

grafting provide polarity and reactivity to PP. These were reported in these studies to be non-homopolymerizable monomers and were assumed to graft on PP as single monomeric units.

However, the SEC, ^1H NMR and MAS ^{29}Si NMR measurements indicate that VTES, in the absence of PP, is susceptible to homopolymerization. VTES yields products with molecular weights as high as 50 kg/mol when initiated by BPO or AIBN (Table 2.2). The VTES monomer was recovered as such in the SEC analysis, indicating that cross-linking via $-\text{Si}-\text{O}-\text{Si}-$ links formed upon hydrolysis with water present in THF (SEC solvent) and subsequent condensation of the VTES ethoxy groups did not occur. The high molecular weights observed with SEC, therefore, indicate true homopolymerization of VTES. ^1H NMR (Figure 2.6a) indicates a C=C conversion of approx. 40% upon homopolymerization of VTES using BPO as initiator. The appearance of a new peak around 1.6 ppm similar to that of the aliphatic H_α atoms in polyolefins suggests the formation of $-\text{CH}-$ groups upon the polymerization of VTES. The conversion of C=C is in agreement with the amount of aliphatic $-\text{CH}-$ groups formed. The VTES polymer was completely soluble in various solvents. This again excludes the possibility of cross-linking by hydrolysis and condensation of ethoxy groups. The MAS ^{29}Si NMR spectra in Figure 2.7 confirm the polymerization of VTES. The chemical shift of the Si atom next to the C=C bond in VTES is -60 ppm. It represents the T^0 structure, where T^x corresponds to $\text{Si}(\text{CH}=\text{CH}_2)(\text{OR})_{3-x}(\text{O}-\text{Si}\equiv)_x$. If the Si atom of this VTES is attached to a long saturated hydrocarbon chain, the ^{29}Si peak of the T^0 structure shifts to -45 ppm [34]. The MAS ^{29}Si NMR spectrum of the polymerized VTES in Figure 2.7b indeed shows a new signal at -45 ppm. The spectrum shows no signals at the chemical shift positions of T^1 , T^2 and T^3 structures resulting from cross-linking via hydrolysis and condensation of the ethoxy groups.

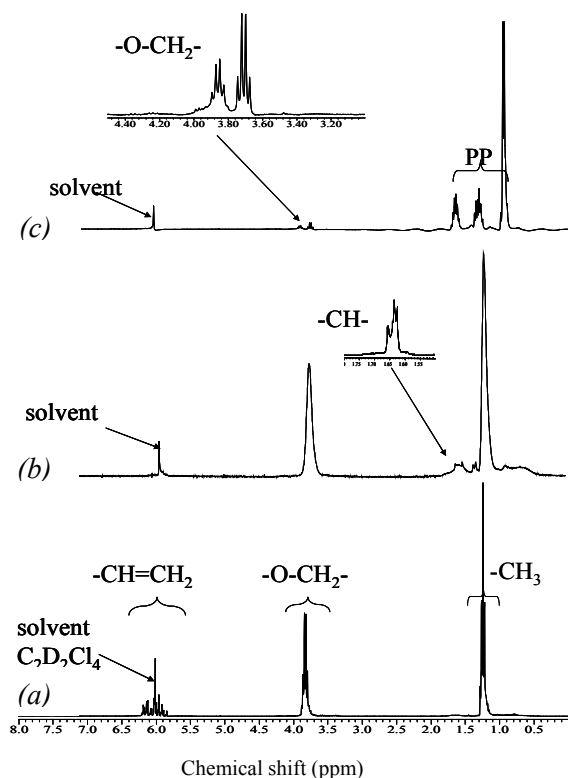


Figure 2.6: ^1H NMR spectra of (a) VTES monomer, (b) VTES homopolymer, (c) PP-g-VTES monomer (13 wt%).

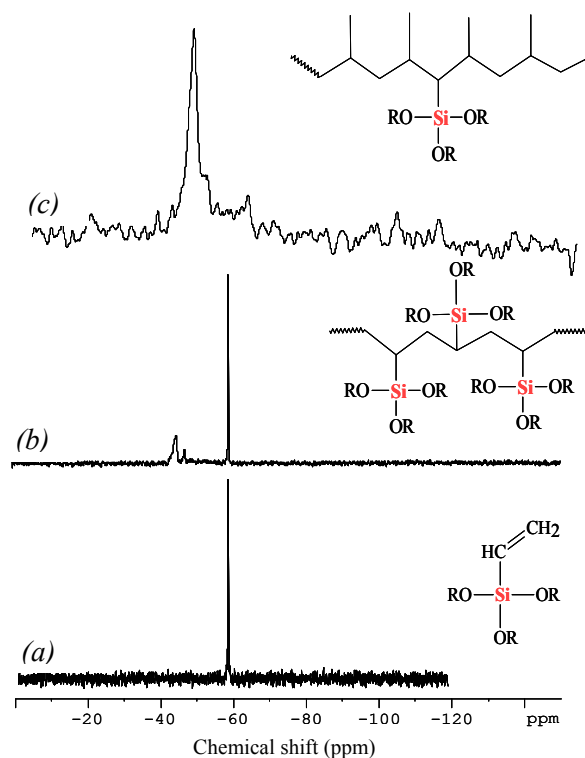


Figure 2.7: MAS ^{29}Si NMR spectra of (a) VTES monomer, (b) VTES homopolymer, (c) VTES-g-PP monomer (10 wt%).

The high molecular weight found with SEC, the appearance of a new peak at 1.6 ppm and the saturation of the C=C bonds visible in the ^1H NMR spectra, and the absence of any other signals than that of T^0 in the MAS ^{29}Si NMR spectra proves that VTES polymerizes along the vinyl unit. Based on these findings, a reaction mechanism for the grafting is proposed in Scheme 3. Part of the VTES may graft as monomeric unit on PP (*structure III*), similar to the grafting of maleic anhydride [29]. The other part of VTES grafted as monomeric unit on PP may then propagate to yield a polymer of VTES grafted on PP (*structure IV*). The other part of the VTES may homopolymerize (*structure II*) and graft on the PP chain by coupling. The grafting of VTES involving homopolymerization is similar to that observed for styrene [10].

Table 2.2: Molecular weight (kg/mol) of polymerized VTES obtained by SEC

Initiator	M_n	M_w
LPO	6.4	13.1
BPO	8.8	17.4
AIBN	9.1	22.4
DTBP	17.0	68.7

2.3.1.2.2 VTES grafting on PP

The structure of VTES-grafted PP was identified and the degree of grafting was quantified taking into account that VTES may homopolymerize during SSM. Figure 2.6c shows the ^1H NMR spectra of PP with 13 wt % grafted VTES. The quartet at 3.76 ppm (the inset shows the magnified region of interest) indicates the presence of ethoxy groups attached to the Si-atom. Although no new peaks indicating the formation of aliphatic C-H groups were observed, the two quartets suggest two different structures and the broadening of peaks indicate that VTES may have grafted as a polymer. No resonances of vinyl H-atoms were observed (6 ppm), implying that there is no unreacted monomer in the grafted PP. The MAS ^{29}Si NMR spectra in Figure 2.7c of modified PP confirm the presence of grafted VTES. When VTES is attached to a long hydrocarbon chain, the ^{29}Si peak of the T^0 structure shifts to ~ -45 ppm [34]. Therefore, the peak at -45 ppm in the spectrum can be assigned to VTES grafted on PP backbone. The susceptibility of VTES to homopolymerize as explained earlier, however, does not completely rule out the possibility of VTES grafting as monomer as well as polymer to PP.

The PP pores are a few micrometers in size and can act as mini-reactors inside the PP-macroparticles, in which VTES can be dispersed and homopolymerize. The VTES can diffuse in the pores of PP or even in the amorphous phase of PP and may graft or homopolymerize there. The solubility parameter of PP is $7.43 (\text{cal/cc})^{1/2}$ and that of VTES is $8.0 (\text{cal/cc})^{1/2}$, so it is expected that the VTES can be absorbed in the amorphous phase of PP during solid-state modification. Moreover, the grafting can take place at the interface between VTES-filled pores and PP, i.e. at the PP surface. However, it was found that grafting during SSM mainly takes place in the amorphous phase of PP. ^1H spin lattice relaxation times in the rotating frame, $T_{1\rho}\{\text{H}\}$, were measured for the amorphous and crystalline phase of PP via cross polarization to ^{13}C and ^{29}Si nuclei in the solid-state. $T_{1\rho}\{\text{H}\}$ for the crystalline and amorphous phase of the original PP were found to be 35 ms and 4 ms, respectively. In case of VTES-grafted PP, it was found that $T_{1\rho}\{\text{H}\}$ for the grafted VTES was also around 4 ms, indicating that grafting took place in the amorphous phase of PP. This restricts the chances of homopolymerization of VTES, since the diffusion of the monomer is limited by the absorption in the amorphous phase of PP.

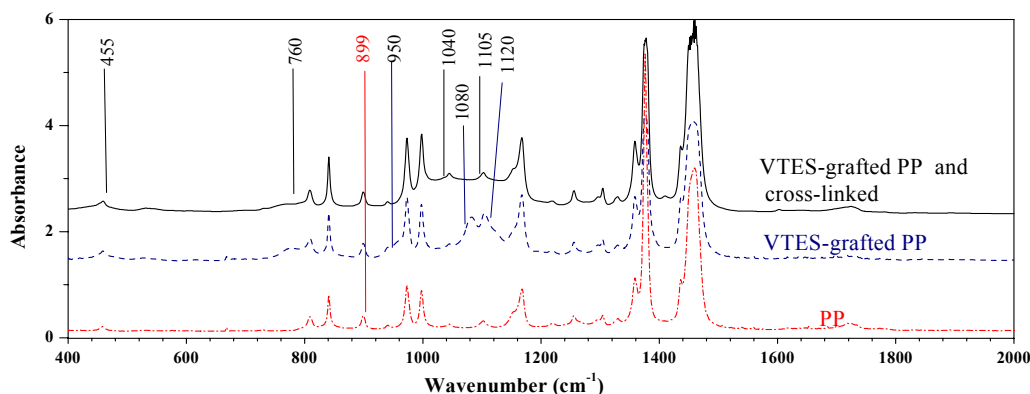


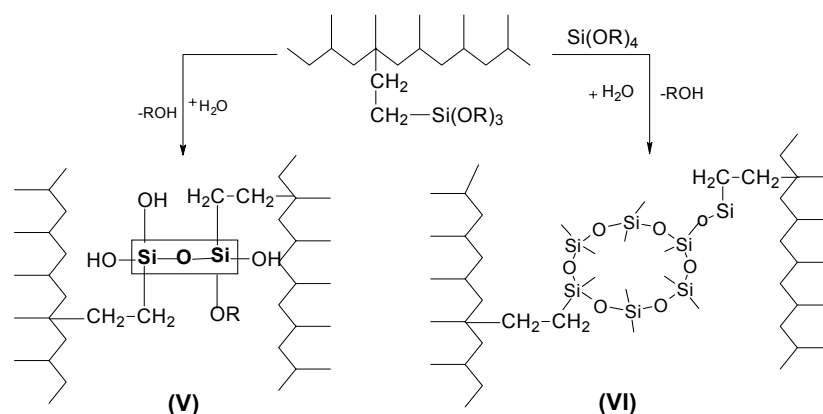
Figure 2.8: FT-IR spectra of VTES-grafted PP and cross-linked VTES-grafted PP.

The FT-IR spectrum in Figure 2.8 illustrates the grafting of VTES on PP via SSM. The band assignment details are given in Table 2.1. The strong Si-O-C stretching [35] vibration bands at 1120 cm^{-1} and 1080 cm^{-1} accompanied by a shoulder at 950 cm^{-1} are indicative for the grafting of VTES on PP. The band at $\sim 1600\text{ cm}^{-1}$ corresponding to vinyl groups $\text{CH}=\text{CH}_2$ of VTES is absent, so no unreacted monomer is present. The cumulative area under the 1120 cm^{-1} and 1080 cm^{-1} peaks is considered to be proportional to the amount of grafted VTES. The area of $-\text{CH}_3$ rocking band of PP at 899 cm^{-1} was used as reference.

The overlapping of various bands in the FTIR-spectrum assigned to the H-atoms attached to different carbons and the similarity in the structures formed make it difficult to identify whether VTES is grafted as monomer or polymer on the PP backbone. The rapid transfer of the tertiary hydrogen abstracted from PP to the free VTES radical may prevent the homopolymerization of VTES during modification of PP via SSM as explained by Rätzsch et al. for grafting of maleic anhydride in the concentration regime of interest [10]. This may suggest that VTES is grafted as a single monomeric unit on PP. The combined results from ^1H NMR and MAS ^{29}Si NMR on modified PP via SSM show that VTES is grafted as monomer on the PP backbone, even though VTES was found to homopolymerize by a free radical reaction mechanism in the absence of PP. No homopolymer of VTES was found in the pores of PP and $\text{T}_{1\rho}\{\text{H}\}$ experiments showed that VTES is present in the amorphous phase of PP. Therefore, VTES is most likely grafted as a monomeric unit on PP (*structure III* shown in Scheme 2) in the concentration range of VTES used for SSM.

2.3.1.2.3 Cross-linking of grafted monomer

The cross-linking of VTES-grafted PP, obtained via melt modification, is a well-studied process used in the cable industry [36]. The cross-linking of VTES-grafted PP involves the formation of Si-O-Si linkages similar those in silica formation (see Scheme 4). The grafted VTES undergoes hydrolysis followed by condensation in the presence of water vapor at 120°C forming (Si-O-Si) linkages, which cross-link the grafted PP chains. This reaction is usually carried out at $\sim 100^\circ\text{C}$ under a nitrogen atmosphere in a closed container. 3 moles of water per mole of grafted VTES is used for the cross-linking reaction. Moreover, if grafted VTES becomes part of the *in-situ* formed silica during the sol-gel reaction, the surface then resembles the cross-linked structures.



Scheme 4: Cross-linking (V) of PP through silane during solid-state modification; sol-gel reaction (VI) of making *in-situ* silica particles in premodified (VTES grafted) PP.

The FT-IR spectrum in Figure 2.8 shows the splitting of the bands at 1040 and 1080 cm^{-1} originating from the stretching of linear (aliphatic) Si-O-Si linkages formed during cross-linking [37]. The shift of transverse optical (νTO) components towards a lower wavenumber also indicates the presence of linear structures [38] as in cross-links. The 950 cm^{-1} band from Si-O-CH₂CH₃ stretching vibrations disappeared due to hydrolysis. Water treated VTES-grafted PP did not dissolve in commonly used good solvents for PP, such as xylene, toluene, tri-chlorobenzene, even when heated up to temperatures as high as 200°C , which is another indication that grafted VTES formed cross-links between the PP chains.

2.3.1.2.4 Silica synthesis in VTES-grafted PP

First, VTES is grafted on PP. The VTES monomer was dispersed in the pores and subsequently diffused into the amorphous phase of PP and then the initiator was added slowly. At the end of a reaction, the reactor was heated to higher temperatures to deactivate all the undecomposed initiator. Subsequently, the VTES-grafted PP was kept under vacuum at about 80°C for 24 hrs to remove all unreacted monomer and then stored in airtight desiccators to avoid absorption of moisture. The second step involves the sol-gel process. This can be combined with the first step in the same set-up or separately under other experimental conditions. This step is similar to the formation of silica described in Section 2.3.1.1.

The FT-IR spectrum in Figure 2.4 also shows the chemical structure of silica synthesized *in-situ* in the VTES-grafted PP. The spectra of PP/silica and (VTES-grafted PP)/silica nanocomposites show the effect of grafting on the chemical structure of silica. The peak at 1060 cm^{-1} assigned to the linear Si-O-Si linkage has shifted towards a lower wavenumber (1040 cm^{-1}) in the spectrum of (VTES-grafted PP)/silica. The peak at 1090 cm^{-1} , attributed to the Si-O-C stretching vibrations, has broadened. The broad shoulder at $\sim 1200\text{ cm}^{-1}$, assigned to νLO , is very weak. All these observations suggest that the grafted VTES takes part in the *in-situ* silica formation. The MAS ^{29}Si NMR spectra in Figure 2.5c show the structure of silica particles formed in the VTES-grafted PP. The grafted VTES (-45 ppm) evolves in the presence of water and TEOS into T^0 and T^1 structures. The Q-type structures ($\text{Q}^2 \sim -92\text{ ppm}$; $\text{Q}^3 \sim -101\text{ ppm}$; $\text{Q}^4 \sim -112\text{ ppm}$) in the spectrum are assigned to the *in-situ* formed silica. From ^1H NMR, FT-IR and MAS ^{29}Si NMR, it can be inferred that grafted VTES becomes part of *in-situ* formed silica. Pre-modification of PP gives better control over adhesion between matrix and *in-situ* formed nano-silica. Grafted VTES has become part of the

in-situ formed silica and, hence, will result in a strong chemical interaction with the polymer matrix.

2.3.2 Efficiency of silane grafting and silica formation

Table 2.3 shows the comparison of the data obtained through the various techniques used to quantify VTES grafting and the silica formed in the pores of PP. The results from TGA, elemental analysis and ^1H solution NMR are consistent, confirming the validity and reliability of these methods to quantify the composition of the nanocomposites. For VTES grafted on PP via SSM, the plot of grafted VTES versus concentration of monomer added shows that the degree of grafting increases initially and forms a plateau above a certain concentration of VTES (see Figure 2.9a). Other researchers observed a similar behavior for grafting non-homopolymerizable monomers [39,40]. The initial increase in grafting efficiency can be attributed to the increased availability of monomers at active PP radical sites with increasing monomer concentration. The solubility of VTES in the amorphous phase determines the grafting efficiency. The grafting efficiency did not decrease at higher concentrations of VTES, although this is commonly observed in easily homopolymerizable monomers such as styrene [10].

Table 2.3: Comparison of amount of monomers grafted/incorporated in iPP after solid-state modification by various analytical methods (wt% is expressed per hundred gram of PP)

TEOS wt%	VTES wt%	TGA wt%	Elemental Analysis wt%	^1H NMR wt%
10	0	2.2		
15	0	4.8	5.8	
20	0	6.5	6.9	
30	0	10.1	10.7	
0	15	1.6		1.5
0	18			2.8
0	20			3.7
0	25	4.2		4.3
10	15	3.6		
15	15	7.4	7.4	
15	20	8.7	8.0	
20	15	6.2		

In the case of the PP/silica nanocomposites the amount of TEOS converted to silica increased with increasing TEOS concentration. Unlike the grafting of VTES on PP, which takes place only in the amorphous phase of PP, silica can also be formed in the pores and is not necessarily restricted to the amorphous phase of PP.

So, the efficiency of silica formation remained constant irrespective of the TEOS concentration as seen in Figure 2.9b. A similar trend was observed for the (VTES-grafted PP)/silica nanocomposites, in which silica particles are formed in the VTES-grafted PP. Figure 2.9b shows the amount of silica formed in VTES-grafted PP as a function of the amount of TEOS added. For simplicity, only data for 2 wt% VTES-grafted PP are shown. It suggests that grafted VTES does not affect the formation of silica.

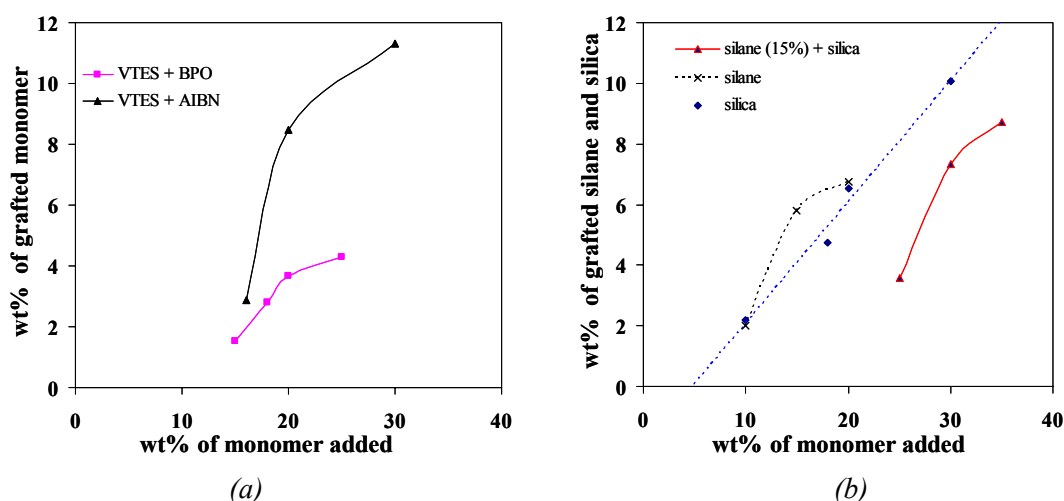


Figure 2.9: Plots of (a) VTES grafted on PP and (b) in-situ formed silica in pores of PP.

2.3.3 Molecular weight of PP in nanocomposites after SSM

Table 2.4 shows the molecular weights of VTES-grafted PP via SSM and PP in nanocomposites determined using high-temperature SEC. The data indicate that no molecular degradation occurred during SSM. The most likely reason is a drop in the rate of β -scission by 5 orders of magnitude as the reaction temperature is reduced from 250°C as used in melt modification to 100°C for SSM as reported by Rätzsch et al. [13]. In addition, the inert nitrogen atmosphere used during the reaction prevents the oxidative degradation.

Table 2.4: Molecular weight of PP nanocomposites (kg/mol) using high temperature SEC

Sample	M_n	M_w
PP as received	106	382
PP + silica (0.5 wt%)	106	387
(VTES-grafted PP) (2 wt%)	100	303
(VTES-grafted PP) (5 wt%)	106	357
(VTES-grafted PP) (10 wt%)	115	325

Based on this, the reaction mechanism is proposed as shown in Scheme 3, in which it is asserted that grafting of PP with VTES might proceed according to a simple ‘chain transfer mechanism’. The reaction temperature used in the present study is 110°C, which is lower as compared to 150°C used in earlier studies [10,41]. This may also have an effect on the molecular weight and the grafting efficiency. In some of these earlier studies, the grafting mechanism for SSM for similar monomers, such as maleic anhydride (MAH)-grafted PP, was proposed based on products of PP grafted in the melt. It was suggested that the grafting reaction takes place after PP chain scission. This assumption was further sustained by the remarkable decrease in molecular weight of the solid-state grafted PP. However, we did not observe such a behavior. This shows that SSM is advantageous over the melt modification process.

2.3.4 Structure and morphology of nanocomposites

The dispersion of the *in-situ* formed silica particles and their size and shape are the most important factors affecting the properties. The morphology of nascent PP powder before modification is also

important in that respect. As described earlier, the size of silica particles can be varied depending on the catalyst used. The base catalyst was used for the synthesis of the PP nanocomposites. ATR FT-IR microscopy was used, along with SEM and TEM, to study the dispersion of *in-situ* formed silica before melt processing the nanocomposites.

2.3.4.1 PP/silica nanocomposites

The FT-IR spectra depicted in Figure 2.10a are from the outer and inner surface of PP/silica nanocomposite macroparticles. The ratio of 899 cm^{-1} band of PP and 1090 cm^{-1} peak assigned to Si-O-Si is taken as the measure of silica formed. The ratio on the surface and inside the powder is approximately same. This suggests that the formation of silica is uniform throughout the macroparticle of PP. TEOS, when mixed with PP powder, is dispersed throughout the particle due to surface tension and capillary effect in the channels. When water is added in the presence of a catalyst, the sol-gel reaction proceeds in the channels and at the surface.

2.3.4.2 (VTES-grafted PP)/silica nanocomposites

Figure 2.10b shows the FT-IR spectrum of VTES-grafted PP powder taken from the reactor and vacuum dried to remove unreacted VTES. The PP powder was characterized in the same way as described above. It is evident that VTES is grafted not only near the surface but also deep in the powder particle. The spectra indicate that grafting occurs even more inside than on the surface (see peaks at 1080 and 1105 cm^{-1}). This may be due to the higher surface tension of VTES and the capillary effect of the pores and preferential absorption of the monomer in the amorphous phase of PP. Also, the grafting reaction is carried at higher temperatures than sol-gel reaction, which enhances the diffusion of monomer into the pores. Figure 2.10c shows that the *in-situ* formed silica-particles are uniformly dispersed in the VTES-grafted PP as well. The grafted VTES may act as nucleus for the growth of silica as observed by Young et al. in the case of Surlyn-silica systems [21].

2.3.4.3 Shape, size and size distribution of in-situ formed silica particles in PP

The size and structure of the silica particles formed in the PP were characterized after oxidation of the nanocomposites in TGA at 800°C. The residue from TGA was characterized with SEM. Silica particles synthesized in the absence of PP were also oxidized to check the effect of oxidation on the particles. By comparing Figure 2.3a and 2.11a (before and after oxidation, respectively), it can be observed that the oxidation of silica does not affect the sizes and the structure of primary particles. Figures 2.11b and 2.11c show the silica formed in PP/silica and (VTES-grafted PP)/silica nanocomposites with typical sizes of 20-100 nm after oxidation.

The morphology as observed by TEM shows a narrow distribution of the particle sizes of silica (see Figure 2.12). The particle sizes near the surface and the center of the PP macroparticle were found to be similar. Figure 2.12a and 2.12b give a magnified view of the *in-situ* formed silica dispersion at the edge and in the center of the PP macro-particle respectively. Silica nano-clusters formed at the edge (ca. 30-100 nm) are similar to those inside the macroparticle PP (ca. 50-100 nm). The morphology was stable even after mixing in a mini-twin screw extruder. Figure 2.12c shows the silica nano-clusters formed in (VTES-grafted PP)/silica nanocomposites. The size and shape of silica formed is comparable to that formed in PP/silica nanocomposites.

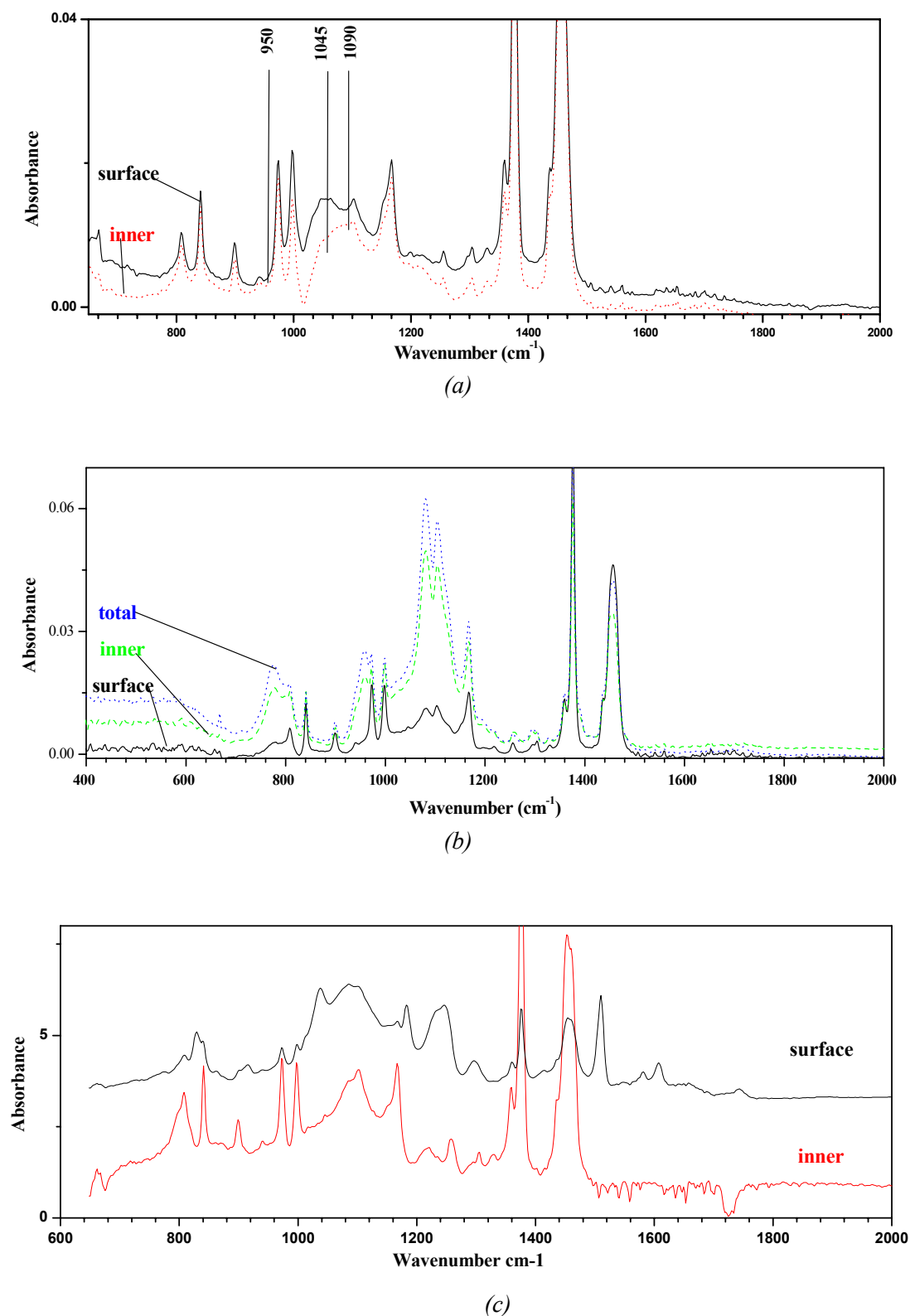


Figure 2.10: FT-IR spectra (in ATR mode) showing the inner and outer surfaces of (a) PP/silica, (b) VTES-grafted PP, (c) (VTES-grafted PP)/silica nanocomposites.

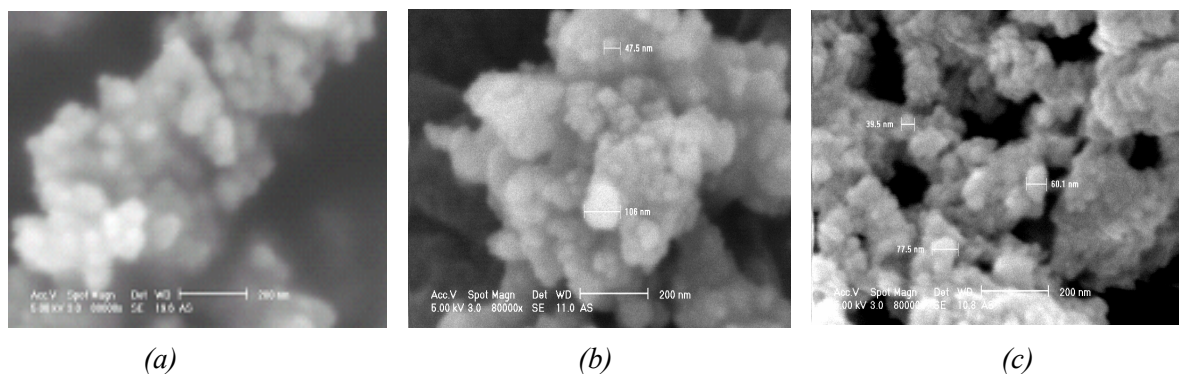


Figure 2.11: SEM micrographs of silica after oxidation in TGA: (a) silica prepared as such, (b) PP/silica nanocomposites, (b) (VTES-grafted PP)/silica nanocomposites.

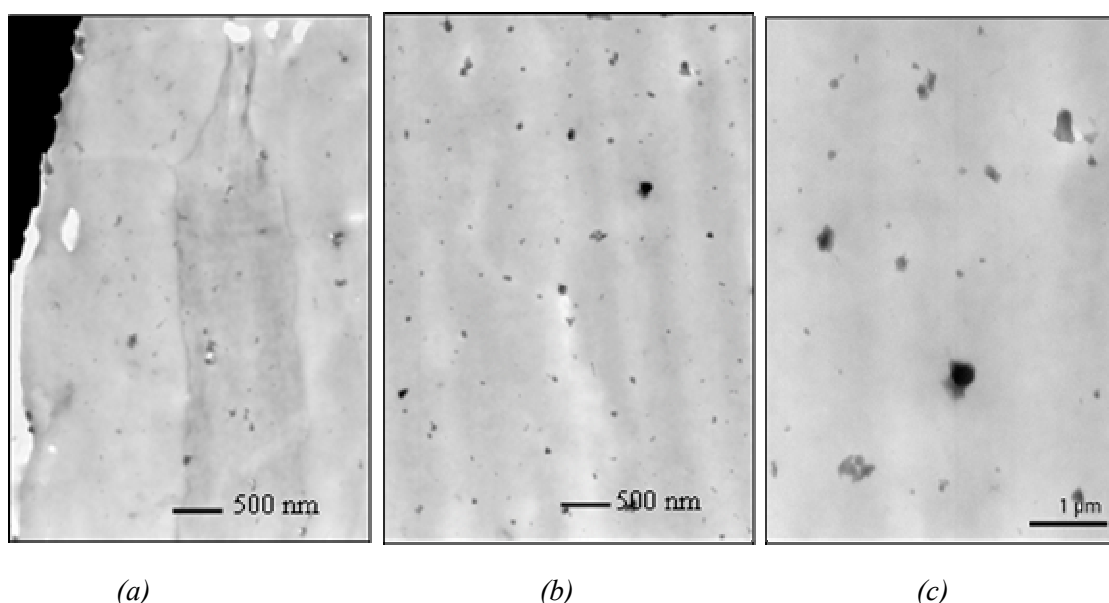


Figure 2.12: TEM micrographs of (in-situ formed) PP/silica nanocomposite (5% of silica): (a) edge of particle, (b) center of a particle and (c) (VTES-grafted PP)/silica nanocomposites.

2.4 Conclusions

PP/silica and VTES-grafted PP/silica nanocomposites were prepared by a new approach combining SSM and a sol-gel chemistry with the idea that grafted monomeric VTES can be incorporated in the silica during the sol-gel reaction to control the interaction. *In-situ* formed silica particles are formed throughout the PP particle. MAS ^{29}Si NMR showed that the silica-like clusters are predominantly Q^3 and Q^4 type, indicating that the *in-situ* formed clusters are highly condensed with some hydroxy groups at the surface. For comparison, bulk polymerization experiments, under the same experimental conditions showed the homopolymerization of VTES to high molecular weight occurs but at limited conversions. This suggests that VTES can be grafted on PP as both monomer and polymer. However, ^1H NMR, MAS ^{29}Si NMR and FT-IR spectroscopy studies demonstrated that VTES grafts on PP as a single monomeric unit which is subsequently incorporated in *in-situ* formed silica. $\text{T}_{1\rho}\{\text{H}\}$ measurements via cross polarization to the ^{13}C and ^{29}Si nuclei in solid-state indicated that VTES was mainly absorbed and subsequently grafted in the amorphous phase of PP and not at the VTES-filled pore-PP interface. The ATR-FTIR microscopy results of PP

macroparticles suggest that VTES is grafted not only on the surface of PP macro-particles but also inside the macro- and sub-particles. No molecular degradation of PP was observed in the nanocomposites and the degree of grafting was comparable to grafting degree in the melt. Particles of 50-150 nm with a near-spherical shape were observed by TEM and SEM. The observation that silica particles are uniformly dispersed proves that the new approach is advantageous over conventional methods.

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Chapter 3

The influence of selective physisorption on the rheology of PP/silica nanocomposites^{*}

Abstract

Confinement and surface effects provided by nano-particles have shown to have macroscopic effects and although the effect on viscosity has not been studied extensively, it is expected that it will also be affected by the presence of nano-particles. We observed an unusual decrease in the dynamic and steady-state viscosity of in-situ prepared silica nano-particle filled polypropylene nanocomposites. With varying filler concentration both the dynamic and the steady-state viscosity show a distinct minimum. This new additives approach challenges the standard molecular weight dependence of viscosity-rule for polymers by altering the melt structure through “selective physisorption of high molecular weight polymer chains” on the surface of the nano-fillers, thereby dramatically reducing the viscosity of the system. This particular rheological behavior is accompanied by strong effects on the crystallization behavior during flow and this has major implications for processing and for the final product properties. However, this effect disappears when there is strong chemical interaction between nano-filler and polymer matrix.

3.1 Introduction

Adding solid particles to a fluid will increase the viscosity of that fluid, at least, that is the usual expectation. For low volume fractions of particles added to Newtonian fluids, this increase of the viscosity is described by the well-known expression of Einstein [1]. With some modifications, this model has been also used for polymeric fluids and it has been confirmed experimentally many times [2]. However, recently Siegel and Mayo [3] described how mesoscopic physics can change the macroscopic properties of materials in a different way.

Nano-particles provide a high surface area and induce confinement effects. The confinement is relative motion of polymer chains (chain dynamics). At low temperatures and lower relaxation times polymers do not show any confinement, because movements occur at a length scale smaller than the distance between the nano-particles [4]. Effects due to the addition of nano-particles to a material can cause change in various properties which can not be explained by traditional models. Table 3.1 gives an overview of the literature on rheology of nano-particle filled polymers. Although interpretations are different, all the observations made in these studies point to unusual effects that nano-scale particles can cause on macroscopic scale.

^{*} Part of this chapter is reproduced from: Jain, S.; Goossens, J.G.P.; Peters, G.W.M.; van Duin, M.; Lemstra, P.J. “Nanoscale events with macroscopic events: Rheology of PP/silica nanocomposites” to be submitted to *Nature Materials*

Table 3.1: Overview of literature reporting on special rheological behavior of nano-particle filled polymers. Various mechanisms have been proposed to explain this behavior.

System		Effect	Explanation
Dilute solution			
Otsuba [5]	Suspension of silica particles in polyacrylamide (PMMA) solution	Decrease in viscosity at lower concentration; viscosity unstable at higher concentration	Adsorption of polymer chains on silica surface
Semi-dilute solution			
Horigome and Otsuba [6]	Polystyrene beads in polymer solution	Increase in viscosity with Newtonian behavior at low shear rate	Flocculation by polymer chains with associated functional groups forming bridges that constantly form and break due to thermal energy
Vignaux-Nassiet et al. [7]	Polybutadiene/ (micrometer sized) silica emulsions	Increase in viscosity	Adsorption of polybutadiene on silica surface
Melt			
Robert et al. [8]	Blends of silica clusters with linear poly(dimethylsiloxane)	Decrease in viscosity	Molecular scale particles less than 0.7 nm cause plasticizing effects.
Mackay et al. [9]	Suspended polystyrene (PS) nano-particles in linear PS	Decrease in viscosity	Local changes in free volume
Xie et al. [10]	PVC/CaCO ₃	Slight decrease in viscosity	Ball bearing effect
Paul et al. [11]	Nylon/clay	Decrease in viscosity	Molecular degradation and slip at wall
Zhang and Archer [12]	PEO/silica	Increase in viscosity	Adsorption of PEO on silica
Hoffman et al. [13]	Polyamide-12/ layered silicate	Increase in viscosity	Superstructure in the melt
Ottinger et al. [14]	HDPE/calcite	Massive increase in shear viscosity	Particle-particle and particle-matrix interactions

In contrast to the increase in the viscosity reported in some of these studies, we observed for low filler loadings a strange rheological behavior. Surprisingly, no one has considered filled systems with nano-particles, i.e. in the range of radius of gyration of polymers, for which overlapping and bridging effects don't apply. The main objective of this study is to determine how the nanoparticle-polymer interactions affect the macroscopic rheological properties and the flow-induced crystallization of nanocomposites melts.

3.2 Experimental

3.2.1 Materials

Porous isotactic polypropylene (PP) having a porosity of $S_{BET(Kr)} = 0.0678 \text{ m}^2/\text{g}$ was supplied in powder form by Euro-Sabic, The Netherlands. The number-average and weight-average molecular weight of PP was 60 and 380 kg/mol, respectively, with polydispersity index of 6.2. The powder was free of antioxidants and stabilizers and stored at -10°C in the dark. Tetraethoxy orthosilicate (TEOS) and ammonium hydroxide (28% NH_3) were obtained from Aldrich Chemicals.

3.2.2 Sample preparation

PP/silica nanocomposites were prepared in-situ using the new procedure as described in Chapter 2, Section 2.3.1.1. The samples were extruded in a twin-screw mini-extruder with a residence time of 5 min at 190°C ; the rotor speed was set at 50 rpm. The extruded samples were melt-pressed in sheets at 180°C and 100 bar for about 10 min using a Collin Compression Press. From these sheets 8 mm and 25 mm diameter discs were cut for rheological measurements.

3.2.3 Measurement of surface area and pore volume

The specific surface area and pore volume (distribution) of silica samples prepared in the absence of PP were investigated by measuring the full nitrogen adsorption isotherm at -195.7 to -195.8°C . These measurements were performed at DSM Research using an automated surface area and pore structure analyzer, the ASAP 2010 from Micromeritics. Samples were evacuated for 16 hours at 120°C in the analysis equipment prior to the adsorption measurements. BET surface area was calculated using the linear form of the BET equation. Excellent linearity was found for these samples when using the part of the nitrogen adsorption isotherm from $p/p^0 = 0.06$ to 0.20 . The molecular cross-section of N_2 was taken as 0.1620 nm^2 . For one sample, the adsorption isotherm of Ar was measured at the temperature of liquid Ar, -185.5°C [15].

3.2.4 Rheology

Small-amplitude oscillatory shear studies using frequency sweeps at a constant strain were conducted on a strain-controlled rheometer (RMS 800, Rheometric Scientific). Most studies were performed at 180°C , although linear viscoelastic measurements were conducted over the range of 160 – 200°C for time-temperature superposition. Some of the measurements were repeated on another strain-controlled rheometer (ARES rheometer, Rheometric Scientific) at 170°C for checking the reproducibility.

3.2.5 Small-Angle X-ray Scattering (SAXS)

In-situ flow-induced crystallization studies with simultaneous SAXS/WAXS experiments were performed on the DUBBLE beamline (BM 26B) at the European Synchrotron Research Facility (ESRF), Grenoble, France using a wavelength of $1.218 \text{ \AA}$. SAXS patterns were recorded using a gas-filled multiwire two-dimensional (2D) detector [16] with a sample-to-detector distance of 2.5 m. For calibration of the SAXS detector, the scattering pattern from an oriented specimen of wet collagen (rat-tail tendon) was used. The experimental data were corrected for background scattering, i.e. subtraction of instrumental factors.

A Linkam shear cell, Type CSS 450, was used to apply a well-defined thermo-mechanical history to the materials. The nanocomposites were mounted between two X-ray transparent windows (Kapton) in the form of $500 \text{ }\mu\text{m}$ thick discs. The gap distance between the two windows was adjusted to the desired sample thickness. The crystallization temperature used was 135°C so that the nucleation and crystal growth times under quiescent conditions are sufficiently long. The following experimental temperature profile was used: (1) heating from room temperature to 220°C at 30 K/min ; (2) holding the samples at 220°C for 5 min to eliminate memory effects or any residual crystallinity before the test; (3) cooling at 30 K/min to 135°C ; (4) holding the temperature at 135°C ($\pm 1^\circ\text{C}$) for 30 min for X-ray measurements. Two-dimensional SAXS images were collected continuously: before, during and after cessation of the applied shear. First, three frames were collected immediately after reaching 135°C . Next, a shear rate of 60 s^{-1} was applied for 5 s. 200 SAXS images were collected of which the acquisition time for the first 40 frames was 3 s, for the next 80 frames it was set at 6 s and for the remaining 80 frames it was 10 s with a dead time of 5 s. The scattering images represent the average scattering intensity from the scattering structures across the thickness of the sample. A quantitative analysis of the orientated fraction is performed using the procedure described by Somani et al. [21]. The total scattered intensity (I_{total}), due to the contribution of all the scatters in a sample, is deconvoluted into two components: $I_{\text{unoriented}}$, which is due to scattering from the randomly distributed

crystalline lamellae, and I_{oriented} , which comes from the oriented lamellae. To evaluate the orientation of the patterns an orientation fraction P_{or} was introduced as the ratio between the intensity due to the oriented crystals and the intensity from isotropic crystals (for an objective comparison both intensities were integrated over an equal angular area): $P_{\text{or}} = I_{\text{oriented}}/I_{\text{isotropic}}$.

3.2.6 Solid-state NMR

The measurements were carried out statically using a Bruker DMX spectrometer operated at 500 MHz. The transverse relaxation time (T_2) was measured using a two pulse sequence $90^\circ\text{-}\tau\text{-}180^\circ\text{-}\tau\text{-aq}$ with a variable τ time starting from $\tau = 2\mu\text{s}$. The 90° pulse length was $5\mu\text{s}$ and the repetition time was 3 seconds. The relaxation decay was characterized by 33 data points at appropriately chosen times and the data were analyzed by a non-linear least-square fitting.

3.3 Results

PP/silica nanocomposites were prepared in-situ via a combination of solid-state modification and sol-gel reaction leading to silica particles that are well dispersed in the PP matrix. The concentrations of silica nano-particles used are quite low compared to previous studies. The geometry of silica nano-particles also influences the interaction between PP-silica nano-particles. The mesoporous spherical silica nano-particles of the order of 20 nm with a narrow polydispersity were synthesized in-situ and confirmed by X-ray scattering and TEM. The TEM picture in Figure 3.1 shows the silica aggregates prepared via sol-gel reaction in the absence of PP. Porosity measurements on silica prepared via the sol-gel method (in the absence of PP) shows the meso- and micro-porous nature of these particles. The BET surface area amounts to $140\text{ m}^2/\text{g}$. The pore size distribution shows a clear maximum in the mesopore range at a pore radius of approx. 10 nm.

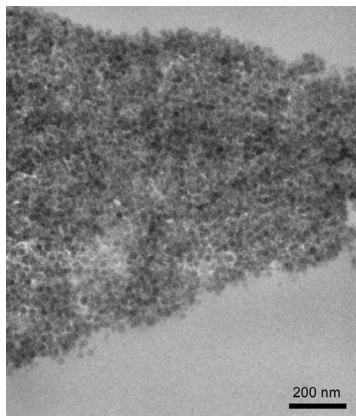


Figure 3.1: TEM of aggregates of silica prepared (ex-situ) by the sol-gel reaction.

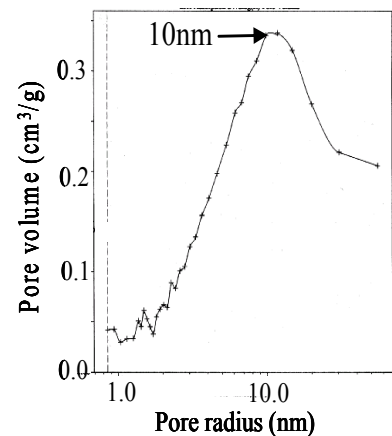
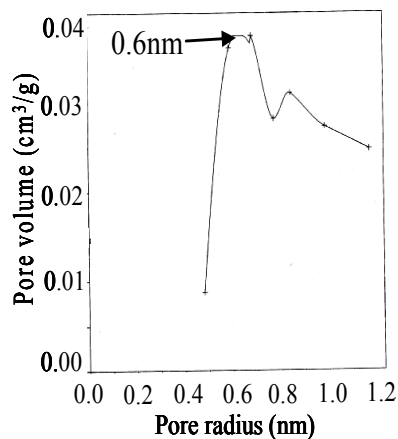


Figure 3.2: Horvath-Kawazoe differential pore volume plots for silica particles prepared via a sol-gel reaction.

The estimated total micropore volume (pores $< 2\text{ nm}$ diameter) is about $0.02\text{ cm}^3/\text{g}$. The distribution (maximum at approx. 0.6 nm) strongly suggests that pores smaller than 0.4 nm radius are present but cannot be entered (also not by Ar). The pore size distribution is shown in Figure 3.2. The average pore diameter calculated from $2V/S$, where V is the total pore volume and S is the BET surface area, 5.9 nm. A maximum is also observed at 10 nm, but this may be attributed to inter-particle pores as silica was in the form of aggregates rather than pores inside individual particles.

Since PP is apolar, it has less affinity to silica. However, the hydroxy groups at the silica surface and the porosity of silica nano-particles may induce some interaction between PP and silica.

3.3.1 Rheological behavior of PP/silica and UHMWPE/silica nanocomposites

An oscillatory, small strain experiments were performed to measure the linear viscoelastic properties, henceforth termed as dynamic properties. Applying a sinusoidal strain rate: $\dot{\gamma} = \gamma_0 \omega \cos \omega t = \dot{\gamma}_0 \cos \omega t$ and measuring the stress $\tau = \tau_0 \sin(\omega t + \delta)$ yields two dynamic moduli, the in-phase or storage (elastic) modulus (G') and the out-of-phase or loss (viscous) modulus (G''):

$$G'(\omega) = \frac{\tau_0''}{\gamma_0} \quad \text{and} \quad G''(\omega) = \frac{\tau_0'}{\gamma_0}$$

The moduli are a function of the frequency ω only. The dynamic viscosity, η' , and elastic part of the linear viscoelastic viscosity, η'' , are defined as:

$$\eta'(\omega) = \frac{G''}{\omega} \quad \text{and} \quad \eta''(\omega) = \frac{G'}{\omega}$$

The magnitude of complex viscosity is given by:

$$|\eta^*| = (\eta' + \eta'')^{1/2} = \left[\left(\frac{G''}{\omega} \right)^2 + \left(\frac{G'}{\omega} \right)^2 \right]^{1/2} = \frac{1}{\omega} |G^*|$$

The linear range of the viscoelastic behavior was determined by varying the strain level. Figure 3.3 shows the strain dependence of the storage modulus for a 0.3 wt% silica filled PP and pure PP. The PP/silica nanocomposites show a linear behavior over the whole range of strains (γ) applied. PP, on the other hand, shows a deviation from linear behavior above 25% strain. Therefore, all small-amplitude oscillatory shear experiments were conducted in the interval $0.1\% < \gamma < 10\%$.

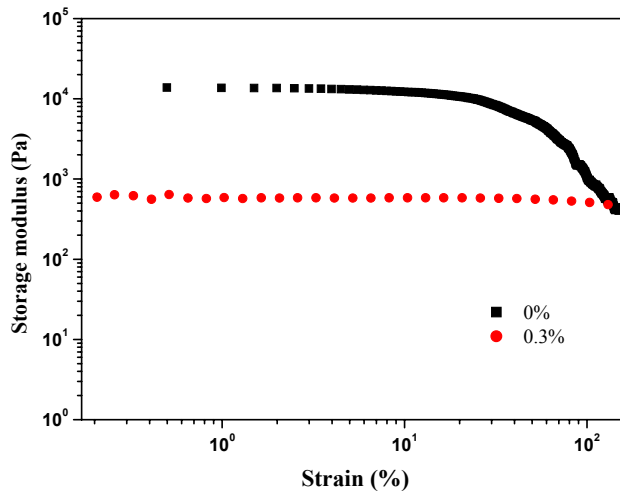


Figure 3.3: Strain dependence of the storage modulus at a frequency of 1 rad/s measured at 180°C. The legend indicates wt% of silica in PP.

Dynamic properties

The dynamic properties of the nanocomposites were measured for a range of silica concentrations. No time-temperature superposition was applied in order to avoid the crystallization or the formation of nuclei, if any, at lower temperatures. Figure 3.4a shows the dynamic viscosity curves for different compositions. The first striking observation is the large decrease (> 10 times!!) in the

dynamic viscosity with increasing silica concentration up to approx. 0.5 wt%, after which it increases again. This is made even clearer in the Figure 3.4b which gives the low frequency values of the dynamic viscosity as a function of silica concentration. Notice that the frequency range over which the dynamic viscosity is constant correlates with the decrease in the dynamic viscosity. The reciprocal value of the frequency at which the transition from a constant to a decreasing dynamic viscosity takes place is a characteristic relaxation time for the given composite. This relaxation time decreases considerably (~ 10 times) with increasing silica concentration up to approx. 0.5 wt%, after which it increases again.

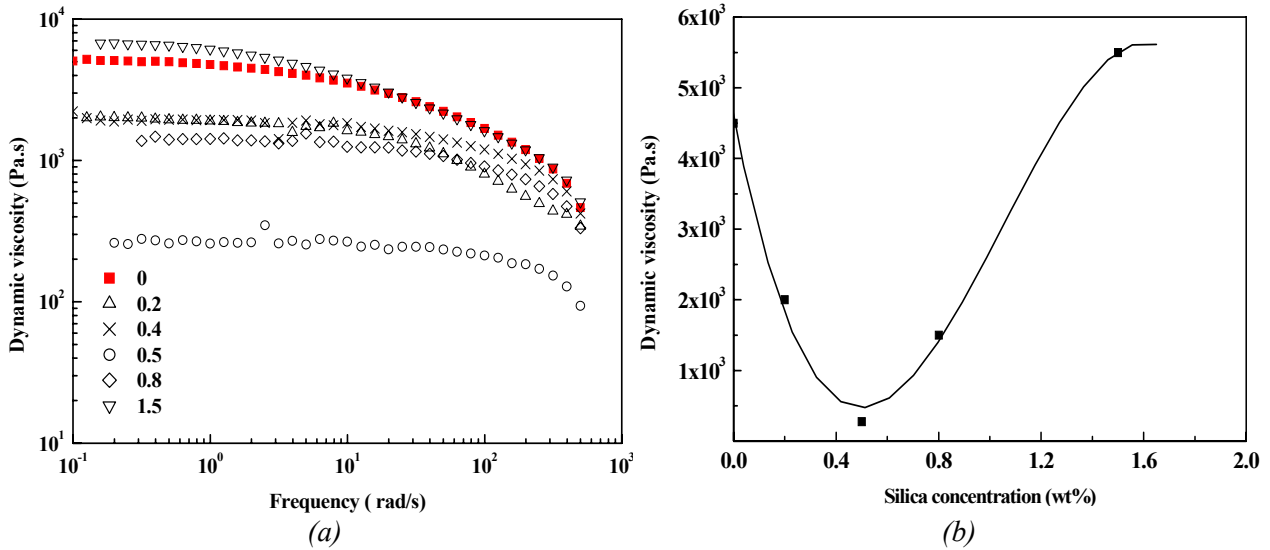


Figure 3.4: (a) Dynamic viscosity against frequency at 1 % strain and at $T=180^\circ\text{C}$. In all figures the legends indicate the wt% of silica in PP, (b) Decrease in the dynamic viscosity ($\omega=1$ rad/s) with increasing silica concentration. The line is to guide the eye.

The phase angle δ expresses the way the material behaves for given frequency ($\delta = 90^\circ$: Newtonian fluid, viscous behavior, $\delta = 0^\circ$ elastic behavior). The phase angle is defined as:

$$\delta = \tan^{-1} \left(\frac{G''(\omega)}{G'(\omega)} \right).$$

For linear polymers, the phase angle is close to 90° in the plateau or zero-shear rate regions. For the PP/silica nanocomposites, the phase angle remains constant over wide range of frequencies as depicted in Figure 3.5a, especially for concentrations where the viscosity shows a minimum. It is seen that the 0.5 wt% composite behaves most viscous, while the pure PP and the 1.5 wt% composite behave most elastic, i.e. the phase angle drops immediately and has the lowest values of all compositions over the full experimental frequency range.

Figure 3.5b shows the storage modulus of the PP/silica nanocomposites. Note the appearance of an intermediate elastic plateau in the low frequency region for the intermediate concentrations. This points to additional relaxation processes. These plateaus shift to lower frequencies with increasing silica concentration. This was also observed by Vignaux-Nassiet et al. [7] for polybutadiene/ (micrometer-sized) silica emulsions. In this context, the presence of an intermediate plateau indicates that polymer chains are adsorbed on the silica nano-particles and the shift of the intermediate plateau towards the lower frequencies suggests the increase in thickness of adsorbed layer with increasing silica nano-particles concentration.

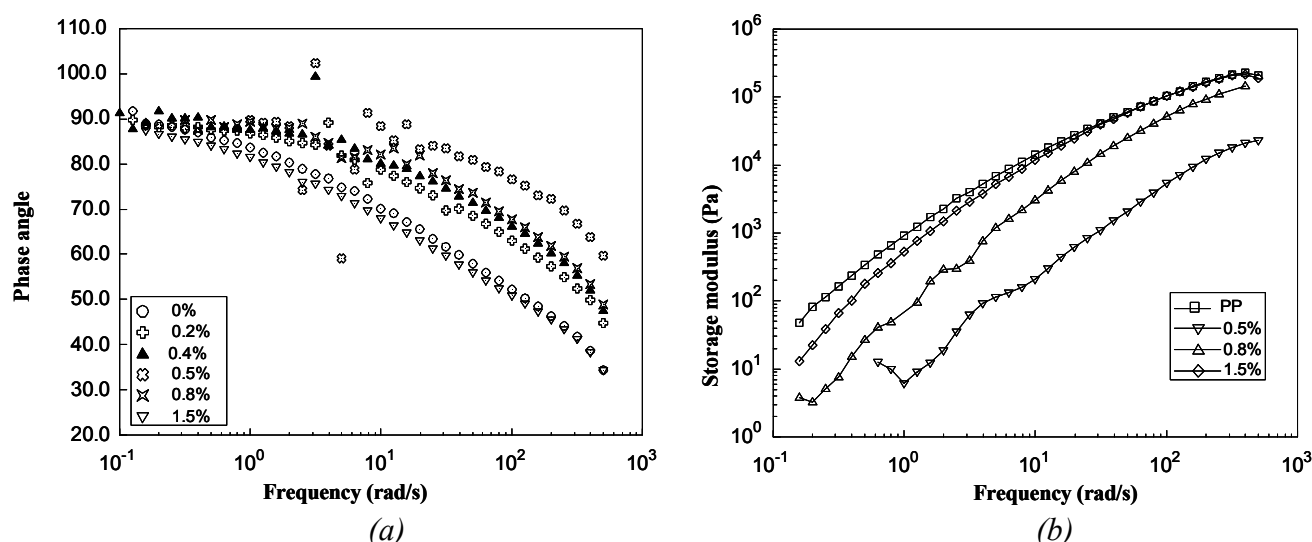


Figure 3.5: (a) Phase angle vs. frequency, (b) Storage modulus vs. frequency at 180°C

Steady-state properties

Steady-state viscosities were measured at the same temperature at which the dynamic properties were measured ($T = 180^\circ\text{C}$) and the results are given in Figure 3.6a. The same trends are observed as found in the dynamic measurements: a large drop in viscosity up to concentrations of 0.5 wt% and the corresponding decrease in the characteristic relaxation time. These results reveal one more feature. For all PP/silica nanocomposites except for the one with 0.5 wt% silica, the shear thinning behavior is followed by shear thickening which changes into shear thinning again at higher shear rates.

For pure polymer melts a simple, yet empirical relation between the steady shear viscosity $\eta(\dot{\gamma})$ and the dynamic viscosity $\eta^*(\omega)$, called “Cox-Merz rule”, often holds fairly well at high shear rates [17]. This rule states that the shear rate dependence of the steady-state viscosity $\eta(\dot{\gamma})$ is equal to the frequency dependence of the linear viscoelastic viscosity $\eta^*(\omega)$. It is clear that the PP/silica nanocomposites prepared in this study do not obey the Cox-Merz rule (see Figure 3.6b). This deviation is typical for suspensions of solid spherical particles. In the limiting case of spheres without inter-particle interactions, only Brownian (thermal) and hydrodynamic forces influence the flow behavior. At sufficiently low shear rates, the motions of the particles are dominated by Brownian motion and equilibrium structures that exist at rest are preserved during flow. As a result, the viscosity does not change with shear rate and the contribution of Brownian forces to the viscosity is maximum.

When increasing the shear rate than certain level, the particle motion is dominated by convective motion and the contribution of the Brownian motion to the viscosity gradually decreases until only a contribution of the hydrodynamic forces remains. The reduction of the Brownian contribution leads to a reduction in the viscosity and a shear thinning region develops. At still higher shear rates, the Brownian contribution becomes negligible but an increase of the hydrodynamic effects can either compensate, causing a pseudo-Newtonian high-shear plateau, or overcompensate, resulting in a shear-thickening zone. Otsuba [18] observed a similar behavior for the steady-state viscosity, as observed in the present study, in a suspension of styrene-(methyl acrylate) copolymer particles in poly(vinyl alcohol) and water with Triton X-100 (ethoxylated octylphenol) as a surfactant. The shear-thickening behavior appeared when the time scale of extension of poly(vinyl alcohol) is

shorter than that of desorption from the particle. This may explain why the nanocomposites with 0.5 wt% silica nano-particles neither show a shear thinning nor thickening behavior.

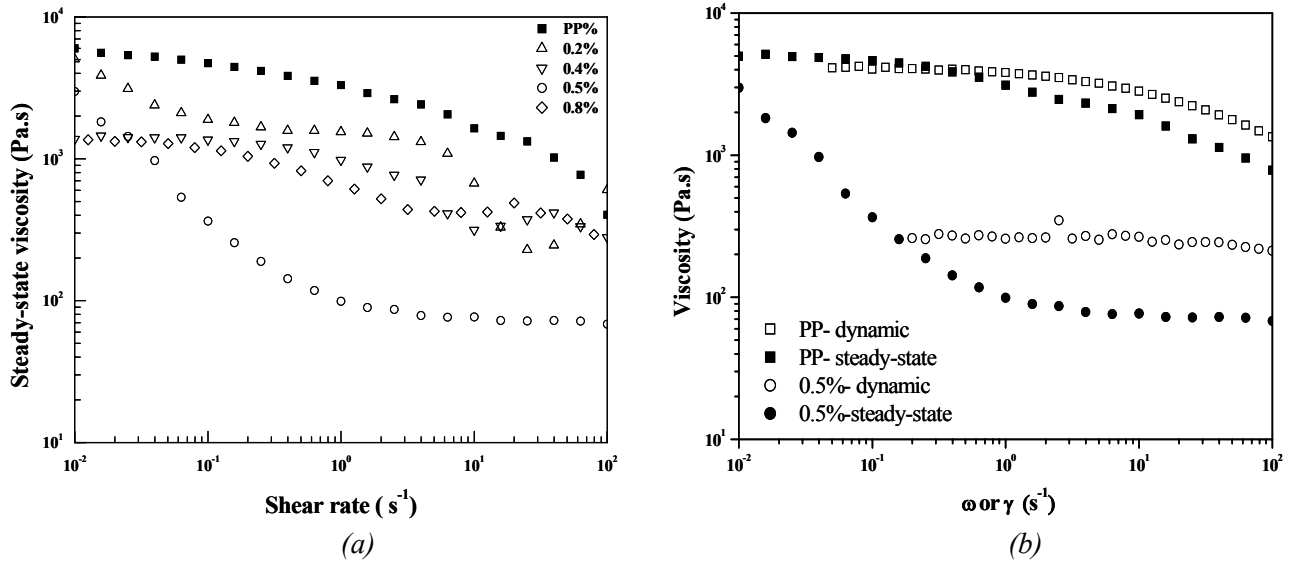


Figure 3.6: Steady-state viscosity against frequency of PP/silica nanocomposites at 180°C. Note the shear thinning behavior at intermediate shear rates (~ 10 - 100 s^{-1}) followed by shear thickening at higher shear rates ($> 100 \text{ s}^{-1}$). Pure PP does not show such a behavior.

3.3.2 Shear-induced crystallization

It is well known that the crystallization of semi-crystalline polymers can be strongly influenced by flow. The additional effect of nano-particles on this behavior is not known. Therefore, in addition to the rheological experiments, shear-induced crystallization experiments were performed. Flow orients molecules, especially the high molecular weight chains. The extent of orientation depends on the strength of the flow, the molecular weight (distribution) and the temperature. It is thought that this behavior is mainly determined by high-end tail of the molecular weight distribution (MWD) [19,20,21]. The flow-induced crystallization (FIC) was studied using SAXS. Figure 3.7 shows a series of two-dimensional SAXS patterns of pure PP and PP nanocomposites at $T = 135^\circ\text{C}$, before and after application of shear. The SAXS images at time $t = 0$ correspond to the amorphous melt and they show a weak, diffuse scattering, indicating the absence of any detectable structure or orientation. The scattering patterns obtained after 180 s clearly show the appearance of meridional maxima for the pure PP and all composites except for the one with 0.5 wt% silica. These maxima, observed after applying shear, originate from oriented structures in the melt. The intensity of the maxima grows as a function of time due to growth of oriented lamellar structures and after a certain time reaches a plateau.

These measurements show that the flow-induced crystallization of nanocomposites is faster compared to pure PP and the nanocomposite with 0.5 wt% silica *does not show any orientation*. The calculated orientated fractions are shown in Figure 3.8. The orientation fraction increases with the incorporation of silica nano-particles until at a certain concentration it starts to decrease, then goes through a minimum and finally increases again. The increase in orientation in case of PP/silica nanocomposites can be attributed to the locally increased stress due to the presence of the particles.

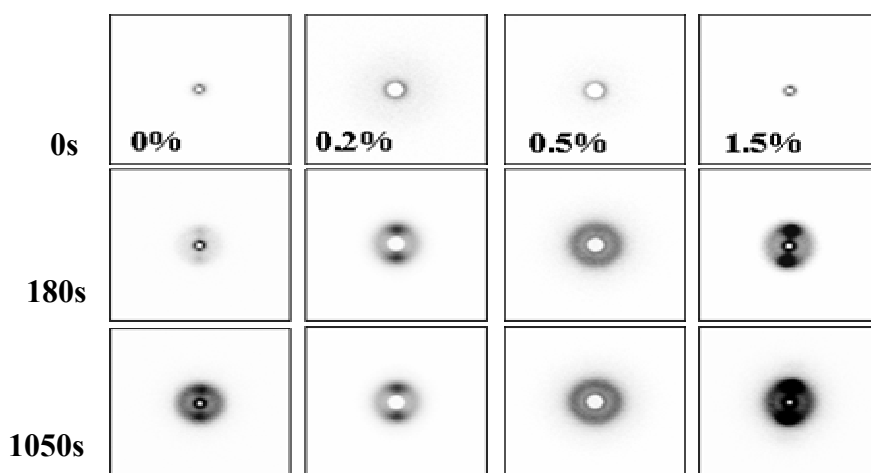


Figure 3.7: 2-D small-angle X-ray Scattering (SAXS) images of PP/silica nanocomposites after shearing at 60 s^{-1} for 5 s at 135°C .

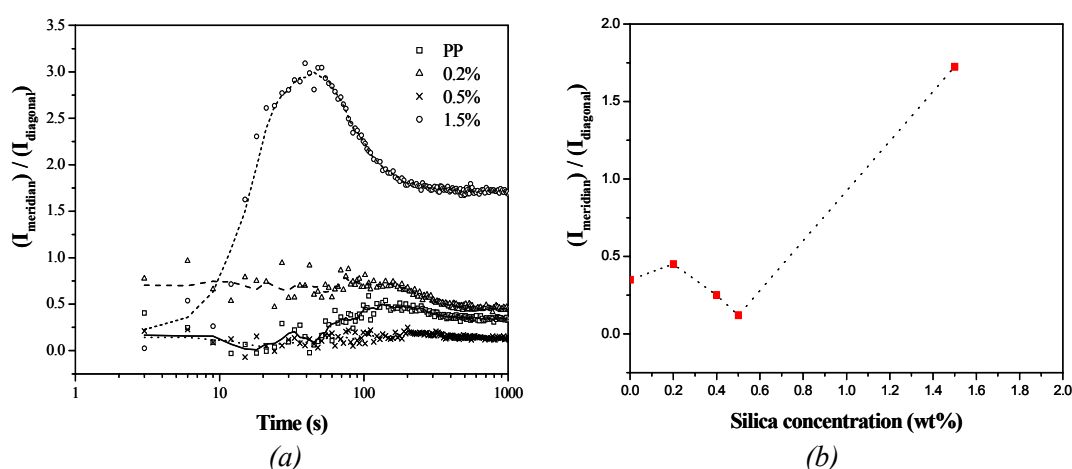


Figure 3.8: (a) The orientation fraction as a function of time, calculated from the ratio of the scattered intensity from the oriented structures to the total scattered intensity. (b) The orientation fraction plotted as a function of silica concentration.

3.4 Discussion

In previous studies on the rheological behavior of nanocomposites, a decrease in dynamic viscosity was attributed to either molecular degradation [11,22] or slip between the polymer chains and the filler particles [23]. Another possibility might be slip at the wall, caused by the nano-particles, when using a capillary rheometer. The molecular weight of the PP in the nanocomposites, determined by HT-SEC, after the rheological measurements, 350 kg/mol, is more or less equal to that before the measurements (380 kg/mol) and that of pure PP (380 kg/mol) indicating that no molecular degradation occurred. There is only a marginal chain scission during the nanocomposite preparation and during the rheological measurements but in view of the effect on the viscosity this is negligible. All rheological measurements were conducted using a plate-plate configuration in contrast with the capillary rheometer used in previous studies. This minimizes the possibility of slip at the wall during measurements. Moreover, measurements were performed by changing the gap (thickness) between the plates (viz. 0.5 mm and 1 mm) and same results were obtained. This rules out that the observed viscosity effect is due to the slip at wall. The results are reproducible and consistent even when measurements were conducted at different temperatures and instruments. This implies that the

effect of silica nano-particles on the viscosity is neither due to the molecular degradation nor due to slip at the wall.

3.4.1 A hypothesis based on selective physisorption and interparticle distance

As mentioned earlier, a few explanations have been given in the literature for the observed drop in the viscosity of nano-particle filled polymer melts. Mackay et al. [9], for instance, blended organic nano-particles, synthesized by intramolecular cross-linking of single polystyrene chains, with linear polystyrene. The blend viscosity was found to decrease and they attributed the decrease in the viscosity to excluded free volume induced around each nano-particles based on the work of Merkel et al. [24]. Merkel et al. found that fumed silica-induced disruption of polymer chain packing and an accompanying subtle increase in free volume through which molecular transport occur. However, it is found in this study that the silica nano-particles act as nucleating agent (see Chapter 4). PP chains start to crystallize on silica surface, so the depletion layer cannot be formed around silica. Therefore, silica does not increase the free volume in the system. Hence, the explanation based on increase in free volume does not hold for the decrease in the viscosity, at least, in this case. Robert et al. [8] explained the small drop in the viscosity of blends of small silica clusters (radius ≈ 0.35 nm) with linear polymers by means of a plasticization effect. However, in the present study, the size of nano-particles is more than 10 nm, so similar arguments do not hold. The explanation proposed in this study is based on selective physisorption of polymer chains on the filler particles.

Several studies reported that preferential adsorption occurs with respect to the molecular weight distribution, especially in polymer solutions [25,26]. For porous adsorbents, the adsorption increases with molecular weight in the range of comparatively low molecular weights. Kolthoff et al. [27] found that the intrinsic viscosity decreases with the adsorption of a copolymer of styrene and butadiene on carbon black. The authors considered this as evidence for preferential adsorption of large chains. When low molecular weight polymer was added to the solution, the viscosity remained constant, but when a high molecular weight fraction was added the viscosity decreased appreciably. Similar effects were observed for polymer melts. El'tekov [28] studied the adsorption of polystyrene on carbon black and concluded that the larger molecules are preferentially adsorbed in the equilibrium state.

Non-polar adsorbates such as PP interact with solid surfaces such as silica via London or dispersive forces only. The adhesion energy (W_a^d) of the adsorbate on the surface is simply given by the geometrical mean value of the surface energy of the adsorbate ($\gamma_L^d = \gamma_L$ for a non-polar substance) and of solid (γ_s^d), hence $W_a^d = 2\sqrt{\gamma_L^d \gamma_s^d}$. Silica has a high surface energy, accounting for its particular adsorption characteristics [29], so it is highly probable to have PP chains adsorbed on the silica surface in the melt. It is also worth to mention the results by Grundke et al. [30] who showed that glass fibers are immediately completely wetted after immersing the fibers in a non-polar PP melt. Chemically modified PP to enhance the polarity and unmodified PP showed similar wetting kinetics with untreated glass fibers. This indicates that PP chains adsorb on glass fibers. A similar assumption is made in this study since silica is similar in structure, surface tension, and polarity to glass fibers.

3.4.1.1 Selective physisorption of high molecular weight polymer chains

There are a number of studies which refer to the preferential adsorption of the high molecular weight chains on silica surface in polymer melts. Maurer [31] reported the adsorption of high molecular weight polyethylene on porous aerosil silica. This was demonstrated by molecular weight distribution of HDPE fractions by GPC. There is a similarity in the system studied by Maurer and the present study as HDPE is similar to PP. Aerosil is also a silica and is similar in structure and surface characteristics to the in-situ prepared silica in this study. This is a strong evidence to have similar physisorption in the PP/silica nanocomposites as well. Furthermore, Vignaux-Nassiet et al. [7] reported similar observations for polybutadiene/(micrometer sized) silica systems, which have similarities with PP/silica nanocomposites prepared in this study. Apart from these studies, this opinion is also shared by Felter et al. [32] who studied the effect of the molecular weight distribution of poly(vinyl chloride) on adsorption.

Considering the adsorption characteristics of silica, the mesoporous nature of the in-situ formed silica and the wetting behavior of glass by PP along with various studies which showed the preferential adsorption of high molecular weight polymer chains on silica surfaces, it is highly probable that the high molecular weight part of PP (i.e. the long chains) adsorb preferentially on the silica nano-particles and the low molecular weight PP forms the surrounding matrix. The experimental evidences observed are discussed here to support the hypothesis.

a. Fractionation of molecular weight

The molecular weight of the PP in PP/silica nanocomposites was determined using HT-SEC. The nanocomposites can be dissolved in xylene, which is a good solvent for PP. If chains are adsorbed on the silica surface, only free chains will dissolve initially and subsequently more and more adsorbed chains will desorb and dissolve in the solvent. In a simple experiment, PP/silica nanocomposites were dissolved in xylene at 140°C for 30 min. and the dissolved PP was separated. The molecular weight of this separated PP was $M_n = 30$ and $M_w = 90$ kg/mol. The complete dissolution of same nanocomposites yields the molecular weight of parent PP, i.e. $M_n = 90$ and $M_w = 350$ kg/mol. This is a clear indication of adsorption of high molecular weight PP chains on the silica surface. Moreover, the observed decrease in the molecular weight is in the same order of magnitude to which viscosity decreased. Figure 3.4 shows that the zero-shear viscosity of 0.5 wt% silica nanocomposites is 250 Pa.s, whereas for pure PP it is approx. 5000 Pa.s. The simple relation for the dependence of the viscosity of polymer on the molecular weight is given by: $\eta \propto (M_w)^{3.4}$. Using the M_w of PP (~ 350 kg/mol), the M_w of the matrix having a viscosity of 300 Pa.s would be approx. 130 kg/mol which is close to the values obtained by HT-SEC.

b. T_2 relaxation time from solid-state NMR

Solid-state NMR magnetization relaxation experiments can provide information about molecular motion. In this work, so-called proton T_2 measurements were used. Proton T_2 relaxation is the decay of the transverse magnetization components after an excitation pulse. In polymers, this decay is basically a dephasing, brought about by mainly the dipolar proton-proton interaction. Different mobility of polymer chains result in different effective dipoles. Thus, phases with different mobility will relax at different rates.

In case of the PP/silica nanocomposites the magnetization decay had to be fitted with a short and a long T_2 -time, representing a rigid and mobile PP phase which can be attributed to the adsorbed and

bulk PP chains respectively. Upon increasing the amount of silica the shorter T_2 -time increases, which indicates that more immobilized material is present in the melt. It is neither easy nor straightforward to deconvolute the PP/silica relaxation spectra and to fit these with different T_2 times corresponding to different mobility. Therefore, in this study we concentrate on the longer T_2 – times corresponding to the mobile phase. However, it must be stressed here that, unlike in some other studies especially on rubber and carbon black, a clear short T_2 -time corresponding to the immobilized layer due to adsorption, was not observed in this study.

Kendall and Sheliker [33] explained the effect of adsorption on the entanglements in the melt. Figure 3.9 illustrates the distinction between the state of an adsorbed and a bulk molecule. The asymmetry introduced by the surface changes the state of adsorbed molecules. The segmental random walk is prevented by the solid surface, so a spherical distribution of the molecular segments is no longer possible. Intermingling of separate chains will also be restricted which reduces the probability of finding another molecule next to the filler. Thus, the surface effectively induces a segregation of the molecules in its vicinity and reduces the entanglement density due to demixing.

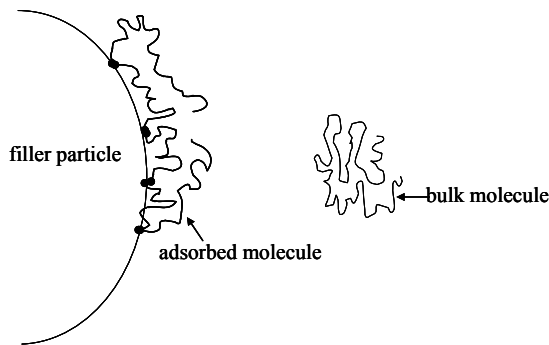


Figure 3.9: Configuration of a polymer molecule adsorbed on the filler surface and a molecule in the bulk. The adsorbed molecules experience a reduced entanglement density due to the surface asymmetry.

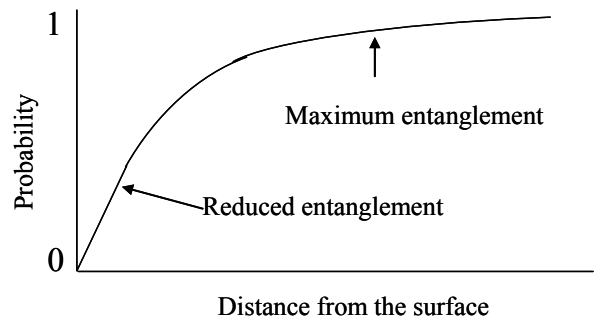
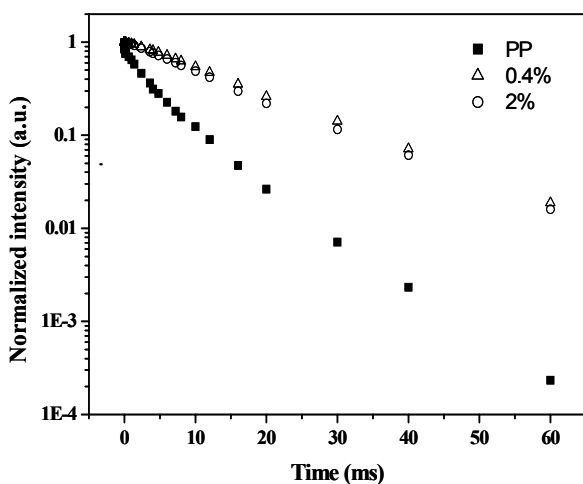
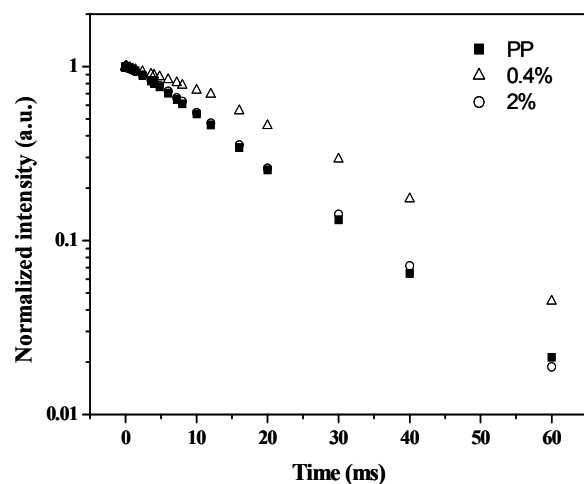


Figure 3.10: Schematic representation of the probability of finding another molecule which may increase the entanglement density.



(a)



(b)

Figure 3.11: T_2 relaxation spectra for PP and PP/silica reconstructed from MAS solid-state NMR measurements. The longer relaxation time indicates a higher mobility or less entangled melt for silica-filled PP compared to pure PP at (a) 170°C and (b) 190°C.

As shown in Figure 3.11a, the longer T_2 relaxation time, which represents the mobile phase, becomes longer confirms that the mobility of chains in the bulk is increased. This can be explained when high molecular weight PP chains are adsorbed at the silica surface and segregated from the bulk matrix, leaving the latter mainly composed of shorter chains. It is also important to note that with increasing temperature the mobility of pure PP also increases and the T_2 -time becomes closer to that of PP/silica nanocomposites as shown in Figure 3.13b. This indicates the decreasing effect of adsorption with temperature.

c. Additional relaxation process observed in storage modulus

The appearance of an intermediate plateau in the elastic modulus G' with increasing silica concentration (Figure 3.5b) points to additional relaxation processes next to relaxation of the pure polymer. The intermediate plateau shifts to a lower frequency with increasing silica concentration. Vignaux-Nassiet et al. [7] observed a similar relaxation behavior in the lower frequency region for polybutadiene/micron-sized silica systems. Pham et al. [34] also observed multiple relaxation modes in PMMA lattices with 1% acid comonomer in de-ionized water containing PEO and attributed the additional relaxation to an adsorbed layer of PEO on the PMMA latex particles.

Vignaux-Nassiet et al. fitted their rheological measurements using a modified Palierne model. The standard Palierne model is not suitable when the dispersed phase is non-deformable. Assuming that polymer chains are adsorbed on the silica surface, a modified dispersed phase made up of elastic silica nano-particles with a viscoelastic coating of polymer chains can be introduced in the Palierne model. The adsorbed chains are assumed to be pinned on the silica surface on the time scale of the rheology experiments. The modified Palierne model contains three adjustable parameters, i.e. the thickness (δ) of the adsorbed polymer layer, the interfacial tension (α), and the relaxation time ν of the Doi-Kuzuu theory. Vignaux-Nassiet et al. established that the thickness (δ) of the adsorbed layer is independent of the concentration of the silica particles and depends only on the molecular weight of the polymer matrix. The relaxation time in which the elementary relaxation time is the same as the matrix chain and molecular weight dependence can be defined.

In line with other observations, the additional relaxation time observed at lower frequencies in storage modulus (G') strongly supports the proposed hypothesis of adsorption of PP chains on the mesoporous silica nano-particles. Although there is not a clear plateau for the 0.5 wt% composite, it seems that the G' curve tends to a lower level than the other composites indicating that the polymer chains forming the matrix have a lower average molecular weight than the pure polymer. The shift of the intermediate plateau to lower frequencies in G' suggests that a large fraction of high molecular weight chains are adsorbed with increasing nanoparticle concentration.

d. Shear-induced crystallization

The flow-induced crystallization experiments revealed the absence of high molecular weight chains in the bulk. The orientation goes through a minimum with increasing silica nanoparticle concentration, following the drop and increase in the dynamic and steady-shear viscosity. The decrease in orientation indicates that the matrix is mainly composed of the low molecular weight part of the polymer.

The shear rate of 60 s^{-1} used in these experiments compares with the critical shear rate at which the critical orientation molecular weight, M^* ($\sim 300\,000 \text{ g/mol}$), does not vary significantly as found by

Somani et al. [21]. However in the present study, the orientation is strongly influenced, indicating that the considerable amount of high molecular weight PP chains is adsorbed on the porous silica nano-particles. However, beyond a certain concentration, when the interparticle distance become less than the radius of gyration, as will be discussed in following section, the bridging effects and chain entanglements overcome the demixing. This does not allow segregation of adsorbed high molecular weight chains from the low molecular weight bulk, which eventually increases the viscosity and hence the stresses experienced during flow increase. Moreover, the reversibility of the adsorption allows desorption of high molecular weight polymer chains and hence orientation is again observed at higher silica nanoparticle concentration. The effect of silica particles on the final orientation is described in more detail in Chapter 6.

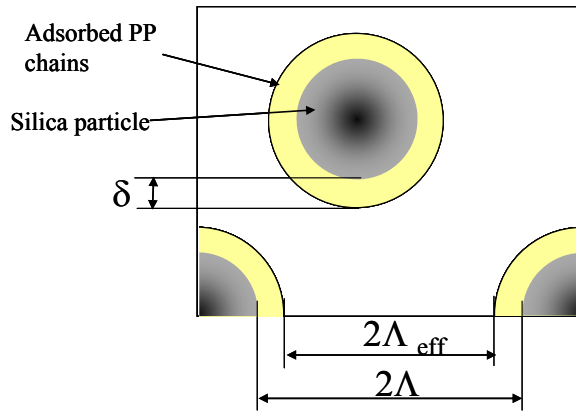


Figure 3.13: Schematics of the core-shell morphology with silica as a hard core and the adsorbed PP chains as a soft shell which increases the effective volume fraction of fillers.

3.4.1.2 Inter-particle distance

An important parameter of nanoparticle-polymer systems is the inter-particle distance between the nano-particles. The selective physisorption coupled with the inter-particle distance governs the rheological behavior of PP/silica nanocomposites. The adsorption of high molecular weight PP chains on the silica surface results in a core-shell morphology with silica forming a hard core and the adsorbed high molecular weight chain PP forming a soft shell. This increases the effective volume fraction of filler particle and reduces the inter-particle distance. This is schematically shown in Figure 3.13.

This effective interparticle distance can be estimated using the simple relation

$$\Lambda/r = [\phi_m/\phi]^{1/3} - 1$$

where 2Λ is the inter-particle distance, r is the radius of the silica nano-particles and ϕ_m is the maximum packing fraction of monodisperse spherical particles (~ 0.638 for random packing). The confined polymer shell thickness ' δ ' around silica nano-particles ($r = 7.5$ nm) is determined from the following formula originally derived by Ilre [35]:

$$R_g = \frac{4\pi}{3} \left[\frac{(r + \delta)^3 - r^3}{4\pi r^2} \right]$$

where R_g is the radius of gyration of the adsorbed chains.

The radius of gyration can be calculated using the following expression:

$$R_g = \sqrt{\frac{2N}{3}}b$$

For the PP used in this study, the degree of polymerization, $N = 100,000$, and the segmental length $b = 1.5 \text{ \AA}$. This gives $R_g = 20 \text{ nm}$ and from that it can be derived that $\delta = 7.5 \text{ nm}$.

This effectively increases the particle volume fraction which is given by:

$$\phi_{eff} = \phi \left(1 + 3\delta/r\right)$$

Λ is calculated using this effective volume fraction. Based on this, three different regimes can be defined: $\Lambda < R_g$, $\Lambda \approx R_g$, $\Lambda > R_g$. This is schematically depicted in Figure 3.14. At a low volume fraction, (below 0.5 vol%, i.e. below 2 % effective volume fraction in the present case), the distance 2Λ between nano-particles is more than 50 nm and system is in $\Lambda > R_g$ regime ($2R_g \approx 40 \text{ nm}$) in which the chains do not experience any confinement or bridging effects. As the silica volume fraction increases, above 0.5 vol%, the distance between nano-particles (2Λ) becomes less than 50 nm and system is in $\Lambda \approx R_g$ regime. A further increase in the silica concentrations results in a smaller interparticle distance, i.e. in the order of the radius of gyration ($\Lambda < R_g$). This leads to the first occurrences of chain confinement and bridging effects. As the volume fraction of silica particles increases beyond certain concentration, i.e. 1 vol%, strong bridging and confinement effects occur (see Figure 3.14). These effects result in formation of super-structures in the melt leading to a non-homogeneous matrix and an increase in the viscosity.

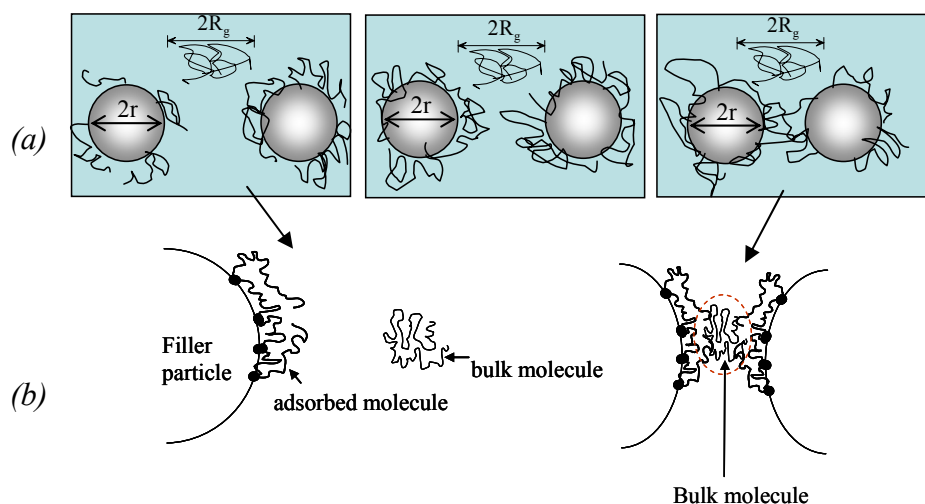


Figure 3.14: (a) Schematics showing the effect of the inter-particle distance on the particle-particle, particle-polymer interactions, (b) schematics showing the adsorption in the melt; when the inter-particle distance is less than the R_g , bridging effect occurs resulting in the network formation.

3.4.1.3 Effect of silane grafting on rheology of PP/silica

To test the hypothesis of selective physisorption, a different approach was explored by modifying the chemical interaction between silica and PP. PP is chemically modified by grafting silane to induce chemical interaction between matrix and in-situ prepared silica (as explained in Chapter 2, Section 2.3.1.2). The strong bonding of silica to the PP may prevent selective physisorption of the high molecular weight chains.

Figure 3.15 shows the dynamic viscosity of the (VTES-grafted PP)/silica nanocomposites. It is assumed that all grafted silane groups are incorporated in the in-situ formed silica. The level of silica is kept at 1 wt% in all the composites measured. The 1 wt% silica system is expected to have a lower dynamic viscosity than PP. However, the chemically linked silica with PP does not allow the selective adsorption and thus separation of low and high molecular weight chains. This eventually results in increase in the viscosity and the advantages obtained without grafting are lost. This evokes the underlying importance of surface characteristic of silica as well as the way it interacts with PP chains. From the viscosity data, it can be observed that the effect of selective physisorption disappears.

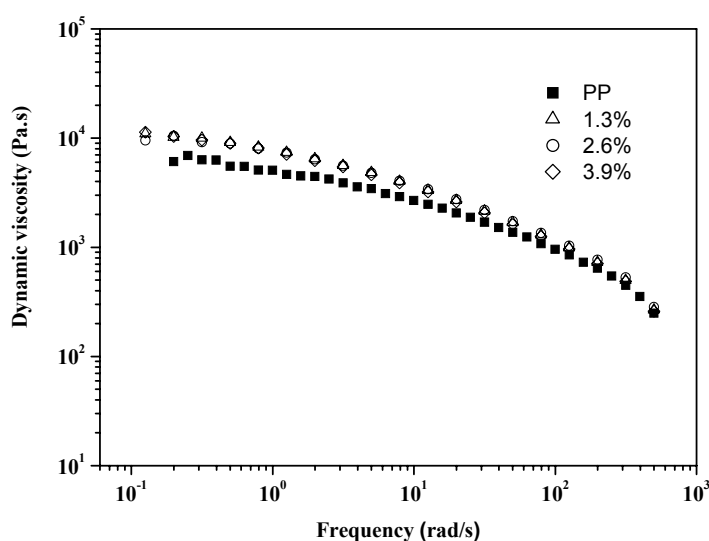


Figure 3.15: Dynamic viscosity against frequency for PP and (VTES-grafted PP)/silica at 180°C. The concentration of silica is 1 wt%. The legends indicate the degree of silane grafting.

3.5 Conclusions

The influence of the concentration of in-situ prepared silica nano-particles in PP on the rheological properties has been discussed. In the low concentration range (c.a. below 1 wt%), a large drop in the viscosity was observed which is attributed to selective physisorption of the high molecular weight polymer chains on the nano-particles. Due to the nucleating effect of silica particles, the hypothesis based on free volume created by nano-particles in polymer melt does not hold in this case. Moreover, it is clear that in the higher concentration range (above 1 wt%), the increase in the viscosity was not caused by hydrodynamic forces but due to physisorption which induces confinement and bridging effects when the inter-particle distance becomes less than R_g . Thus, nano-particles change the macroscopic properties by influencing the mesoscopic physics. These effects were lost when the surface properties of these nano-particles were changed, i.e. when silica particles were chemically grafted to matrix. Furthermore, when grafted or surface treated, the nano-particles form chemical bonding and thus concepts like physisorption may not apply then.

3.6 References

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Chapter 4

Effect of in-situ prepared silica nano-particles on non-isothermal crystallization of polypropylene^{*}

Abstract

The crystallization behavior of PP/silica nanocomposites prepared via solid-state modification and sol-gel reaction is investigated and shows that in-situ formed silica nano-particles act as nucleating agents. The non-isothermal crystallization kinetics is studied using a combined Avrami-Ozawa approach and shows a two-stage crystallization process: the primary stage is characterized by nucleation and spherulitic growth and the secondary stage is characteristic of the perfection of crystals. Silica speeds up the primary stage, resulting in a more narrow lamellar thickness distribution. The crystallization activation energy decreases with increasing silica concentration. The nucleating efficiency of the in-situ prepared silica particles based on the scale as proposed by Lotz and coworkers is found to be 20% and this is higher compared to silica nano-particles as well as other nano-fillers studied. The melting behavior indicates the formation of more perfect crystals with a narrow lamellar thickness distribution and the WAXD patterns show that silica nano-particles induce the formation of crystals with the β -modification in PP at high silica concentration (ca. 5 wt%).

4.1 Introduction

The intrinsic properties of semi-crystalline polymers strongly depend on their crystalline morphology. The difficulty in understanding the deformation behavior of semi-crystalline polymers arises from their two-phase structure, i.e. crystalline and amorphous phase; the latter can be subdivided into a mobile and a rigid amorphous phase. In this respect, the effect on the yield stress and strain hardening has been a subject of extensive discussion. Strobl et al. [1] and Schrauwen et al. [2] reported that the yield stress is dominated by crystallinity and the lamellar thickness, while the strain hardening modulus depends mainly on the crystallization conditions and the resultant entanglement density of the amorphous phase. The state of the entangled amorphous phase originates from the crystallization process, since it allows the rearrangement of chains during folding and perfection. This “reeling in” of polymer chains during crystallization leads to disentanglement of the chains in the melt or amorphous phase [3,4]. The effect is more pronounced for lower cooling rates, resulting in a lower entanglement density and, consequently, a lower strain hardening modulus. Schrauwen et al. [2] showed that the yield stress at which slip of the lamellae is reached depends on both the crystallinity and lamellar thickness. The influence of the crystallization kinetics on the tensile modulus and yield stress of PP was also studied by Dasari et

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al. [5]. However, at higher strains the entangled amorphous phase dominates the deformation process. A higher entanglement density implies that a higher stress is generated when the material is stretched. These findings can even have a more pronounced effect for polymers with nucleating additives [6].

In this context, it is important to note that the crystallization kinetics and crystalline morphology of semi-crystalline polymer and their nanocomposites strongly depend on the processing conditions [7,8,9]. Filler particles can influence the flow and hence the crystallization behavior and final morphology during processing. Industrial polymer processes, such as extrusion, injection molding, and film blowing, are operated under dynamic, non-isothermal conditions. To study effect of the complex thermal conditions during processing, it is important to investigate the behavior of PP-nanocomposites during non-isothermal crystallization [10]. Although numerous investigations were carried out on isothermal crystallization of PP with nucleating agents and of PP-nanocomposites [11,12,13,14], very few studies have been published related to non-isothermal crystallization of PP-nanocomposites [15,16,17]. Isothermal crystallization is an idealized crystallization process for which the Avrami theory has been well established to describe the crystallization kinetics [18,19]. Effects of cooling rate and thermal gradients within the sample are avoided in the isothermal crystallization process, whereas conditions are continuously changing in non-isothermal crystallization. The Avrami equation was thus modified to accommodate the changing temperature (environment) [20,21]. Ozawa [22] extended this modified equation by incorporating the effect of the cooling rate, assuming that the crystallization proceeds at a constant cooling rate. Despite these corrections, these methods can still not fully explain the non-isothermal crystallization kinetics of polymers. Liu et al. [23] developed a new approach by combining the Avrami and Ozawa theories and described successfully the non-isothermal crystallization of poly(ether ether ketone) (PEEK).

Table 4.1 summarizes the findings of some studies on PP-nanocomposites and PP with nucleating agents. Commercial nucleating agents are mostly used in a concentration range below 1 wt%, above which their efficiency reaches a plateau level. This is one of the reasons for concentrating at low filler levels (only up to 1 wt%) in this study. However, the mechanism of nucleation and the concomitant morphology during crystallization are different.

As mentioned earlier, the amorphous phase of PP in these nanocomposites is also affected during processing [24]. It was observed in PP/talc (micrometer-sized fillers) systems that the nucleating effect of the filler lowers the glass transition temperature (T_g) resulting in an amorphous phase with a higher mobility [25]. This stresses the importance to study the effect of filler particles on the entanglement density and mobility of the amorphous phase, since it strongly influences the strain hardening behavior [26,27]. The presence of an interfacial layer between the bulk polymer and the filler surface has been well established by various techniques. Therefore, it can be expected that nano-particles with their surface-to-volume ratio may influence the bulk properties [28].

In this chapter the effect of in-situ formed silica nano-particles on the non-isothermal crystallization kinetics of PP/silica nanocomposites was investigated using the combined Avrami-Ozawa approach as developed by Liu et al. [23]. The crystallization activation energy was evaluated using Kissinger's method [29]. The nucleation efficiency based on the efficiency scale as proposed by Lotz and coworkers [30] was also calculated to investigate the effectiveness of the silica nano-particles as nucleating agents. The melting behavior of PP/silica nanocomposites was

studied by DSC to probe the recrystallization and phase transformation/modifications in these PP-nanocomposites. X-ray diffraction studies were carried out to examine the presence of various crystal forms to explain the DSC melting curves. Further, the effect of silica nano-particles on the T_g and the amorphous phase was investigated by DMTA.

Table 4.1: Crystallization and nucleation data for PP-fillers and nucleating agents from literature

Researchers	Filler (% wt)	T_c (°C)	NE [#] (%)
<i>Quian et al. 2004 [21]</i>	0	117	0
PP/nano SiO ₂ (typical size ~ 16 nm)	1	117.1	0.5
	3	118.8	8.5
	5	121.2	20
<i>Lin et al. 2004 [22]</i>	0	117.5	0
PP/nano-CaCO ₃ (typical size ~ 40-60 nm)	2	116.1	0
	5	116.8	0
	10	120.8	16
<i>Kristiansen et al. 2003 [11]</i>	0	113	0
PP/DMDBS [bis(3,4-dimethyl benylidene sorbitol)*]	0.2	113.5	2
	0.4	120	28
	0.6	127	56
<i>Gui et al. 2003 [18]</i>	0	117	0
PP/NA-40 (12H-dibenzo[d,g][1,3,2] dioxaphosphocin, 2,4,8,10-tetrakis (1,1-dimethylethyl)-6-hydroxy-6-oxide, sodium salt)*	0.2	127	47
	0.4	128	52
	0.6	128	52
<i>Xu et al. 2002 [20]</i>	0	110	0
PP/montmorillonite	3	115	17.8
<i>Pozsgay et al. 2002 [17]</i>	0	112	0
PP/montmorillonite (vol%)	2	117	19
	5	119	27
	10	121	34.6
<i>Lotz et al. 1994 [32]</i>			
PP		114.2	0
DBS (dibenzylidene sorbitol)*	0.4	123.2	41
Talc	1	121.4	32
9-fluorene carboxylic acid*	1	119.4	18
2,4 (bis butylamine) 6-hydroxy, 1, 3, 5 triazine*	1	115.5	5

4.2 Experimental Section

4.2.1 Materials

PP/silica nanocomposites were prepared by the sol-gel method according to the procedure described in Chapter 2, Section 2.3.1.1.

4.2.2 Differential Scanning Calorimetry (DSC)

The crystallization and melting characteristics and the nucleation efficiency were investigated with a TA Instruments Q1000 Differential Scanning Calorimeter, calibrated with Indium for temperature and heat of fusion. Sample weights between 3-4 mg were used in standard aluminum pans. The analysis was conducted under standard heating and cooling rates of 10 K/min unless otherwise indicated.

[#] NE= Nucleation efficiency, calculated based on the scale defined in this study (with lower limit for pure PP and upper limit as 138°C)

* commercial nucleating agents

The non-isothermal crystallization kinetics of these samples were investigated by conducting cooling scans at rates of 2.5, 5, 10, 15, 20 and 30 K/min from 225°C to 30°C. Prior to the heating or cooling scan, the samples were held at 225°C for 5 min under a nitrogen atmosphere to erase the thermal history and prevent self-seeding of PP.

The nucleating efficiency is based on the calorimetric efficiency scale proposed by Lotz and coworkers [30] and is calibrated according to their procedure based on the ‘ideally’ nucleated polymer attainable via self-nucleation. Two limits need to be defined: the lower limit refers to the virgin (non-nucleated) PP and the upper limit to an optimally self-nucleated PP. The nucleation efficiency is a percentage of the range defined by the two limits and is expressed as:

$$NE = \frac{T_c - T_{c1}}{T_{c2max} - T_{c1}} \times (100) \quad [4.1]$$

where T_{c1} and T_{c2max} are the crystallization temperatures of the virgin and best self-nucleated PP respectively. T_c is the crystallization temperature obtained in the presence of the nucleating agent (in this case silica). NE is thus equal to 0 for a non-nucleating action and 100 for the optimum efficiency.

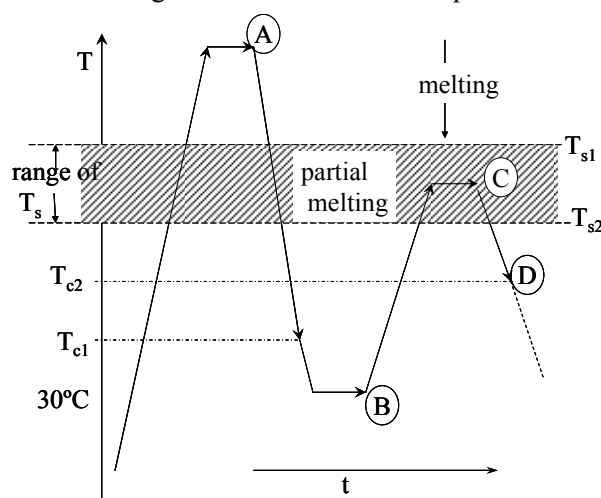


Figure 4.1: Self-nucleation procedure as applied in DSC runs (Fillon et al.[35]), A) complete melting of sample to eliminate most nuclei, B) creation of standard state, C) self-nucleation step, and D) crystallization step.

Self-nucleation is carried out by heating PP to a selected temperature (T_s). When T_s is located in the melting range, the formation of stabilized polymer crystal fragments occur. The self-nucleation procedure as applied in DSC is shown in Fig. 4.1. The heating and cooling rates are 10 K/min. The procedure essentially involves four steps.

Step A. Complete melting of samples at 225°C for 10 min to erase the previous thermal history. This step eliminates most of the nuclei.

Step B. Creation of the standard state (T_{c1}) by cooling from (A) at the predetermined cooling rate of 10 K/min to some low temperature T_1 ($\sim 30^\circ\text{C}$). The crystallization takes place at the lower limit of the crystallization range (T_{c1}). T_{c1} depends on the molecular characteristics of the polymer and decides the lower limit of the nucleation efficiency scale.

Step C. This self-nucleation step is carried out by heating the sample to selected temperatures (T_s). When T_s is located in the partial melting zone (hatched section in Figure 4.1), the formation of stabilized crystals occurs. The concentration of these crystals increases with decreasing T_s and reaches saturation at $T_s = T_{s2}$.

Step D. This second crystallization step is achieved by subsequent cooling of the samples which give crystallization peaks at temperature T_{c2} ($T_{c2} \geq T_{c1}$). An increase in T_c indicates an increase in the concentration of nucleation sites induced by the self-nucleation process.

The virgin sample crystallizes at T_{c1} , whereas the best nucleated sample crystallizes at the higher crystallization temperature T_{c2max} . These two extremes decide the limits of the nucleation efficiency scale.

4.2.3 Dynamic Mechanical Thermal Analysis (DMTA)

Glass transition temperatures and moduli of the samples were measured using a TA Instruments Q800 Dynamic Mechanical Analyzer fitted with a single cantilever geometry. Samples were extruded in a twin-

screw mini-extruder with a residence time of 5 min at 190°C; the rotor speed was set at 50 rpm. The extruded samples were compression into sheets and bars of approx. 20 x 15 x 1 mm size were cut from the sheets. For the measurements, the samples were cooled to -80°C and held for 5 min and subsequently heated to 100°C at 2 K/min. A dynamic force of 0.25 N oscillating at 1 Hz and an amplitude of 0.2 µm was used.

Three parameters, i.e. the loss modulus, the intensity of the transition (S) and full-width at half maximum (FWHM), were used to evaluate the glass transition temperature and the state of the amorphous phase in the samples. The maximum of the peak in the loss modulus curve is taken as the T_g [31]. The intensity of the transition (S) is an important parameter reflecting the mobility and content of the amorphous phase [25]. It is defined as:

$$S = \frac{E_a' - E_d'}{E_d'} \quad [4.2]$$

where E_a' and E_d' denote the storage modulus before and after the transition respectively. A higher S means a higher mobility (less entanglement) or a higher content of amorphous phase. The full-width at half maximum (FWHM) refers to the homogeneity of the amorphous phase. A higher value of FWHM compared to base materials implies a higher inhomogeneity.

4.2.4 X-ray characterization

Wide-angle X-ray diffraction (WAXD) experiments were performed at the DUBBLE (Dutch-Belgium) Beamline at the European Synchrotron Research Facility (ESRF), Grenoble, France. An X-ray wavelength of 1.218 Å was used. The WAXD patterns were detected by a curved linear microstrip-gas chamber detector. The diffraction angles at the WAXD detector were calibrated with Si-powder. The nanocomposites films/powder were put in aluminum pans, and placed in a THMS600 hot stage (Linkam Scientific Ltd.) for X-ray diffraction. Data were recorded while the sample was heated and cooled at 10 K/min. Diffraction patterns were taken at every 6 s, which corresponds to one diffraction pattern for each degree Celsius in the temperature ramp.

4.3 Results and discussion

The silica nano-particles used in this study are in-situ formed silica particles of 30-80 nm as observed by transmission electron microscopy. The silica particles have hydroxy-groups on its surface as found by MAS ^{29}Si NMR [32]. PP/silica nanocomposites with silica concentrations up to 5 wt% were used in this study, but we concentrate on the concentration range up to 1 wt%.

4.3.1 Crystallization Behavior

As an example, the crystallization curves of PP and PP/silica nanocomposites at a cooling rate of 10 K/min are shown in Figure 4.2. The increase of the crystallization peak temperature (T_c) upon increasing silica concentration shows the nucleating effect of silica on PP crystallization up to 1 wt% silica, while at higher concentration it decreases slightly and remained constant. Table 4.2 shows the crystallization data at various cooling rates. As the cooling rate is increased, T_c is shifted to lower temperatures for both PP and PP/silica nanocomposites, which may be due to a larger supercooling at higher cooling rates. However, these measurements were performed on pressed films of 3-4 mg. Temperature gradients in the samples are therefore not expected. At a given cooling rate, T_c of silica-filled PP is higher than that of pure PP. The overall crystallization time (t_c) was calculated using the following equation:

$$t_c = \frac{(T_s - T_e)}{\beta_0} \quad [4.3]$$

where β_0 is the rate of cooling, T_s is the initial crystallization temperature and T_e is the final crystallization temperature.

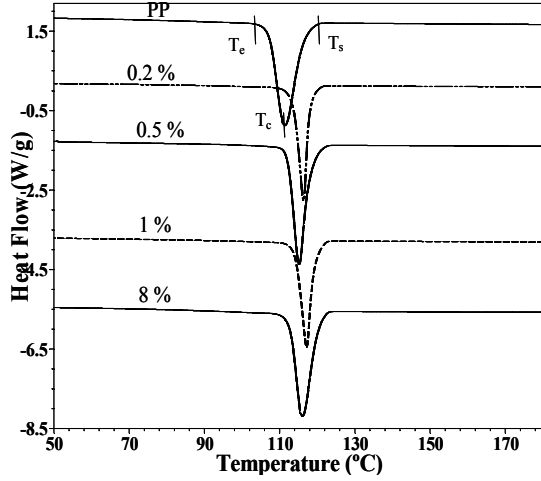


Figure 4.2: DSC cooling scans of PP/silica nanocomposites at 10 K/min as a function of silica concentration.

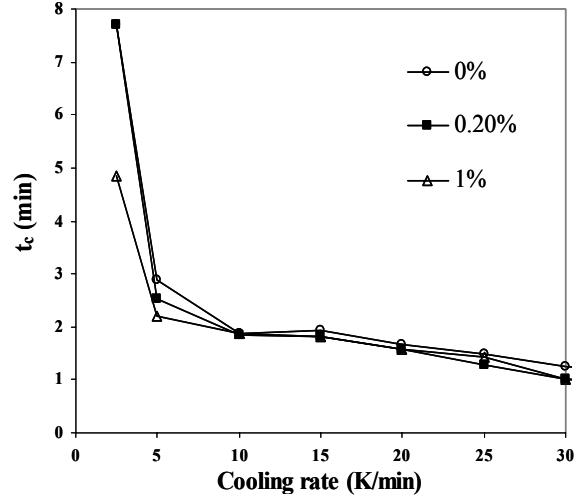


Figure 4.3: Overall crystallization time as a function of cooling rate of PP/silica nanocomposites for different silica concentrations.

Figure 4.3 shows that t_c is reduced upon increasing the silica concentration at different cooling rates. The increase in T_c and the decrease in t_c are typical characteristics of nucleation-controlled polymer crystallization [33]. Usually, homogeneous nucleation starts spontaneously below the melting temperature [30] and proceeds rather slowly, whereas the heterogeneous crystallization starts as soon as the crystallization temperature is reached. The crystallization of silica nanoparticles filled PP proceeds mainly via heterogeneous nucleation by the silica nano-particles. However, two regions, i.e. below and above approx. 1 wt% of silica, can be distinguished showing a different nucleating effect.

The DSC-experiments at different cooling rates apparently show the same trend, except at the highest cooling rate, i.e. 50 K/min. Since for each cooling rate a different sample was used, the repeatability seems sufficient to draw conclusions. The apparent decrease in nucleation and growth rate may be related to a different nucleation mechanism and the physisorption of high molecular weight PP chains on silica. In the lower concentration range, the segregation of high and low molecular weight chains influence the chain diffusion to the crystal growth front, while at higher concentrations of silica, segregation is not possible as explained in Chapter 3 and also later in Chapter 6. Also, at higher concentrations, silica start agglomerating which effectively reduces the surface area and hence nucleation sites.

The degree of crystallinity at the peak melting temperature is calculated from the melting heat [ΔQ_m (J/g)] obtained from the DSC measurements according to the following equation (after normalization for the PP concentration):

$$X_c = \frac{\Delta Q_m}{\Delta H_o} \quad [4.4]$$

where $\Delta H_o \approx 207.1$ J/g is the melting enthalpy of 100% crystalline PP [26]. The degree of supercooling and the degree of crystallinity of PP/nanocomposites with varying silica levels at a cooling rate of 10 K/min are listed in Table 4.3. The degree of supercooling is defined as the difference in the melting peak temperature (T_m) and crystallization peak temperature (T_c) at a given cooling rate. The nucleating effect of silica results in a shifting of T_c to higher temperatures,

whereas T_m is hardly affected indicating that the supercooling (ΔT) is reduced with increasing silica concentration followed by a slight increase at high concentrations. When a polymer crystallizes with less supercooling, it crystallizes uniformly.

From the DSC curves, the values of the relative crystallinity at various cooling rates can be calculated. The relative crystallinity as a function of time can be defined as:

$$X_t = \frac{X_t}{X_\infty} = \frac{\int_{t_0}^t \left(\frac{dQ}{dt} \right) dt}{\int_{t_0}^{t_e} \left(\frac{dQ}{dt} \right) dt} \quad [4.5]$$

where (dQ/dt) represents the rate of heat flow, X_t and X_∞ represent the crystallinity at time t and at the end of crystallization process (t_e) respectively. For the non-isothermal crystallization process, the crystallization time t and temperature T are related as:

$$t = \frac{|T_0 - T|}{\beta_0} \quad [4.6]$$

where T is the temperature at time t and T_0 is the initial temperature when crystallization starts. Thus, the relative crystallinity as a function of temperature (T), $X_c(T)$, is defined as:

$$X_c(T) = \frac{\int_{T_s}^T \left(\frac{dQ(T)}{dT} \right) dT}{\int_{T_s}^{T_e} \left(\frac{dQ(T)}{dT} \right) dT} \quad [4.7]$$

The relative crystallinity as a function of time and temperature for PP and PP/silica nanocomposites is shown in Figure 4.4. All curves have similar sigmoid shapes. The curvature of the upper parts (near complete crystallization) in the plots is due to the spherulitic impingement in the later (growth) stages of crystallization. From the curves in Fig. 4.4, the half-life times to reach 50% crystallinity ($t_{1/2}$) of non-isothermal crystallization are calculated and listed in Table 4.3 for a cooling rate of 10 K/min. It can be observed that PP in the presence of silica is nucleated at higher temperatures and crystallizes faster. The faster increase of the initial stages of crystallization also signifies the nucleating effect of silica nano-particles.

4.3.2 Non-isothermal crystallization kinetics

The Avrami equation [18,19] was developed for isothermal crystallization, although it has also been used to study non-isothermal processes [34], and reads as:

$$X_t = 1 - \exp(-Z_t(T)t^n) \quad [4.8]$$

where n is the Avrami exponent, $Z_t(T)$ the Avrami rate constant and X_t the relative crystallinity at time t . Although the physical meaning of Z_t and n cannot be related to the non-isothermal case in a simple way, their use provides further insight into the kinetics of non-isothermal crystallization. For non-isothermal processes, the Avrami crystallization rate $Z_t(T)$ has been corrected [20,35], since the temperature is constantly changing during the measurements.

Table 4.2: Non-isothermal crystallization behavior of PP/silica nanocomposites at different cooling rates

silica content cooling rate K/min	0%		0.2%		0.5%		1.0%		1.5%		2.0%	
	Ts °C	Te °C	Ts °C	Te °C	Ts °C	Te °C	Ts °C	Te °C	Ts °C	Te °C	Ts °C	Te °C
2.5	126	118	107	118	129	118	125	121	118	122	116	123
5	123	115	108	116	125	113	122	118	116	118	114	120
10	120	111	102	114	124	106	116	115	107	117	111	116
15	119	113	90	123	123	113	117	115	109	115	106	114
20	118	112	85	124	112	112	116	114	106	114	103	113
25	118	111	85	121	111	111	115	114	108	115	102	107
30	119	106	82	119	110	110	115	112	105	113	110	103
50	116	105	76	115	115	108	112	110	102	111	100	107

Table 4.3: Crystallization and melting data for PP/silica nanocomposites at a scanning rate of 10 K/min

Sample	T _c		T _m		ΔT (°C)	$t_{1/2}$ (sec)	X _c (%)	Φ
	Peak (°C)	ΔH_c (J/g)	Peak (°C)	Peak (°C)				
PP	111.4	99.7	158.7	158.7	47.4	64.5	48.1	1.00
PP/silica 0.2%	114.2	102.3	159.8	159.8	45.5	55	49.4	0.89
PP/silica 0.5%	117.6	106.5	160.3	160.3	42.7	51	51.5	0.78
PP/silica 1.0%	116.2	110.2	160.2	160.2	44.0	46	53.7	0.74
PP/silica 1.5%	114.6	100.7	160.0	160.0	45.4	32	49.4	0.71

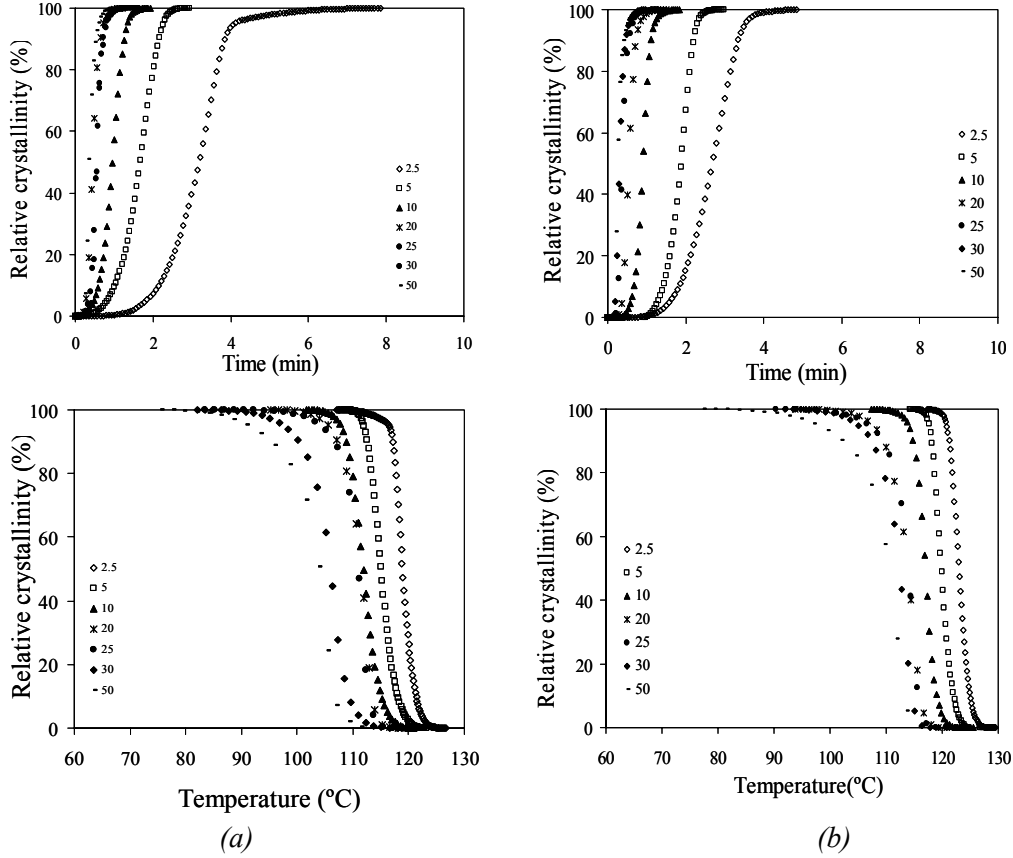


Figure 4.4: Relative crystallinity versus time and temperature for non-isothermal crystallization: (a) PP and (b) PP/silica (1 wt%) at different cooling rates indicated by the legends.

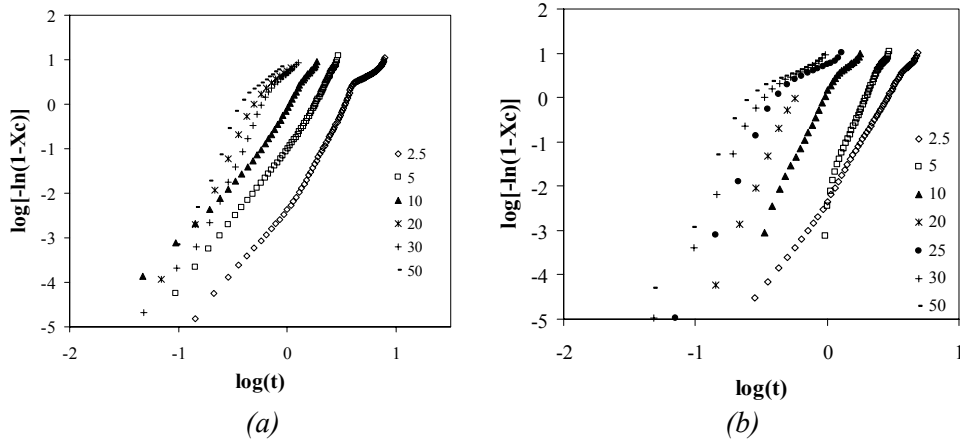


Figure 4.5: Avrami plots of $\log[-\ln(1-X_t(t))]$ versus $\log t$ for non-isothermal crystallization of (a) PP and (b) PP/silica (1 wt%).

Thus, the parameter characterizing the non-isothermal crystallization kinetics has been defined as:

$$\log Z_c = \frac{\log Z_t}{\beta_0} \quad [4.9]$$

where β_0 is the cooling rate.

According to equation (4.8), $\log[-\ln(1-X_t(t))]$ is plotted against $\log t$ in Figure 4.5. The values of the Avrami exponent n and the rate parameter Z_c (i.e. corrected Z_t) can be determined from the slope and intercept of the curves, respectively. It is interesting to note that the curves for PP show only one region (secondary crystallization is less significant) as also observed by He et al. [36] and Xu et al. [15]. At high cooling rates, a prominent saturation at higher relative crystallinity is observed.

However, for the silica nano-particle filled PP, each curve shows two regions: the so-called primary stage followed by the secondary stage (plateau region). This has not been reported in earlier studies on PP-nanocomposites [15]. Such a behavior was reported for PEEK by Liu et al. [23] (non-isothermal crystallization), Cebe et al. [34] and Lee et al. [37] (isothermal crystallization).

The change from primary to secondary stage occurs near relative crystallinity of approx. 70%. Each regime gives different values for n (n_1 and n_2) and Z_c (Z_{c1} and Z_{c2}) as listed in Table 4.4. For PP, the two stages are not distinct; however, for comparison with the nanocomposites, the curves are split into two stages: below and above 70% crystallinity. The average values of the Avrami exponent for PP are $n_1 = 4.4$ and $n_2 = 2.1$, although the value of exponent n is more scattered than for isothermal crystallization. Here, $n_1 \approx 4$ correspond to three-dimensional spherical growth and homogeneous nucleation in the primary crystallization stage as normally observed in the case of an isothermal crystallization process; $n_2 \approx 2$ can be assigned to one-dimensional linear growth and athermal nucleation. For PP/silica nanocomposites, $n_1 \approx 7$ corresponds to solid-sheaf growth from the silica nano-particles (as usually observed in cold-crystallization of polymers [23]) and thermal nucleation during the primary stage. The initial increase and then decrease of the Avrami exponent n with the increasing cooling rate stresses the importance of thermal nucleation in the non-isothermal crystallization process, whereas $n_2 \approx 2$ implies that, during the secondary stage, the crystallization proceeds similar to PP with one-dimensional growth and athermal nucleation.

Generally, the secondary stage is related to crystal perfectioning, which is triggered by the initially poorly crystallized molecules or small, imperfect and metastable crystals. However, selective adsorption of PP-chains at the silica particle surface may also play a role. The decrease in n_2 with increasing cooling rate corresponds to a faster or more effective secondary stage, which is mainly caused by the slower primary stage and more imperfect crystals. PP/silica nanocomposites have lower n_2 values than PP at a given cooling rate suggesting a faster secondary crystallization for PP/silica nanocomposites. The primary stage in PP/silica nanocomposites has already proceeded towards higher crystallinity and more perfect crystals than in PP due to the nucleating effect of silica nano-particles. This perfectioning step results in lamellar thickening [2]. The “reeling in” of polymer chains also occurs during the secondary stage. Schrauwen et al. [2] reported that this results in a more disentangled amorphous phase which eventually affects the mechanical properties. The faster cooling rates are especially important as they closely mimic the most industrial molding processes. If the secondary crystallization is not complete, the product will continuously crystallize over time in turn changing the properties of the product.

The Avrami equation for the non-isothermal crystallization is valid for the early stages of the crystallization only and does not account for the secondary crystallization of the polymers [23,33]. The observations from these Avrami plots are indicative, but not conclusive. The Avrami equation was extended by Ozawa [22] to develop a simple method to study the non-isothermal crystallization kinetics, assuming that the crystallization occurs at a constant rate. According to Ozawa, the relative crystallinity at temperature T , X_T , can be calculated using following equation:

$$1 - X_T = \exp \left[- \frac{K(T)}{|\beta_0|^m} \right] \quad [4.10]$$

where $K(T)$ is a function related to the overall crystallization rate and m is the Ozawa exponent that depends on the dimensions of crystal growth.

Table 4.4: Values of n , Z_c and $t_{1/2}$ for PP and PP/silica nanocomposites obtained from Avrami analysis for non-isothermal crystallization process

Sample cooling rate K/min	0%			0.2%			1.0%		
	primary n_1	stage Z_{c1}	secondary n_2	primary n_1	stage Z_{c1}	secondary n_2	primary n_1	stage Z_{c1}	secondary n_2
2.5	3.7	0.13	1.8	4.7	0.08	2.1	4.6	0.13	4.0
5	3.4	0.64	4.9	5.2	0.64	4.2	7.3	0.37	5.4
10	2.9	0.95	3.5	7.8	1.05	2.8	6.3	1.03	3.2
15	7.0	1.17	2.6	8.4	1.18	2.3	11.3	1.13	2.9
20	4.7	1.17	2.2	10.7	1.14	2.6	6.4	1.02	2.0
30	4.6	1.07	2.4	6.0	1.21	1.8	6.2	1.26	1.8
50	4.7	1.12	2.0	5.7	1.17	1.7	6.3	1.20	1.5

Table 4.5: The values of b and $F(T)$ at different degrees of crystallinity calculated from Figure 4.6

$X_c(t)$ (%)		20	40	60	80
Pure PP	b	1.24	1.22	1.19	1.18
	$F(T)$	7.56	9.31	10.46	11.35
PP/silica 0.2%	b	1.2	1.26	1.3	1.13
	$F(T)$	7.41	8.96	9.66	11.3
PP/silica 1%	b	1.09	1.16	1.14	1.21
	$F(T)$	6.58	8.6	9.51	10.45

However, Ozawa's theory neglects the slow secondary crystallization and the dependence of folded chain length (lamellar thickness) on the temperature [38]. Cakmak and Lee [39] and Kim et al. [10] showed that the Ozawa analysis cannot describe the non-isothermal crystallization of polymers in general.

As the degree of crystallinity is related to the cooling rate β_0 and the crystallization time t (or temperature T), the relation between β_0 and t needs to be defined for a given degree of crystallinity. Consequently, Liu et al. [23] derived a new kinetic equation for non-isothermal crystallization by combining equations (4.8) and (4.10) yielding:

$$\log Z_t + n \log t = \log K(T) - m \log \beta_0 \quad [4.11a]$$

$$\log a = \log F(T) - b \log t \quad [4.11b]$$

where $F(T) = \left[\frac{K(T)}{Z_t} \right]^{1/m}$ and b is the ratio of the Avrami to Ozawa's exponent, i.e. $b=n/m$. $F(T)$

refers to the cooling rate at unit crystallization time when the system has a definite degree of crystallinity. This designates that $F(T)$ has a physical and practical meaning for non-isothermal crystallization processes, as Z_t has in the isothermal case. Therefore, the Avrami-equation and Ozawa-equation are combined as suggested by Liu et al. [23].

According to equation (4.11), a plot of $\log \beta_0$ against $\log t$ at particular degree of crystallinity gives a straight line with slope $-b$ and intercept $F(T)$. Figure 4.6 shows the plots of $\log \beta_0$ versus $\log t$ at various degrees of crystallinity and values of b and $F(T)$ are calculated for each particular crystallinity and listed in Table 4.5. At a certain degree of crystallinity, a high value of $F(T)$ indicates that a high cooling rate is needed to reach this degree of crystallinity in a unit time, which is related to the difficulty of the crystallization process for that particular material. Table 4.5 shows that the value of $F(T)$ increases with increasing crystallinity. However, the $F(T)$ values of PP/silica nanocomposites at a particular degree of crystallinity are lower and the increase in $F(T)$ with the degree of crystallinity is slower than for PP. This demonstrates that the silica nano-particles facilitate the crystallization process. The combined Avrami/Ozawa equations are able to describe the non-isothermal crystallization process for PP and PP/silica nanocomposites.

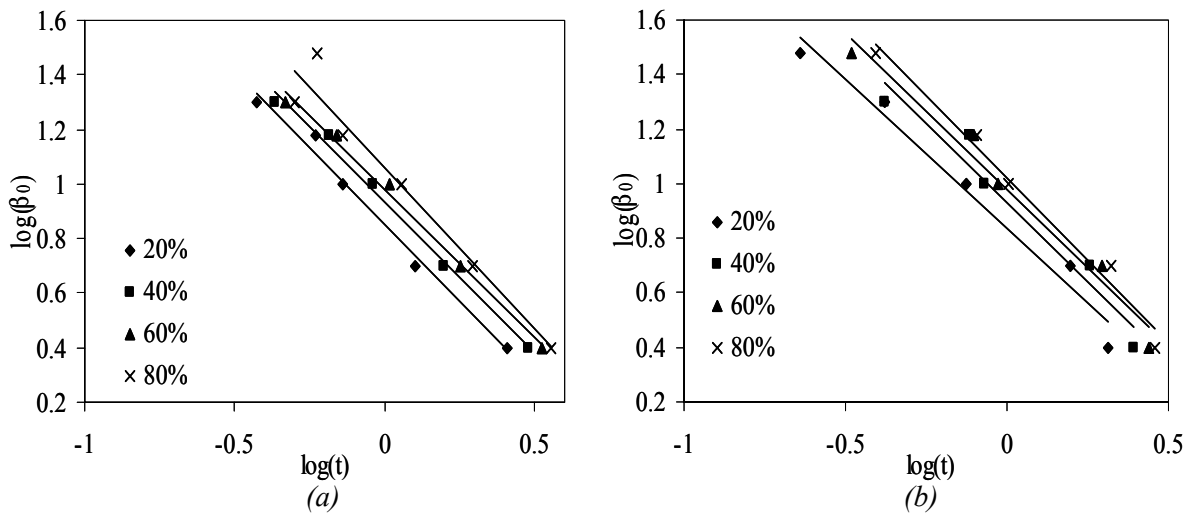


Figure 4.6: $\log \beta_0$ versus $\log t$ from the combined Avrami-Ozawa equation as proposed by Liu et al. for (a) PP and (b) PP/silica (1 wt%).

The cooling rate greatly influences the non-isothermal crystallization process. Considering this, Kissinger [29] proposed a method to determine the activation energy of crystallization by calculating the variation of crystallization temperature as a function of the cooling rate:

$$\frac{\left[\ln \left(\frac{\beta_0}{T_c^2} \right) \right]}{\left(\frac{1}{T_c} \right)} = \frac{-\Delta E}{R} \quad [4.12]$$

where β_0 is the cooling rate, R is the gas constant and T_c is the peak crystallization temperature.

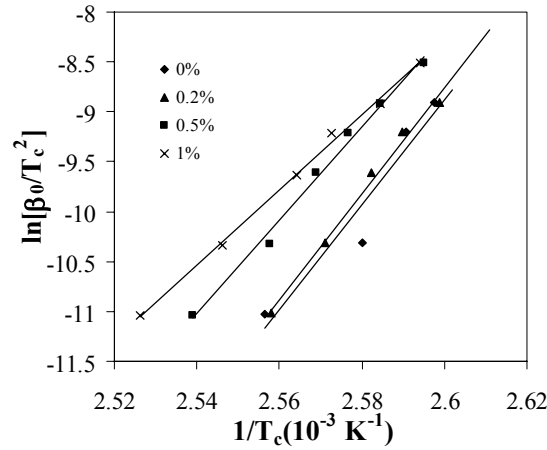


Figure 4.7: Kissinger plot for calculating the non-isothermal crystallization activation energies for PP and PP/silica nanocomposites.

The crystallization activation energies (ΔE) calculated from the slopes of $\ln[\beta_0/T_c^2]$ vs. $1/T_c$ (Figure 4.7) are: 435, 322, 324, 311 and 358 kJ/mol for 0, 0.2, 0.5, 1 and 1.5 wt% silica, respectively, in PP nanocomposites. The ΔE of the nanocomposites reduced as the silica concentration is increased up to 1 wt%, after which it displays a marginal increase, which implies that the silica nano-particles influence the PP chains to crystallize easier and accelerate the crystallization rates during the non-isothermal crystallization process.

4.3.3 Nucleation efficiency

Figure 4.8a illustrates the evolution of the crystallization temperature for PP after self-nucleation carried out at different temperatures and a fixed cooling rate (10 K/min). From these curves, T_{c2max} for the best self-nucleated PP is chosen as the upper limit and T_{c1} of the virgin PP is taken as the lower limit. The higher temperature T_{c2max} is 138.6°C, whereas the lower temperature T_{c1} is 110.6°C, i.e. a difference of 26.9°C defines the efficiency scale.

The nucleating efficiency of silica nano-particles is plotted in Figure 4.8b as a function of silica concentration. It shows a nucleating efficiency close to 20%, which is comparable to most commercially available nucleating agents, independent of the underlying mechanism of nucleation. The in-situ prepared silica nano-particles show a higher nucleation efficiency compared to other PP-nanocomposites, including silica (see Table 4.1). The nucleating efficiency, however, saturates above 0.5 wt% up to 1 wt% followed by a slight decrease at higher concentrations. As explained earlier, the reason for the apparent decrease in nucleation efficiency is not clear at this moment and

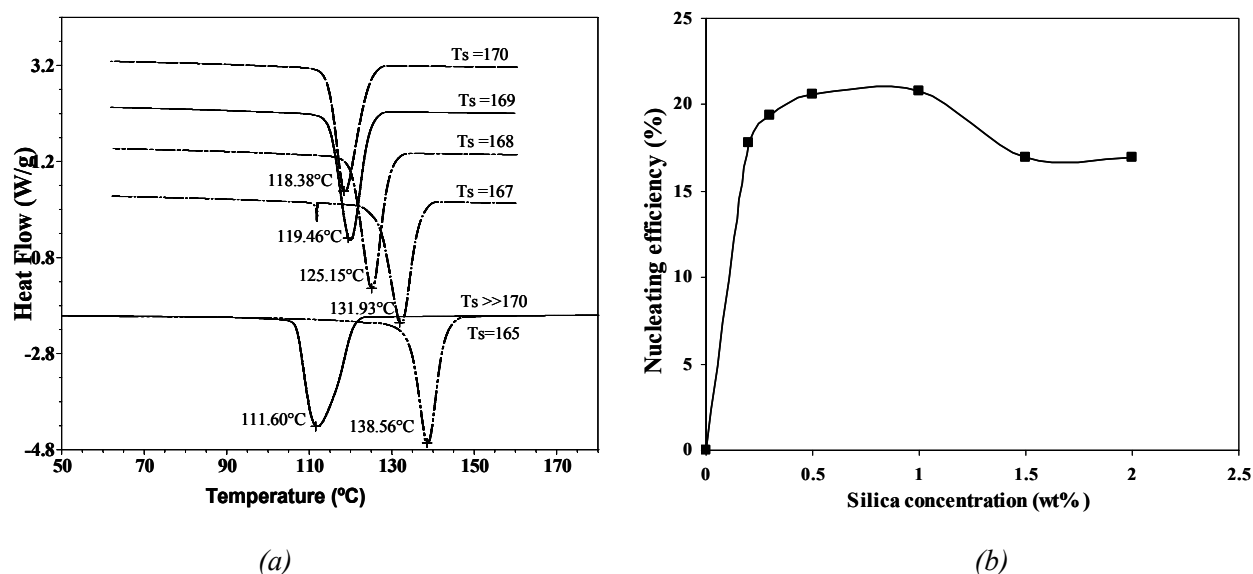


Figure 4.8: (a) Cooling curves of PP after melting and annealing for 5 min at various T_s to determine the nucleation efficiency scale, (b) nucleating efficiency as a function of silica concentration in PP/silica.

4.3.4 Melting behavior of PP/silica nanocomposites

The melting behavior of the PP/silica nanocomposites reflects the nucleating effect of silica. The narrow melting peaks exhibited by the PP/silica nanocomposites during heating (Figure 4.9a) suggest that more perfect lamellae are formed with a narrow size distribution. The melting temperature (T_m) is constant but the width of melting peaks, which reflects the lamellar thickness distribution, reduces considerably compared to pure PP. The values of T_m , onset of T_m and start (T_{sm}) and end (T_{em}) of the melting endotherms are listed in Table 4.6. All samples show a melting temperature of 160°C, which is the characteristic melting point of the α -phase of PP [21]. A small melting peak at approx. 145°C is observed, although not clear in Figure 4.9a for low silica concentrations, but for nanocomposites with high silica loading it is more prominent. This can be assigned to PP-crystals with the β -modification, which is known to have a melting temperature near 145°C [40]. This is also confirmed by the X-ray diffraction studies.

The melting enthalpy is almost constant for all the samples, which implies that the crystallinity remains constant. However, from the crystallization enthalpy, the PP/silica nanocomposites show a slightly higher crystallinity than PP. The recrystallization process may explain this discrepancy. The completion of melting of the β -modification and the start of the recrystallization in the α -modification (α') occurs simultaneously; the peak due to recrystallization reaches a maximum at 150°C during the melting process. The area of the peak from the β -modification in the silica-nucleated PP is too small to deconvolute from the peak area due to the recrystallization-melting process of α -form during heating with a standard heating rate.

In case of PP/silica nanocomposites with higher silica loadings in the range of 2-5 wt%, the amount of crystals with the β -modification induced by the silica nano-particles is higher (Figure 4.9b). The peak at 145°C corresponding to the β -modification is persistent even after repeated melting and crystallization. This signifies that it is not the effect of thermal treatment or melt memory, but a true effect of the silica nano-particles on the PP crystal modification.

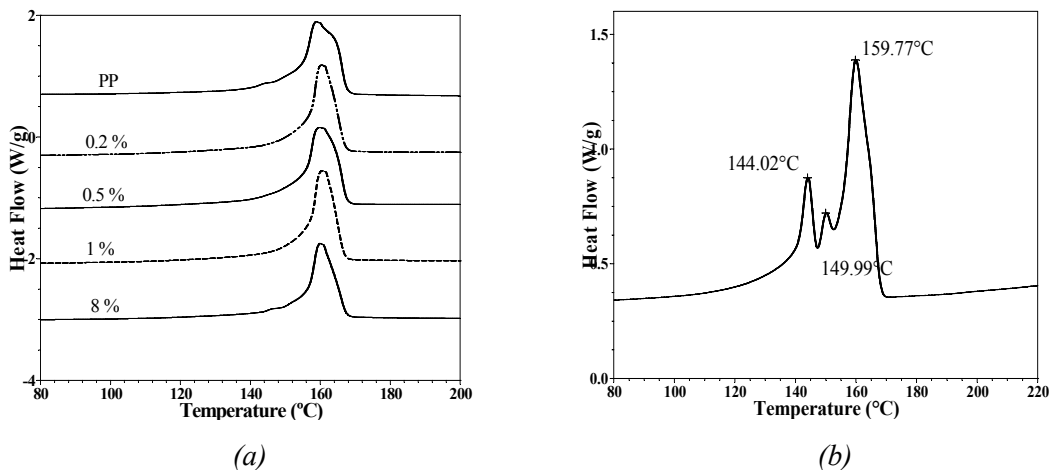


Figure 4.9: (a) Melting curves of PP/silica nanocomposites with 10 K/min, (b) melting curve of PP-nanocomposite with 5 wt% silica to demonstrate the formation of the β -modification.

Table 4.6: Melting data for PP/silica nanocomposites at a heating rate of 10 K/min

Silica	T_m	$T_{m \text{ onset}}$	DH_m	T_{sm}	T_{em}	ΔT	X_{cm}
wt%	(°C)	(°C)	(J/g)	(°C)	(°C)	(°C)	%
0.0	158.7	153.7	91.1	171.0	99.9	71.1	44.0
0.2	159.8	154.7	92.4	169.4	114.5	54.9	44.8
0.5	160.3	156.1	91.8	169.9	127.8	42.1	44.6
1.0	160.2	156.2	93.0	168.7	125.7	43.0	43.4
8.0	159.8	155.6	75.1	170.7	121.8	48.9	39.4

In-situ X-ray diffraction confirms the presence of small β -crystallites in the PP/silica nanocomposites. The WAXD patterns recorded during cooling from the melt (225°C) to 70°C at 10 K/min are shown in Figure 4.10. The peak near $2\theta = 16^\circ$ is the characteristic peak of the 300 plane of the β -unit cell [41]. This peak is extremely small in the case of PP, but significantly more intense for silica-nucleated PP. The peaks in the WAXD pattern appear as soon as the material attains an appreciable degree of crystallinity. The appearance of these peaks relates well to the crystallization temperature calculated from DSC cooling curves. The crystallization temperature for PP/silica (1 wt%) nanocomposites is 120°C, whereas for PP it is 110°C.

4.3.5 Effect of silica on the glass transition temperature

As explained earlier, the peak in the loss modulus curve is used to determine the glass transition temperature and the intensity of transition calculated from the change in the storage modulus before and after glass transition is used to determine the mobility of the amorphous phase. Figure 4.11a shows the evolution of the loss modulus, which is always higher for the PP/silica nanocomposites than for PP and Figure 4.11b shows the storage modulus curves. The storage modulus shows a sharp decrease for the PP/silica nanocomposites at the glass transition temperature, but it is still higher than for pure PP.

The T_g for PP/silica nanocomposites is observed at temperatures below that of PP (see also Table 4.7). This can be explained with the help of the DSC results: the silica acts as a nucleating agent leading to the formation of more perfect crystals along with a faster crystallization of PP (Table

4.3). Perfectioning of crystals results in a more disentangled amorphous phase which has a higher mobility in the PP/silica nanocomposites, resulting in a slightly lower T_g value. The effect is more evident from the drop in elastic modulus and is further explained by using the intensity of the transition.

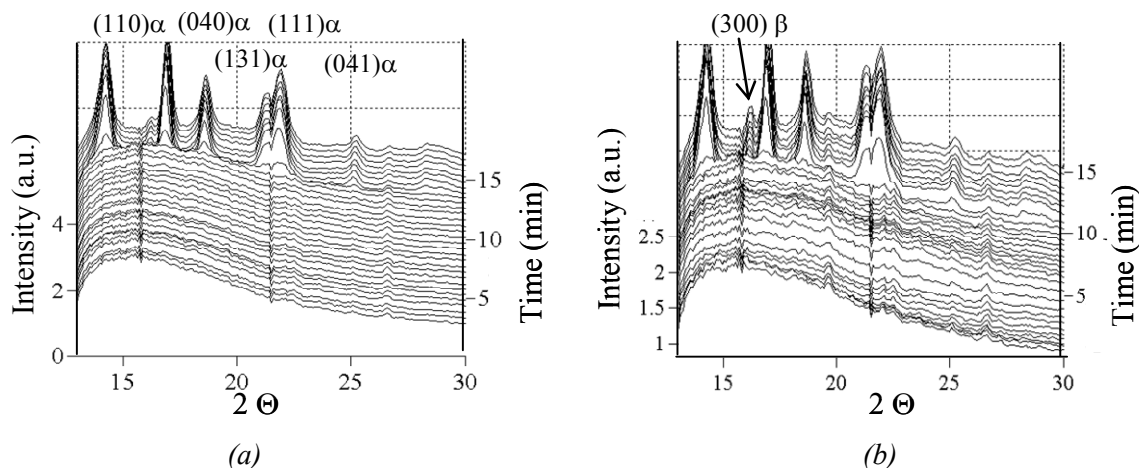


Figure 4.10: WAXD patterns of (a) PP and (b) PP/silica (1 wt%) obtained during cooling from 225 to 70°C at 10 K/min.

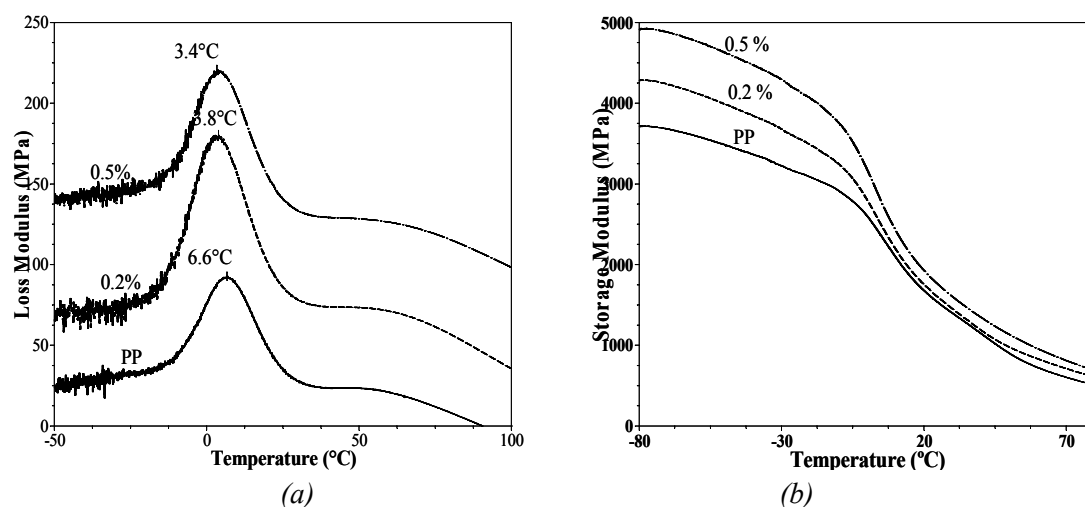


Figure 4.11: The mechanical data for PP/silica nanocomposites as a function of silica concentration: (a) loss modulus as a function of temperature, (b) storage modulus as a function of temperature.

Table 4.7: T_g , intensity of transition and full width at half maximum (FWHM) of PP/silica nanocomposites measured by DMTA

silica %	T_g (°C)	S %	FwHM (°C)
0	6.6	1.25	38.1
0.2	3.8	1.71	40.7
0.5	3.4	1.82	39.8

The intensity of the transition (S) is higher for the PP/silica nanocomposites than for PP (Table 4.7). This indicates that the nanocomposites have a less entangled (more mobile) amorphous phase; however, DSC data shows that the nanocomposites have a higher crystallinity than PP. The higher intensity of the transition suggests that the drop in elastic modulus above the glass transition is higher which in turn indicates a more disentangled amorphous phase. Díez-Gutiérrez et al. [25] observed that the T_g as well as the intensity of the transition decreased for PP/talc composites. According to them, the addition of a rigid phase reduces the mobility of polymer chains, which results in lower values of the transition. The concentration of talc used in their work was 40 wt%, whereas in the present work the silica concentration is less than 1 wt% implying that the concentration of the rigid amorphous phase is not high enough to affect the mobility of the polymer chains. However, the kinetics of crystallization in the presence of fillers does affect the amorphous phase. The inhomogeneity of the amorphous phase and a high mobility imparts the flexibility to the matrix, which results in a decrease of storage modulus at higher temperatures even with a higher crystallinity. An increase of the FWHM values of T_g was observed for PP/silica nanocomposites. This was expected considering the nucleating effect of silica, which results in faster crystallization and more perfect crystals but in a more inhomogeneous amorphous phase than for PP.

4.4 Conclusions

The crystallization behavior as studied with DSC shows that in-situ prepared silica nano-particles via the sol-gel method in PP/silica nanocomposites act as nucleating agents. The nucleating efficiency of the silica particles is 20% in the low concentration range and is comparable to the most commercial nucleating agents, and higher compared to silica nano-fillers as well as other nano-fillers studies by other researchers. The nucleation efficiency saturate above certain concentration (c.a. 1 wt%) of silica. The apparent decrease in nucleation and growth may be related to a different nucleation mechanism and the physisorption of high molecular weight PP chains on silica. In the lower concentration range, the segregation of high and low molecular weight chains influence the chain diffusion to the crystal growth front, while at higher concentrations of silica, segregation is not possible. Also, at higher concentrations, silica start agglomerating which effectively reduces the surface area and hence nucleation sites.

The combined Avrami-Ozawa equation approach provides a convenient tool to study the non-isothermal crystallization process for these systems. The non-isothermal crystallization kinetics proceeds in two-stages: the primary stage is characterized by nucleation and spherulitic growth and the secondary stage is characteristic of the perfection of crystals. Silica, being a heterogeneous nucleating agent, accelerates the primary stage, resulting in more perfect crystals. This is also evident from the melting behavior of the PP/silica nanocomposites. The crystallization activation energy decreases with increasing silica concentration in the PP/silica nanocomposites. The melting curves indicate the formation of more perfect crystals and a narrow distribution of crystal size and lamellar thickness. The WAXD patterns show that silica nano-particles induce the β -modification in PP. Though at low levels, the presence of some crystals with the β -modification may help in improving the mechanical properties, especially the impact toughness of the PP.

4.5 References

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- [41] The X-ray wavelength used is 1.218 Å.

Chapter 5

Non-isothermal crystallization of in-situ prepared (VTES-grafted PP)/silica nanocomposites*

Abstract

PP/silica nanocomposites were prepared via solid-state modification (SSM) by grafting VTES on PP and then making in-situ silica nano-particles which are chemically attached to PP through VTES. The effect of grafting is investigated in this study. SSM results in non-random grafting on VTES on PP, which eventually affects the crystallization behavior of PP, resulting in non-uniform crystals compared to solution-grafted PP. The crystallization temperature increases with degree of grafting, indicating heterogeneous nucleation for (VTES-grafted PP)/silica nanocomposites. The overall crystallization time is longer for (VTES-grafted PP)/silica nanocomposites. Silica is grafted via VTES to multiple PP chains which results in network formation or in other words silica become nodal points for PP cross-links. The effect of fractionation due to non-random grafting of VTES on PP is also observed from the melting behavior which shows multiple and broad melting peaks.

5.1 Introduction

Various mechanisms on improving the toughness of semi-crystalline polymers have been proposed in literature [1,2,3]. Addition of rubbery fillers improves the toughness but at the expense of the modulus [2]. On the other hand, the use of hard inorganic fillers (1-10 microns) improves the impact toughness of semi-crystalline polymers, while at the same time maintaining the high modulus [4,5,6]. Typical fractions of the particulates used for improvement of the properties are as high as 20 vol% and consequently the processibility is strongly affected.

Moreover, the properties of composites are strongly dependent on the nature of the filler-polymer interface. The interfacial strength affects delocalization of the stresses during macroscopic deformation of the composites. It was recently shown that debonding is an important mechanism in improving the toughness of micro-particle filled polymers, because it allows plastic stretching of ligaments between the debonded particles [7]. Thio et al. [8] suggested that a weak interaction/adhesion between filler particles and the polymer matrix is desirable for toughening. Several methods have been explored to control the filler-matrix interface. Addition of interfacial or coupling agents is one of the most commonly used methods [13]. For example, silica particles can be chemically grafted on PP using vinyl triethoxy silane (VTES), which acts as an interfacial agent between the silica particles and PP matrix as described in Chapter 2. Reactive compatibilization is another method of controlling the interface. Grafting of polar monomers on an apolar matrix

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material is also an important alternative for increasing interaction between filler and matrix. PP, being a non-polar semi-crystalline polymer, is often modified by grafting polar monomers, such as alkoxy silane, maleic anhydride, methyl methacrylic acid (MMA) [13]. The grafted polar groups increase the interfacial interaction between fillers and matrix leading to stronger adhesion in case of PP/filler systems as explained in Chapter 2.

Polymer nanocomposites with hard fillers (10-100 nm) exhibit properties comparable to, or even superior to, those of traditional composites at substantially lower filler content ($< 2\%$) and, thus, enabling a greater retention or in some cases even improvement of the processability. Although much progress has been made in the field of nanocomposites in the last two decades, the challenge is still to achieve single-particle dispersions resulting in isotropic properties. Several preparation methods, such as melt compounding [9,10], solution blending [11], melt intercalation, *in-situ* intercalative polymerization [12] were developed to prepare PP-nanocomposites. However, the aforementioned problem of dispersion is not fully solved. In some of recent studies coupling agents or functional co-polymers for dispersion of nanoparticles were used [13]. These modifiers change the interface between polymer matrix and fillers which makes it difficult to understand the mechanism of improvement in the toughening.

In this study, nanocomposites were obtained by in-situ preparation of silica nano-particles in PP which was grafted in solid-state with vinyl triethoxy silane (VTES) (see Chapter 2, Section 2.3.1.2). This approach provides a simple, solvent-free process. The reaction temperature is low, which avoids degradation of PP and yields the final product in powder form which can be directly processed. Grafting of polar monomers on PP via solid-state modification (SSM) has widely been studied [14] and predominantly takes place in the amorphous phase of PP. This results in a non-random distribution of grafted VTES monomer and blocks of grafted VTES are formed. Melt or solution grafting, on the other hand, takes place in the homogeneous phase because the crystalline part is absent and this results in random grafting. Figure 5.1 shows the schematics of the non-random grafting via SSM against random grafting in solution or melt.

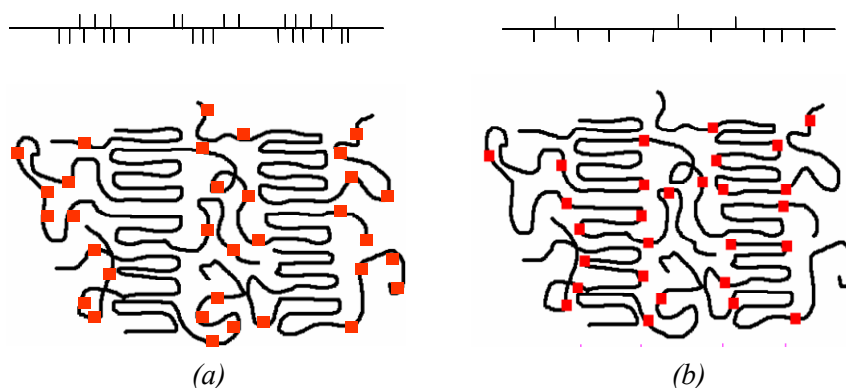


Figure 5.1: Schematics showing (a) non-random grafting of VTES on to PP via solid-state modification (SSM) and (b) random grafting in solution. (■) represents the VTES monomer.

The manner in which the monomer grafts (random versus non-random) on the PP backbone will affect the composition of the amorphous phase, the chain configuration and hence the crystallization kinetics as well as the morphology of PP. Non-random grafting may result in clustering imposing constraints on the mobility of PP chains. This has far reaching implications on the crystallization

kinetics and hence the developed structure and morphology. Schrauwen et al. [15] and Strobl et al. [16] found that the properties of semi-crystalline polymers are governed by the degree of crystallinity, lamellar thickness and entanglement density. In this respect, it is important to investigate the effect of the interfacial agent on these two factors as it can have opposite effects on the properties.

Industrial polymer processes, such as extrusion, injection molding, and film blowing, are performed under dynamic, non-isothermal conditions which strongly affect crystallization. Liu et al. [17] developed a new approach by combining the Avrami and Ozawa theories and described successfully the non-isothermal crystallization of poly(ether ether ketone). This approach was used in the previous chapter to analyze the non-isothermal crystallization kinetics of PP/silica nanocomposites. The effect of VTES grafted on PP on the crystallization behavior and kinetics, crystallization activation energy, and nucleation efficiency of (VTES-grafted PP)/silica nanocomposites are investigated in this chapter. The melting behavior as studied by DSC and X-ray diffraction gives more insight in the chemical and lamellar thickness distribution, and the crystal modification formed in the presence of this additional component.

5.2 Experimental

5.2.1 Materials

Porous isotactic polypropylene (PP) having a porosity of $S_{BET(Kr)} = 0.0678 \text{ m}^2/\text{g}$ supplied by Euro-Sabic, The Netherlands, was used. The number-average and weight-average molecular weight of PP was 60 and 380 kg/mol respectively. The powder was free of antioxidants and stabilizers and was stored at -10°C in the dark. Vinyl triethoxy silane (VTES) and tetraethoxy orthosilicate (TEOS) were obtained from Aldrich Chemicals. α - α' -Azoisobutyronitrile (AIBN), from Merck Schuchardt, Germany, and dibenzoyl peroxide (BPO), from Fluka Chemicals were used as free-radical initiators.

5.2.2 Synthesis of nanocomposites

Grafting via SSM

Chemical modification of PP by grafting VTES (VTES-grafted PP) and (VTES-grafted PP)/silica nanocomposites were performed according to the procedures described in detail in Chapter 2, Section 2.3.1.2. The first step in the synthesis of (VTES-grafted PP)/silica nanocomposites was grafting of VTES on isotactic PP via (SSM). First, VTES was dispersed in the pores and subsequently migrated to the amorphous phase of PP at the reaction temperature, then the initiator was added at the desired rate. At the end of the reaction, the reactor was heated to higher temperatures to decompose all residual initiator. Subsequently, the modified PP was kept under vacuum at approx. 80°C for 24 hrs to remove all unreacted monomers and was then stored in airtight desiccators to avoid absorption of moisture.

Grafting via solution blending

VTES was solution-grafted on PP. The same reaction parameters as used for the SSM method were applied. First, PP was dissolved in xylene at 130°C at 10 wt% concentration and then slowly cooled down to 100°C at which the PP remained soluble for at least 4 hrs. Then, a mixture of VTES and AIBN (2 wt%) was added slowly with continuous stirring and the reaction was carried for 2 hrs. After the reaction was completed, the solution was cooled down and xylene was removed subsequently by drying under vacuum.

5.2.3 Differential scanning calorimetry (DSC)

The crystallization and melting characteristics were investigated with a TA Instruments Q1000 Differential Scanning Calorimeter with sample weights between 3-4 mg. The analysis was conducted under standard heating and cooling rates of 10 K/min unless otherwise indicated. Prior to the heating or cooling scan, the samples were held at 225°C for 5 min under a nitrogen flow to erase the thermal history and to prevent self-seeding of PP. The non-isothermal crystallization kinetics of these samples were investigated by conducting cooling scans at 2.5, 5, 10, 15, 20 and 30 K/min from 250°C to 30°C .

5.2.4 X-ray characterization

Wide-angle X-ray diffraction (WAXD) experiments were performed at the DUBBLE (Dutch-Belgium) Beamline at the European Synchrotron Research Facility (ESRF), Grenoble, France. An X-ray wavelength of 1.218 Å was used. The WAXD patterns were detected by a curved linear microstrip-gas chamber detector. The WAXD detector was calibrated with Si powder. The films/powder were put in aluminum pans and placed in a THMS600 hot stage (Linkam Scientific Ltd.). Data were recorded while the sample was heated and cooled at 10 K/min. Diffraction patterns were taken every 6 s, which corresponds to one pattern for each degree of temperature variation.

Theory of crystallization

The theory of isothermal crystallization kinetics is well developed and has been experimentally verified [18]. However, for non-isothermal crystallization, the heating rate is always changing which required a modification of the theories developed for isothermal crystallization. Here, the corresponding methods for calculation and interpretation are briefly summarized.

From the DSC curves the values of the relative crystallinity at various cooling rates can be calculated. The relative crystallinity as a function of time can be defined as:

$$X_t = \frac{X_t}{X_\infty} = \frac{\int_{t_0}^t \left(\frac{dQ}{dt} \right) dt}{\int_{t_0}^{t_e} \left(\frac{dQ}{dt} \right) dt} \quad [5.1]$$

where (dQ/dt) represents the heat flow, X_t and X_∞ represent the crystallinity at time t and at the end of crystallization process t_e , respectively. For the non-isothermal crystallization, the crystallization time t and the temperature T are related as:

$$t = \frac{|T_0 - T|}{\beta_0} \quad [5.2]$$

where β_0 is the cooling rate, T is the temperature at time t and T_0 is the initial temperature when crystallization starts. Thus, the relative crystallinity as a function of the temperature $X_c(T)$ is defined as:

$$X_c(T) = \frac{\int_{T_s}^T \left(\frac{dQ(T)}{dT} \right) dT}{\int_{T_s}^{T_e} \left(\frac{dQ(T)}{dT} \right) dT} \quad [5.3]$$

Similar studies as used for PP/silica nanocomposites were carried to investigate the effect of VTES grafting on non-isothermal crystallization kinetics of (VTES-grafted PP)/silica nanocomposites. The Avrami equation [18] for isothermal crystallization with the Jeziorny correction [19] reads as:

$$X_t = 1 - \exp(-Z_t(T)t^n) \text{ and } \log Z_c = \frac{\log Z_t}{\beta_0} \quad [5.4]$$

where n is the Avrami exponent, $Z_t(T)$ the Avrami rate constant and X_t the relative crystallinity at time t . As noted in the previous study, the modified Avrami equation is valid for the early stages of the crystallization, therefore the approach used by Liu et al. [17] by combining Avrami and Ozawa equation is used to study the non-isothermal crystallization behavior:

$$\log Z_t + n \log t = \log K(T) - m \log \beta_0 \quad [5.5a]$$

$$\log a = \log F(T) - b \log t \quad [5.5b]$$

where $F(T) = \left[\frac{K(T)}{Z_t} \right]^{1/m}$ and b is the ratio of the Avrami to Ozawa's exponent, i.e. $b=n/m$. $F(T)$

refers to the cooling rate at unit crystallization time when the system has a definite degree of

crystallinity. This designates that $F(T)$ has a physical and practical meaning for non-isothermal crystallization processes, as Z_t has in the isothermal case.

5.3 Results and discussion

The manner in which a monomer is grafted on backbone of polymer chains affects the crystallization behavior. Clustering of grafted monomers, position of grafting, and the distance between the grafted monomers, are some of the factors which may play a significant role in determining the crystallization kinetics and the structure. A random and non-random grafting can be achieved by grafting VTES in solution and via SSM, respectively. The results are divided into two parts. First, the effect of grafting on PP (without silica) is studied and later in second part, the study is extended for (VTES-grafted PP)/silica nanocomposites.

5.3.1. VTES-grafted PP without silica

VTES-grafted PP with 3 different degrees of grafting, viz, 1.5 wt%, 2.6 wt% and 3.9 wt% in solution and 0.7 wt%, 2.3 wt% and 3.2 wt% via SSM, were used in this study. The degree of grafting was estimated according to the procedure described in Chapter 2. Three different levels were used depending on the inter-group distance which will be elaborated in subsequent sections.

5.3.1.1 Molecular structures of random versus non-random VTES-grafted PP

Solution grafting of VTES results in random grafting on PP. The maximum degree of grafting used in this study is 3.9 wt%, which gives an average spacing of more than 15 nm between two grafted VTES molecules. Therefore, these isolated grafted monomers due to random grafting cannot form clusters. However, in the case of non-random grafting, sequences of VTES are formed. These polar sequences can interact with each other and due to these cooperative effect form clusters which are rich in grafted VTES. This imparts some elasticity to the materials, like cross-linking of a rubber, which is observed from the rheological behavior shown in Figure 5.2. The storage modulus is a measure of the elasticity of polymers in the melt. The increase in storage modulus, especially in the lower frequency range, reflects the increase in elasticity. The slope of curves becomes less than 2 in the lower frequency range and at higher degrees of grafting, the modulus even reaches a plateau implying network formation. However, it is not easy to distinguish the increase in modulus at lower frequencies is only due to clustering or also due to long chain branching of PP during grafting. The dynamic viscosity of VTES-grafted PP shown in Figure 5.2b is comparable to pure PP which indicates that there is marginal chain scission but in comparison to the increase in modulus at lower frequencies it is less. Therefore, it can be assumed that the observed increase in elasticity is due to a clustering effect rather than the long chain branching of PP. This will affect the crystallization and structure development during processing. The following sections concentrate on the effect of grafting on the crystallization behavior.

5.3.1.2 Effect of grafting on the crystallization behavior of PP: solution grafting versus SSM

VTES-grafted PP in solution

Figure 5.3a depicts the cooling curves of VTES-grafted PP in solution and via SSM with varying degrees of grafting. The crystallization temperature increases slightly with increasing VTES grafting which implies that the grafted VTES-groups act as nucleating sites. During the cooling, material start to crystallize, the peak temperature during crystallization is taken as the crystallization

temperature (T_c). The peak crystallization temperature (T_c), onset of crystallization (T_{onset}) and starting temperature for crystallization (T_s) are listed in Table 5.1. A high degree of grafting may constrain the crystallization. It can be noted from Table 5.1 that the onset of crystallization does not vary with the degree of grafting. The crystallization peak is narrower compared to pure PP, which indicate a uniform crystal size. Solution grafting results in a more random structure. The average distance between two grafted monomer units can be calculated from the bond length ($C-C \approx 1.53 \text{ \AA}$) and angle ($\alpha \approx 34^\circ$) of the PP backbone. This gives the distance between two grafted monomers: 40 nm, 25 nm and 15 nm for 1.5 wt%, 2.6 wt% and 3.9 wt% respectively. These numbers are less than the usually observed lamellar thickness (10 - 20 nm) for PP. Thus, solution grafting will hardly affect the chain folding process and, hence, the crystallization proceeds as for pure PP, except at higher degrees of grafting, i.e. 3.9 wt%.

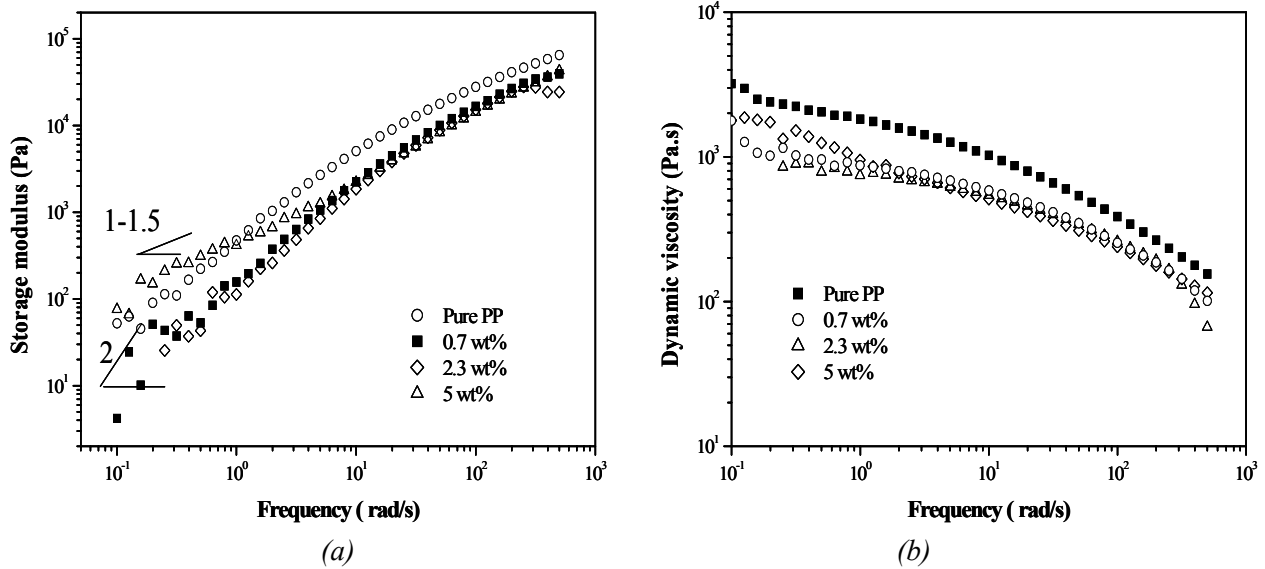


Figure 5.2: (a) Storage modulus and (b) dynamic viscosity of VTES grafted on PP via SSM measured at 180°C at constant strain of 1%.

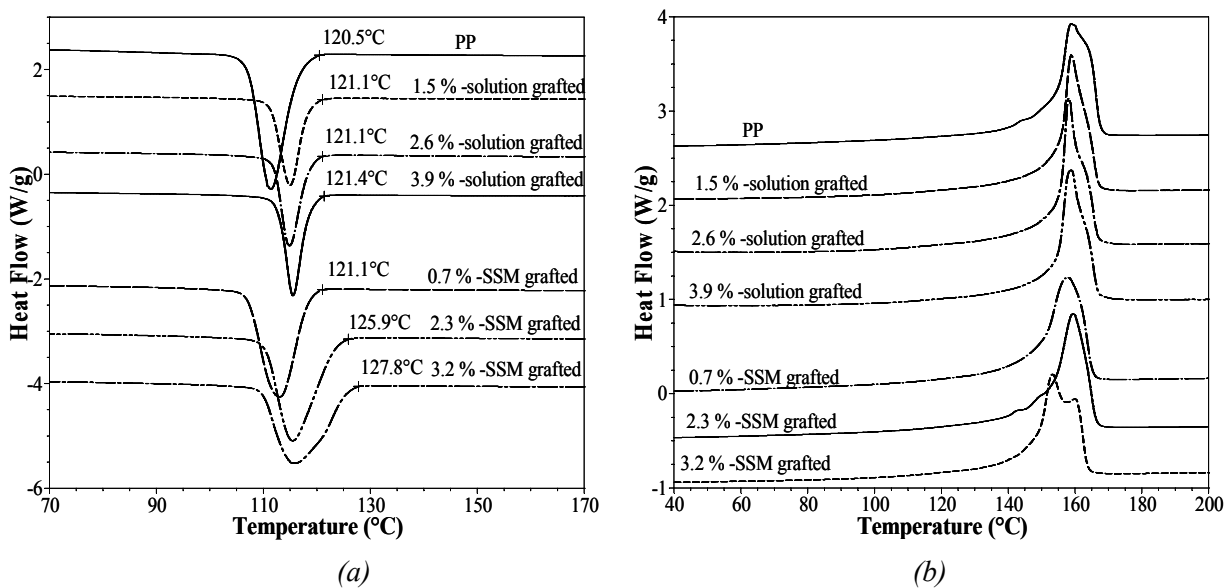


Figure 5.3: (a) Crystallization and (b) melting behavior of SSM and solution-grafted VTES on PP showing the effect of random versus non-random grafting. For all measurements the heating and cooling rate was 10 K/min. The heating curves are measured after the crystallization rates as shown in (a).

Similar observations are made from the melting behavior of solution-grafted PP. The melting endotherms in Figure 5.3b show narrow melting peaks for all the solution-grafted PP. Moreover, the melting peaks are narrower compared to pure PP indicating the formation of crystals with a more uniform crystal size (see also Table 5.2). The melting curve of the 3.9 wt% grafted PP system exhibits a small shoulder at slightly higher temperatures. The main melting peak is slightly lower than those observed for lower degrees of grafting with an additional shoulder at higher temperatures suggesting formation of thicker lamellae along with thinner lamellae. In general, the melting behavior is not different from PP illustrating that grafting in solution yields random structures. As a result, chain folding or fractionation and eventually the crystalline structure and morphology are not affected.

PP grafted via SSM

VTES grafting via SSM mainly takes place in the amorphous phase which result in non-random grafting. Consequently, fractionation of chain segments with different degrees of grafting can take place, what may yield a distribution of crystal sizes. The cooling curves of SSM-grafted PP with an increasing degree of grafting are shown in Figure 5.3a. There is a clear effect of grafting on crystallization. The crystallization temperature (T_c), onset of crystallization (T_{onset}) and starting temperature of crystallization (T_s) are listed in Table 5.1. The crystallization peaks are broad in the case of SSM-grafted PP compared to solution-grafted samples, which suggests the formation of non-uniform crystalline size/structures. At higher degree of grafting (3.2 wt%), the crystallization exotherm can be deconvoluted into two peaks: a main peak corresponding to $T_c = 115^\circ\text{C}$ and a shoulder peak at approx. 120°C . The two peaks are better separated at higher cooling rates (see Table 5.1). This is probably due to the formation of sequences of VTES grafts. These fractionated chains of different length crystallize at different degrees of undercooling (and temperatures in case of non-isothermal crystallization) which broaden the crystallization peak. The onset of crystallization is also at a higher temperature in case of SSM-grafted PP.

A complex behavior is expected to result in broad or multiple melting peaks for PP during heating at a constant rate (10 K/min). The melting behavior of VTES-grafted PP via SSM is also shown in Figure 5.3b along with the solution-grafted PP system. As expected, the curves show a broadening and splitting of the melting peak with increasing degree of grafting in case of VTES-grafted PP via SSM. The data of the melting peaks are listed in Table 5.2.

The effect of non-random grafting is twofold; first, the non-random grafting results in fractionation and segregation of VTES-rich PP with different compositions forming imperfect crystals of different size and, secondly, the shift to lower melting temperatures suggest the formation of crystals with thinner lamellae due to exclusion of the grafted VTES from the crystal lamellae. In the case of random grafting, the average distance between the grafted monomers is larger than the usually observed lamellar thickness; however, in the case of non-random grafting, where VTES-rich phase has spacing less than 10 - 20 nm, multiple melting peaks were observed during melting. Moreover, it is possible that clustering of grafted VTES induces formation of crystals with a different modification such as β -modification or imperfect α -modification. These modifications show a rather low melting temperature, for instance β -modifications melts at approx. 147°C

Table 5.1: Non-isothermal crystallization behavior of VTES-grafted PP via solution and SSM at different cooling rates

Solution-grafted									
degree of grafting 1.5%				2.6%				3.9%	
rate	T _s	T _c	T _e	T _{onset}	T _{c2}	T _s	T _c	T _e	T _{onset}
K/min	°C	°C	°C	°C	°C	°C	°C	°C	°C
2.5	127.9	119.5	112.4	124.1	--	128.8	119.7	104.2	123.9
10	123.2	114.9	102.2	118.2	--	124.4	114.8	100.9	118.4
15	120.9	113.5	102.5	116.8	--	120.2	112.1	92.7	116.5
20	120.8	112.2	97.9	115.8	--	119.9	110.7	92.6	115.6
25	119.8	111.0	93.3	114.8	--	118.6	111.2	92.6	115.1
30	118.5	110.3	91.0	114.4	--	117.6	110.3	95.9	114.3
40	116.9	108.8	87.2	112.8	--	117.1	109.0	86.4	113.9
50	115.8	107.5	85.5	111.9	--	116.1	107.6	82.8	112.2

SSM-grafted									
degree of grafting 0.7%				2.3%				3.2%	
rate	T _s	T _c	T _e	T _{onset}	T _{c2}	T _s	T _c	T _e	T _{onset}
K/min	°C	°C	°C	°C	°C	°C	°C	°C	°C
2.5	128.3	119.2	105.0	124.0	--	134.1	121.7	107.1	128.8
10	121.9	112.9	94.4	118.3	--	127.7	115.5	97.2	123.5
15	122.6	112.1	91.2	118.7	--	125.5	112.7	95.4	120.2
20	120.4	109.9	91.9	117.2	--	124.4	111.7	92.3	118.6
25	120.4	109.8	87.2	116.5	--	122.6	108.7	90.8	117.1
30	118.0	109.5	80.3	114.4	--	122.0	105.2	89.8	117.3
40	118.0	107.2	74.0	114.5	--	119.9	105.1	84.3	114.1
50	117.0	105.6	74.7	113.6	--	118.9	103.2	76.7	112.7

T_s: the starting temperature of crystallization peak, T_c: end temperature of crystallization peak, T_e: the peak crystallization temperature, T_{c2}: additional crystallization peak or shoulder temperature
T_{onset}: onset of crystallization. The details are explained in Chapter 4.

Table 5.2: Melting behavior of VTES-grafted PP via solution and SSM at 10 K/min after crystallization at different cooling rates

Solution-grafted									
degree of grafting					2.6%				
1.5%					3.9%				
rate	T _m	T _s	T _e	T _{onset}	T _m	T _s	T _e	T _{onset}	T _{md}
K/min	°C	°C	°C	°C	°C	°C	°C	°C	°C
2.5	159.5	98.0	174.1	156.4	160.1	108.8	170.3	156.7	
10	158.1	108.3	173.8	154.9	158.7	97.8	172.8	155.0	
15	157.2	99.7	174.0	153.7	158.2	97.0	170.7	154.4	
20	157.2	96.4	173.2	153.4	158.6	96.7	171.8	154.0	
25	157.8	102.1	173.5	153.7	158.6	99.5	172.9	153.6	
30	156.9	111.3	171.2	153.5	157.5	97.0	173.4	153.4	162.5
40	157.2	97.1	172.3	153.1	158.1	106.9	170.6	153.0	162.2
50	156.9	93.3	174.1	152.7	157.7	104.2	171.7	152.7	163.2

SSM-grafted									
degree of grafting					2.3%				
0.7%					3.2%				
rate	T _m	T _s	T _e	T _{onset}	T _m	T _s	T _e	T _{onset}	T _{md}
K/min	°C	°C	°C	°C	°C	°C	°C	°C	°C
3	158.8	9363.0	168.3	150.3	155.0	90.1	168.8	149.5	142.0
10	157.7	105.6	168.5	148.4	153.1	90.5	169.4	147.7	159.6
15	156.2	99.5	168.9	149.3	156.1	101.0	169.8	150.5	162.6
20	156.7	97.6	170.0	148.2	155.8	93.3	171.8	149.6	162.4
25	155.7	82.0	171.7	148.6	162.3	98.0	171.8	149.7	155.6
30	161.2	98.3	170.8	147.2	159.9	95.2	168.4	146.4	151.7
40	161.4	93.1	171.6	148.4	162.3	100.0	171.4	148.8	154.9
50	131.6	102.5	170.9	148.1	162.2	95.2	172.0	148.4	154.7

T_s: the starting temperature of melting peak, T_e: end temperature of melting peak, T_m: the peak melting temperature, T_{m2} & T_{m3}: additional melting peaks or shoulder temperature
T_{onset}: onset of melting peak.

5.3.1.3 Non-isothermal crystallization kinetics of VTES-grafted PP

The crystallization curves of 2.6 wt% solution-grafted and 2.3 wt% SSM-grafted PP have been constructed using the equations (5.1) and (5.3) and are shown as examples in Figure 5.4. The curves show a similar sigmoidal shape as observed for PP. However, the onset of crystallization is higher and the time for maximum crystallization is shorter at a given cooling rate for VTES-grafted PP. The effect of cooling rates on crystallization is stronger in case of VTES grafting via SSM. This can be expected due to segregation and fractionation of PP chains due to non-random grafting as discussed above.

From these curves, $\log [-\ln(1-X_c)]$ was calculated using equation (5.5) and plotted against $\log t$ in Figure 5.5. The values of the Avrami exponent n and rate parameter Z_c calculated from these plots are listed in Table 5.3. A two-stage crystallization is observed in the VTES-grafted PP. The curves can be divided in three different zones depending on the degree of crystallinity, namely: below 10% - 1st primary crystallization, between 10 and 70 % - 2nd primary crystallization and above 70% - secondary crystallization. The values of n are indicative for the change in slope and do not represent the actual growth dimension which has practical significance in isothermal crystallization.

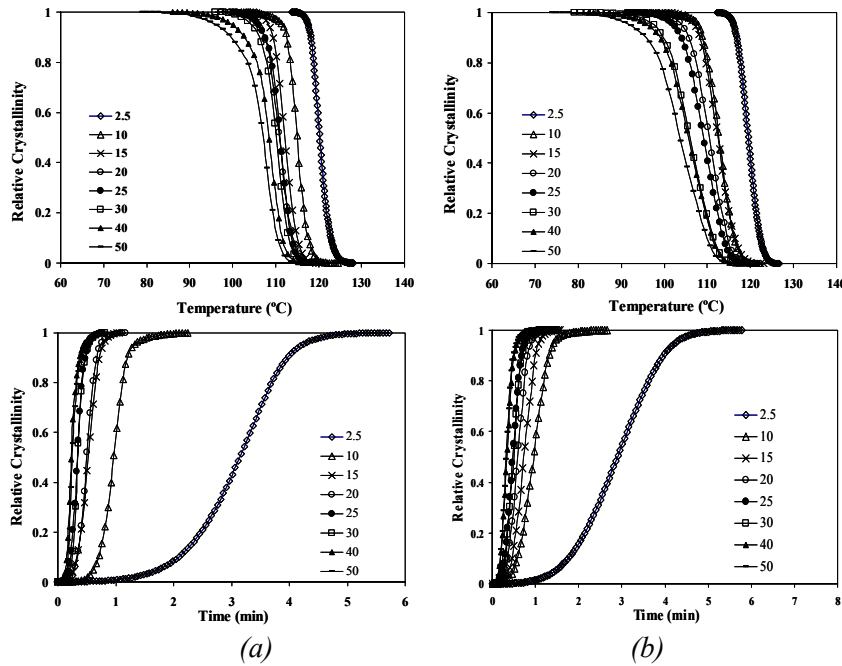


Figure 5.4: Relative crystallinity as a function of time and temperature for the non-isothermal crystallization process: (a) 2.6 wt% PP grafted in solution and (b) 2.3 wt% PP grafted via SSM.

In case of solution-grafted PP, the values of n vary more than for SSM grafting, n_1 and n_{12} are significantly different for solution-grafted PP than SSM, at least in the lower cooling rate range. While the average values of PP are $n_1 = 4.4$ and $n_2 = 2.1$ (see Chapter 4), for solution-grafted PP the average value of n_1 is between 4 and 5, except for the degree of graft of 2.6 wt% system in which n_1 varies from 2 to 4, whereas the values of n_{12} are almost same for all samples. The secondary crystallization is mainly controlled by diffusion of polymer chains. The n_2 for solution-grafted PP is higher ($\sim 1.5 - 3.5$), which implies that the secondary crystallization proceeds slower than for PP and demonstrate the decreased mobility of chains due to the grafted VTES. On the other hand, PP grafted via SSM shows more uniform values of n . Moreover, there is no difference between n_1 and n_{12} and is in the same range for the various degrees of grafting. However, n_2 is even higher ($\sim 2.5 -$

4) than for the solution-grafted PP. This implies that the secondary crystallization proceeds even slower due to a lower mobility which is related to the non-random grafting during SSM. The segregation of VTES-rich and non-grafted chains results in smoother 3-dimensional growth in the primary stage; however, clustering of these VTES grafted sites makes secondary crystallization difficult.

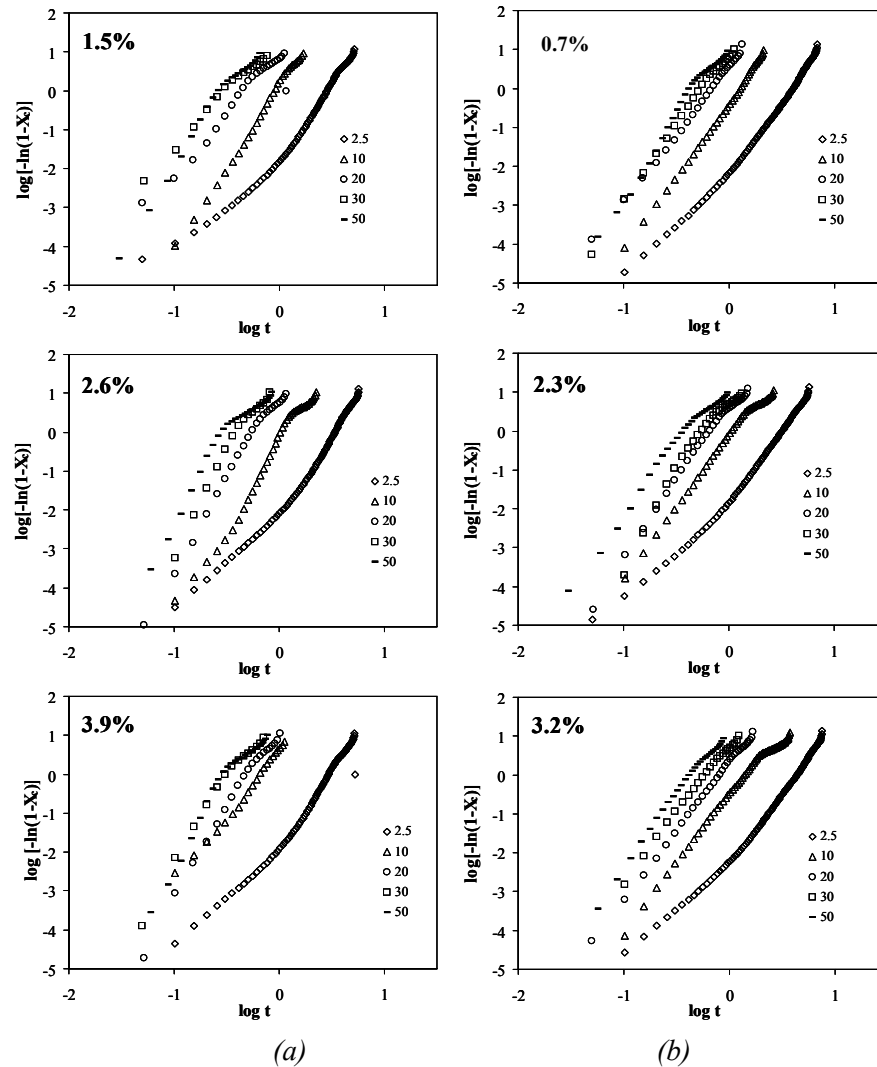


Figure 5.5: Avrami plots of $\log [-\ln(1-X_c)]$ versus $\log t$ for non-isothermal crystallization kinetics of VTES-grafted PP with varying degree of grafting via (a) solution method and (b) SSM.

This becomes even more clear when the Liu approach is used which combines the Avrami and Ozawa equations [6]. The plots of $\log \beta_0$ versus $\log t$ at various degrees of crystallinity are shown in Figure 5.6. The values of b and $F(T)$ are calculated from these plots according to equation (5.5) and listed in Table 5.4. $F(T)$ values increases with increasing crystallinity for all samples. Higher values of $F(T)$ at a certain degree of crystallinity means that a higher cooling rate needed to reach this crystallinity in unit time indicating the difficulty of the crystallization.

For solution-grafted PP, $F(T)$ does not vary with increasing degree of grafting. The highest degree of grafting is 3.9 wt% at which the average distance is larger than lamellar thickness, so chain folding will not be affected. However, such a high degree of grafting does decrease the mobility which is reflected in the n_2 value for secondary crystallization.

Table 5.3: Values of n and Z for VTES-grafted PP via solution and SSM from modified the Avrami analysis for non-isothermal crystallization process

solution graft											
1.5%						2.6%					
rate	1 st primary	2 nd primary	secondary	1 st primary	2 nd primary	secondary	1 st primary	2 nd primary	secondary	1 st primary	2 nd primary
K/min	n_1	Z_{c1}	n_2	Z_{c2}	n_1	Z_{c1}	n_2	Z_{c2}	n_1	Z_{c1}	n_2
2.5	3.03	0.15	4.75	0.10	4.03	0.21	2.92	0.20	4.47	0.15	3.36
10	5.05	0.95	6.06	0.98	2.11	1.09	4.10	1.00	5.17	1.06	2.85
15	3.50	1.11	4.11	1.16	3.11	1.70	3.04	1.06	4.61	1.20	2.22
20	5.27	1.19	4.74	1.16	2.39	1.41	2.66	1.06	4.21	1.19	1.89
25	3.81	1.15	4.38	1.19	2.52	2.03	4.19	1.13	4.69	1.17	2.31
30	5.83	1.22	4.86	1.17	2.56	1.58	2.89	1.11	3.33	1.15	1.84
40	3.80	1.13	3.88	1.14	1.67	1.91	3.90	1.12	4.11	1.14	2.33
50	5.74	1.16	4.69	1.13	1.77	1.52	4.73	1.13	4.38	1.13	2.15
SSM grafted											
0.7%						2.3%					
rate	1 st primary	2 nd primary	secondary	1 st primary	2 nd primary	secondary	1 st primary	2 nd primary	secondary	1 st primary	2 nd primary
K/min	n_1	Z_{c1}	n_2	Z_{c2}	n_1	Z_{c1}	n_2	Z_{c2}	n_1	Z_{c1}	n_2
2.5	3.28	0.14	3.73	0.12	4.39	0.15	3.12	0.19	3.75	0.17	3.84
10	3.65	0.90	3.88	0.91	4.14	0.77	3.66	0.97	3.81	0.98	1.88
15	3.24	1.01	3.86	1.04	4.01	1.24	3.91	1.05	3.87	1.06	3.12
20	3.24	1.04	3.91	1.08	3.45	1.51	4.12	1.10	3.67	1.08	2.33
25	3.85	1.07	4.35	1.09	2.83	1.55	3.68	1.10	3.46	1.09	2.57
30	3.91	1.08	4.07	1.09	2.59	1.61	5.59	1.16	3.59	1.08	2.01
40	3.83	1.11	4.23	1.10	2.31	1.82	3.91	1.11	3.18	1.08	2.20
50	4.90	1.09	5.15	1.10	2.17	1.54	4.65	1.12	3.40	1.07	1.94
3.2%											
rate	1 st primary	2 nd primary	secondary	1 st primary	2 nd primary	secondary	1 st primary	2 nd primary	secondary	1 st primary	2 nd primary
K/min	n_1	Z_{c1}	n_2	Z_{c2}	n_1	Z_{c1}	n_2	Z_{c2}	n_1	Z_{c1}	n_2
2.5	3.28	0.14	3.73	0.12	4.39	0.15	3.12	0.19	3.75	0.17	3.84
10	3.65	0.90	3.88	0.91	4.14	0.77	3.66	0.97	3.81	0.98	1.88
15	3.24	1.01	3.86	1.04	4.01	1.24	3.91	1.05	3.87	1.06	3.12
20	3.24	1.04	3.91	1.08	3.45	1.51	4.12	1.10	3.67	1.08	2.33
25	3.85	1.07	4.35	1.09	2.83	1.55	3.68	1.10	3.46	1.09	2.57
30	3.91	1.08	4.07	1.09	2.59	1.61	5.59	1.16	3.59	1.08	2.01
40	3.83	1.11	4.23	1.10	2.31	1.82	3.91	1.11	3.18	1.08	2.20
50	4.90	1.09	5.15	1.10	2.17	1.54	4.65	1.12	3.40	1.07	1.94

For VTES grafted on PP via SSM, $F(T)$ is higher. For PP, $F(T) = 7.5$ at 20% and 11.3 at 80%, whereas the values for PP grafted via SSM are 7.6 and 16.6, respectively. From the values of $F(T)$, it is evident that non-random grafting of VTES via SSM results in more constrained crystallization stage, especially during the secondary crystallization where diffusion of polymer chains plays an important role. This in turn also affects the amorphous phase which remains more entangled.

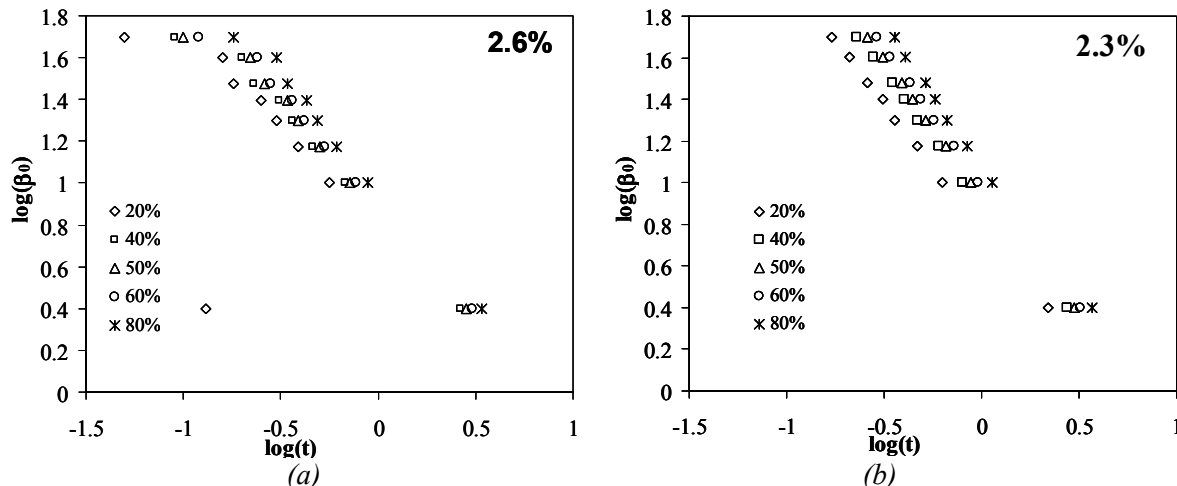


Figure 5.6: Log β_0 versus log t plots obtained from combined Avrami-Ozawa approach for VTES-grafted PP with varying degree of grafting via (a) solution method and (b) SSM.

Table 5.4: The values of b and $F(T)$ at various degree of crystallinity calculated using the combined Avrami-Ozawa equations for VTES grafted on PP via solution and SSM

	$X_c(t)\%$		20%	40%	50%	60%	80%
Solution grafted	1.5	b	1.18	1.13	1.13	1.14	1.18
		$F(T)$	5.23	7.02	7.64	8.22	9.81
	2.6	b	1.05	1.06	1.08	1.08	1.12
		$F(T)$	5.57	6.81	7.39	7.94	9.32
	3.9	b	1.05	1.07	1.07	1.08	1.10
		$F(T)$	5.00	6.18	6.66	7.16	8.33
SSM grafted	0.7	b	1.14	1.13	1.14	1.15	1.17
		$F(T)$	7.64	10.07	10.99	12.00	14.24
	2.3	b	1.18	1.22	1.24	1.25	1.28
		$F(T)$	6.15	8.17	9.16	10.20	12.57
	3.2	b	1.18	1.20	1.20	1.21	1.23
		$F(T)$	7.82	10.85	12.15	15.74	16.61

5.3.2. (VTES-grafted PP)/silica nanocomposites

SSM was used to prepare PP/silica nanocomposites with the PP being grafted with VTES (see Chapter 2, Section 2.3.1.2). The crystallization behavior of solution and SSM-grafted VTES on PP investigated using DSC illustrated the effect of random and non-random grafting. The crystalline structure is affected by the way in which monomer is grafted on the polymer backbone. This is important for investigating the effect of the VTES in PP/silica nanocomposites, since it is used to impart chemical interaction between silica particles and PP matrix.

5.3.2.1 Molecular structure of (VTES-grafted PP)/silica nanocomposites

A schematic of the molecular structure of the product obtained by making in-situ silica particles in VTES-grafted PP via SSM is shown in Figure 5.7. The grafted VTES groups which are located close to the pore walls of the PP can be incorporated in the silica during synthesis, whereas the

grafted VTES which is in the amorphous PP phase inside the sub-particles is probably not incorporated. This makes the system more complex and these nanocomposites will have influence of the crystallization behavior of PP/silica nanocomposites and VTES-grafted PP via SSM.

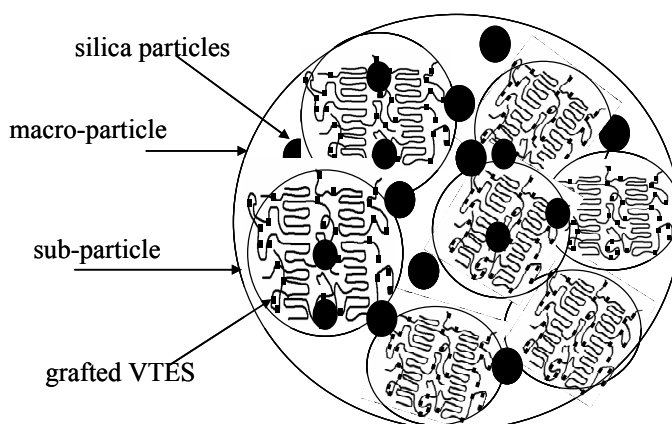


Figure 5.7: Schematics of the chemical micro-structure of (VTES-grafted PP)/silica nanocomposites prepared via SSM and sol-gel reaction. Some of the in-situ prepared silica particles contain grafted VTES.

In addition, because the silica particles are connected via grafted VTES to numerous PP chain, a network is formed. In other words, silica particles become nodal points for PP cross-links as was shown by the rheological measurements. Figure 5.8 shows the dynamic viscosity of PP/silica nanocomposites with and without VTES grafting onto PP. PP/silica nanocomposites with 2 wt% silica have a slightly higher viscosity, but the viscoelastic behavior is similar to that of PP. The samples with the higher degree of grafting, i.e. 5 wt%, are used to illustrate the effect of cross-linking. Indeed, the typical viscoelastic behavior of cross-linked materials with a very high viscosity at lower frequencies with a strong shear thinning at higher frequencies is observed. This additional cross-linking effect due to incorporation of grafted VTES units in the silica particles result in a further reduction of the mobility of PP chains. This will have a strong effect on the crystallization behavior of these nanocomposites. In the subsequent sections, the crystallization of the (VTES-grafted PP)/silica nanocomposites with varying degrees of VTES grafting at a given silica concentration is discussed.

5.3.2.2 Crystallization behavior of (VTES-grafted PP)/silica nanocomposites

The DSC cooling curves of (VTES-grafted PP)/silica nanocomposites are shown in Figure 5.9. The crystallization temperatures (T_c) observed for a given silica concentration show that with increasing VTES grafting the crystallization temperature (T_c) increases. In the previous section, we showed that grafting of VTES alone also increases the T_c . However, the increase in T_c is not high enough to be conclusive of the additional nucleating effect due to grafting of silica. The crystallization data at various cooling rates are listed in Table 5.5. T_c shifts to lower temperatures with increasing cooling rate as also observed for the PP/silica nanocomposites discussed in Chapter 4. The data also show that the onset of crystallization sets in at higher temperatures with increasing VTES concentration. However, the overall crystallization times are slightly longer than non-grafted PP/silica nanocomposites. The silica nano-particles are attached to the PP chains through VTES and decrease the mobility and diffusivity of the polymer chains, resulting in longer time to complete the crystallization process, especially at low cooling rates. Although crystallization starts at a higher temperature due to the additional nucleating effect of VTES, the constraints on the mobility of the

PP chains results in a retarded growth. At slower cooling rates, the non-grafted and 2.6 wt% grafted systems exhibit a higher T_c and a lower crystallization time. The crystallization proceeds spontaneously as the temperature reaches the crystallization temperature, which is characteristic for heterogeneous nucleation, as also reported in Chapter 4 for PP/silica nanocomposites without grafting.

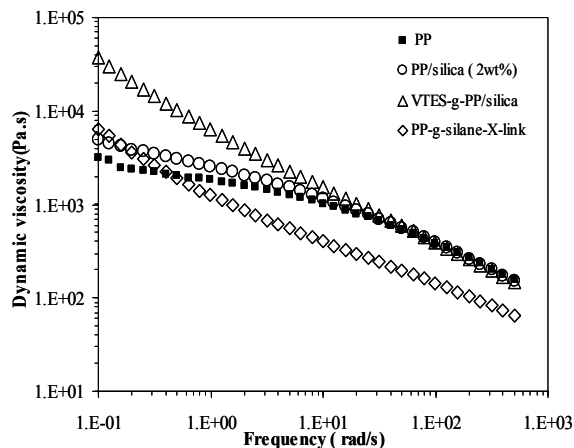


Figure 5.8: Dynamic viscosity of (VTES-grafted PP)/silica nanocomposites. The increase in viscosity for silica-grafted PP indicates network formation.

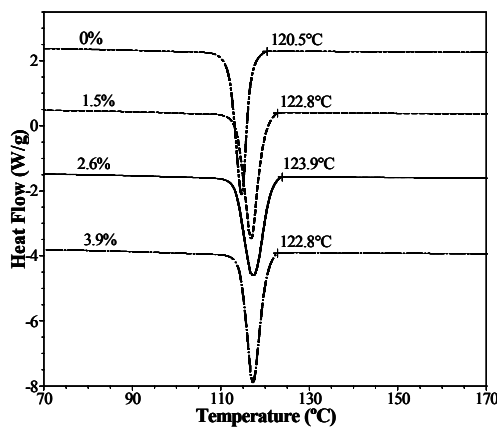


Figure 5.9: The cooling exotherms of (VTES-grafted PP)/silica at a rate of 10 K/min. The running parameter shows the percentage of VTES grafted on PP. The concentration of silica is 1 wt%.

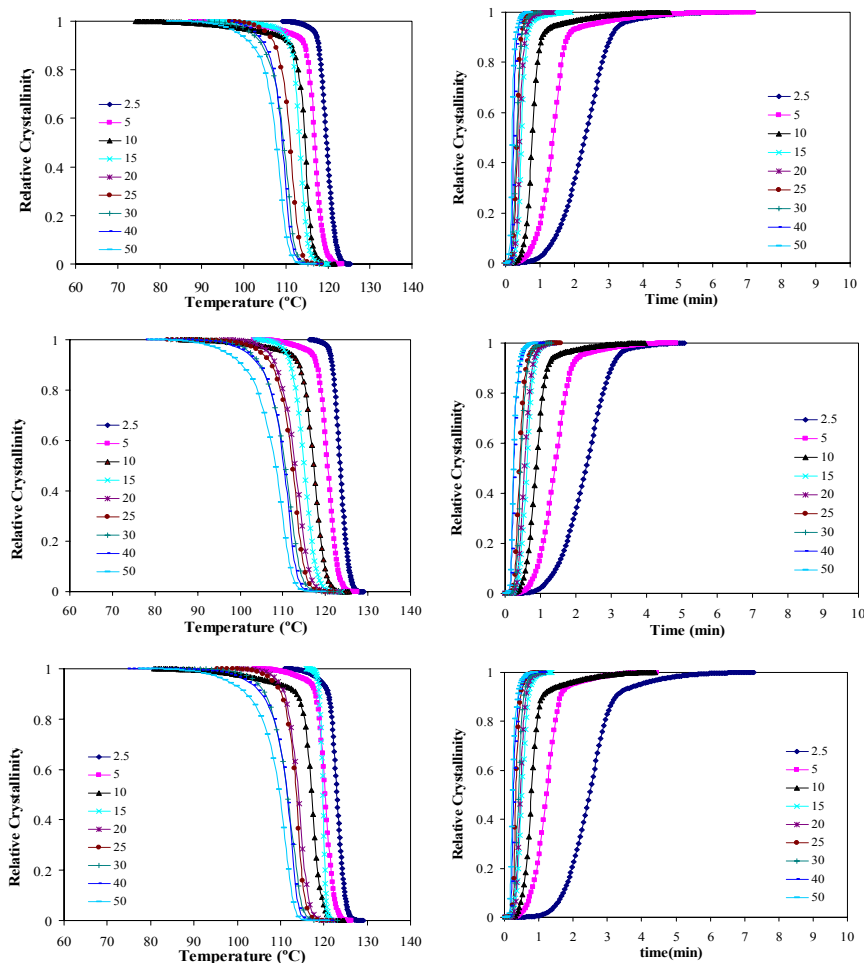


Figure 5.10: Relative crystallinity versus time and temperature for the non-isothermal crystallization process (top) PP/silica; (middle) (VTES-grafted PP) /silica (2.6 wt%); (bottom) (VTES-grafted PP)/silica (3.9 wt%).

Table 5.5: Non-isothermal crystallization behavior of VTES-grafted PP/silica nanocomposites at different cooling rates

degree of grafting rate °C/min	0				1.5%				2.6%				3.9%			
	T _s (°C)	T _c (°C)	T _e (°C)	t _s (min)	T _s (°C)	T _c (°C)	T _e (°C)	t _s (min)	T _s (°C)	T _c (°C)	T _e (°C)	t _s (min)	T _s (°C)	T _c (°C)	T _e (°C)	t _s (min)
2.5	125.3	119.2	109.3	6.45	128.9	122.4	110.4	7.41	129.0	123.1	116.2	5.07	129.1	122.6	110.9	7.28
5	123.4	116.6	87.4	7.2	126.1	119.5	95.5	6.11	127.4	120.4	103.2	4.83	126.3	120.2	104.1	4.44
10	121.6	114.6	74.2	4.75	124.7	116.7	81.7	4.3	125.6	117.3	86.1	3.96	124.7	117.1	80.8	4.4
15	120.1	113.3	91.7	1.89	122.9	115.4	100.9	1.47	123.7	115.2	103.6	1.36	122.5	115.7	101.8	1.39
20	119.6	112.1	92.0	1.38	122.4	114.2	93.0	1.47	123.3	113.7	95.7	1.38	123.1	114.7	100.2	1.15
25	118.6	111.3	96.7	0.88	122.1	114.2	87.6	1.38	122.0	113.1	83.0	1.56	121.8	114.2	95.4	1.06
30	117.9	110.2	87.7	1.01	120.5	112.3	86.3	1.14	123.2	112.2	83.7	1.32	123.1	113.2	87.6	1.19
40	118.2	109.3	95.9	0.56	119.4	112.5	72.6	1.17	119.4	111.4	77.9	1.04	121.6	112.8	74.9	1.17
50	116.7	108.1	82.3	0.72	119.4	111.4	72.0	1.01	119.5	110.2	78.9	0.85	120.8	111.8	77.8	0.91

Table 5.6: Values of n and Z_c for PP/silica and VTES-grafted PP/silica nanocomposites from the modified Avrami analysis for non-isothermal crystallization process

degree of grafting rate K/min	0%				1.50%				2.60%				3.90%			
	primary stage n_1	primary stage Z_{c1}	secondary stage n_2	secondary stage Z_{c2}	primary stage n_1	primary stage Z_{c1}	secondary stage n_2	secondary stage Z_{c2}	primary stage n_1	primary stage Z_{c1}	secondary stage n_2	secondary stage Z_{c2}	primary stage n_1	primary stage Z_{c1}	secondary stage n_2	secondary stage Z_{c2}
2.5	3.9	0.23	1.8	0.60	4.4	0.17	1.8	0.56	3.9	0.23	2.4	0.49	5.2	0.13	2.0	0.50
5	3.7	0.71	1.0	1.03	3.9	0.74	1.3	0.89	4.4	0.69	1.5	0.94	4.2	0.78	1.5	0.71
10	4.5	1.05	0.9	1.07	4.8	1.04	1.1	1.06	4.5	1.01	1.0	1.05	4.9	1.09	1.1	1.04
15	4.5	1.21	1.1	1.10	5.6	1.22	2.4	1.15	4.8	1.15	2.6	1.10	4.5	1.19	2.3	1.11
20	5.1	1.23	1.8	1.09	5.2	1.18	2.1	1.10	5.1	1.15	2.0	1.07	6.3	1.24	2.4	1.10
25	5.5	1.27	2.0	1.10	5.7	1.22	1.6	1.09	5.0	1.16	1.4	1.07	5.6	1.24	1.9	1.09
30	5.2	1.21	1.9	1.08	5.5	1.20	1.9	1.08	5.5	1.14	1.9	1.06	6.0	1.17	2.1	1.07
40	5.1	1.18	2.2	1.08	5.6	1.22	1.5	1.08	4.6	1.15	1.6	1.06	6.5	1.22	1.5	1.05
50	4.6	1.15	1.6	1.06	5.5	1.17	1.3	1.05	4.3	1.11	1.7	1.05	4.4	1.11	1.6	1.05

The crystallization curves in Figure 5.10 for VTES-grafted nanocomposites show a similar sigmoidal shape as observed for PP/silica nanocomposites. However, T_{onset} is higher for 2.6 wt% (VTES-grafted PP)/silica nanocomposites and also the time for crystallization is shorter at a given cooling rate. As the degree of grafting is further increased from 2.6 to 3.9 wt%, the trend reverses, showing a lower crystallization temperature and a slightly higher crystallization time. The slopes of the crystallization curves in Figure 5.10 indicate that in case of PP/silica and 3.9 wt% (VTES-grafted PP)/silica nanocomposites the crystallization proceeds slower than in the case of the 2.6 wt% (VTES-grafted PP)/silica nanocomposites. This suggests an increasing nucleating effect of silica nano-particles in the presence of modified VTES grafted PP.

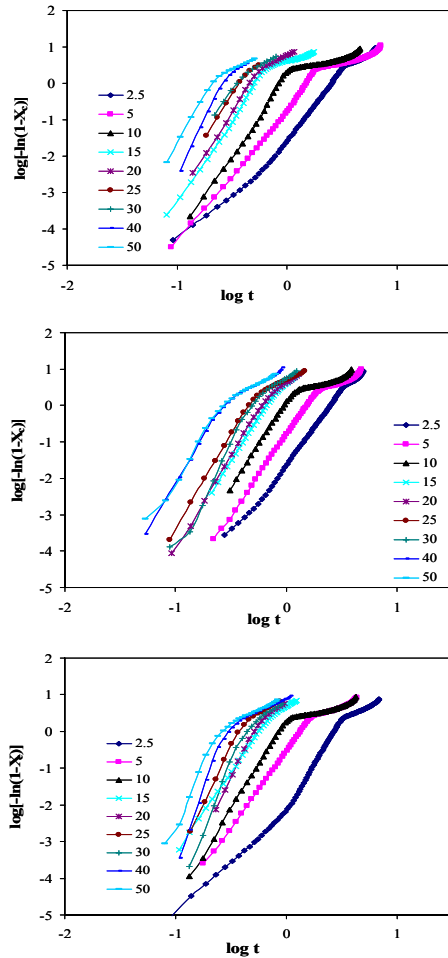


Figure 5.11: Avrami plots of $\log[-\ln(1-X_c)]$ versus $\log t$ for non-isothermal crystallization kinetics of (a) PP/silica, (b) 2.6 wt% (VTES-grafted PP) /silica, (c) 3.9 wt% (VTES-grafted PP)/silica.

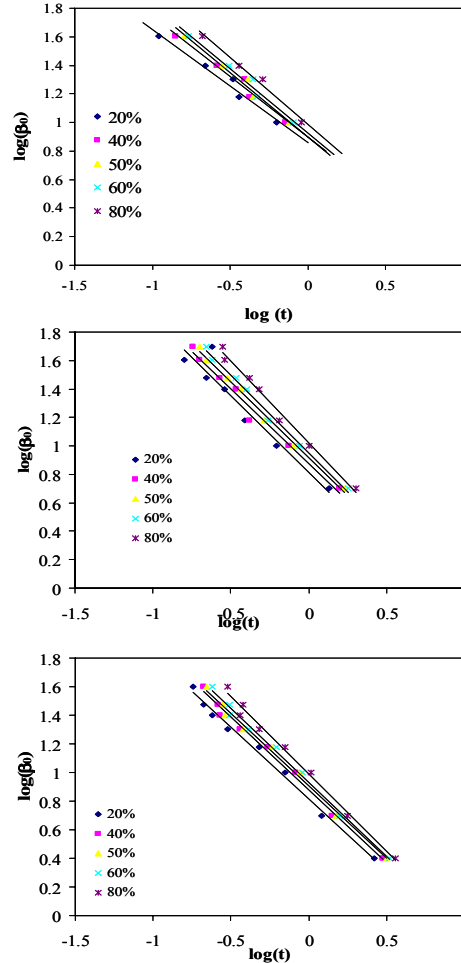


Figure 5.12: $\log \beta_0$ versus $\log t$ from the combined Avrami-Ozawa equations for (a) PP/silica, (b) 2.6 wt% (VTES-grafted PP) /silica, (c) 3.9 wt% (VTES-grafted PP)/silica.

5.3.2.3 Non-isothermal crystallization kinetics of (VTES-grafted PP)/silica

From the crystallinity curves, plots of $\log[-\ln(1-X_t(t))]$ against $\log t$ are constructed using equation (5.4) in Figure 5.11. Table 5.6 lists the values of the Avrami exponent n and rate parameter Z_c calculated from these plots. A two-stage crystallization is also observed in the (VTES-grafted PP)/silica nanocomposites. The two-stage character is more pronounced for the PP/silica and 3.9 wt% (VTES-grafted PP)/silica nanocomposites than for the 2.6 wt% (VTES-grafted PP)/silica nanocomposites. The average value of Avrami exponent n_1 is 4.7 for PP/silica and (VTES-grafted PP)/silica nanocomposites which implies that the crystallization proceeds in a similar way and that

the grafting does not affect the rate of crystallization, at least in the primary stage. In the earlier section on VTES-grafted PP via SSM, it was also observed that crystallization in the primary stage proceeds smoothly. However, the average values of n_2 for PP/silica and (VTES-grafted PP)/silica are 1.4 and 1.9 respectively, which indicates that during the secondary stage the crystallization proceeds slower in case of VTES-grafted nanocomposites. The plots of $\log \beta_0$ versus $\log t$ at various degrees of crystallinity are shown in Figure 5.12. $F(T)$ values increases with increasing crystallinity for all samples. Below 50% crystallinity, $F(T)$ values for the (VTES-grafted PP)/silica nanocomposites are smaller than for PP/silica nanocomposites demonstrating the ease with which the VTES-grafted PP crystallizes in the early stages of crystallization. However, above 60% crystallinity, the trend reverses, i.e. the $F(T)$ values for PP/silica nanocomposites are smaller than for (VTES-grafted PP)/silica nanocomposites, implying that the crystallization proceeds more difficultly in (VTES-grafted PP)/silica nanocomposites. As mentioned earlier, above 70% crystallinity, the secondary crystallization prevails and diffusion of chains becomes more difficult due to constraints imparted by the grafting and cross-linking. An increase of $F(T)$ values for (VTES-grafted PP)/silica nanocomposites thus signifies that the chains are more entangled in the amorphous melt.

Table 5.7: Values of b and $F(T)$ versus degree of crystallinity

wt%	$X_c(t)\%$	20	40	50	60	80
0.0	b	0.93	0.95	0.97	0.98	1.04
	$\log F(T)$	0.86	0.89	0.90	0.92	0.99
1.5	b	0.92	0.96	0.94	1	1.05
	$\log F(T)$	0.84	0.9	0.93	0.96	1.02
2.6	b	0.96	0.98	1.02	1.05	1.06
	$\log F(T)$	0.81	0.88	0.91	0.94	1.01
3.9	b	0.97	1.00	1.01	1.03	1.08
	$\log F(T)$	0.82	0.88	0.90	0.93	0.99

5.3.2.4 Effect of VTES grafting on the melting behavior of the nanocomposites

Figure 5.13 shows the melting endotherms of (VTES-grafted PP)/silica nanocomposites. An enhanced nucleation usually results in more perfect and uniform crystal formation. However, due to non-random grafting multiple melting peaks can be observed, which can be deconvoluted in multiple peaks. Nevertheless, the size distribution is less broad compared to VTES-grafted PP via SSM. This may be due to the nucleation effect of silica nano-particles. In the non-isothermal crystallization kinetics part, the primary crystallization is prominent up to 2.6 wt% grafted VTES, but at higher VTES grafting levels secondary crystallization is dominating.

The isothermal crystallization behavior was investigated in various temperature regimes to get insight in the melting behavior for (VTES-grafted PP)/silica nanocomposites. Figure 5.14 shows the melting endotherms at 10 K/min after isothermal crystallization at the temperature indicated next to the curve. The melting behavior is similar to the one observed by Lotz and coworkers for PP. However, the temperatures at which the transitions are observed are higher than those reported by Lotz and coworkers due to the nucleating effect of silica nano-particles in VTES-grafted PP. The melting behavior itself is not so surprising, especially for PP, the details of which are reported elsewhere [20]. Varga et al. [20] showed that if PP is cooled below a critical temperature (T^*) a considerable amount of crystals with the β -modification during melting and recrystallization can be present. The melting endotherms after isothermal crystallization at $T^* = 123^\circ\text{C}$ in this study are

similar to that observed in PP when crystallized at $T^* = 105^\circ\text{C}$ by Varga et al. It shows that for (VTES-grafted PP)/silica nanocomposites, more stable β -nuclei are formed and T^* shifts to higher temperatures. The segregation of chains due to non-random grafting also helps in the early stage of crystallization. The increase in T^* also signifies that more stable α -crystals can be formed even at relatively high temperatures, and hence when (VTES-grafted PP)/silica nanocomposites are quenched, they are already in the stable crystal form compared to neat PP. The presence of the β -modification in the quenched samples is also confirmed by the WAXD-pattern (inset in Figure 5.14a). This in essence shows that the β -phase formed is not an effect of thermal treatment, but is due to the presence of silica particles.

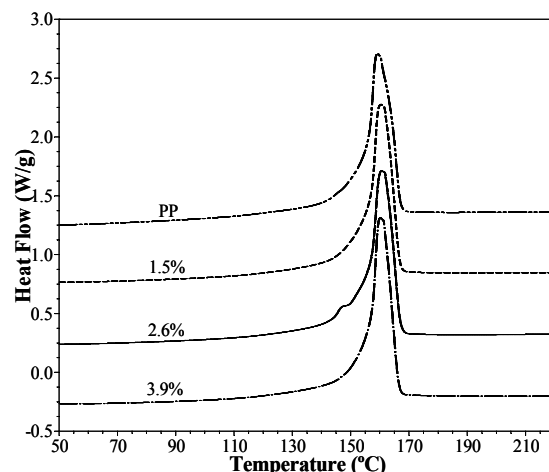


Figure 5.13: Melting behavior of (VTES-grafted PP)/silica nanocomposites in standard mode as a function of degree of grafting at 10 K/min.

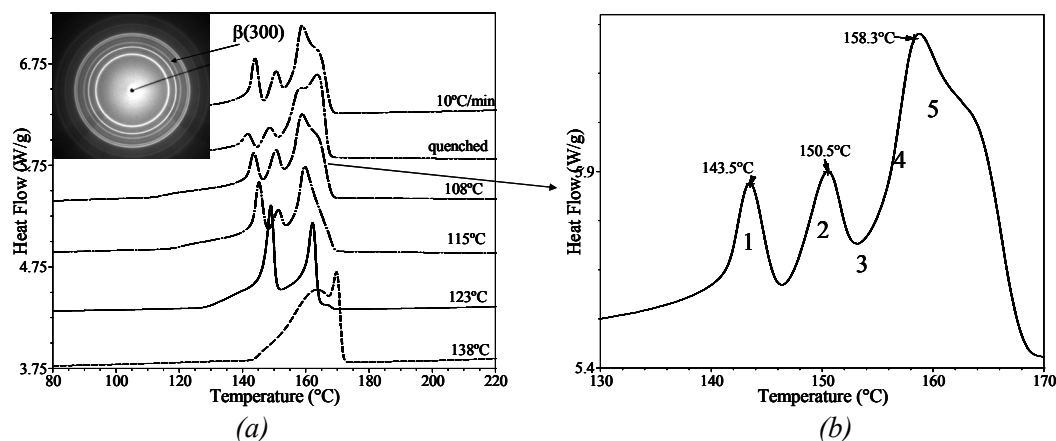


Figure 5.14: (a) Melting endotherm after isothermal crystallization and 2-D WAXD pattern of VTES-grafted PP/silica nanocomposites (inset). (b) Details of multiple melting peaks: (1) melting of β -phase with partial recrystallization, (2) melting of β -phase recrystallized in the process, (3) crystallization of α -phase on $\alpha\beta^*$ latent seeds, (4) melting of α -phase, (5) melting of α -phase produced in recrystallization/melting process

5.4 Conclusions

SSM results in non-random grafting of VTES on PP and hence clustering which can interact with each other. This is also proved from the increase in storage modulus in rheological measurements. Non-random grafting of VTES segregates the VTES-rich PP. This results in broadening of crystallization peaks which can be deconvoluted in at least two peaks originating from crystal with

different crystal size. Random grafting of VTES on PP via solution blending does not show these effects. Segregation and clustering in VTES-grafted PP via SSM decrease the mobility and hence diffusivity of chains. Thus, secondary crystallization becomes more difficult and the amorphous phase remains more entangled. $F(T)$, which is an indication of difficulty in crystallization, has higher values for PP grafted via SSM than via solution blending and the difference increases at higher crystallinity. This supports our argumentation that the chain mobility decreases markedly for PP grafted via SSM.

The crystallization temperatures increase with increasing degree of grafting, indicating heterogeneous nucleation in case of (VTES-grafted PP)/silica nanocomposites. However, the increase is not significantly different compared to VTES-grafted PP. The total time of crystallization goes through a minimum at 2.6 wt% grafting. The silica nano-particles are attached to the PP chains through VTES thus decreasing the mobility and diffusivity of the polymer chains, so it takes a longer time to complete the crystallization process especially at lower cooling rates. The rheological experiments show that when silica is grafted via VTES to multiple PP chains a network is formed or in other words silica particles become nodal points for PP cross-links. The two-stage crystallization process is more prominent in nanocomposites without grafting and at very high grafting levels, indicating the nucleation and growth is faster in case of intermediate grafting. $F(T)$ values are lower during the initial crystallization stages with increasing degree of grafting; however, it shows a sharp increase at higher crystallinity due to constraints in the PP chains by grafting bulky VTES monomers (and possibly incorporated silica particles in them). This has been also observed for VTES-grafted PP via SSM without silica. The melting curves show β -modification and imperfect α -crystals due to grafting which results in irregular chain folding. The effect of fractionation due to non-random grafting of VTES on PP is also observed from melting curves which shows multiple and broad melting peaks.

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Chapter 6

Rheological behavior of PP/silica nanocomposites and the consequences for flow-induced crystallization^{*}

Abstract

The flow-induced crystallization (FIC) behavior of silica nano-particles filled PP nanocomposites has been studied using in-situ SAXS and WAXD. These nanocomposites show for a specific range of concentration of nanoparticles a large drop in the viscosity. This viscosity drop is caused by the selective adsorption of high molecular weight chains on the nano-fillers. This provides the possibility of creating a more uniform and isotropic morphology in a product. The X-ray scattering study shows that orientation is minimal at 0.5 wt% silica concentration, at which the viscosity also shows a minimum. In comparison to pure PP, the flow-induced crystallization is faster in silica-filled PP, but this enhancement is lowest for the concentration where the orientation is minimal. The fraction of oriented material measured with SAXS shows a maximum in time for all silica concentrations, but the effect is higher in case of a high silica concentration. For all samples, a finite orientation level is detected with SAXS. However, TEM and FT-IR dichroic ratio measurements on samples made in simple shear flow and injection molded bars show a uniform and isotropic structure. This can be explained by a three-stage crystallization process: 1) formation of row nuclei due to orientation, especially in local areas between the particles where stress maxima (i.e. orientation maxima) occur; 2) relaxation of row nuclei or destruction of the orientation due to local inhomogeneous, unsteady flow into unoriented row nuclei and/or a number of primary point nuclei; 3) growth from these nuclei in random directions. The three-stage mechanism helps in the random growth of crystals reducing the anisotropy in the material.

6.1 Introduction

Flow causes orientation of the molecules in a polymer melt. The orientation depends on the shear stress levels during the flow. The shear rate is a measure of the stress when the viscosity is constant. Figure 6.1a shows a schematic picture of the increase in orientation with increase in shear rate assuming that the viscosity of the polymer melt is constant. It is well understood that molecular orientation enhances flow-induced crystallization (FIC) by formation of row nuclei or, for high enough stress levels, shish structures followed by the growth of non-isotropic crystals. Various studies performed on the structure and morphology resulting from FIC indicate that the molecular weight and molecular weight distribution strongly influence FIC [1], especially the high end tail of the molecular weight distribution [2]. The high molecular weight polymer chains have a long relaxation time, so the oriented structures once formed remain intact in the melt. Somani et al. [2]

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showed that when a polymer melt with a broad molecular weight distribution is subjected to a particular shear rate only the longer chains orient (with different degrees of extension) with a low end cutoff at a critical molecular weight (M^*). This critical molecular weight does not vary significantly above a shear rate of 60 s^{-1} . Moreover, the relaxation time for the shorter chains is too short to form long-lasting extended structures.

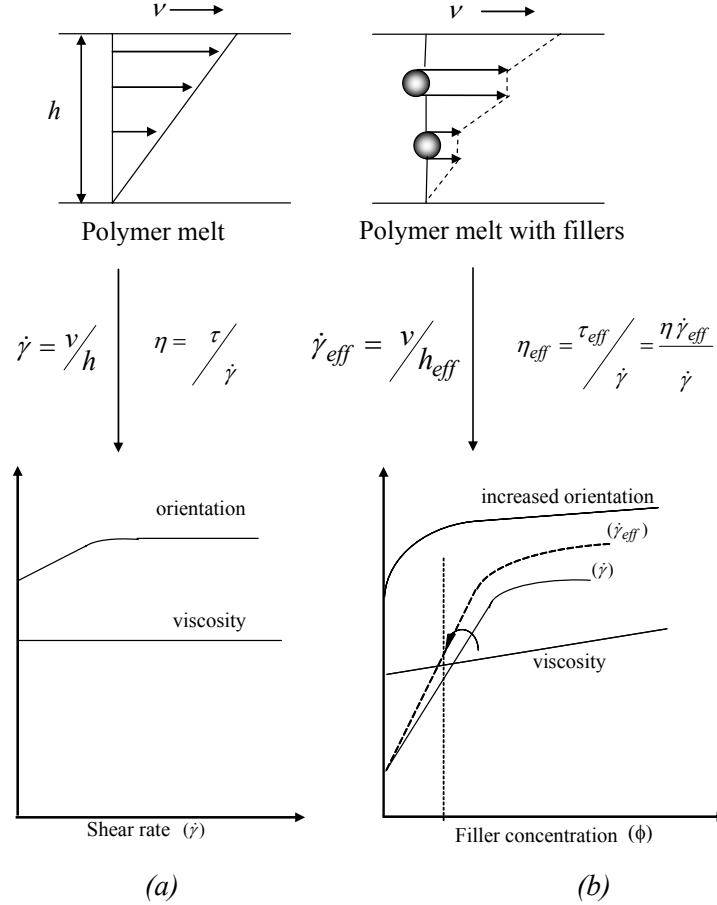


Figure 6.1: The effect of flow on orientation in (a) polymer melt and (b) particulate-filled polymer melt. Particulate-filled polymers exhibit an increase in the viscosity and shear stresses resulting in higher orientation (see Hwang et al. [5]).

The acceleration of the crystallization kinetics can be attributed to the increase in the nucleation rate caused by orientation of the polymer chains. In most polymer processing steps not only shear, but also elongational flow occurs. Both the elongational and shear stresses induce anisotropy in the structures that develop during processing.

Polymers are often modified by adding inorganic hard fillers [3]. It has been shown that conventional fillers, such as talc, calcium carbonate, glass beads, may improve the properties, but at the expense of processing properties, i.e. increase in the viscosity [4,5]. This is one of the main concerns, since the concentration of filler particles is usually high ($\sim 20 \text{ vol\%}$).

The suspension of particles in polymers can lead to complex rheological behaviour. As an example, Figure 6.2 illustrates the computed distribution of stresses in a shear flow for a particle-filled polymer melt. For a macroscopic shear flow, a local elongational flow is generated between the separating particles. Also other high stress regions occur when two particles approach each other.

The presence of elongational flows between separating particles leads to highly non-uniform and anisotropic micro-structures. Figure 6.1b schematically elucidates the increase in local orientation for a macroscopic shear flow, as a result of an increase in shear rate between particles [5]. FIC of polymer/nano-filler systems, which can improve the properties and retain the processability, is not well studied. The nano-particles, which have sizes in the range of the radius of gyration of polymer molecules, may also influence the orientation under flow similar to micro-particles.

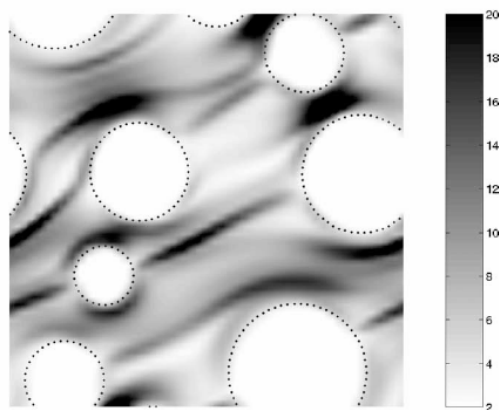


Figure 6.2: Distribution of stresses in a macroscopic shear flow in the presence of hard particles for a polymer melt. The scale on the right hand shows the increasing stress with intensity (Figure is reproduced from Hwang et al. [5]).

In Chapter 3, the decrease of the viscosity by addition of nano-fillers in polypropylene in the low concentration regime (< 2 wt%) was reported. This drop in the viscosity is not due to a local change in the free volume as suggested by Mackay et al. [6], since silica particles act as nucleating agents for PP [7], preventing the formation of a free volume layer around the particles. The origin of the viscosity drop lies in a nano-scale phenomenon, namely selective physisorption of the long polymer chains on the nano-particles surface. This leaves a polymer matrix with relatively low molecular weight chains and a lower entanglement density. The adsorption of polymer chains on (hard silica) particles results in a core-shell like morphology of a particle with deformable soft shell of adsorbed high molecular weight chains on hard-core silica. For a given shear rate, the matrix viscosity decreases and hence the stresses in the nanocomposite melt.

This chapter is focused on the flow-induced crystallization of PP/silica nanocomposites in the lower concentration regime, typically below 2 wt%. In contrast to an increased orientation on addition of fillers, the orientation is reduced and the resultant crystalline structures will be more isotropic, giving enhanced control over the properties of polymeric materials. The effect of nano-fillers on the orientation and crystallization kinetics as a result of shear flow is investigated using in-situ SAXS-WAXD measurements and related to the observed drop in the viscosity. Furthermore, to show that the concepts demonstrated in simple shear flow also apply under processing conditions, the orientation in injection molded bars was characterized using FT-IR spectroscopy. The final structure of the nanocomposites plays an important role in the development of the properties (see Chapter 7).

6.2 Experimental Section

6.2.1 Materials and nanocomposite preparation

Porous isotactic polypropylene (PP) having a porosity of $S_{BET(Kr)} = 0.0678 \text{ m}^2/\text{g}$ was supplied in powder form by Euro-Sabic, The Netherlands. The number-average and weight-average molecular weight of PP were 60 and 380 kg/mol, respectively. The powder was free of antioxidants and stabilizers and stored at -10°C in the dark. Tetraethoxy orthosilicate (TEOS) and ammonium hydroxide (28% NH_3) were obtained from Aldrich Chemicals. The PP/silica nanocomposites were prepared according to the procedure described in Chapter 2, Section 2.3.1.1.

6.2.2 X-ray measurements

In-situ flow-induced crystallization with simultaneous SAXS/WAXS experiments were performed on the DUBBLE beamline (BM 26B) at the European Synchrotron Research Facility (ESRF, Grenoble, France) using a wavelength of $1.218 \text{ \AA}$. SAXS patterns were recorded using a gas-filled multiwire two-dimensional (2D) detector [8] with a sample-to-detector distance of 2.5 m. For calibration of the SAXS detector, the scattering pattern from an oriented specimen of wet collagen (rat-tail tendon) was used. The WAXD patterns were collected simultaneously with a curved linear microstrip-gas chamber detector. The diffraction angles at the WAXD detector were calibrated with Si-powder. WAXD images were collected with the same acquisition cycles as used for SAXS. The experimental data were corrected for background scattering, i.e. subtraction of instrumental factors. The experimental set-up is shown in Figure 6.3a.

A Linkam shear cell, Type CSS 450, was used to apply a well-defined thermo-mechanical history to the materials. The nanocomposites were mounted between two X-ray transparent windows (Kapton) in the form of $500 \text{ }\mu\text{m}$ thick discs. The gap distance between the two windows was adjusted to the desired sample thickness. The crystallization temperature used was 135°C so that the nucleation and crystal growth times under quiescent conditions are sufficiently long. The following experimental temperature profile was set (see also Figure 6.3b): (1) heating at 30 K/min from room temperature to 220°C ; (2) holding the samples at 220°C for 5 min to eliminate memory effects or any residual crystallinity before the test; (3) cooling at 30 K/min to 135°C ; (4) holding the temperature at 135°C ($\pm 1^\circ\text{C}$) for 30 min for X-ray measurements. Two-dimensional SAXS images were collected continuously: before, during and after cession of the applied shear. First, three frames were collected immediately after reaching 135°C . Next, a shear rate of 60 s^{-1} was applied for 5 s. 200 SAXS images were collected of which the acquisition time for the first 40 frames was 3 s, for the next 80 frames it was set at 6 s and for the remaining 80 frames it was 10 s with a dead time of 5 s.

Analysis of SAXS data

The two-dimensional SAXS data were treated using the FIT2D program of Dr. Hammersley of ESRF. For quantitative analysis of orientation, the procedure described by Somani et al. [2] was followed. The total scattered intensity (I_{total}) due to the contribution of all the scatters in the samples can be deconvoluted into two components: $I_{\text{unoriented}}$, which is due to scattering from the randomly distributed crystalline lamellae, and I_{oriented} , from the oriented structures. Here it has to be pointed out that I_{oriented} comes only from scattering in the meridian direction as no equatorial streaks were observed. For a more precise analysis, the contribution of isotropic crystals in I_{oriented} was subtracted. For both I_{oriented} and $I_{\text{unoriented}}$, the q-value at which the maximum intensity occurs was used to calculate the long period $L=2\pi/q$ (I_{max}) of the oriented (kebabs) and the isotropic crystals.

To evaluate the orientation of the patterns an orientation parameter P_{or} was introduced as the ratio between the intensity due to the oriented crystals and the intensity from isotropic crystals (for an objective comparison both intensities were integrated over an equal angular area):

$$P_{\text{or}} = I_{\text{oriented}} / I_{\text{unoriented}} \quad (6.1)$$

The relative crystallinity (X_c) can be calculated using the following approximation:

$$X_c = [I(t) - I(0)] / [I(s) - I(0)] \quad (6.2)$$

where $I(0)$, the total scattered intensity at $t = 0$, can be attributed to the amorphous, non-crystalline melt. Then, the increase in total scattered intensity, $I(t) - I(0)$, is directly proportional to the crystallinity. $I(s)$ is the

steady-state value at the end of crystallization, when the total scattered intensity reaches a plateau. From the time evolution of the relative crystallinity, a kinetic parameter $t_{1/2}$ can be determined. The half-time $t_{1/2}$ is the time at which the relative crystallinity reaches 50% of its final value.

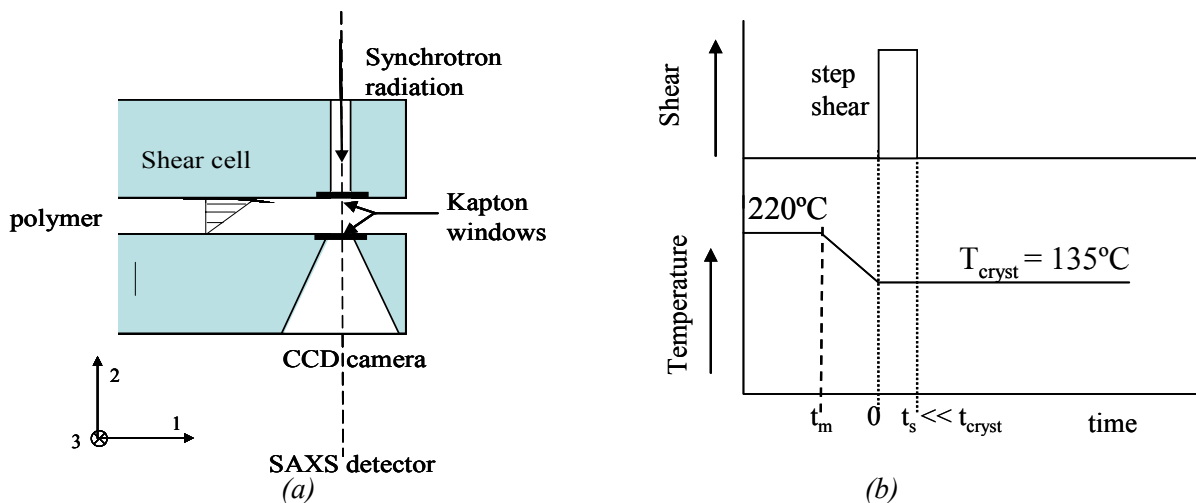


Figure 6.3: (a) Schematics of the shear cell used for in-situ SAXS-WAXS. The flow (1), velocity gradient (2) and vorticity (3) axes are indicated, (b) experimental protocol for the flow-induced crystallization experiments. T_{cryst} is the isothermal crystallization temperature at which shear is applied and crystallization proceeds. t_m is the annealing time in the melt to erase the melt history, t_s is the shear time

6.2.3 Transmission electron microscopy

Transmission electron microscopy was performed using a Jeol JEM 2000 FX microscope, operated at 80 kV. The sheared samples were trimmed at low temperatures (-20°C), at 10 mm from the center of the sample, i.e. at the circular rim at which the X-rays passed through. Next, the samples were treated during 46 hours with a ruthenium tetroxide (RuO_4)-solution [9]. Ultra thin sections (~ 70 nm) were obtained at room temperature using a Reichert Ultracut E microtome. The microtome direction was perpendicular to the flow direction.

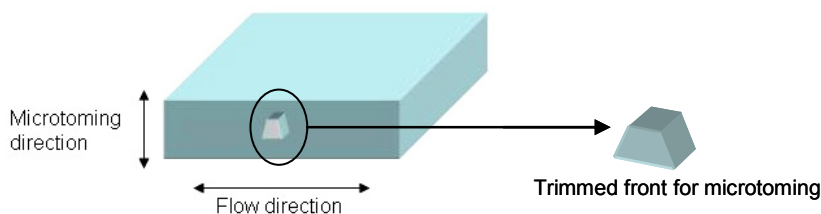


Figure 6.4: Schematics illustrating the direction of view for TEM along with flow direction and microtome direction.

6.2.4 FT-IR measurements

The characterization of the orientation behavior of specific chromophoric groups of a polymer molecule can be achieved by means of linear polarized light Fourier-transform infrared (FT-IR) spectroscopy [10]. If the molecules in a sample are distributed at random, polarized radiation passing through the samples will give a normal FT-IR absorption spectrum. If the molecules in the sample are oriented and polarized light passes through the sample with the transverse electric (TE) wave parallel to the flow direction, the molecular vibrations rotated into the flow direction absorb more energy, whereas those rotated perpendicular to the stretch direction will absorb less energy compared with a non-oriented sample. As a measure for the level of orientation, the dichroic ratio D of a specific absorption band can be calculated. D is defined as the ratio of absorption measured parallel to the flow direction ($A_{\parallel}$) and that measured perpendicular to the flow direction ($A_{\perp}$).

$$D = A_{\parallel} / A_{\perp}$$

For perfectly oriented material the dichroic ration (D_0) can be obtained as $D_0 = 2 \cot^2 \phi$, where ϕ is the transition moment angle of the vibration, i.e. the angle between the transition moment vector M of the measured absorption and the chain axis of the polymer. For perfectly oriented materials, ϕ may be calculated directly from the dichroic ratio. When only a certain fraction f of the molecules is perfectly oriented and the

remaining molecules are randomly oriented, the relationship between f and the dichroic ratio is then expressed as:

$$f = \frac{(D_0 + 2)(D - 1)}{(D_0 - 1)(D + 2)}$$

The incomplete orientation can also be treated by assuming that all the molecules have an average angle θ with the flow direction. The relationship between f and average angle θ has been obtained as

$$f = \frac{(3 \langle \cos^2 \theta \rangle - 1)}{2}$$

The fraction, f , is often called Hermans' dichroic function. If the molecules are fully oriented parallel to the flow direction, f will be 1, if a random orientation exists f will be 0 and if the molecules are fully orientated perpendicular to the flow direction f will be -0.5 . Generally, it is not easy to evaluate ϕ accurately and hence to evaluate f . Instead, a dichroic orientation function f_D , which satisfactorily gives the qualitative changes in molecular orientation, is defined and expressed as

$$f_D = \frac{D-1}{D+2}$$

To determine the molecular orientation in the injection molded bars, 25 μm thick couples were cut at various positions, viz. gate, middle and end from the centre of the bar, using a Reichert Ultracut E microtome. To determine the orientation, line mapping experiments were performed over the thickness of the bar using a Biorad FTS 6000 FT-IR spectrometer coupled to a Biorad UMA 500 infrared microscope. The IR radiation was polarized using a rotatable wire-grid polarizer. The microscope was equipped with a broad-band MCT detector. Spectra were recorded with a resolution of 4 cm^{-1} , co-adding 50 scans. In this work, the vibration at 998 cm^{-1} is used for the determination of the orientation function. The transition moment angle for this vibration is 18° [11].

6.3 Results

6.3.1 Rheology of PP/silica nanocomposites

The nanocomposites prepared in this study are special in the sense that, within a specific concentration range, the addition of nano-particles results in a drop in the viscosity (see Figure 6.5a), which results in lower shear stresses in the matrix. It is anticipated that the decrease in local shear stress will lower the orientation under flow. The inset in Figure 6.5a shows the viscosity as a function of silica concentration.

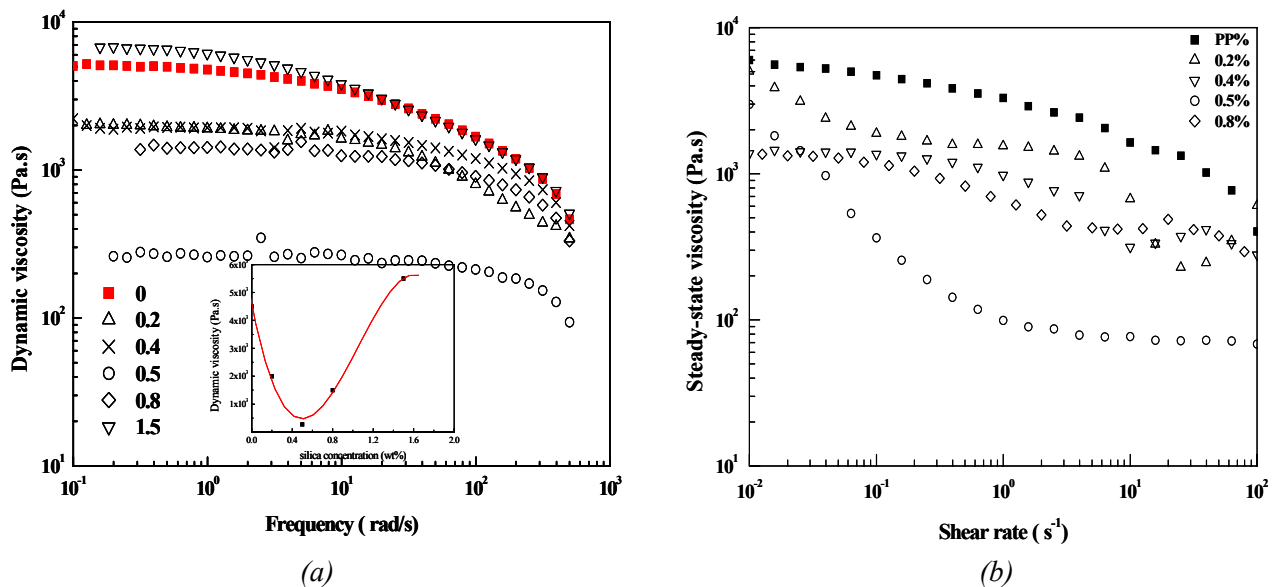


Figure 6.5: (a) Dynamic viscosity measured as a function silica concentration at 180°C and (b) steady-state viscosity as a function silica concentration at 180°C.

The selective physisorption of long chains on the nano-particle surfaces results in a segregation, with shorter chains in the matrix and longer chains adsorbed on the particles. This segregation leads to a less entangled matrix filled with deformable core-shell particles, i.e. a shell of adsorbed high molecular weight chains around the hard core-silica particles. The short chain matrix thus has a lower viscosity which also leads to a lower composite viscosity in contrast to the normally observed increase in the viscosity of particle-filled polymers. The dynamic and steady state viscosity of PP/silica nanocomposites measured as a function of filler concentration is shown in Figure 6.5. In the lower concentration regime (less than 1 wt%) the viscosity is reduced and exhibits a minimum at approx. 0.5 wt% silica. At higher concentrations confinement and bridging effects, apart from hydrodynamic forces, start playing a role which results in an increase in the viscosity of composites, as explained in Chapter 3.

6.3.2 Flow-induced crystallization

Under quiescent conditions crystalline structures grow isotropically which gives a uniform SAXS intensity distribution over the whole azimuthal angular range. When flow is applied, chain orientation results in faster crystallization. Oriented molecules form row nuclei or shishes and kebabs grow on these row nuclei. The kebab-like crystalline structure shows an intensity distribution in meridional direction observed as two lobes due to anisotropic growth in one direction. Figure 6.6 shows a series of two-dimensional SAXS patterns of PP and PP nanocomposites at a crystallization temperature of 135°C before and after application of shear of 60 s⁻¹ for 5 s. The SAXS images at time $t = 2$ sec correspond to the amorphous melt and consists of weak diffuse scattering indicating the absence of any detectable structure or orientation. The intensity of discrete scattering in the consecutive images becomes stronger due to growth of lamellar structures in the melt. After a certain time, when maximum space filling is attained, the intensity becomes constant and no significant changes can be observed in the scattering patterns.

For comparison, the scattering patterns for quiescent crystallization of PP are also shown in the first column. No oriented structures are developed under quiescent condition at 135°C. The scattering pattern shows an isotropic intensity distribution suggesting the formation of isotropic crystalline structures. The intensity continues to increase up to 1800 s, after which the experiments were stopped indicating that the crystallization process may not be completed.

Upon shearing, orientation and structure development can be observed in PP at approx. 180 s (see column 2). The appearance of two strong lobes along the meridian shows the development of anisotropic structures. This is typical for FIC. Furthermore, the scattering intensity increases more rapidly along these lobes than the surrounding unoriented structures suggesting that the oriented row nuclei are more than point nuclei. The scattering intensity increases continuously and saturates before 1800 s. The FIC for unfilled PP has been well studied and similar features were observed in previous studies [1].

For filled systems, however, the effect of FIC has not been studied. In the view of the observed reduction in the viscosity, it is not surprising to find a very specific trend for orientation under flow as a function of filler concentration.

- The scattering patterns obtained after 30 s clearly show, for 0.2 wt% and 0.4 wt% silica (column 3 & 4), the appearance of meridional maxima. PP with 0.2 wt% silica shows a stronger and more intense development of the meridional maxima as a function of time than the 0.4 wt% silica

system. The overall intensity for both 0.2 wt% and 0.4 wt% reaches a plateau after 480 s in contrast with the sheared pure PP, of which the intensity continuously increases.

- For 0.5 wt% silica (column 5), the intensity shows a uniform distribution as a function of azimuthal angle over the whole experimental time. The total intensity reaches a plateau after approx. 480 s, which is comparable to the 0.2 wt% and 0.4 wt% samples. No meridional maxima are observed.
- In case of 0.8 wt% silica (column 6), which still has a lower viscosity than PP, the evolution of the scattering patterns is different from those of the 0.2 and 0.4 wt% silica samples. The meridional maxima appear after 90 s and the intensity of these maxima does not grow as rapidly as observed for other compositions (except 0.5 wt%). However, the overall intensity increases indicating the development of isotropic crystalline structures.
- A further increase in concentration, i.e. 1.5 wt% silica, results in a higher viscosity. The strong meridional lobes (maxima) appear immediately and grow rapidly. The intensity of the lobes is much higher than for pure PP and all other compositions. This indicates a strong effect of flow.

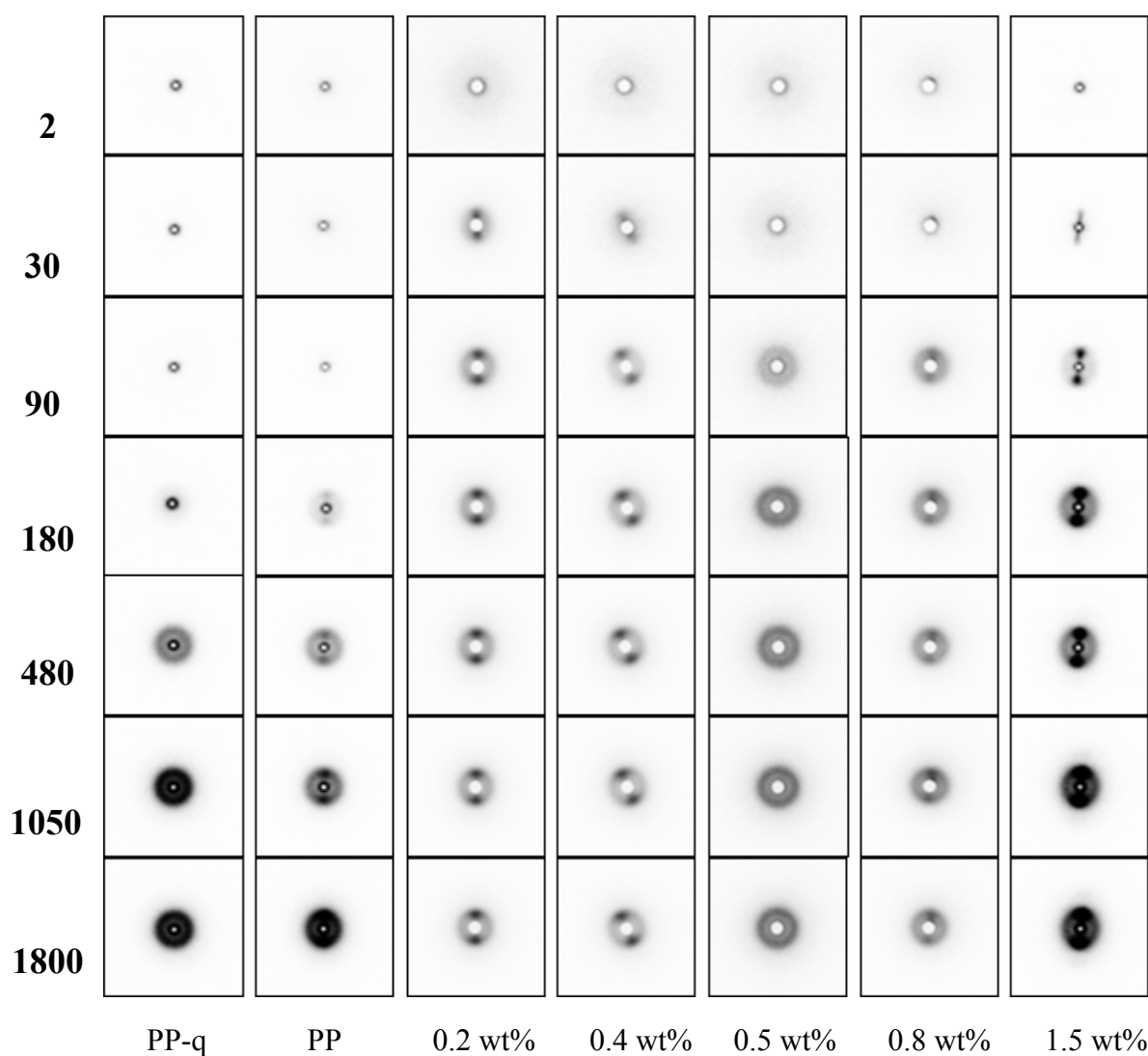


Figure 6.6: 2-D SAXS patterns of sheared PP/silica nanocomposites. The time (sec) is indicated on the left. The images in the first row are just before the application of shear. For comparison, patterns of quiescently crystallized PP are shown in the first column.

The images show some general features:

First, crystallization is accelerated by the shear flow. Figure 6.7a shows the total normalized intensity as a function of time for the PP and PP/silica nanocomposites. The intensity reaches the plateau faster when shear is applied. Moreover, the overall crystallization rate increases with increasing silica concentration. In Chapter 4, it was reported that the silica nano-particles act as nucleating agent for PP. However, it is not possible to attribute the increase in crystallization only to the nucleating effect of silica. There is a much stronger nucleating effect due to flow. Figure 6.7b shows the crystallization half-life time ($t_{1/2}$) of PP/silica nanocomposites under shear and quiescent conditions. The $t_{1/2}$ under quiescent condition was obtained from isothermal crystallization experiments performed at 135°C using standard DSC (TA Instruments Q1000). For all concentrations considered, the $t_{1/2}$ of quiescent crystallization is considerably higher than for FIC. The $t_{1/2}$ behavior clearly shows an enhanced nucleation effect due to flow on PP and silica-filled PP at high concentrations (1.5 wt %).

Second, the effect of flow on orientation with increasing silica concentration. Normally flow orients especially the high molecular weight polymer molecules creating extra nuclei thus accelerating the overall crystallization kinetics. An exception is seen for the lower concentration regime, i.e. below 1.5 wt% silica, where the crystallization behavior is slowed down in comparison to the sheared pure PP. The meridional maxima which are a measure of orientation become less strong as the silica concentration increases and goes through a minimum at 0.5 wt% silica (See also Figure 6.8).

Third, the most oriented pattern is observed at a concentration of 1.5 wt% silica. However, 1.5 wt% is not the limit to have maximum orientation; the orientation can still increase with increasing silica concentration. The increase in the orientation along the meridian under shear in case of PP/silica nanocomposites at higher silica concentrations can be attributed to the higher experienced shear stresses due to presence of the particles and the smaller inter-particle distances. The particles start to interact with each other and the viscosity increases again and the orientation increases sharply and instantly.

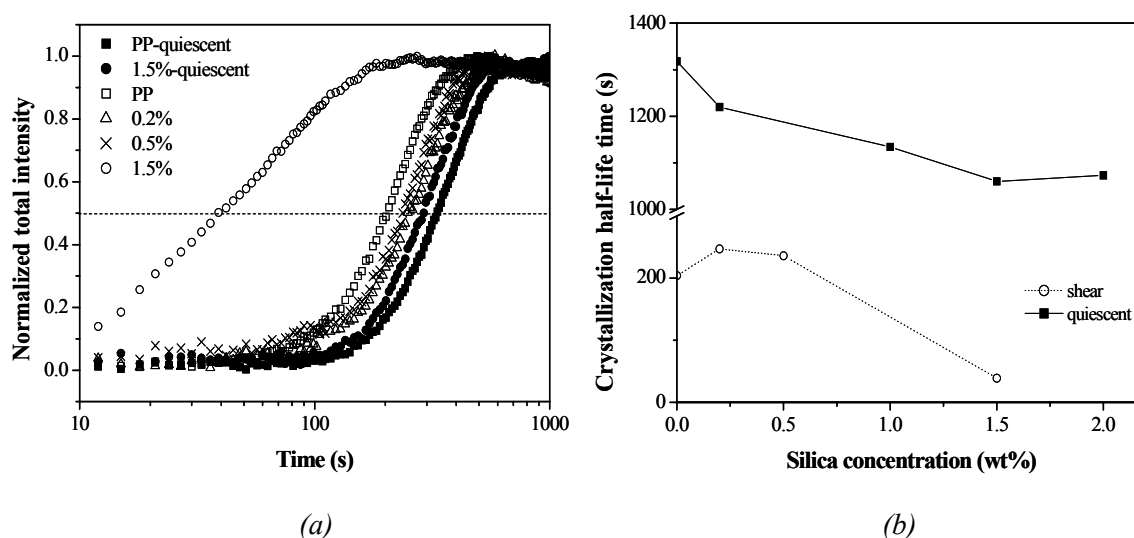


Figure 6.7: (a) Normalized intensity taken as relative crystallinity, (b) half-life time of crystallization as a function of silica concentration. The dashed line shows the half-life time calculated from FIC data from Figure 6.7a. The half-life time under quiescent condition is determined from DSC experiments.

Figure 6.8 shows the time evolution of the normalized total scattered intensity to maximum scattering, the normalized azimuthal angle independent component (unoriented) and the normalized intensity of the oriented structures with varying concentration of silica nanoparticles. In general, the total intensity rises rapidly after the application of shear. However, for 0.2 wt% and 0.5 wt% silica, the increase in the intensity is slower than for sheared pure PP. The initial period of the crystallization during which formation of nuclei takes place is important, since its characteristics subsequently determine the overall crystallization behavior (see also Figure 6.7). The difference in total and unoriented intensities reflects the orientation.

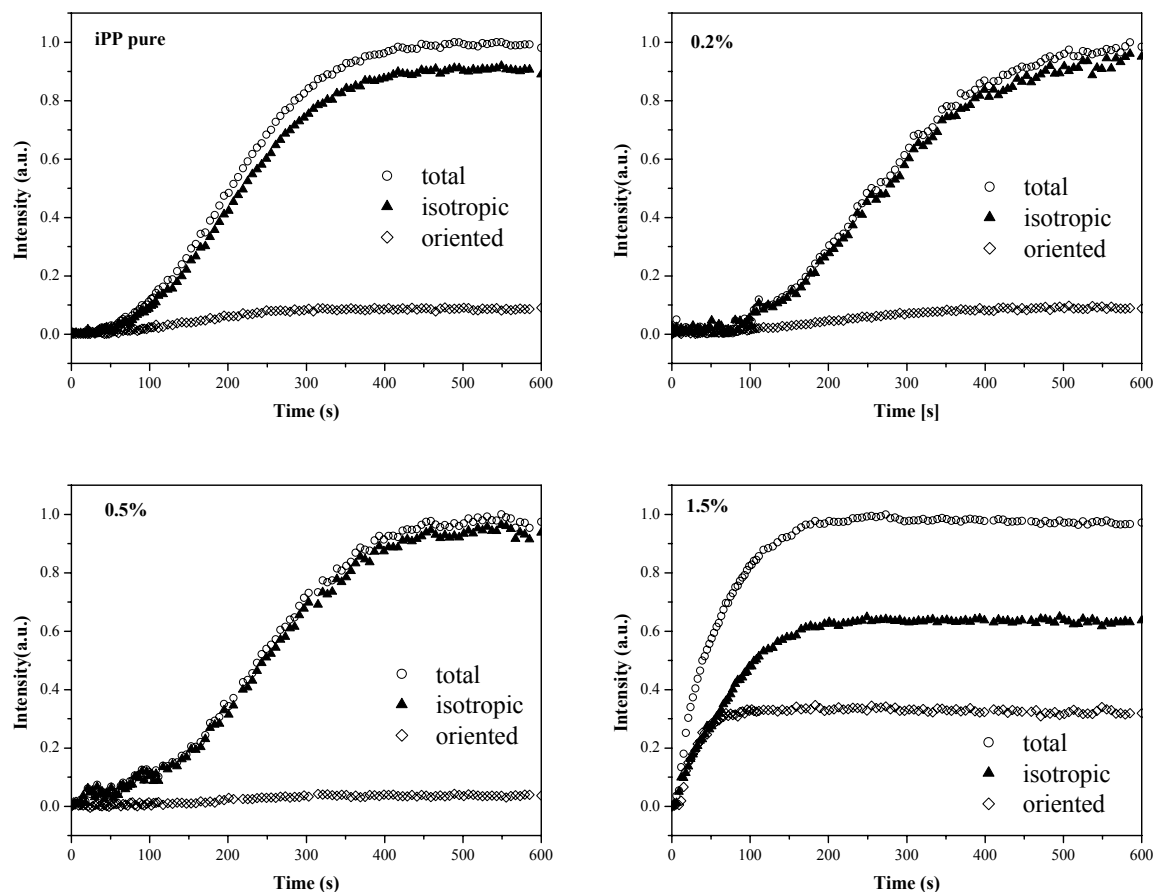


Figure 6.8: Time evolution of total scattered intensity, the azimuthal angle-independent component (unoriented) and the intensity of the oriented structures for PP/silica nanocomposites with varying concentration of silica nanoparticles. The plots are constructed from 2-D SAXS images.

For clarity, the orientation fractions ($I_{\text{oriented}}/I_{\text{isotropic}}$) for the different systems calculated from these results are shown in Figure 6.9. Compared to pure PP, the scattered intensity attributed to oriented structure is relatively high for the 1.5 wt% silica system and the orientation fraction is more than 5 times higher. For 0.5 wt% silica, the orientation intensity is negligible. The orientation goes through maximum due to random growth of crystals after an initial induction period.

Figure 6.10 shows the long period calculated from the scattering patterns of oriented and unoriented structures. It is essential here to note the time scales of the appearance and growth of oriented and random structures. Oriented structures initially have a large long period, but as more lamellae are

inserted the long period of oriented structures starts to decrease and reaches a plateau almost at the same level as that of randomly grown crystals.

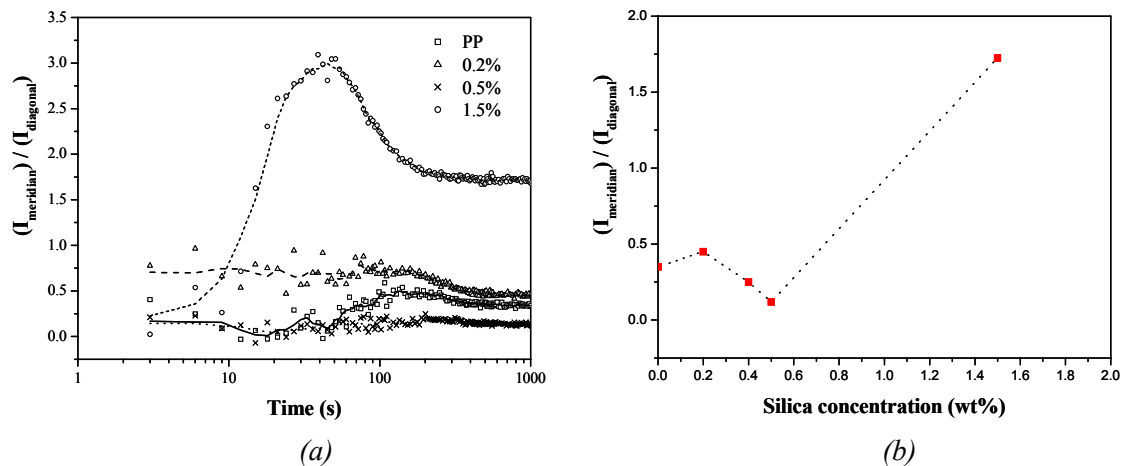


Figure 6.9: (a) Orientation fraction ($I_{\text{oriented}}/I_{\text{total}}$) calculated from Figure 6.8; (b) plot of orientation as a function of silica content.

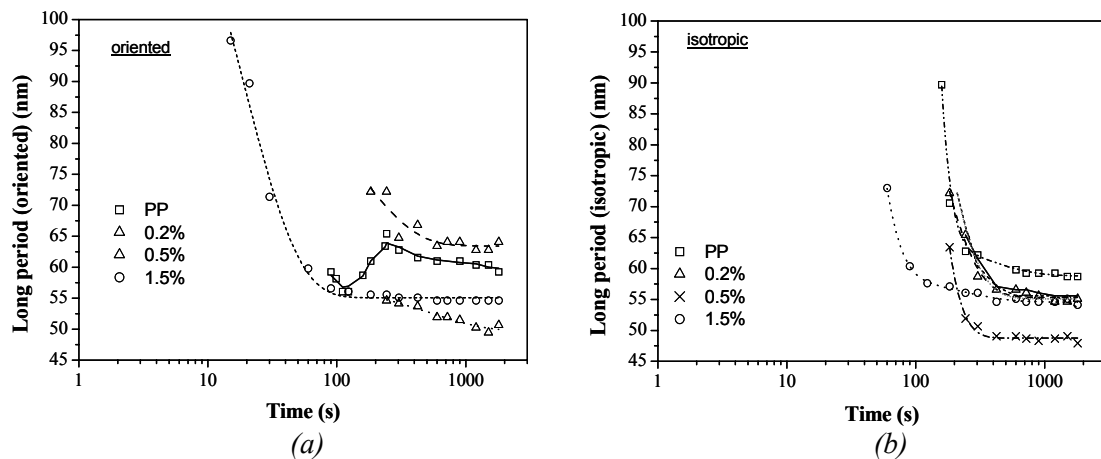


Figure 6.10: Long period calculated from SAXS. Note the time scale of the change in long period of oriented and random structures.

In-situ synchrotron WAXD experiments also confirm the orientation behavior as observed by SAXS. Figure 6.11a shows the schematic representation of shish-kebab structures formed during flow-induced crystallization and respective reflections in WAXD patterns. The chain axis (c) and neutral axis (b) are also shown. The reflections corresponding to different crystalline phases and axes are given elsewhere [12,13]. Figure 6.11b shows the 1-D WAXD intensity profiles of PP/silica nanocomposites after shear. The peak at $2\theta = 12.1^\circ$ originates from the (110) reflection of the unit cell of the α -crystal form of PP. The intensity of the (110) reflection relative to other reflections, for instance (040), in WAXD gives a qualitative indication of orientation. It is known that detectable structures in the WAXD appear later than SAXS; however, once developed they can be easily probed. Another important reflection is (300) originating from the β -crystal modification. The intensity of this reflection with respect to the (130) reflection of the α -modification increases in case of PP/silica nanocomposites and is higher in the 0.5 wt% silica system than for the other concentrations. The significant amount of the β -modification in case of 0.5 wt% silica system indicates that silica nano-particles nucleate the β -modification and is stronger even when there is no orientation. Although the amount of β -phase is higher, the α -phase is more prominent in the

nanocomposites at higher concentration of silica nano-particles than in the pure PP due to strong orientation.

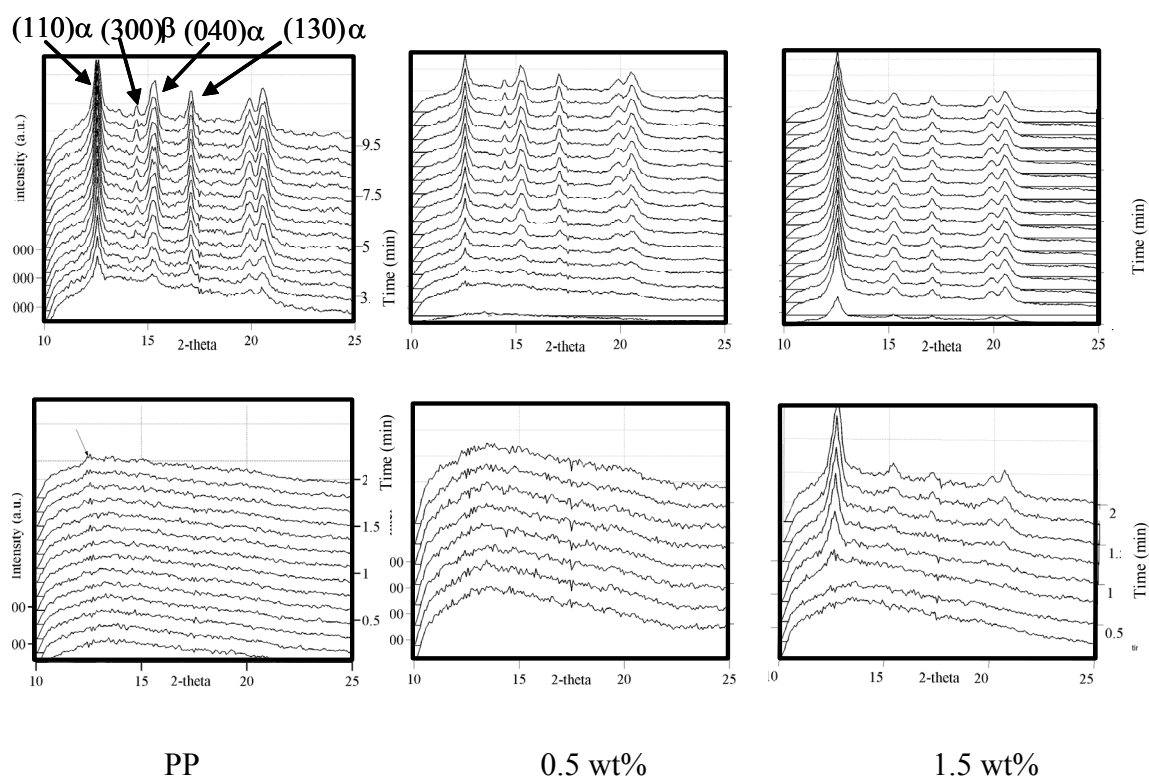
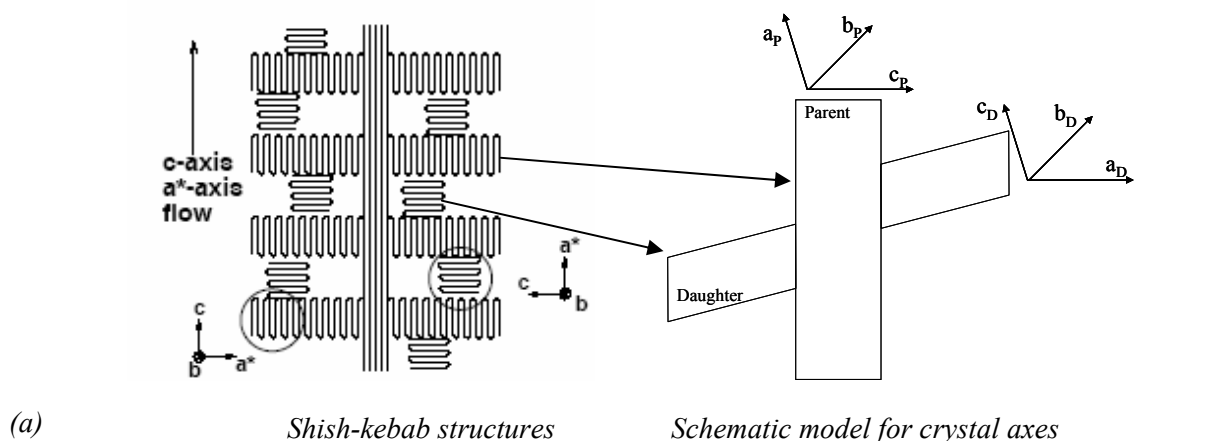


Figure 6.11: (a) Schematics showing the flow direction and alignment of the lamellae in the oriented crystallites that explains the diffraction patterns in Figure 6.11b, (b) Development of 1-D WAXD patterns during and after shearing. The (110) reflection at $2\theta = 12.1^\circ$ is taken as the reference for orientation. PP shows the appearance of (110) reflection after 100 s, whereas in case of 0.5 wt% silica it appears after 3 min. On the other hand, for 1.5 wt% silica system, orientation is developed within 30 s. All samples show the presence of the β -modification, the relative amount is higher in silica-filled PP.

6.4 Discussion

6.4.1 Rheology

When the long chains are preferentially adsorbed on the silica nano-particles, the short chains form the matrix which has a lower viscosity than the unfilled melt. The amount of adsorbed polymer

chains and the inter-particle distance between filler particles are the two important parameters which govern this effect. This was discussed in detail in Chapter 3. The increasing amount of adsorbed long chains with increasing filler concentration is significant, especially due to the broad molecular weight distribution of the used PP. The experimental findings discussed in Chapter 3 suggest that the long polymer chains which are important for the orientation are adsorbed, encapsulating the silica nano-particles and shorter chains forming the matrix as shown schematically in Figure 6.12a. Although the shear rate is effectively increased because of presence of particle, the viscosity drops which due to segregation effect reduces the effective shear stress which in turn reduces the orientation under flow as anticipated in Figure 6.12b. At high silica concentration, the high effective local shear stresses can only give a high local orientation, if long enough chains are available [5].

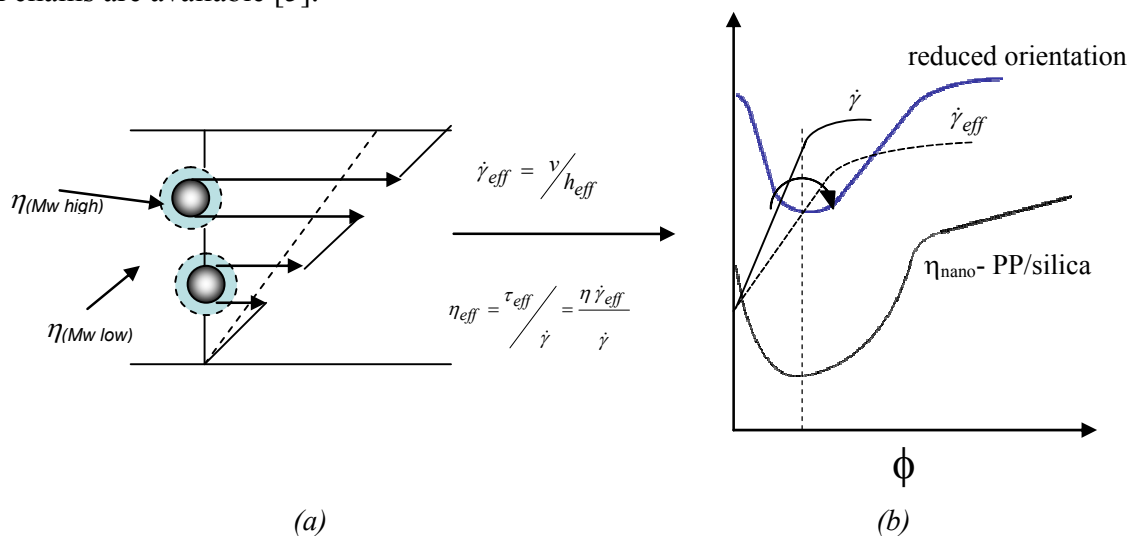


Figure 6.12: The deformable shell (of adsorbed long PP chains) and hard core (silica nano-particles) morphology results in decreased shear stresses due to segregation effects which reduces the orientation under flow.

6.4.2 Structure development under flow

Both the growth of oriented lamellae and randomly distributed lamellae is expected to occur in the melt after cessation of shear. The FIC experiments show that the orientation decreases and goes through a minimum at 0.5 wt% silica (at 0.2 wt% silica the orientation is slightly higher than for pure PP). The orientation follows the same trend as the viscosity and there is no orientation in case of 0.5 wt% silica. Figure 6.13 shows TEM micrographs of PP and PP/silica composites after FIC. The growth of lamellae in pure PP is perpendicular to the flow direction. This is usually observed in FIC. In the presence of silica nano-particles, the lamellae start to grow more randomly, although at 0.2 wt% silica concentration a significant amount of lamellae grow perpendicular to the flow direction. When the silica concentration is further increased to 0.5-0.8 wt%, due to less orientation, the number of random lamellae increases. This is a clear evidence of less orientation under flow and a more random growth of lamellae for PP/silica nanocomposites compared to pure PP.

6.4.3 Orientation fraction

The analysis of the 2-D SAXS images gives information on the crystallization kinetics and orientation. It was found that for 0.2 wt% and 0.5 wt% silica systems, the crystallization rate after flow is slower and the half-life times are higher than for pure PP (see Figure 6.7). This is in

agreement with the observation that also the orientation decreases in this concentration regime. Besides, the adsorption of chains slows down the crystallization due to partial demixing from the matrix. Even though silica acts as a nucleating agent, these two effects, i.e. less orientation and adsorption of long chains, slow down the overall crystallization compared to pure PP after flow.

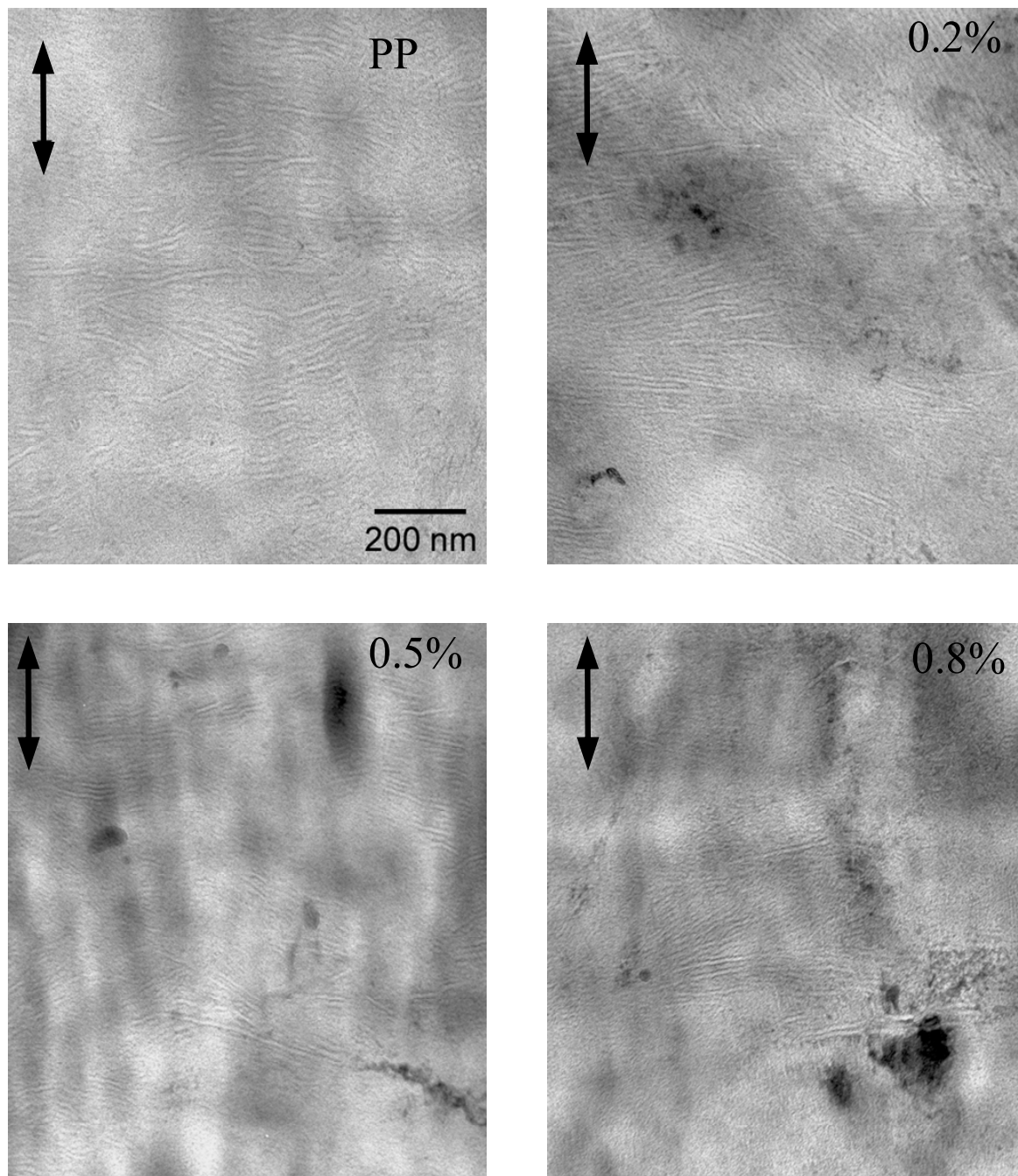


Figure 6.13: TEM micrographs of FIC after complete crystallization showing the direction of lamellar growth. The arrows indicate the flow direction.

In all cases, the orientation goes through maximum with time except in case of 0.5 wt% silica (see Figure 6.9a). As described above, due to random growth of lamellae on the oriented structures, the orientation fraction decreases rapidly. The time scales of the appearance of random structures and the maximum in the orientation are comparable. Figure 6.14 shows the orientation and long period

from oriented structures in the same plot. The onset of the reduction in orientation or formation of random structures shift to shorter times with increasing silica content, emphasizing two effects: the increase in crystallization rate due to orientation and, the nucleating effect of silica nano-particles. Another reason could originate from the preferential adsorption of longer PP chains on silica surface. The adsorbed chains may orient under shear and start to crystallize but the matrix is of shorter chains and crystallize faster and hence lamellae (predominantly from the unoriented chains) start to grow earlier on the oriented structures compared to pure PP which has longer chains entangled with shorter chains in the matrix. A similar observation can be made from the normalized intensity curve shown in Figure 6.7.

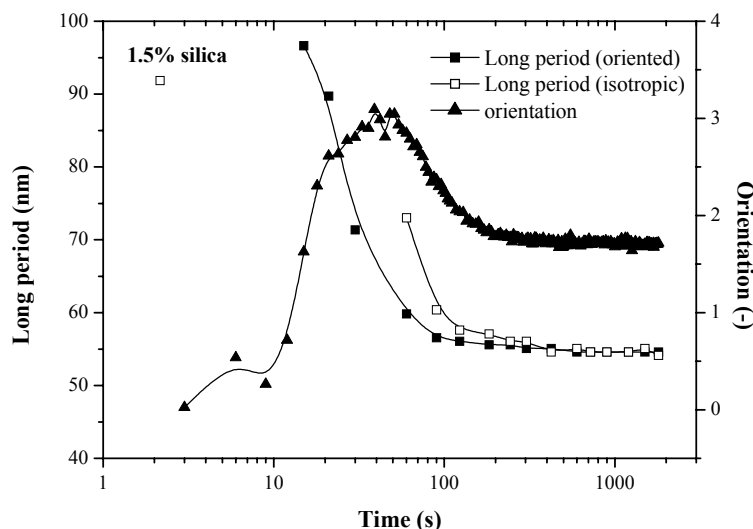


Figure 6.14: Long period and orientation as a function of time for 1.5 wt% silica.

6.4.4 Crystallization kinetics under flow

The Avrami approach was applied in order to obtain more information about the crystallization mechanism. The following kinetic equation for space filling is used:

$$X_t = 1 - \exp(-Kt^n)$$

where n is the Avrami exponent related to the type of nucleation, K the Avrami rate constant and X_t the space filling (or relative crystallinity) at time t obtained from the normalized total scattering intensity of SAXS. The slope of the linear dependence $\log(-\ln(1-X_t))$ vs. $\log(t)$ gives directly the value of the Avrami exponent n .

The Avrami plots obtained from these data are shown in Figure 6.15. In case of quiescent crystallization of PP, the slope of the curve n is 2.6 which is close to 3 signifying a three-dimensional growth. It must be noted that the crystallization is not completed in 30 min at 135°C. The crystallinity obtained is less than 50% of the crystallinity, if let to completion at the same temperature. In case of FIC of PP, two distinct regimes can be observed: 1) up to 5% relative crystallinity, $n = 1$, i.e. one-dimensional crystal growth which can be assigned to the formation of oriented structures in the melt due to the applied shear pulse; 2) beyond 5% relative crystallinity, $n = 2.5$ which indicates 2- and 3-dimensional crystal growth. This is expected in case of flow-induced crystallization of PP.

Similar effects are observed for the PP/silica nanocomposites, but with varying degree of impact.

- For instance, for 0.5 wt% silica system no orientation is observed, i.e. no initial growth of oriented one-dimensional structures, but a steep increase with a small plateau and then $n = 2.5$. A quiescent crystallization of PP also shows $n = 2.5$.
- On the other hand, the 0.2 wt% silica system shows initially an oriented growth ($1 < n < 2$) with scattered points, followed by 3-dimensional growth.
- The 1.5 wt% silica system shows a different behavior; the initial slope immediately after the step shear is $n \approx 2.5$ and at a later stage $n = 1$ is observed.

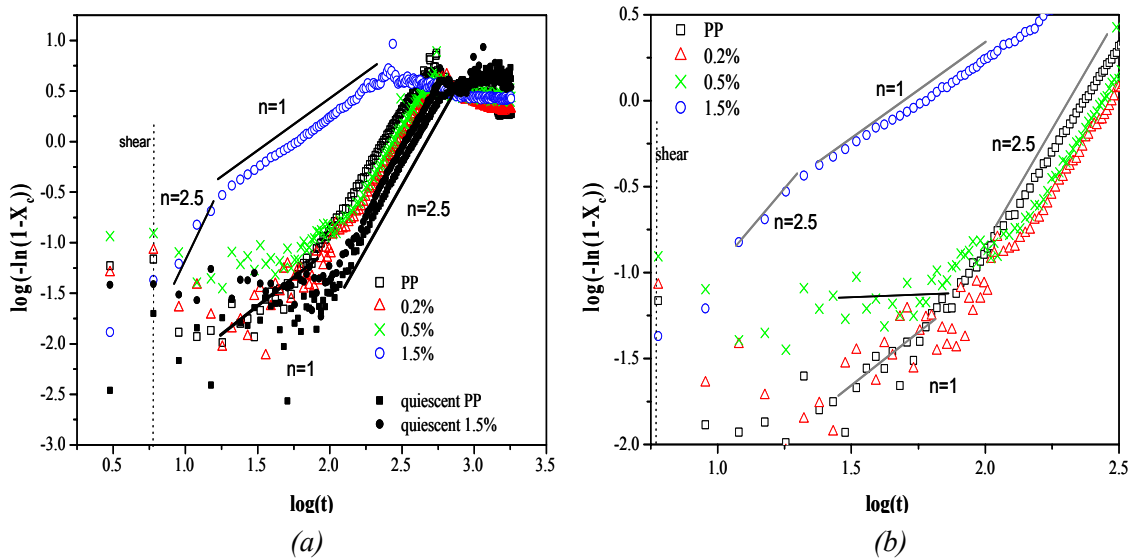


Figure 6.15: (a) Avrami plot of pure PP under quiescent and flow-induced crystallization conditions. $n \sim 1$ signifies a one-dimensional growth; $n \sim 3$ suggests a three-dimensional growth. (b) Magnified view of the initial crystallization period immediately after application of shear.

However, the initial slope $n = 2.5$ in case of 1.5 wt% silica, is at a very early stage of crystallization and the points are scattered. A possible explanation for such a behavior is that the isothermal crystallization can start at this temperature (135°C) due to the nucleating effect of silica nanoparticles even before the shear pulse is applied and this may result in 3-dimensional growth. However, the time of crystallization and the relative crystallinity do not show a clear trend with increasing silica concentration. The half-life time of crystallization shown in Figure 6.7 for the 1.5 wt% silica system is relatively high, so the nucleation effect due to the silica particles may not be the only possibility. Moreover, when a shear pulse is applied, the presence of silica nano-particles may result in more sites for the development of oriented structures with smaller or no shish, which eventually shows a three-dimensional growth at the start followed by one-dimensional growth and, in the last stage, three-dimensional growth ($n = 2.6$). The adsorption of the long PP chains and the three-stage crystallization process can explain this behavior. At higher silica concentrations, the molecules orient due to high stresses, but since the shorter chains have a smaller relaxation time the row-nuclei structures break down to point-nuclei structures.

The three-stage crystallization process is schematically shown in Figure 6.16: 1) The PP chains adsorbed on the silica nano-particles orient under shear, especially in local areas between the particles, where stress maxima (i.e. orientation maxima) occur resulting in more oriented structures, called row nuclei; 2) destruction of the orientation due to the local inhomogeneous, unsteady flow and the formation of unoriented row nuclei and/or primary point-nuclei structures due to break

down of row-nuclei structures; 3) growth from these disoriented nuclei. Such a behavior was reported for shear-induced crystallization of poly(butylene terephthalate) [14]. Thus, the nucleating effect of silica and the combination of formation and relaxation of row nuclei to primary point nuclei results in $n = 2.5$ in the initial period of crystallization immediately after shear. These point nuclei grow in number from the oriented structures, which yields $n = 1$ and this is also the reason for very high orientation (5 times higher than PP). When the crystallization reaches the third stage, it would be expected that more homogeneous crystalline structures develop compared to PP. In case of pure PP, the crystallization proceeds in two stages: the first stage with $n = 1$, due to oriented structures and later two- and three-dimensional epitaxial growth of kebabs on the oriented structures. This two-stage crystallization for pure PP usually results in oriented structures in injection-molded or extruded samples, however, flow-induced crystallization for silica-filled PP (in the higher silica concentration regime) does not follow this behavior and gives a more homogeneous, isotropic morphology.

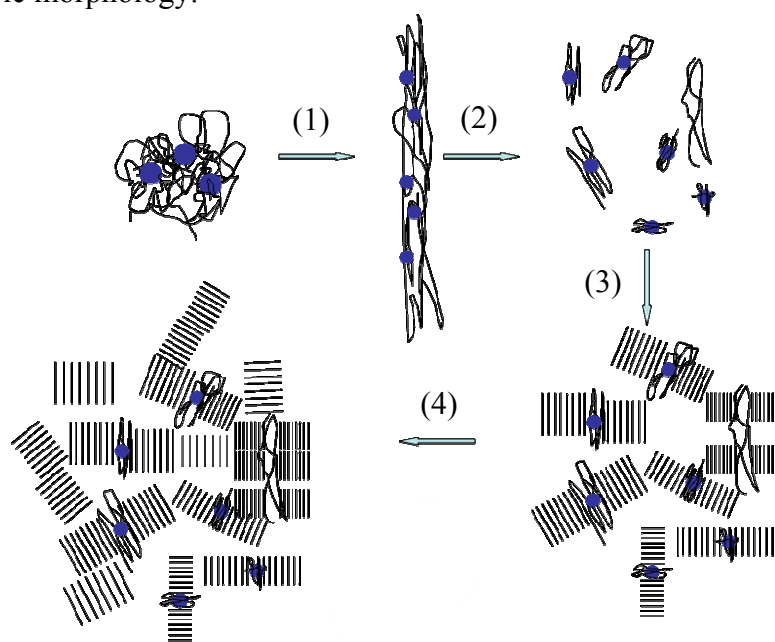


Figure 6.16: Schematics of FIC for 1.5 wt% silica; (1) flow-induced row nuclei; (2) formation of primary point nuclei due to relaxation and destruction of the row nuclei, (3) growth of the disoriented nuclei; (4) final isotropic structure.

6.4.5 Orientation in injection molding

Injection molding of polymers is a complex flow while FIC is performed under simple shear flow. However, the concepts developed in FIC under high shear may be applied to injection molding conditions. For this reason, injection molded bars were characterized for orientation using FT-IR dichroic ratio measurements. Less orientation is anticipated in case of 0.5 wt% silica. This can be seen in Figure 6.17, which depicts the orientation fraction (f) along the thickness on injection molded bars of PP and silica-filled PP. Not only 0.5 wt%, but also 2 wt% silica systems show less orientation.

One would expect more orientation in injection-molded samples of nanocomposites with a higher viscosity that show a higher initial orientation in FIC. For this reason, injection-molded samples of 2 wt% silica are compared with pure PP over the length of injection molded bars. Moreover, the

three stage-crystallization process explained above which indicate less orientation in the end (completion) can only be verified by characterizing the orientation of injection-molded bars. The results from FT-IR measurements on pure PP and 2 wt% are shown in Figure 6.18 and show no orientation in the flow direction. These observations are in agreement with the TEM micrographs after FIC experiments (Figure 6.11). The TEM pictures show that lamellae grow in a random fashion even after FIC, minimizing the effect of orientation, if any, during the flow. The final structures and morphology characterization of the processed material, along with in-situ SAXS-WAXD results, confirm that the orientation is less pronounced and, that the non-uniformity and anisotropy in the structure is reduced in case of silica-filled PP.

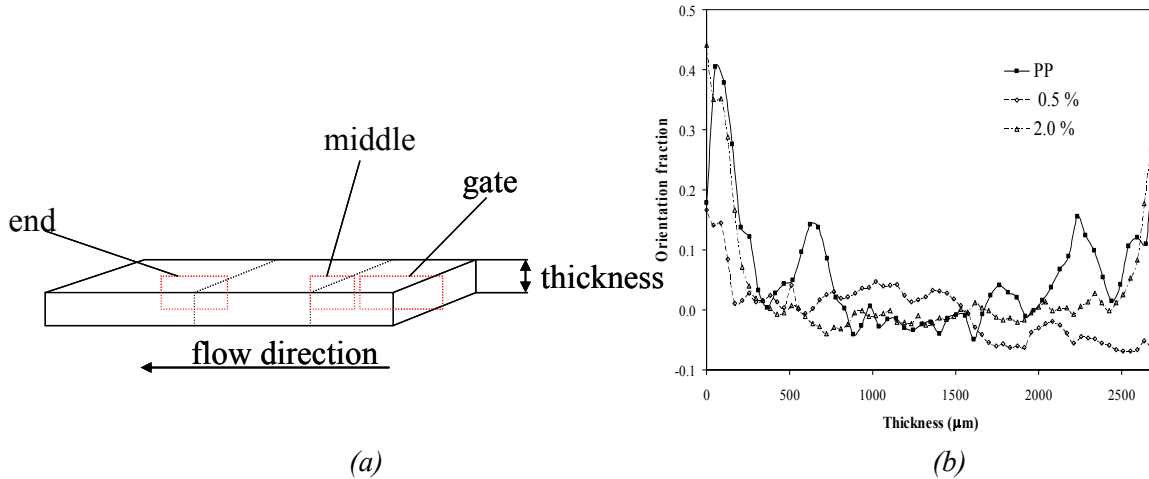


Figure 6.17: (a) Schematic of injection-molded bar and the places of measurements. (b) Orientation fraction calculated from FT-IR dichroic ratio measurements on the middle of the injection molded bars.

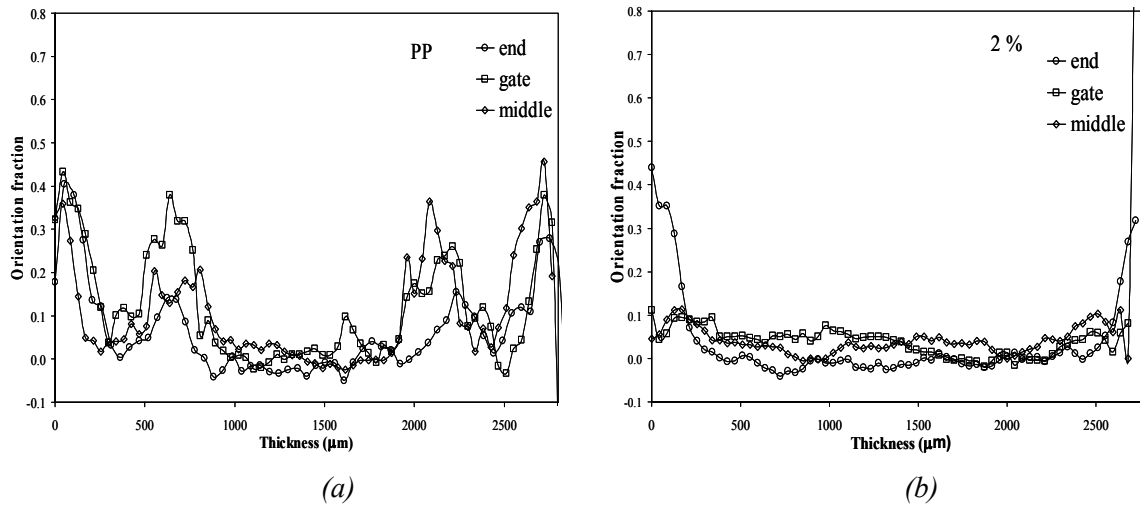


Figure 6.18: Orientation fraction calculated from FT-IR dichroic ratio measurements for (a) PP and (b) PP/silica (2 wt%). The legends on the graphs indicate the position of the section taken from the injection molded bar.

6.5 Conclusions

Flow-induced crystallization of silica nano-particles filled PP in the lower concentration regimes (up to 1.5 wt%) have been studied using in-situ SAXS-WAXD. The silica nano-particles reduce the viscosity of composites in a specific range of concentration by selective adsorption of longer chains. The drop in viscosity reduces the effective shear stress and hence the orientation during the flow.

SAXS analysis showed that orientation fraction shows minimum at 0.5 wt%, following the same trend as the viscosity. The adsorption of longer chains results in demixing of longer chains from the shorter chains which form the matrix. These short polymer molecules do not orient under flow. The Avrami analysis showed a one-dimensional growth in the early stage of crystallization upon application of shear in case of PP and higher silica concentrations. For the 0.2 wt% and 0.5 wt% systems where the orientation is less, the Avrami exponent is 2.6, which suggests a three-dimensional growth as observed in quiescent crystallization conditions. At high silica concentration, the viscosity increases and thus the effective shear stresses are also higher which results in a higher orientation. However, TEM and FT-IR dichroic ratio measurements show no orientation in FIC and injection molding even at higher silica concentration of 2 wt%. This is explained on the basis of three-stage crystallization process, in which initially during flow, row nuclei are formed, which further form primary point nuclei from which lamellae grow in a random direction overruling the orientation effect due to flow. This is usually observed when crystallization is carried under less supercooling and this may have been aided by the nucleating effect of silica nano-particles and the adsorbed polymer molecules. Although the exact mechanism is difficult to predict at this stage, these polymer adsorbing nano-fillers show promising aspects with respect to reduction in non-uniformity and anisotropy in the structure and morphology of the final processed products.

6.6 References

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Chapter 7

Mechanical properties of PP/silica nanocomposites*

Abstract

The mechanical properties of PP/silica nanocomposites prepared via an in-situ sol-gel reaction were studied. For these nanocomposites, three different regimes can be distinguished depending on the concentration of silica nano-particles. In Regime I, i.e. silica concentrations below 1.0 wt%, a layer of selectively adsorbed high molecular weight chains is present on the silica particles, resulting in a phase separated matrix of relatively low molecular weight chains with no overlap of between the adsorbed layers. Regime II is defined as the silica concentration range in which the adsorbed layers overlap, leading to interaction and bridging effects, but the viscosity is still influenced by the adsorption process. In Regime III, the silica particles form aggregates and the increased interactions also increases the viscosity. For low silica concentrations, i.e. Regime I, the Young's modulus increased slightly in comparison to pure PP due higher crystallinity, while the effect of orientation was less. The yield stress, which mainly depends on the lamellar thickness, also showed a minimum. The strain hardening modulus, which mainly depends on the mobility of the amorphous phase, also followed a similar trend. At the higher silica concentrations in Regime II, the Young's modulus exhibited a higher increase. The yield stress is also higher, because the particles influence the flow-induced crystallization process, leading to a higher crystallinity and larger lamellar thickness. The adsorbed chains interact and form bridges, thus the amorphous phase becomes more entangled ultimately leading to an increase in strain hardening modulus. In Regime III, the mechanical properties are negatively influenced by the particle aggregation, which results in stress localization. A stability analysis of the necking process supported the behavior in these regimes. The impact strength increased slightly in Regime I, but in Regime II increased more than 1.5 times to that of PP.

7.1 Introduction

Inorganic particle-filled nanocomposites of semi-crystalline polymers have been explored extensively in recent years, especially to improve the mechanical properties, notably modulus and tensile strength [1,2,3,4]. The preparation of nylon-clay nanocomposites by Toyota researchers [5] sparked an enormous interest amongst researchers to make high-performance nanocomposites with low-cost inorganic fillers. Nano-particles may influence the properties of the polymer matrix in different ways, especially for semi-crystalline polymers. The exact mechanism for improvement of the properties for these systems has not been clarified.

*This chapter is partly reproduced from “ Mechanical properties of PP/silica nanocomposites” to be submitted to Macromolecules.

Various mechanisms for improving the toughness of semi-crystalline polymers have been proposed in literature, especially by the addition of rubbery particles. The brittle-to-ductile transition (BDT) of polymer composites depends on both extrinsic and intrinsic properties. The intrinsic variables include mechanical properties of matrix, interfacial adhesion, and morphological parameters, such as the volume fraction, size and shape of the dispersed particles, while the extrinsic variables include specimen geometry, strain rate, and temperature. Wu [6] developed a theory for both amorphous and semi-crystalline polymers in which the interparticle ligament thickness [7] is the single parameter controlling the BDT. Below a critical value, which is related to the entanglement density of the polymer, the interparticle ligament thickness becomes too small to allow for craze formation and a transition to shear yielding occurs. A further function of the particles is to induce cavitation and thereby change the stress state in the matrix around the particle. Argon and co-workers proposed a stress delocalization by using inorganic hard fillers. Unlike for rubbery particles, debonding, i.e. the separation of the filler particles from the polymer matrix, should be allowed for to facilitate shear yielding of the matrix [8].

Most studies on nanocomposites reported an increase in stiffness but a lower toughness compared to the matrix polymer, although Argon and co-workers found that some particle-filled systems display an improvement in toughness [1]. They explained this by a combined deformation mechanism of crack deflection and shear yielding that allows local plastic stretching of the polymer matrix material around the particles after debonding [9], though the experimental results on which the mechanism was based are strongly dominated by flow. The typical filler concentrations used were as high as 20 vol%, which invariably deteriorates the processibility.

The final properties are the resultant of not only the composition, but also of the processing or shaping step, in many cases underestimated or even neglected. During the shaping process, polymers are subjected to complex flow fields with large temperature and pressure gradients in short times, resulting in large variations in properties throughout the entire product. It is well known that especially the high molecular weight part is very sensitive to flow due to the long relaxation times. The addition of particles usually leads to an increase of the viscosity. Moreover, the increase in shear stresses inculcates local shear heating that may result in molecular degradation. Further, the addition of particles may result in the development of anisotropic structures during processing.

In the previous chapters, the synthesis, characterization, rheology, and the crystallization behavior of a new class of polypropylene/silica nanocomposites prepared via an in-situ sol-gel reaction in porous structures were described that show a significant drop in the viscosity and a much lower orientation during of the polymer chains during shear-induced crystallization. This anomalous behavior was attributed to selective physisorption of the higher molecular weight PP chains on the silica particles, leaving a matrix composed of shorter chains, as explained in detail in Chapter 3. The matrix also becomes less entangled. Figure 7.1 shows the schematic representation of the structure formed in these PP/silica nanocomposites. Three regimes depending on the silica concentration, resultant morphology, and molecular structures due to adsorption and increasing can be identified:

Regime I: The characteristic feature of this regime is the demixing and resultant reduced entanglement. The viscosity is reduced and inter-particle distance is sufficiently large to avoid inter-particle interactions or bridging effects. The longer molecules adsorbed on the surface

experiences less entanglement than molecules in the bulk because its proximity at surface has introduced asymmetries as schematically shown in Figure 7.6.1b. This reduces overall entanglement density of the materials. The molecular segments can no longer be spherically distributed as in the bulk because segmental random walk is prevented by the solid-surface. Intermingling of separate chains is also restricted by the surface so that probability of finding another chain next to the filler is small and may only arise in the bulk of polymer. Such a behavior has been described by Kendall and Sherliker for LDPE/aerosil silica [10].

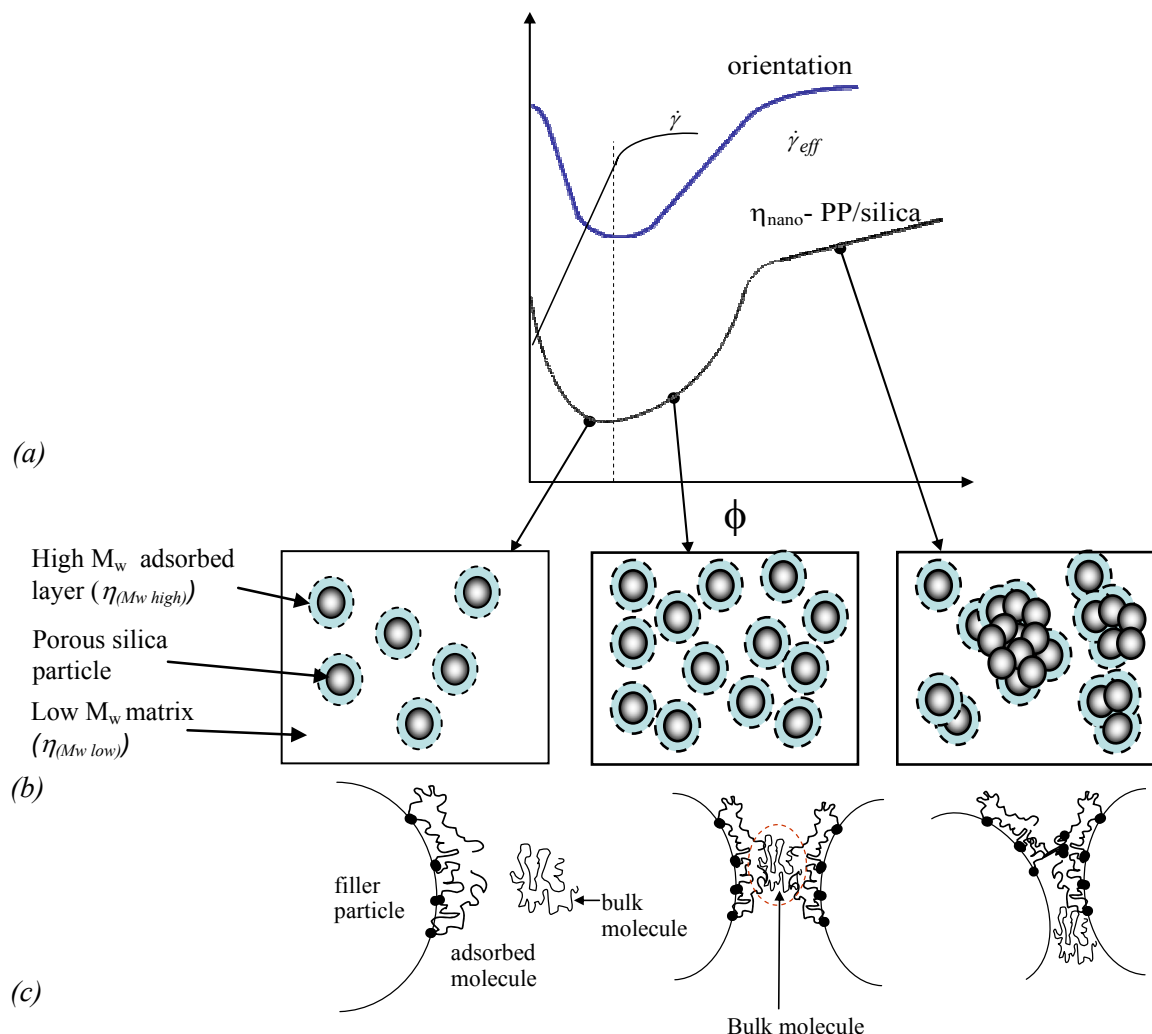


Figure 7.1: (a) Schematic showing the decrease in viscosity and orientation as a function of filler concentration, (b) Schematic morphology of PP/silica nanocomposites, showing the adsorbed layer around mesoporous silica nano-particles and less entangled matrix of low molecular weight PP, (c) A reduced entanglement of nanocomposites is because polymer molecule bound to a filler particle experiences reduced entanglement due to the surface asymmetry.

Regime II: This regime has intermediate range of concentration of silica at which particles are sufficiently apart to have adsorption effect, but the distance between particles subsequently decreases with increasing adsorbed layer thickness which results in intermingling of adsorbed molecules with free molecules in bulk. This regime is characterized by increase in viscosity after minimum. Despite small interactions, the entanglement density may be still lesser than pure polymer matrix due to restricted movement of adsorbed chains.

Regime III: This regime can be identified at high concentration of silica at which particles are close to each other forming aggregates, thus affecting adsorption and subsequently increasing the entanglement density due to intermingling and bridging effects. Increased interactions and bridging also increases the viscosity.

Depending on the size and shape and concentration of the fillers, nanocomposites fall under one of these three regimes. Of special interest are Regimes I and II, in which low molecular weight polymer chains form the matrix, thus reducing the viscosity in turn improving the processability. On the contrary it is well known that solid-state strength of material depends on the molecular weight of matrix, especially M_n . The effect of molecular orientation and morphology developed during the flow on the final properties is investigated keeping in mind the changing distribution of molecular weight in different regimes. Therefore, it can be anticipated that the three concentration regimes may exhibit different mechanical properties.

7.2 Experimental Section

7.2.1 Materials

Porous isotactic polypropylene (PP) having a porosity of $S_{BET(Kr)} = 0.0678 \text{ m}^2/\text{g}$ was supplied in powder form by Euro-Sabic, The Netherlands. The number-average and weight-average molecular weight of PP was 60 and 380 kg/mol, respectively. The powder was free of antioxidants and stabilizers and stored at -10°C in the dark. Tetraethoxy orthosilicate (TEOS), and ammonium hydroxide (28% NH_3) were obtained from Aldrich Chemicals.

7.2.2 Nanocomposite preparation

PP/silica nanocomposites were prepared according to the procedure described in Chapter 2, Section 2.3.1.2. TEOS was used as the precursor for making silica nano-particles by a sol-gel reaction. TEOS was added to the PP powder in the reactor and stirred for about 30 min at 60°C . After dispersing TEOS, a mixture of water (molar ratio of TEOS/ H_2O was 1:5) and ammonium hydroxide (1 wt% on TEOS) was added slowly at a controlled rate to the reactor with continuous stirring under a nitrogen atmosphere. Ammonium hydroxide acts as catalyst and being base avoids aggregation of silica particles. The sol-gel reaction was carried out in the reactor at 60°C for 3 hrs with continuous stirring. The thus formed sol in PP is then gelled at 80°C for 5 hrs under a nitrogen atmosphere and dried under vacuum at 100°C for 24 hrs, and subsequently stored at -4°C in the dark.

7.2.3 Optical Microscopy

Optical microscopy is performed using a Zeiss Axioplan 2 Microscope. The samples were microtomed in couples with a thickness of approx. $5 \mu\text{m}$ using a Cryo-microtome, Leica RM 2165, at -20°C . Samples were cut perpendicular to the flow direction. The half sections were cut along the thickness and examined along the thickness direction, i.e. perpendicular to the flow direction.

7.2.4 DMTA

Glass transition temperatures and moduli of the samples were measured using a TA Instruments Q800 Dynamic Mechanical Analyzer fitted with a single cantilever geometry. Samples were extruded in a twin-screw mini-extruder with a residence time of 5 min at 190°C ; the rotor speed was set at 50 rpm. The extruded samples were compression molded into sheets and bars with a size of approx. $20 \times 15 \times 1 \text{ mm}$ were cut from the sheets. For the measurements, the samples were cooled to -80°C and held for 5 min and subsequently heated to 100°C at 2 K/min. A dynamic force of 0.25 N oscillating at 1 Hz and amplitude of $0.2 \mu\text{m}$ was used.

7.2.5 Crystallinity and lamellar thickness

The degree of crystallinity and lamellar thickness are measured by two different techniques: (a) wide- and small-angle X-ray scattering experiments and (b) DSC experiments.

Wide- and small-angle X-ray scattering experiments

The details of wide- and small-angle measurements under flow are given in Chapter 6, Section 6.2.2. The degree of crystallinity is calculated from WAXD profile using:

$$X_{cs} = \frac{C - A}{A}$$

where C is the area of crystalline peaks and A is the area of the amorphous halo [11,14]. The long spacing D_c is calculated as described in Chapter 6. The average lamellar thickness (L_c) is calculated from

$$L_c = D_c \cdot X_{cs}$$

The average lamellar thickness (L_T) is also measured from the TEM micrographs of the samples of the flow-induced crystallization experiments.

DSC experiments

The crystallization and melting characteristics were investigated with a TA Instruments Q1000 Differential Scanning Calorimeter, calibrated with Indium for temperature and heat of fusion. Sample weights between 3-4 mg were used in standard aluminum pans. The analysis was conducted under standard heating and cooling rates of 10 K/min unless otherwise indicated. The crystallization temperature (T_c) measured during cooling, peak melting temperature (T_m) during heating and the degree of crystallinity (X_{cm}) calculated from the melting enthalpy are reported for all the samples in Table 7.1.

7.2.6 Mechanical properties

Samples were prepared by first extruding the nanocomposites for homogenization and then injection molding into dumbbell shaped tensile bars for tensile testing and thick impact test bars for the impact tests. Injection molding is performed on mini-injection molding machine (DSM). The dimensions of impact test bars were 60 x 12 x 4 mm and for the tensile bars were 90 x 12 x 2 mm.

○ High speed tensile impact test

Injection molded impact test bars were notched to a depth of 2 mm. Impact tensile tests were performed at a speed of 1 m/s, using a Zwick Rel Hydraulic tensile testing machine. The piston displacement and force were measured at a sample rate of 2.5 MHz using a piezoelectric force transducer. The impact toughness energy was calculated by integrating the area under measured force-displacement curves and divided by the initial cross-sectional area behind the notch. At least 5 specimens were measured and the average is reported. All the tests were performed at room temperature.

○ Tensile strength

Tensile tests were performed on a Zwick Z010 tensile testing machine at a constant speed of 10 mm/min. Dumbbell-shaped tensile specimen were used for the tests. Engineering stress-strain curves were constructed from the measured force, clamp displacement, and initial cross-sectional area of the middle of the specimen.

7.3 Results and discussion

The prepared PP/silica nanocomposites display different morphologies with increasing filler concentrations and three regimes can be distinguished. The parameters controlling the intrinsic behavior, i.e. the Young's modulus, yield stress, and strain hardening modulus, are first discussed which may help to understand and relate the observed impact properties with the underlying morphology, and to get the insight in the effect of segregation of the high molecular weight by selective adsorption on the particles coupled with possible changes in entanglement density on the solid-state properties. The intrinsic properties can be investigated by focusing the tensile behavior at low extension rates (~ 10 mm/min).

7.3.1 Intrinsic deformation behavior

The Engineering stress-strain curves show how the materials behave intrinsically under the applied deformation. The tensile deformation behavior of PP and PP/silica nanocomposites is shown in Figure 7.2a. The typical behavior for semi-crystalline polymers is observed: an initial elastic region followed by reaching the yield stress after which strain softening and strain hardening occurs. In Chapter 4, it was reported that the crystallinity increases with increasing silica concentration, due to the nucleating effect of silica. This should result in an increase of the yield stress, which is indeed observed from the intrinsic deformation behavior of PP/silica nanocomposites with silica concentrations larger than 1 wt% (Regime II). In line with this, the strain softening behavior, which is mainly attributed to the amorphous phase [12], decreases. However, at the lower silica concentrations (Regime I), the yield stress and extent of the strain softening do not follow the expected trend.

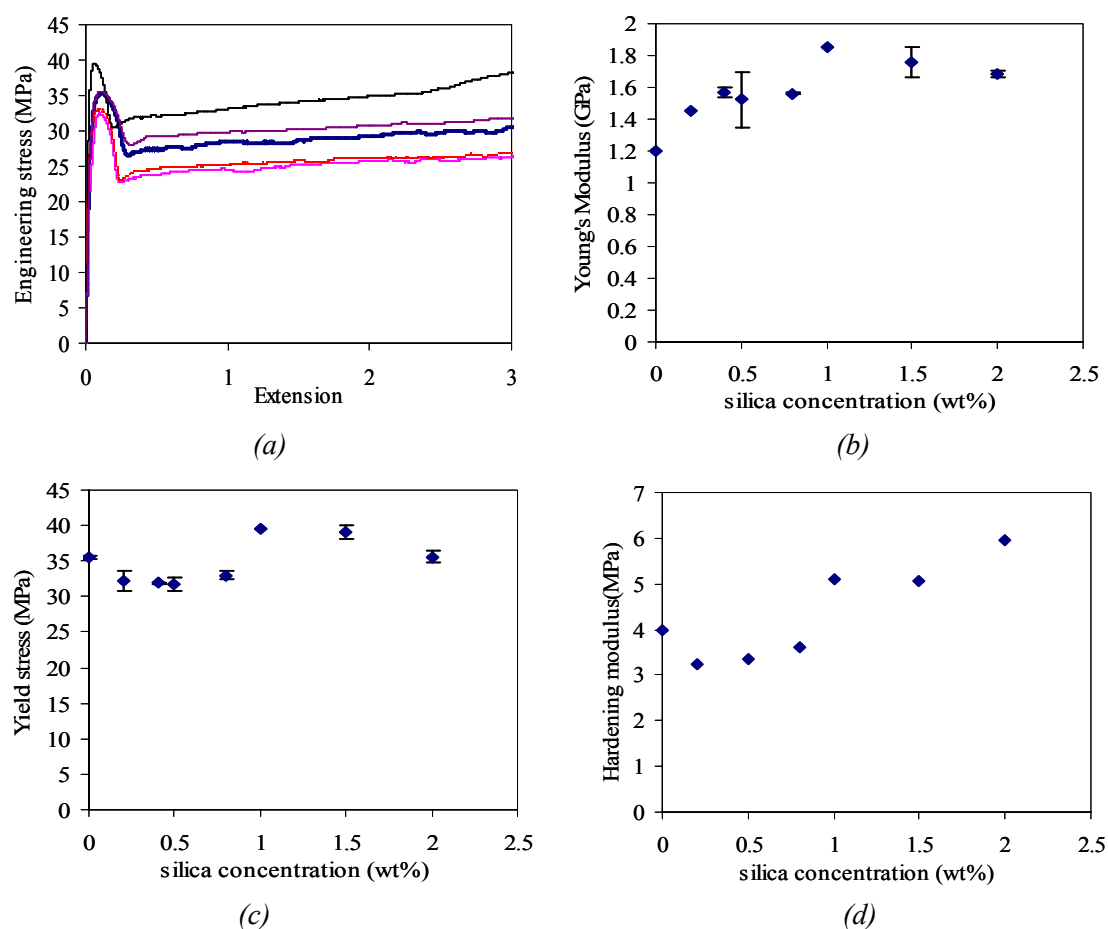


Figure 7.2: Overview of the mechanical properties as a function of silica concentration: a) stress-strain behavior, b) Young's modulus, c) yield stress, and d) strain hardening modulus. The tensile tests were performed at 10 mm/min.

7.3.2 Young's modulus

The Young's moduli calculated from the stress-strain curves are plotted as a function of silica concentration (Figure 7.2b). The modulus increases with the increasing silica concentration due to the reinforcing effect of the silica particles. However, at a silica concentration of 1 wt% (Regime II), a sudden incremental increase is observed, which can only partially be explained by an increase

in crystallinity due the nucleating activity of the particles. It is more likely that the effect of particles on the flow-induced crystallization is responsible for this sudden increase. Since the samples for tensile testing were prepared by the injection molding process, the molecules have experienced a complex flow field with varying shear stresses as a function of time and position in the tensile bar.

The resulting morphologies after injection molding for PP and PP with different silica concentrations are shown in Figure 7.3. In all samples, white surface layers with highly oriented material can be observed. Especially for pure PP and the 0.5 wt% system these extend to far inside the sample. Further, it can be observed that in the core regions of the sample an isotropic, spherulitic morphology is present with a large size distribution reflecting the strong temperature gradients experienced. The thickness of the layered structure is much smaller for the other concentrations (0.8 and 2.0 wt%), but more interestingly the spherulites are much smaller, while also the size distribution is very narrow. These observations are in agreement with the results from Chapter 6, which showed in this particular concentration range a very strong flow-induced orientation and the formation of row nuclei, which either relax or break down to a large number of point nuclei, on which upon ceasing flow lamellae grow in random directions. The strong nucleation also yields a higher crystallinity, finally resulting in a higher modulus.

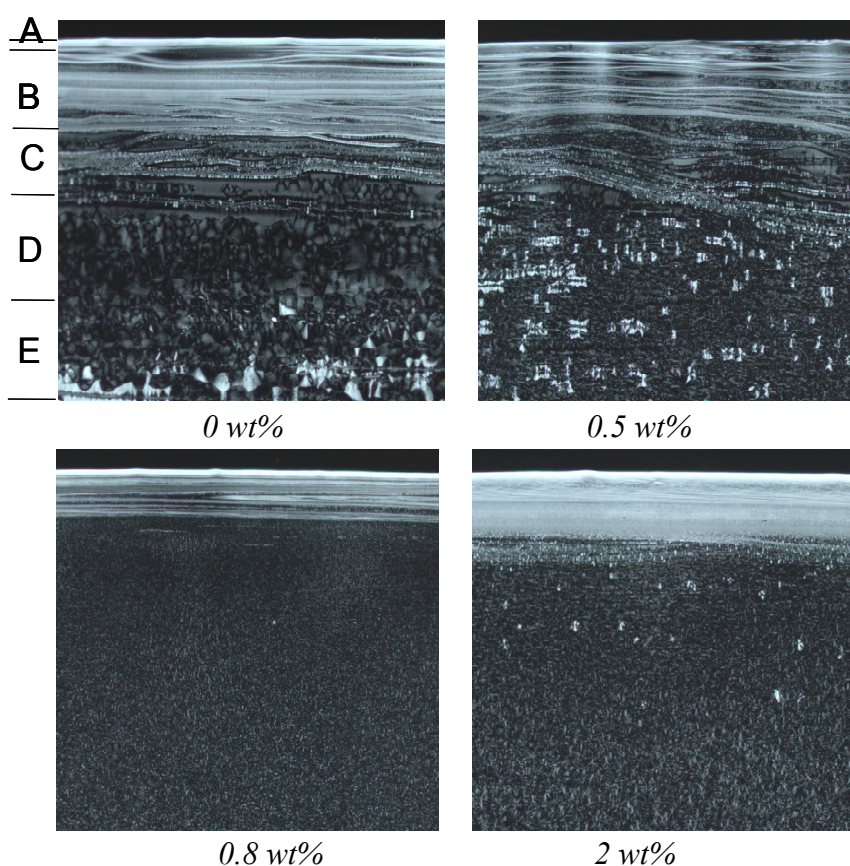


Figure 7.3: Optical micrographs of injection molded bars of PP and PP/silica nanocomposites showing the layered structures, which can be identified as: (A) skin layer, (B) shear layer, (C) fiber-like layer, (D) fine-grained layer, and (E) isotropic core.

7.3.3 Yield stress

The yield stress is calculated from the stress-strain curves for the nanocomposites and plotted as a function of silica concentration in Figure 7.2c. The nanocomposites exhibit a different trend in the

low and high concentration regimes. Although the crystallinity increases, the yield stress decreased up to 0.8 wt% silica (Regime I) and increased again in Regime II. Previous studies showed that the lamellar thickness is the controlling parameter for yielding in PP [13]. Schrauwen et al. [14] later verified that the yield stress mostly depends on the lamellar thickness, but also to some extent on the crystallinity. The models used in these studies assume that yielding involves a thermal activation of screw dislocations with the Burger's vector parallel to the chain axis.

In Regime I the yield stress slightly drops compared to pure PP. Table 7.1 lists the lamellar thickness along with the melting and crystallization temperature and crystallinity of the nanocomposites with increasing silica concentration. The lamellar thickness is calculated by two different and independent methods, namely SAXS and TEM. The lamellar thickness decreases as the concentration of silica increases (up to 1 wt%).

Table 7.1: Crystallization and melting data of PP/silica nanocomposites

Material	T _c [°C]	T _m [°C]	X _{cm} [%]	X _{cs} [%]	D _c [nm]	L _s [nm]	L _T [nm]
PP	111.4	159.7	48.1	30.25	60	18.2	16
0.2 wt%	114.2	160.8	49.4	23.38	55	12.9	12
0.5 wt%	117.6	155.5	51.5	19.2	45	8.6	10
2 wt%	114.6	165.4	59.4	55.61	54	30.0	2 wt%

X_{cm} and X_{cs} are degree of crystallinity calculated from DSC and X-ray scattering experiments respectively.

D_c and L_s are Long spacing and lamellar thickness calculated from X-ray scattering experiments.

L_T is lamellar thickness measured from TEM micrographs.

Results of the isothermal crystallization under quiescent conditions as studied by DSC are pictured in Figure 7.4 and show that the crystallization rate increases, while the half-life times of crystallization at a particular temperature decrease in the lower concentration regime with increasing silica concentration. However, in Regime II the half-life times are longer than in Regime I. This can be explained by segregation of the long and short chains due to selective adsorption, characteristic for Regime I. In addition to the nucleating effect of silica, the shorter, less entangled chains crystallize faster than the adsorbed high molecular weight chains. However, less orientation results in less nuclei formation thus FIC is slower in Regime I as shown in Figure 7.5. These two phenomena effectively increase the crystallization time in Regime I, predominantly forming an unstable α -phase with thinner lamellae leading to a lower yield stress in Regime I.

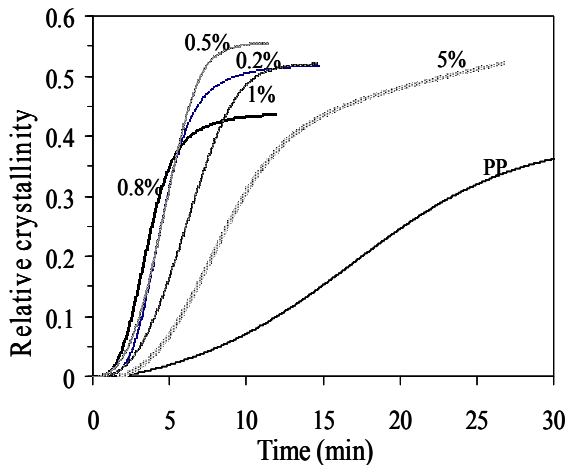


Figure 7.4: Isothermal crystallization of PP/silica nanocomposites at 130°C.

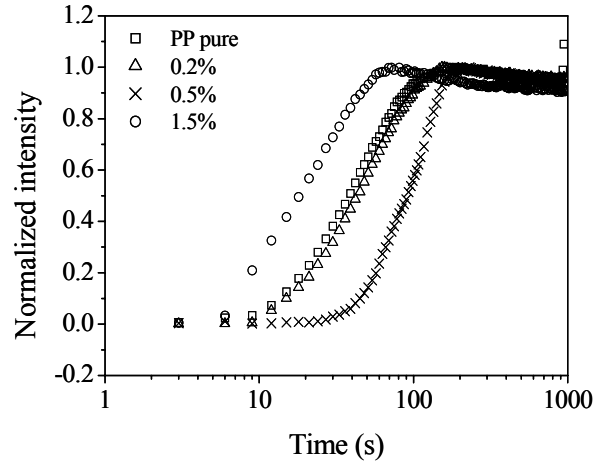


Figure 7.5: Isothermal crystallization at 130°C after application of a shear pulse of 60 s⁻¹ for 5 s.

In Regime II, the yield stress is higher because of the presence of thicker lamellae. Further, the inter-particle distance after adsorption become less and the entanglement density increases. The increase in viscosity increases the shear stresses during flow and orientation induced nuclei accelerates crystallization (see also Figure 7.5). The nucleating effect due to silica and flow results in faster crystallization and thicker lamellae; also the confinement due to the lower inter-particle distance forces the chains to form thicker lamellae instead of growing in length.

7.3.4 Strain hardening modulus

The strain hardening modulus depends on the mobility of the amorphous phase and is clearly not related to the degree of crystallinity [14]. It is generally accepted that after yielding the crystallites no longer form physical cross-links, and chain movement through the crystals is possible. As a consequence, a similar relationship between the entanglement density and the strain hardening modulus for semi-crystalline polymers can be applied as used for amorphous polymers. The crystalline domains only have a damping effect, but no real entropic contribution at large strains.

The strain hardening modulus can be determined from an expression derived by Haward [15], which relates the hardening parameter, the stress level during neck propagation and the elongation factor in the neck. For thermoplastics in general, $\sigma_0 > 3G$ and under these conditions strain localization is favored and necking may be expected. Once necking is established, with a constant load or drawing stress, the ratio of the strain in any section of the test piece passing through the neck first accelerated and then slows down as strain hardening takes over. Haward assumed that during this process the local strain rates in the adjacent sections of the necking area are equal. Based on this, Timmermans [16] proposed the following equation:

$$\frac{(\sqrt{3} + \mu)\sigma_0}{\sqrt{3}G} = \frac{\lambda_D - \frac{1}{\lambda_D^2}}{1 - \frac{1}{\lambda_D}}$$

where λ_D is the natural draw ratio and μ is the pressure coefficient (0.23 for PP). Since in an established neck the strain rate develops slowly, the natural draw ratio is assumed to be approximately equal to the elongation factor in the neck during neck propagation. The natural draw ratio is determined by taking the ratio of change in the cross-sectional area in the middle of the tensile bar before and after the test. The value of hardening modulus for pure PP is 3.96 MPa, which is close to the values obtained by Schrauwen [17] (3.9 MPa) and Haward [15] (4.1 MPa) for PP with similar molecular weights.

The strain hardening modulus is shown in Figure 7.2d as a function of silica concentration. Also two regimes are observed similar to the yield stress. As described earlier in the low silica content regime, hydrodynamic effects are not active, demixing due to adsorption of longer chains yields a less entangled matrix, and the low dynamic viscosity reduces effective strains on the molecules. The short chain matrix is less entangled and the flow-induced crystallization in this case is slower (Figure 7.5). Chains from the amorphous phase can disentangle and crystallize even when quenched as in injection molding. This is also supported by DMTA measurements.

At higher silica concentrations, the entanglement density increases, since the inter-particle distance becomes less than the radius of gyration and particles start feeling each other and effectively demixing does not occur, thus the adsorbed chains can intermingle with molecules in bulk. The

non-isothermal crystallization behavior of the in-situ prepared silica/PP nanocomposites and its influence on the entanglement density and strain hardening was described in Chapter 4. The nucleating effect of silica and higher stresses experienced by PP due to higher viscosity results in formation of multiple nuclei. This eventually increases the crystallization rate, resulting in a more entangled melt. This eventually leads to an increase in strain hardening modulus.

Figure 7.6 shows the loss modulus and storage modulus curves as a function of temperature for increasing silica content. The peak in loss modulus is considered as the glass transition temperature (T_g). The intensity of the transition (S) is an important parameter reflecting the mobility and content of the amorphous phase [18]. It is defined as:

$$S = \frac{E_a' - E_d'}{E_d'}$$

where E_a' and E_d' denote the storage modulus before and after the transition, respectively. A higher S means a higher mobility (less entanglement density) or a higher content of amorphous phase. The trend in the intensity of transition and T_g can be observed from the values of listed in Table 7.2. Figure 7.6 and the values suggest the marginal decrease in T_g and a steep decrease in intensity of transition in Regime I. This indicates that the amorphous phase is less entangled. These two reasons, i.e. a less entangled amorphous phase and a low molecular weight matrix compromising a major part of the amorphous phase, result in a lower strain hardening modulus at low silica loadings.

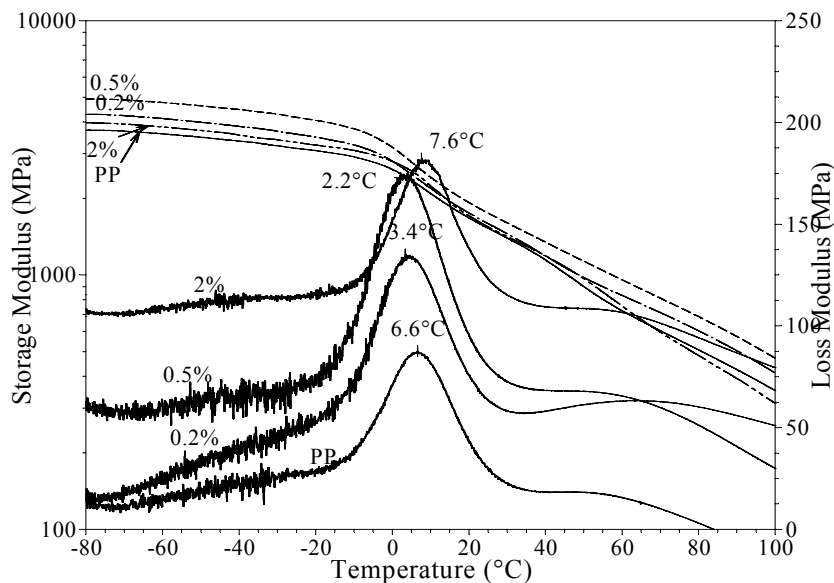


Figure 7.6: Storage and loss modulus of PP/silica nanocomposites measured as a function of temperature.

Table 7.2: The values of glass transition temperature (T_g) and intensity of transition (S) obtained from DMTA measurements

silica %	T_g (°C)	S %
0	6.6	1.25
0.2	3.4	1.71
0.5	2.2	1.82
1.0	5.3	1.10
2.0	7.6	1.51

7.3.5 Stability of neck drawing (σ_0/G)

The existence of stable necking is important, since it reflects the ability of the neck to bear the transferable load. Schrauwen [17] described the importance of the stability of neck drawing for improvement in mechanical properties, especially for toughness. The plot of σ_0/G versus λ_d shows three regions: In region I ($0 < \sigma_0/G < 3$), the material deforms homogeneously and then fails at the ultimate limit defined by the following equation at arbitrary values of κ_t assumed in the model.

$$\frac{\sigma_0}{G} = \frac{\lambda_d^2 - \frac{1}{\lambda_d}}{\kappa_t - 1}$$

In region II, the material forms a stable neck upon drawing, whereas in region III the neck starts at the yield point, but a stable neck cannot be formed and the material fails. For increasing value of σ_0/G , necking rupture followed by strain delocalization gives a brittle failure.

σ_0/G as a function of silica concentration is shown in Figure 7.7. As expected, σ_0/G also shows different behavior in different regimes. For pure PP, σ_0/G is approx 10. In Regime I, the strain hardening modulus is lower than that of pure PP, thus σ_0/G is higher (~ 12), and follows exactly the inverted trend as that of the strain hardening modulus. The higher σ_0/G than pure PP indicates unstable necking, which may result in brittle failure. The impact strength in Figure 7.8, however, shows a slight improvement compared to PP, which may be due to the influence of the filler particles. This will be discussed in a later section.

In Regime II, the strain hardening modulus increases with increasing silica concentration. This eventually lowers σ_0/G till a third regime, where aggregation of silica particles occurs. The lower the value of σ_0/G , the more ductile the material is. For example, σ_0/G is 1.5 times lower for 2 wt% silica compared to pure PP. In this regime, the nanocomposites are expected to be tougher. The impact strength indeed increases by a factor of more than 1.5 compared to PP. A further increase in silica concentration, viz. 5 wt%, results in silica aggregates which eventually decrease σ_0/G and imparts brittleness to the nanocomposites.

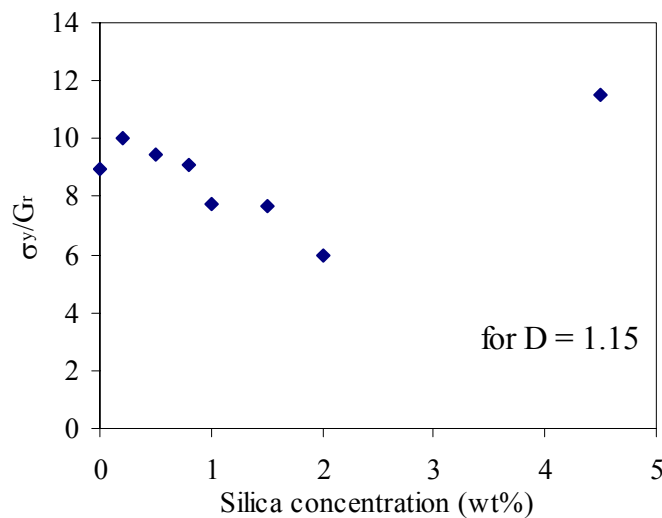


Figure 7.7: Stability of the neck calculated using method described by Schrauwen et al. [17].

7.3.6 Impact fracture

The impact strength depends on several parameters such as the intrinsic deformation behavior, the entanglement density affecting cavitation, plasticity, flow-induced orientation and the stress distribution around the nano-particles. The effect of orientation and stress distribution are mainly due to the processing and particles respectively, whereas the other factors are inherent to the specific material and depend in its molecular structure and morphology.

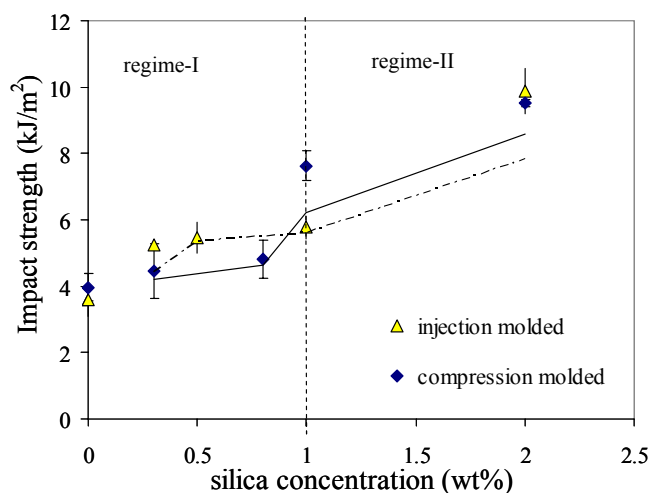


Figure 7.8: The high-speed tensile impact strength as a function of silica concentration calculated from load-displacement curves.

Figure 7.8 shows that the impact strength increases with the incorporation of silica nano-particles. Besides the change in intrinsic deformation behavior due to adsorption and demixing (in Regime I) of the high molecular weight chains, crack deflection, local plastic deformation after debonding and nano-particle mobility [19] may be reasons for improvement. The debonding process, which is the separation of the filler particles from the polymer matrix, is thus affected in these particle-filled composites [20]. It was shown in the past that debonding is an important mechanism in improving toughness of particle-filled semi-crystalline polymers, because it allows plastic stretching of ligaments between the debonded particles. The silica particles have no chemical interaction with PP matrix, so debonding from the matrix is easier, resulting in delocalization of the impact stresses.

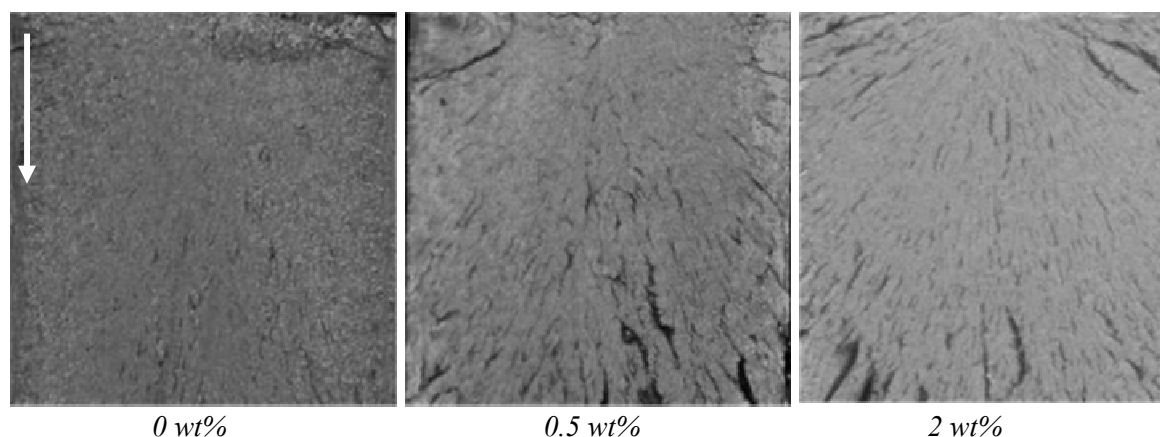


Figure 7.9: Optical micrographs of fracture surfaces of the impact tests of PP/silica nanocomposites. The arrow shows the direction of impact.

The optical micrographs of the fracture surfaces in Figure 7.9 show the crack propagation and fracture behavior during impact in the nanocomposites. PP samples show an abrupt fracture, thus exhibiting a brittle behavior due to localization of stresses during fracture. For the PP/silica nanocomposites, the crack propagates in the forward direction along the impact force. Crack propagation is facilitated due to plastic stretching of the matrix in the presence of silica nanoparticles. The stress delocalization and crack propagation improves the impact strength of PP in presence of silica nano-particles.

7.4 Conclusions

Three different regimes in PP/silica nanocomposites depending on the concentration of silica nanoparticles, adsorption and demixing and inter-particle distance were identified. Regime I, below 1 wt% silica, is characterized by adsorption and demixing of longer chains which reduces the viscosity. Regime II is above 1 wt%, where the inter-particle distance after adsorption is smaller, but no particle aggregates are formed and is marked with increase in the viscosity. Regime III is at high concentration of silica, typically 5 wt%, where silica particles form agglomerates. Regime III behaves like conventional particle-filled composites.

The mechanical behavior of the PP/silica nanocomposites also exhibits these regimes. In the lower concentration Regime I, the Young's modulus increases slightly due a higher crystallinity, while the effect of orientation is less. The yield stress, which mainly depends on the lamellar thickness, also goes through a minimum. The demixing due to adsorption, the slower crystallization kinetics under flow due to the low viscosity are the main factors which lower the yield stress. The strain hardening modulus, which mainly depends on the entanglement density of the amorphous phase, also follows a similar trend. However, the reason for the decrease in strain hardening modulus lies in the selective adsorption and demixing of the longer PP chains, which results in a less entangled matrix. The decrease in T_g and the intensity of the transitions observed from DMTA also suggests the same.

At higher silica loading in Regime II, the Young's modulus exhibits a higher increase, because of the higher orientation at the wall due to flow. The fact that the crystallinity and the lamellar thickness increase results in an increase in the yield stress. The inter-particle distance decreases and particles start to feel each other. Moreover, the adsorbed chains can interact and form bridges, thus the amorphous phase becomes more entangled, while also the T_g increases, ultimately showing up as an increase in the strain hardening modulus.

The stability of the neck analysis via (σ_o/G) also shows a different behavior in these regimes. The higher σ_o/G ratio observed in Regime I leads to a decrease in toughness. σ_o/G decreases with increasing silica content in Regime II, which indicates toughening of nanocomposites; however, at even higher concentrations (Regime III), σ_o/G starts to decrease imparting brittleness. As expected from the σ_o/G behavior, the impact strength increases slightly in Regime I, but in Regime II increases more than 1.5 times to that of PP. The improvement is also aided by crack-deflection and debonding of the particles from the matrix, which allow plastic stretching and hence stress delocalization. In conclusion, with these new types of PP/silica nanocomposites it is possible to control the structure development (morphology) during the flow and the molecular structure

(arrangement). Controlling these two parameters, the mechanical properties of the final products can be improved with enhanced processing characteristics.

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Appendix A

Impact strength of (VTES-grafted PP)/silica nanocomposites

Abstract

The effect of chemical interaction between nano-filler with the polymer matrix on the mechanical properties is studied for PP/silica nanocomposites. Silica particles improved the impact toughness of PP by a factor of 2 when there is no chemical interaction between particles and matrix. However, for (VTES-grafted PP)/silica nanocomposites, the impact toughness decreased. This indicates that chemical bonding between filler particles and PP matrix results in brittle failure. The brittle failure in (VTES-grafted PP)/silica compared to PP/silica, was also observed from the fractured surfaces. These results support the hypothesis that debonding is necessary for improving the impact toughness.

A.1 Introduction

In Chapter 7, it was shown that the reinforcing and toughening effect of the nanoparticles on the polymer matrix in PP/silica nanocomposites is achieved at rather low filler concentrations in comparison to conventional particulate-filled composites. Some of the recent studies also reported similar observations but they used either coupling agents or co-polymers for dispersion of nanoparticles [1]. The coupling or compatibilizing agents or surface modifiers change the interface between polymer matrix and fillers which makes it difficult to understand the mechanism of improvement in the toughening.

The interface between the inorganic particles and the polymer plays an important role in determining the properties of composites. The interfacial strength is expected to affect the delocalization of stresses during the macroscopic deformation of the composites. It is shown recently that debonding is an important mechanism in improving toughness of micro-particle filled polymers because it allows plastic stretching of ligaments between the debonded particles [2,3]. Thio et al. [3] found that an increase in chemical interaction between the fillers (micrometer-sized glass particles) and the polymer matrix is detrimental for the toughening, since debonding in this case is not possible.

An attempt is made to investigate the effect of chemical interaction between filler and polymer matrix on the toughness. The PP matrix is modified by grafting vinyl triethoxy silane (VTES) via solid-state modification (SSM) and the concentration is kept low enough to ensure that only monomeric units are grafted on PP (see Chapter 2). This grafted VTES then takes part in the sol-gel reaction to make in-situ silica, hence producing a strong chemical interaction between in-situ prepared silica and PP matrix.

A.2 Experimental

The synthesis and chemical characterization of nanocomposite are described in Chapter 2, Section 2.3.1. High speed impact tests were performed under similar conditions as described in Chapter 7. All samples were injection molded as described in Chapter 7.

A.3 Results and discussion

A.3.1 Adhesion between polypropylene and in-situ prepared silica particles

As described in Chapter 2, the adhesion between silica and PP is established by grafting VTES on PP via SSM and then making in-situ silica nano-particles ((VTES-grafted PP)/silica). The schematic representation of the morphology of PP/silica and (VTES-grafted PP)/silica nanocomposites is shown in Figure A.1. In PP/silica nanocomposites, silica particles have no interaction with matrix, whereas in case of (VTES-grafted PP)/silica nanocomposites, particles are chemically bonded to PP backbone at one or several points.

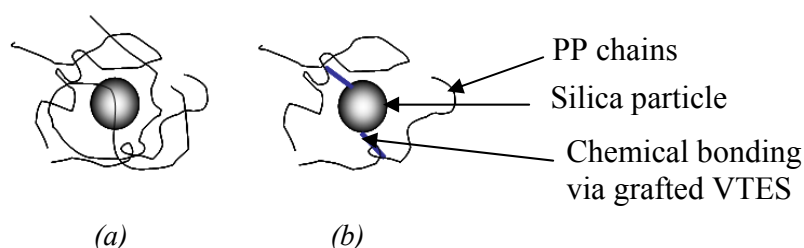


Figure A.1: Schematic diagrams showing the morphology of (a) PP/silica; (b) (VTES-grafted PP)/silica nanocomposites

In Chapter 2, it has been shown that PP/silica and (VTES-grafted PP)/silica nanocomposites have uniform dispersion and silica particle size. For detail see Section 2.3.4. The nanocomposites prepared via this novel method have stable morphology.

A.3.2 High speed tensile impact strength

Values of impact strength determined using high speed tensile impact machine are presented in the Figure A.2. In Chapter 7, it was shown that with addition of silica particles two-fold increment of impact toughness can be achieved. However, (VTES-grafted PP)/silica showed opposite effect. The impact strength of silica filled VTES grafted PP although is same as pure PP but decreased compared to PP/silica. This confirms the hypothesis that weaker interaction between the rigid particles and the polymer matrix lead to higher toughness. Similar observations were made by Thio et al. [3] in case of glass bead filled PP.

Furthermore, the optical micrographs in Figure A.3 show the propagation of fracture during impact test. PP and (VTES-grafted PP)/silica shows brittle fracture, in fact (VTES-grafted PP)/silica shows more brittleness than PP. PP/silica exhibit ductile behavior in which crack propagates in the direction of fracture thus improving the toughness. These observations confirm the hypothesis of debonding and explain the role of compatibilizer or interfacial agent as dispersing agent more than for improvement in mechanical properties.

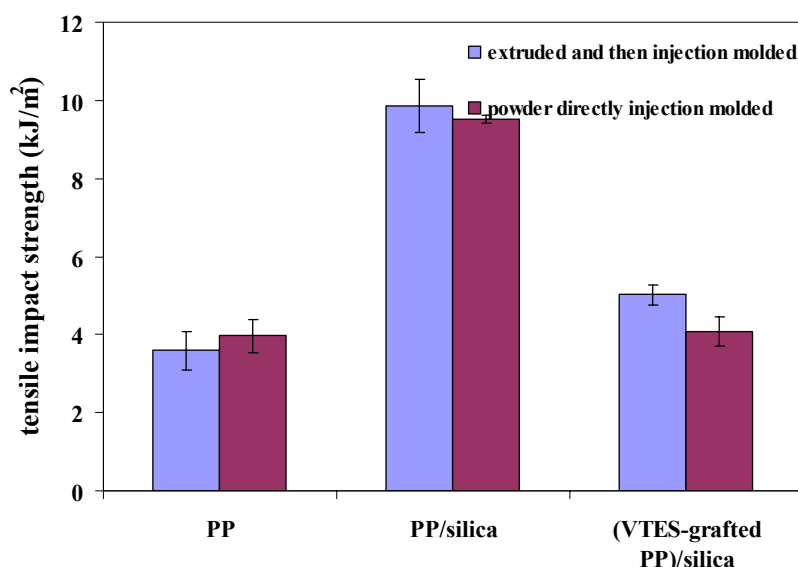


Figure A.2: High speed tensile impact strength normalized by nominal fracture area. In all the composites the concentration of silica is 2 wt% and that of VTES is 2 wt%..

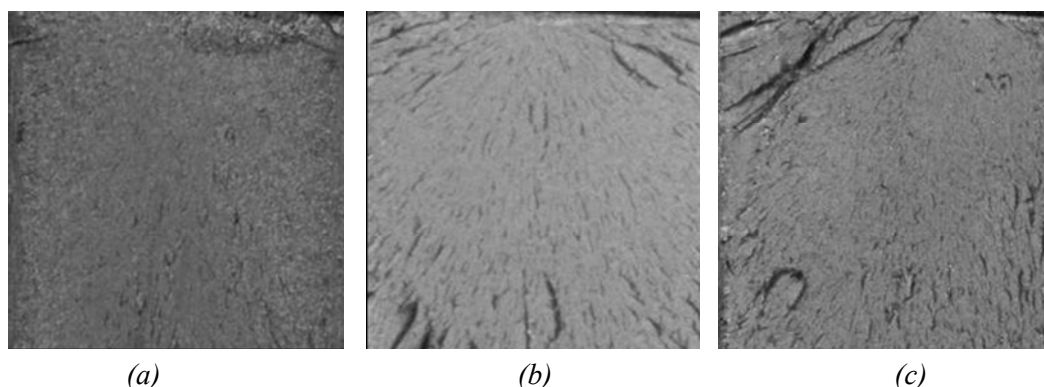


Figure A.3: The optical micrographs of impact fracture samples of (a) PP, (b) PP/silica, (c) (VTES-grafted PP)/silica nanocomposites.

A.4 Conclusions

High speed tensile impact test results illustrate the effect of grafting. Silica particles improve the impact toughness of PP by a factor of 2 when there is no chemical interaction between particles and matrix. However, in case of (VTES-grafted PP)/silica nanocomposites, the impact toughness decreased to a level less than pure PP. This indicates that the grafting which imparts chemical bonding between filler particles and PP matrix resulted in brittle failure. The brittle failure in (VTES-grafted PP)/silica compared to PP/silica, was also observed from the fractured surfaces.

A.5 References

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Appendix B

Rheology of PP/silica fiber and UHMW-PE/silica nanocomposites

Abstract

PP/silica hollow fibers and UHMW-PE/silica nanocomposites were prepared using the same method used for making PP/silica nanocomposites. These nanocomposites display two characteristics: a) in PP/silica hollow fibers, the dimensions and characteristics of fillers are changed, b) in UHMW-PE/silica nanocomposites, the matrix is changed keeping the class of materials same, i.e. polyolefin. Both these nanocomposites exhibit a similar rheological behavior as observed for PP/silica nanocomposites. In the lower filler concentration range both show a drop in the viscosity. The critical concentration, however, depends on the inter-particle distance which is governed by the size and shape of the fillers. The morphology observed by SEM shows that the particles are attached to the matrix which confirms the adsorption on silica. These observations enable to generalize the concept of selective physisorption of long polymer chains on the filler for the reduction in the viscosity of polymeric materials.

B.1 Introduction

PP/silica nanocomposites investigated earlier were prepared in-situ via a combination of solid-state modification and sol-gel reactions leading to well dispersed silica particles in the PP matrix. The experimental approach and synthesis is described in Chapter 2. Two different nanocomposites were prepared using the same approach: a) ultra high molecular weight polyethylene (UHMW-PE)/silica particles and b) PP/hollow silica fibers. The UHMW-PE used has a broad molecular weight distribution and the nascent powder is also porous allowing the preparation of silica particles in-situ by the sol-gel method. Another system prepared contains silica fibers rather than particle in PP. The hollow silica fibers were synthesized using dl-tartaric acid as templating agent in the sol-gel method. By investigating these two systems, the polymer matrix in the first case and the shape, aspect ratio and surface characteristics of the filler in the second case are changed. The dynamic viscoelastic behavior was studied using mechanical spectrometry and the morphology was observed using SEM and TEM.

As anticipated, a similar drop in the dynamic viscosity with fillers in the lower concentration range is observed. Figure B.1 shows the trend and extent of the drop in dynamic viscosity of PP/silica hollow fibers, in comparison to the PP/silica particle system reported in Chapter 3. The following sections discuss the observed effects in more detail

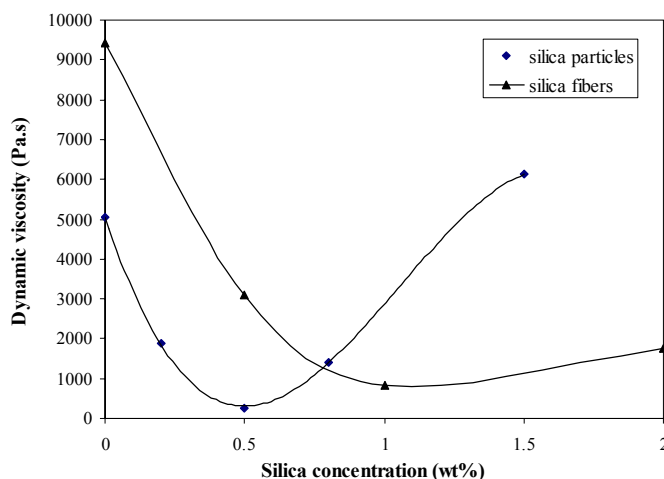


Figure B.1: The dynamic viscosity as a function of filler content at a constant strain of 0.5% and frequency of 1 rad/s. The drop in the viscosity emphasizes the generalization of the selective physisorption concept.

B.2 Experimental section

B.2.1 Materials

The polypropylene (PP) used is the 15M10 homopolymer grade supplied by DSM, Geleen, The Netherlands. It has a melt flow index of 15 g/10 min and high isotacticity (96.5%), a density of 0.905 g/cm³, M_w of 350 kg/mol and $M_w/M_n = 5.6$. The ultra-high molecular weight polyethylene (UHMW-PE) used is STAMYLAN UH034 grade supplied by DSM, Geleen, The Netherlands. It has the following properties: $M_w = 3.3 \times 10^6$ g/mol, $M_w/M_n > 8$, an apparent density of > 0.4 kg/m³, average particle size of 150 μ m, pore size ranging from 500 nm – 10 μ m. The precursor of silica, tetraethoxy orthosilicate (TEOS), dl-tartaric acid, and ammonium hydroxide (28% NH₃) were supplied by Sigma-Aldrich Chemie GmbH, Steinheim, Germany. The materials were used without further purification.

B.2.2 Synthesis of nanocomposites

Silica nanofibers were prepared [1] according to the procedure described by Nakamura [2] and Miyaji [3]. A mixture of 5 ml ethanol and 0.06 g H₂O with 0.02 g of dl-tartaric acid was prepared and 0.73 g of TEOS was added without stirring. The mixture was allowed to stand for 30 minutes. Subsequently, 2 ml of 28 % NH₄OH was slowly added and the solution was again allowed to stand for 30 minutes. The white precipitate was washed with water on a 400-mesh brass test sieve (Sigma-Aldrich Chemie GmbH, Steinheim, Germany) to remove colloidal aggregates. The obtained nanofibers were dried for 12 hrs in a vacuum oven at 100°C. Silica fibers prepared using this procedure are shown in Figure B.2a. The average diameter of the hollow silica fibers is approx. 500 nm and 1-10 microns in length.

Preparation of PP/silica fibre nanocomposites

A 10 wt% solution of polypropylene was prepared by dissolving polypropylene pellets in xylene at 130°C. Silica nano fibers were added and the mixture was stirred for 30 minutes. The solvent was removed at 90°C under vacuum. The morphology of resulting nanocomposites is shown in Figures A.2 b and c.

Preparation of UHMW-PE/silica nanocomposites

UHMW-PE/silica nanocomposites were prepared by the sol-gel method according to the procedure described in Chapter 2, Section 2.3.1.1. TEOS was added to the UHMW-PE powder in the reactor and stirred for approx. 30 min at 60°C. After dispersion of TEOS, a mixture of water and ammonium hydroxide was added slowly at a controlled rate to the reactor with continuous stirring under a nitrogen atmosphere. The molar ratio of TEOS/H₂O was 1:5 and the concentration of NH₄OH was 1 wt % on TEOS. The sol-gel reaction was carried out in the reactor at 60°C for 3 hrs with continuous stirring. The thus formed sol in UHMW-PE is then gelled at 80°C for 5 hrs under a nitrogen atmosphere, dried under vacuum at 100°C for 24 hrs, and subsequently stored at -4°C in the dark. The samples were physically mixed with antioxidants (concentration of 0.5 wt%) using a powder blender before any further treatment and dried under vacuum at 100°C before chemical and structural characterization.

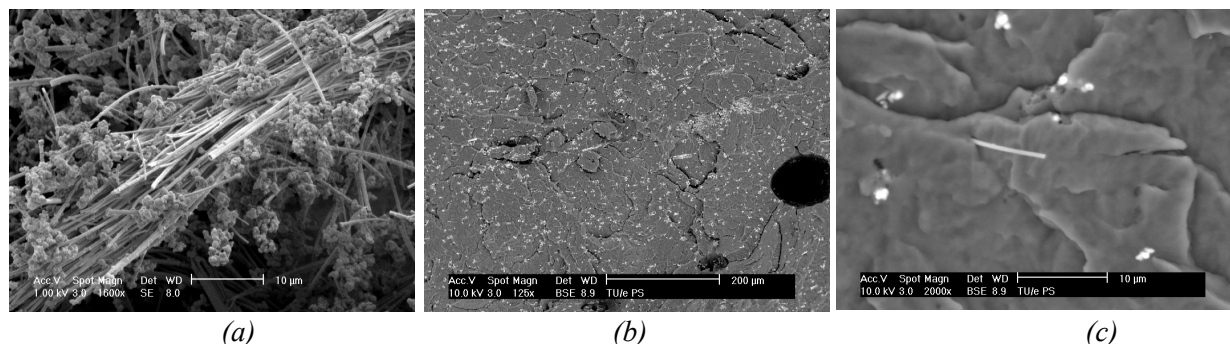


Figure B.2: SEM micrographs of (a) silica fibres prepared using the templating method; (b) and (c) PP-silica fibre composites, using different magnification.

B.2.3 Rheometry

The rheology of the polypropylene samples with and without silica nanofibers was investigated with the help of rheometry. The polypropylene samples were prepared by compression molding at 180°C in a Collin type 300E hydraulic press. Experiments were performed on an Ares Melt Rheometer (Rheometrics Scientific) having a plate-plate geometry with a diameter of 8 mm and a gap of approximately 0.4 mm between the plates. The strain was set in the range of linear viscoelasticity at a temperature of 180°C in the frequency window between 0.1-100 rad/sec. For UHMW-PE/silica nanocomposites experimental temperatures was 160°C, unless otherwise specified.

B.2.4 Scanning Electron microscopy (SEM)

A Philips Field Emission Gun (FEG) Environmental Scanning Electron Microscope (ESEM) was used to determine the exact shape and size of the formed silica particles. A thin layer of chromium was evaporated onto the surface of the samples. This thin layer enhances the conductivity needed to achieve a good contrast. A qualitative measure of the distribution of size of particles was also obtained from the SEM micrographs.

B.3 Results and discussion

B.3.1 Rheology of PP/silica fibers

Dynamic viscosity as a function of frequency of the PP/silica fibers measured at 170°C is shown in Figure B.3. A similar trend is observed as for the PP/silica particles nanocomposites. The concept of selective physisorption is discussed in Chapter 3. The filler concentration range in which a lower viscosity is observed, in case of silica fibers, is wider compared to silica particles. This may be due to the critical concentration at which the inter-particle distance becomes smaller than the radius of gyration is higher for fibers due to the larger dimensions and higher aspect ratio. Additionally, silica fibers are hollow which also may promote the adsorption of longer PP chains. Alike the PP/silica particles composites, the PP/silica fibers system also shows more Newtonian behavior with decreasing viscosity. The storage moduli of these composites from the same measurements are plotted in Figure B.4. At silica concentrations ($\sim 3.5\%$), the inter-particle distance is smaller so particles with adsorbed layers start to interact and form superstructures which change the slope in the storage modulus. Additional relaxation time in case of 1 % fibers at lower frequency is due to the adsorbed layers of longer PP chains on the silica fibers, as also reported in Chapter 3 and by Vignaux-Nassiet et al. [4].

B.3.2 Rheology of UHMW-PE/silica

Similar effects are observed for the UHMW-PE/silica composites. The dynamic viscosities as a function of silica concentrations measured at 150°C (on RMS 800) and 160°C (on ARES) are shown

in Figure B.5. The two set of measurements were performed on two different instruments (mentioned in bracket) to check the reproducibility of the measurements. Different temperatures were selected in this case to study the effect of temperature on the physisorption. As mentioned earlier, the MWD of UHMW-PE is quite broad (> 8), thus selective physisorption is more probable. Figure B.6 shows the storage modulus of the nanocomposites produced from the same measurements. This also indicates a similar observation as found for PP/silica and PP/silica fibers composites.

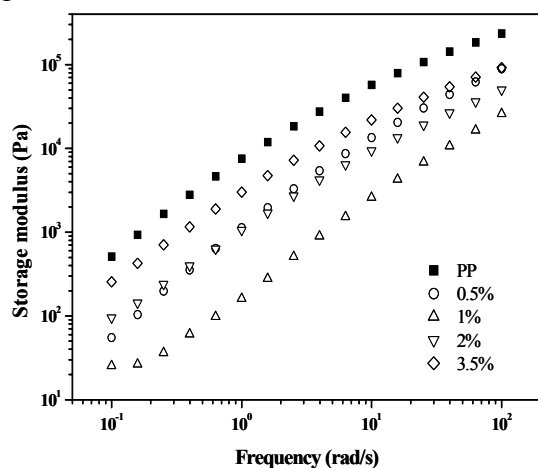


Figure B.3: Dynamic viscosity against frequency at 0.5% strain and 170°C for various filler concentrations.

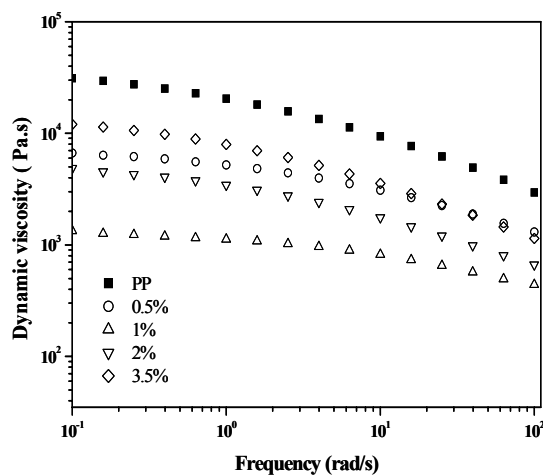


Figure B.4: Storage modulus as a function of frequency measured under similar conditions. Notice the small plateau in storage modulus for 1 wt% fiber loading at lower frequencies.

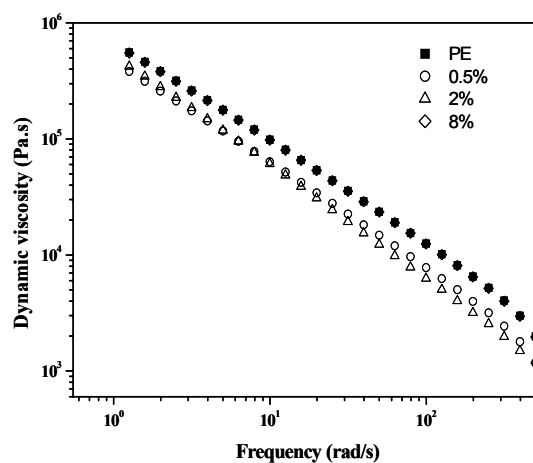
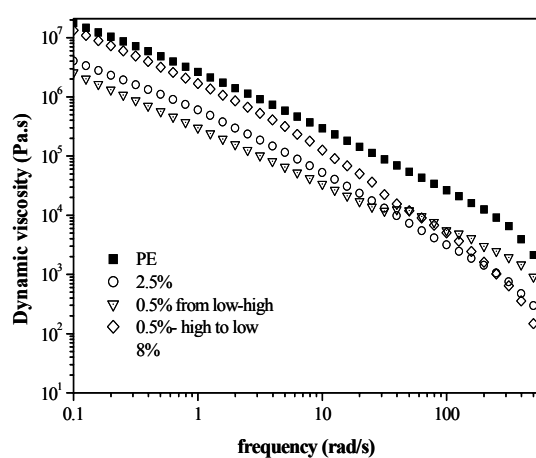


Figure B.5: Dynamic viscosity as a function of frequency of UHMW-PE/silica composites measured at (left) 150°C on RMS 800 and, (right) 160°C on ARES.

B.3.3 Morphology of UHMW-PE/silica- evidence of adsorption

In general, if there is no interaction between filler particles and matrix material, a clear and sharp boundary at the interface is observed in scanning electron micrographs. Moreover, the particles would appear to be loosely located in the matrix. However, if there is a strong interaction between the filler particles and the matrix, the interface becomes diffuse and particles appear as adhered to the matrix. Figure B.7 shows images of UHMW-PE/silica nanocomposites with 8 wt% silica particles. The images are from cold-fractured surfaces of samples. Since there is no chemical interaction between silica particles and UHMW-PE matrix, one would expect the silica particles to be detached from the matrix. However, the silica particles are attached to the matrix, which is a

clear indication of a physical interaction between the particles and the matrix. It is very plausible that UHMW-PE chains are strongly adsorbed onto the silica surface. Note that UHMW-PE is similar in chemical behavior and properties to PP, so similar effects are expected in PP/silica nanocomposites. In other words, this provides additional and direct evidence for the adsorption concept, which is discussed in detail in Chapter 3.

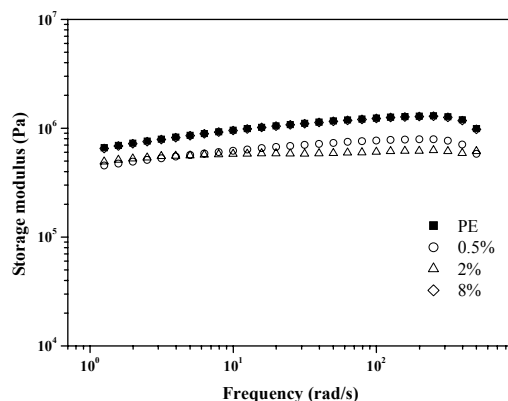


Figure B.6: Storage modulus of UHMW-PE/silica composites measured at 160°C on ARES

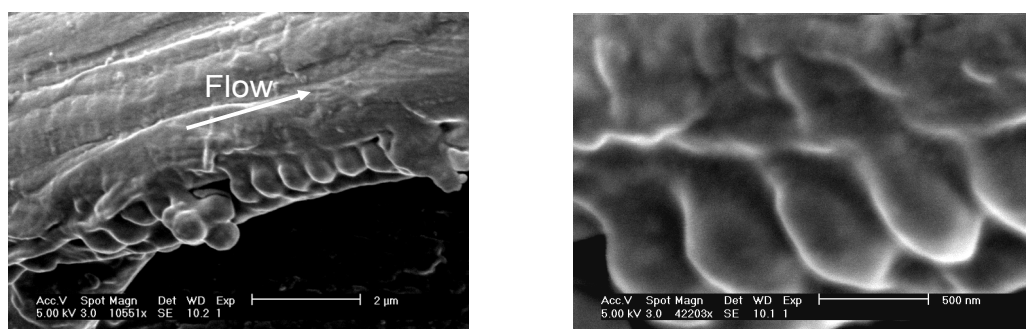


Figure B.7: SEM of cold-fractured surfaces of UHMW-PE/silica nanocomposites. Silica particles are attached to matrix, which indicates the adsorption of UHMW-PE chains on the silica surface.

B.4 Conclusions

Both PP/silica fiber and UHMW-PE/silica particle nanocomposites exhibit a decreases in the dynamic viscosity as observed for PP/silica nanocomposites. Morphology, rheological measurements, and the similar chemical characteristics of the matrix supports that the hypothesis of selective physisorption of long polymer chains on the silica surface, which indicate that the concept can be generalized for polymers of same class and fillers with similar physical and surface characteristics as in-situ prepared silica.

B.5 References

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Technology Assessment

The area of synthetic polymers is seemingly mature in view of the fact that no novel polymers are expected with substantial market potential (conclusion forum discussion IUPAC meeting, July 2004, Paris). This statement holds in particular for polymers used in engineering applications, whereas expectations are still running high for new functional polymers in special applications, e.g. polymers in medicine, electronics (display technologies), albeit in small quantities.

The vast majority of polymers for engineering applications used today were developed in the 1960s and 1970s. Consequently, the focus in the coming decades will be on exploiting the ultimate properties of existing polymers.

The performance of polymeric materials and products does not solely depend on the chemical structure but equally well on the processing step, viz. the orientation of the long molecules induced e.g. in fiber spinning or drawing of films. The performance can also be boosted by the use of additives. In fact, a plastic (product), notably in engineering applications, is often a polymer as base material + additive(s). There are numerous additives and for increasing mechanical properties, notably the stiffness, a distinction is made between particulate matter (usually inorganic filler particles) and fibrous matter, short fibers as well as continuous fibers in composite structures.

In the 1990s, an intriguing concept was put forward by Argon, Bartczak et al. Micrometer-sized inorganic particles dispersed in polymers such as polyethylene (PE) and polypropylene (PP) could increase both the stiffness and toughness, to some extent the holy grail of polymer physics. Recently, Schrauwen (Ph.D. thesis TU/e, 2003) convincingly showed that this synergistic effect was mainly due to flow-induced orientation and not to a specific induced morphology as proposed originally.

Since the start of the 21st century, the focus has been on nano-technology and also in the polymer area expectations are running high concerning the use of nanometer-sized additives, boosting the properties at low loadings. Well-known examples subject of intensive research are carbon nano-tubes (CNTs) and nano-clays. The main problem to date is to disperse these nano-particles homogeneously into the polymer matrix, merely via conventional mixing/blending techniques.

In the thesis a novel approach is used to disperse nano-sized silica particles into polymers. Polypropylene (PP) was chosen as a model material in view of its increasing importance as a cheap and versatile engineering plastic and well-documented properties in literature. Silica nano-particles were grown directly via a sol-gel method in (porous) PP reactor powder. Upon subsequent processing, the melt-viscosity decreased by a factor of 10-20 in the presence of only a minute amount of silica particles, less than 1 wt%, without sacrificing the properties. In fact, the detrimental effect of flow-induced orientation is virtually absent, giving the potential of more dimensionally stable molded products. The effect did not seem to be specific for PP; similar effects were observed for other polymer systems, such as high molecular weight polyethylene.

The viscosity of polymer melts increases strongly with increasing molar mass, doubling the molecular weight increases the viscosity by approx. a factor 10. The properties such as strength, and notably toughness, also increase with increasing molecular weight. Consequently, processing polymers into useful products is often a compromise between on the one hand the ease of processing (low molar mass) and the desired product properties (high molecular weight). The findings in the thesis could provide the basis for a breakthrough technology in the area of polymer processing, viz. more easy processing of higher molecular weight polymers and, maybe, processing of hitherto intractable high molecular weight polymers by using smart additives.

The scientific basis for the observations is related in the thesis to selective physisorption of the high molecular weight tail in the usually broad spectrum of molecular weights, characteristic of synthetic polymers. The concept of “selective physisorption of long polymer chains” proves that the mesoporous nanoparticles can improve the processibility by decreasing the viscosity of nanocomposite melts. The concept of “selective physisorption of long polymer chains” can be extended to various other nanocomposites by changing either the fillers or polymer matrix.

Samenvatting

In de laatste jaren is een grote vooruitgang geboekt in de begripsvorming hoe de eigenschappen van polymeren gerelateerd kunnen worden aan de onderliggende chemische structuur, d.w.z. monomere eenheid, molecuulgewichtsverdeling en de chemische samenstellingsverdeling, en hoe dit het zelforganisatieproces van polymere ketens op verschillende lengteschalen beïnvloedt, in het bijzonder onder afwezigheid van stroming en in de vaste fase. De product eigenschappen zijn echter niet alleen de resultante van de chemische structuur, maar zeker ook van het verwerkingsproces, hetgeen met name in de academische wereld vaak onderbelicht of zelfs verwaarloosd wordt. De meeste synthetische polymeren, de klasse van de thermoplasten, worden verwerkt via de gesmolten toestand (polymere smelt). In het algemeen nemen de eigenschappen zoals taaiheid en sterkte toe met toenemende molecuulmassa maar ook de viscositeit van het gesmolten polymeer neemt zeer sterk toe met toenemende molecuulmassa. De (nul-) viscositeit van gesmolten polymeren schaaft boven een kritische waarde met het (gewichtsgemiddelde) molecuul gewicht tot de macht 3.4; dit betekent dat een verdubbeling van de molecuulmassa leidt tot een verhoging van de smelt-viscositeit met meer dan een decade! In de praktijk betekent dit dat men een compromis zoekt tussen verwerkbaarheid (lage viscositeit) en eigenschappen.

Behalve via de chemische structuur en verwerkingsomstandigheden, bijv. via het al of niet bewust induceren of juist vermijden van moleculaire oriëntatie, kan men de eigenschappen ook sturen via additieven. In feite is een plastic (product) meestal een combinatie van polymeer plus additieven, met name in constructieve toepassingen. In het huidige “nano-technologie” tijdperk is er veel aandacht voor het gebruik van additieven met nanometer dimensies. In deze context is een nieuwe route ontwikkeld voor het maken van nanocomposieten via een combinatie van vaste-stof modificatie in poreuze polypropreen (PP) reactorpoeders en sol-gel reacties van silica precursors. De morfologie van deze reactorpoeders, verkregen via de zogenaamde reactor korrel technologie, bestaat uit korrelstructuren met poreuze kanaaltjes. De reacties worden uitgevoerd in deze kanaaltjes. Twee typen nanocomposieten zijn gemaakt: met en zonder chemische interactie tussen de PP-matrix en de silica-deeltjes. Voor het eerste type werd PP eerst geënt met vinyl triethoxy silaan (VTES) gevolgd door de bereiding van silica nano-deeltjes in de poriën van PP via een sol-gel reactie van tetraethoxy orthosilicaat (TEOS). Voor het tweede type werd alleen de laatste stap gebruikt. Chemische en structurele karakterisatie laten zien dat micro- en mesoporeuze silica nano-deeltjes in-situ werden gevormd met hydroxyl groepen op het oppervlak. Voor de geënte monsters werd VTES niet alleen aan het oppervlak maar ook in het binnenste gedeelte van de PP poederdeeltjes met een uniforme verdeling gevonden. De geënte VTES-groepen bleken geïntegreerd te zijn in de in-situ gesynthetiseerde silica deeltjes. De dispersiegraad van de silica deeltjes in de geënte en niet-geënte monsters was uniform.

De op deze manier geprepareerde silica nano-deeltjes hadden een grote invloed op de reologie, de kiemvormingsstap tijdens kristallisatie en de slagvastheidseigenschappen van polypropreen: een *betere vloeï* samen met een *homogene verdeling van de eigenschappen* over het product. De

reologie van deze speciaal geprepareerde nanocomposieten is bestudeerd. De micro- en mesoporeuze aard van de in-situ gemaakte silica en de hoge oppervlakte-energie leidt tot een voorkeursadsorptie van polymeerketens met een hoge molecuulmassa. Dit resulteert in een segregatie van de ketens met een lage molecuulmassa, die de matrix vormen, en ketens met een hoge molecuulmassa die een aparte laag vormen om de silica deeltjes, resulterend in een kern-schil morfologie. Deze selectieve fysisorptie van de lange PP ketens op de silica deeltjes leidt tot een daling van de viscositeit in het lage silica concentratie gebied, waarin de afstand tussen de deeltjes groter is dan de gyratiestraal van de polymeerketens.

De mechanische eigenschappen van semi-kristallijne polymeren worden bepaald door de lameldikte, kristalliniteit en de verstrengelingsdichtheid van de amorf fase. Vanwege die reden is het kristallisatieproces van groot belang. De niet-isotherme kristallisatiekinetiek van de PP/silica nanocomposieten was bestudeerd met behulp van een gecombineerde Avrami-Ozawa benadering en liet een kristallisatieproces zien bestaande uit twee stadia: het primaire stadium gekarakteriseerd door kiemvorming en groei en een secundair stadium gekarakteriseerd door kristal perfectionering. De kiemvormingseffectiviteit van de in-situ gebaseerd op de schaal zoals voorgesteld door Lotz en medewerkers gemaakte silica deeltjes bedroeg 20% in het lage silica concentratie gebied. Dit is hoger dan voor andere nano-vulmaterialen gerapporteerd in de literatuur.

De vaste-stofmodificatie van PP met VTES resulteerde in een niet-willekeurige verdeling van de geënte VTES-groepen, terwijl enting via oplossing wel een willekeurige verdeling liet zien. Dit werd geïllustreerd door een toename van de elasticiteit door de vorming van clusters van geënte VTES-groepen, die als vernettingspunten fungeren. Twee kristalpopulaties met verschillende lamel diktes worden waargenomen voor de niet-willekeurige enting met VTES leidend tot segregatie van PP ketens met een verschillende chemische samenstelling, d.w.z. plaatsing van de geënte VTES-groepen over de PP ketens. De segregatie en clustering leidt tot een verlaging van de mobiliteit en daarmee samenhangend diffusie van de PP ketens. Dit beïnvloedt met name het secundaire kristallisatieproces, wat inhoudt dat de amorf fase sterker verstrengeld is. Reologische metingen lieten zien dat wanneer de silica deeltjes verbonden zijn door meerdere PP ketens via de geënte VTES groepen, netwerkvorming optreedt.

Het stromingsgeïnduceerde kristallisatiegedrag werd bestudeerd met behulp van in-situ kleine hoek Röntgenverstrooiing (SAXS) en grote hoek Röntgendiffractie (WAXD). Het is algemeen bekend dat stromingsgeïnduceerde kristallisatie bepaald wordt door het gedeelte met een hoge molecuulmassa, maar de verlaging van de viscositeit suggereert een matrix bestaande uit ketens met een lage molecuulmassa. De afwezigheid van de hoge molecuulmassa ketens in de matrix in combinatie met de lagere afschuifsnellheden als gevolg van de lagere viscositeit resulteert in minder oriëntatie onder stroming. Daardoor hebben deze nanocomposieten een meer uniforme en isotrope structuur en morfologie in het uiteindelijke product. De Röntgendiffractie studie liet zien dat de oriëntatie minimaal is bij een silica concentratie van 0.5 gewichtsprocent, wanneer de verlaging van de viscositeit maximaal is. PP met een hogere silica concentratie (boven 1 gew%) lieten een sterkere stromingsgeïnduceerde oriëntatie zien. Transmissie Elektronen Microscopie en FTIR-dichroïsme metingen aan producten gemaakt via spuitgieten lieten echter zien dat een uniforme en isotrope structuur en willekeurige groeirichting optreedt van de kristallen. Om dit gedrag te kunnen verklaren, is een driestaps kristallisatieproces voorgesteld: 1) vorming van rijvormige kiemen als

gevolg van moleculaire, spanningsgeïnduceerde oriëntatie; 2) relaxatie van rijvormige kiemen of destructie van georiënteerde structuren in een aantal primaire puntkiemen; 3) groei vanuit deze niet georiënteerde puntkiemen in willekeurige richting.

De mechanische eigenschappen van de PP nanocomposieten en het effect van de chemische interactie tussen vulmateriaal en de polymeermatrix was bestudeerd als functie van de silica concentratie. De voordelen van verbeterde vloeispanning en een homogene (isotrope) structuurontwikkeling tijdens verwerking op het versterkende effect van de nano-deeltjes zijn bestudeerd. De verbetering van de mechanische eigenschappen, vloeispanning, rekverstevigingsmodulus, Young's modulus en de hoge snelheid slagvastheid zijn bediscussieerd als functie van de silica concentratie. De PP/silica nanocomposieten lieten drie verschillende silica concentratie regimes zien, gebaseerd op de morfologie gevormd door de fysisorptie van de hoog molecuulmassa ketens op de silica deeltjes. In het laag concentratie regime (beneden 0.8 gew%), waarin de deeltjes geen interactie vertonen, nemen de vloeispanning en rekverstevigingsmodulus in beperkte mate af, terwijl de toename van de Young's modulus marginaal is. In het tussenliggende concentratieregime (tussen 0.8 en 2 gew%) beginnen de deeltjes interactie te vertonen en viscositeit neemt toe. De vloeispanning en rekverstevigingsmodulus samen met de Young's modulus nemen sterk toe. Bij nog hogere concentraties treedt aggregatie van de deeltjes op wat resulteert in een gestaag toenemende vloeispanning, rekverstevigingsmodulus en Young's modulus en de materialen laten een brosse breuk zien. Het effect van chemische interactie tussen vulmateriaal en matrix bleek nadelig voor de eigenschappen.

De technologische consequenties van de nieuwe systemen kunnen enorm zijn, maar kunnen nu nog niet voorspeld worden. De verlaging van de viscositeit is in het bijzonder voordelig voor spuitgietoperaties: lagere injectiedrukken en cyclustijden zijn benodigd of, anders gezien, de verwerkingstemperatuur kan worden verlaagd met een betere dimensiestabiliteit. De verwachting is de ontwikkelde materialen een voorname rol kunnen vervullen in precisiespuitgieten van dunwandige producten met hoge eisen op het gebied van dimensiestabiliteit. Vanuit wetenschappelijk oogpunt gezien is het effect van de verlaging van de viscositeit met nano-deeltjes, naar onze mening, het eerste echte "nano-effect" waargenomen op het gebied van synthetische polymeren.

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List of publications

Patents

1. Jain S.H.; Goossens J.G.P.; Peters G.W.M. "Mesoporous nanoparticles as viscosity modifiers for polyolefins". (Appl. EP05101388.6).

Journal publications

1. Jain, Sachin; Goossens, Han; van Duin, Martin; Peters, Gerrit; Lemstra, Piet. "Nanoscale events with macroscopic consequences: Rheology of PP/silica nanocomposites" to be submitted to *Nature Materials* (2005).
2. Jain, S.; Goossens, J.G.P.; van Duin, M.; Peters, G.W.M.; Denka, H.; van Asselen, O.L.J. "Flow-induced crystallization of PP/silica nanocomposites" to be submitted to *Macromolecules* (2005).
3. Jain, S.; Goossens, J.G.P.; van Duin, M.; Govaert, L.; Lemstra, Piet. "Consequences of nano-scale effects: Mechanical properties of PP/silica nanocomposites" to be submitted to *Macromolecules* (2005).
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6. Jain, S.; Goossens, J.G.P.; van Duin, M.; Lemstra, P.J. "Effect of in-situ prepared silica nanoparticles on non-isothermal crystallization of Polypropylene". Accepted in *Polymer*.
7. Garcia, M.; van Vliet, G.; Jain, S.; Schrauwen, B.A.G.; Sarkissov, A.; van Zyl, W.E.; Boukamp, B. "Polypropylene/silica nanocomposites with improved mechanical properties". *Review Advanced Materials and Science* (2004), 6, 169-175.
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1. Jain, S.; Diepens, M.; Goossens, J.G.P.; Lemstra, P.J. "Hollow silica nanofibers and their polymer based Nanocomposite". *Conference proceedings of NANOCOMPOSITES 2004, San-Francisco, CA, USA, September 1-3, 2004*.
2. Jain, S.; Goossens, J.G.P.; van Duin, M.; Lemstra, P.J. "Sol-gel derived hybrid nano-composites via solid-state modification in porous polymers". *Conference proceedings of NANOCOMPOSITES 2004, San-Francisco, CA, USA, September 1-3, 2004*.
3. Jain, S.; Goossens, J.G.P.; van Duin, M.; Lemstra, P.J. "Sol-gel derived hybrid nanocomposites in porous semi-crystalline polymers". *Abstracts of Papers, 228th ACS National Meeting, Philadelphia, PA, United States, August 22-26, 2004*.
4. Jain, S.; Goossens, J.G.P.; van Duin, M.; Lemstra, P.J. "Shear-induced crystallization, rheology and mechanical behavior of PP/silica hybrid nanocomposites". *Abstracts of Papers, 228th ACS National Meeting, Philadelphia, PA, United States, August 22-26, 2004*.
5. Jain, S.; Goossens, J.G.P.; van Duin, M.; Picchioni, F. "Hybrid Nanocomposites via solid-state modification of porous polypropylene." *MODEST 2004 Lyon-Villeurbanne, France August 29 - September 02, 2004*.

6. Jain, S.; Goossens, J.G.P; van Duin, M.; Picchioni, F. "Hybrid Nanocomposites via Solid-State Modification in Porous PP" *MACRO 2004, Paris, France, July 2-6, 2004*.
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1. Jain, S.; Murata, K.; Anazawa, T. "Manufacture of transparent resin composites with high tensile strength". (JP 2002200623)
2. Jain, S.; Murata, K.; Anazawa, T. "Manufacture of thermosetting resin/chain polymer composites having enhanced toughenss". (JP 2002241618)
3. Jain, S.; Murata, K.; Anazawa, T. "Manufacture of thermosetting polymer composites with good mechanical properties". (JP 2002265615)
4. Jain, Sachin; Garcia, Monserrat; Winnubst, Louis; Schipper, Dik; de Groot, Klaas; Blank, Dave H.A. "Friction and wear studies on UHMWPE/silica nanocomposites". To be submitted to *Nature Materials* (2005).
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8. Jana, S.; Jain, S. "Dispersion of nanofillers in high performance polymers using reactive solvents as processing aids", *Polymer* (2001) 42, 6897.
9. Jana, S.C; Jain, S. "Dispersion of nanoparticles in high performance polymers using thermosetting resins." *SPE-ANTECH proceedings*. (2001), 59th (Vol. 2), 2180-2184.

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1. Jain, S.H.; Murata, K.; Anazawa, T. "Improvement in Mechanical Properties by Photopolymerization Induced, Phase Separation and Controlled Morphology, in Polymer Blend of Photocurable/Linear Polymers" *proceedings of PPS-18, Guimaraes, Portugal, June 16-20, 2002*.
2. Jain, S. "Photopolymerization induced controlled phase separation and morphology in binary blends of photocurable/linear thermoplastic Polymers" *proceedings of 7th European Symposium on Polymer Blends, Lyon-Villeurbanne, France, May 27-29, 2002*.
3. Jana, S.; Jain, S. "Improvements in mechanical and thermal properties of PES forming nanocomposites using Fumed silica as filler and epoxy as reactive solvent" *proceedings of PPS-16, Sanghai, China, June 18-23 2000*.
4. Jain, S. "Use of epoxy as processing aid and dispersing agent of nanoparticles in composites of high performance thermoplastic polymers" *GE Plastics, Mt. Vernon, IN, USA, May 12, 2000*.

Curriculum vitae

Sachin Hukamichand Jain was born on April 6th 1976 at Paldhi (Dist. Jalgaon), India. He did his primary school education at Paldhi. After finishing higher secondary school from CBSE (Central Board of Secondary Education) school at Bhusawal and Jalgaon in 1994, he studied Polymer Engineering at the Maharashtra Institute of Technology, University of Pune, Pune India. In the final year, he worked at National Chemical Laboratory, Pune, India on the project “Structure development in Nylon-PP blends with special aspect on in-situ fiber formation during processing”

Later, he went to The University of Akron, Akron, OH, USA to study Master of Science in Polymer Engineering. He received his Masters degree in June 2000 after completion of his Masters thesis entitled, “Nanocomposites of high performance engineering thermoplastics using epoxy as processing aid” under supervision of Dr. Sadhan Jana.

In July 2000, he went to Japan to work at Kawamura Institute of Chemical Research, Dai-nippon Ink and Chemical Inc., Sakura, Chiba prefecture. He spent six months working on polymerization-induced phase separation and in-situ prepared nanocomposites in photocurable polymers.

From July 2001, he started his PhD-study described in this thesis at the Eindhoven University of Technology, Eindhoven, The Netherlands, in the group “Polymer Technology (SKT)”, under supervision of Prof. P.J. Lemstra. During his PhD-study, he completed all six modules of the course ‘Registered Polymer Technologist’ (RPK, Register Polymeerkundige), organized by the ‘National Dutch Graduate School of Polymer Science and Technology’ (PTN, ‘Polymeertechnologie Nederland’).

